

The Coal Mines Regulations, 2011

Source: <https://indiankanoon.org/doc/77873508/>

[Skip to main content](#)

[Legal Document View](#)

[Tools for analyzing structure and cite text of judgments](#)

[Unlock Advanced Research with](#)

[PRISM](#)

[AI](#)

[Integrated with over 4 crore judgments and laws ? designed for legal practitioners, researchers, students and institutions](#)

-

[Know your Kanoon](#)

-

[Doc Gen Hub](#)

-

[Counter Argument](#)

-

[Case Predict AI](#)

-

[Talk with IK Doc](#)

-

...

[Upgrade to Premium](#)

[Cites

0

, Cited by

3

]

Union of India - Act

The Coal Mines Regulations, 2011

UNION OF INDIA

India

The Coal Mines Regulations, 2011

Rule THE-COAL-MINES-REGULATIONS-2011 of 2011

Published on 20 October 2011

Commenced on 20 October 2011

[This is the version of this document from 20 October 2011.]

[Note: The original publication document is not available and this content could not be verified.]

The Coal Mines Regulations, 2011

Published vide Notification No. G.S.R. 773(E), dated 20th October, 2011

Ministry of Labour and Employment

G.S.R. 773(E). - The Following Draft Regulations Namely the Coal Mines Regulations, 2010, which the Central Government Proposed to make, in Exercise of the powers conferred by section 57 of the Mines Act, 1952 (35 of 1952

) and on the recommendations of the Committee constituted under section 12 of the said Act, and in supersession of the Coal Mines, Regulations, 1957, are hereby published as required by sub-section 1 of section 59 of the said Act for the information of all persons likely to be affected thereby and notice is hereby given that the said draft regulations will be taken into consideration after the expiry of a period of three months from the date on which copies of the Gazette in which this notification is published, are made available to the public;

1.

The objections or suggestions, if any, may be addressed to Shri Santosh Kumar Singh, Under Secretary, Ministry of, Labour and Employment, Shram Shakti Bhawan, Rafi Marg, New Delhi - 110001.

2.

Any objection or suggestion, which may be received from any person in respect of the said draft regulations within the period specified above, will be considered by the Central Government.

Chapter-I Preliminary

1. Short title, extent and application.

(1)

These regulations may be called the Coal Mines Regulations, 2011.

(2)

They shall extend to the whole of India.

(3)

They shall apply to every coal mine.

2. Definitions.

- (A) In these regulations, unless the context otherwise requires,-

(1)

"Abandoned Mine Methane (AMM)" means such working as have been abandoned with no intention of working in future;

(2)

"Abandoned Mine Methane (AMM)" means a natural gas recovered from abandoned mines or part thereof;

(3)

"Act" means the Mines Act, 1952;

(4)

"approved safety lamp" and "approved electric torch" mean respectively, a safety lamp or an electric torch, manufactured by such firm and of such type as the Chief Inspector may from time to time, specify by notification in the Official Gazette;

(5)

"assistant-manager" means a person possessing a Manager's Certificate appointed in writing by the owner, agent or manager to assist the manager in the control, management and direction of the mine or part thereof, and who takes rank immediately below the manager;

(6)

"auxiliary fan" means a forcing fan or an exhausting fan used below ground wholly or mainly for ventilating one or more faces forming part of a ventilating district;

(7)

"average output" of any mine, means the average output per month during the preceding financial year of the total output from all excavations within the specified mine boundaries;

(8)

"banksman" means a person appointed to superintend the lowering and raising of persons, tools and materials and to transmit signals at the top of a shaft or incline;

(9)

"booster fan" means a mechanical ventilator used below ground for boosting the whole current of air passing along the intake or return airway of a mine or ventilating district;

(10)

"coal" includes anthracite, bituminous coal lignite, peat and any other form of carbonaceous matter sold or marketed as coal:

(11)

"Coal Bed Methane (CBM)" means a natural gas tapped in a coal seam or adjacent strata;

(12)

"Coal Mine Methane (CMM)" means a natural gas recovered from a coal mine or part thereof;

(13)

"Committee" means a Committee appointed under section 12 of the Act;

(14)

"competent person" in relation to any work or any machinery, plant or equipment means a person who attained the the age of 20 years and who has been duly appointed in writing by manager as a person competent to or perform that work, or to supervise the operation of that machinery, plant or equipment, and Who is responsible for the duties assigned to him, and includes a shot firer;

(15)

"contractor" means an individual, an association of individuals, company, firm, local authority or body who contracts for the services or operations in a mine, and includes a sub-contractor;

(16)

"deep-hole drilling and blasting" means drill holes made more than three metres in depth and used for blasting in an opencast mining operation;

(17)

"designer" means an individual, association of individuals, company or institution who designs a coal mining system, method of coal mining or machinery, plant, equipment, appliance or substances for use in coal mines;

(18)

"discontinued working" means such working in a mine as have been discontinued for some reasons and are inaccessible or rendered inaccessible but are likely to be worked again;

(19)

"District Magistrate" in relation to any mine, means the District Magistrate or the Deputy Commissioner, as the case may be, who is vested with the executive powers of maintaining law and order in the revenue district in which the mine is situated:

Provided that in the case of a mine, which is situated partly in one district and partly in another, the District Magistrate for the Purposes of these regulations shall be the District Magistrate authorised in this behalf by the Central Government;

(20)

"disused working" means such working in a mine where work has been temporarily stopped due to some reasons but which are accessible and include unused working.

(21)

"electrical supervisor" (to copy from IER) shall have the same meaning as defined under the Indian Electricity Rules, 1956.

(22)

"explosive" shall have the same meaning as is assigned to that term in the Indian Explosives Act, 1884 (4 of 1884);

(23)

"face" means the moving front of any working place or the in by end of any gallery, roadway or drift;

(24)

"fiery seam" means a seam in which a fire or spontaneous heating exists in the working below ground or in open cast working lying within the precincts of a mine;

(25)

"financial year" means period of twelve months from the first day of April to the last day of March;

(26)

"flame proof enclosure" means an enclosure in which the parts which can ignite an explosive atmosphere are placed and which can withstand the pressure developed during an internal explosion of an explosive mixture, and which prevents the transmission of the explosion to the explosive atmosphere surrounding the enclosure;

(27)

"Form" means a Form as set out in the First Schedule;

(28)

"gas" includes fume or vapour;

(29)

"gassy seam of the first degree" means coal seam or part thereof lying within the precincts of a mine not being an open cast working whether or not inflammable gas is actually detected in the general body of the air at any place in its working below ground, or when the percentage of the inflammable gas if and when detected, in such general body of air does not exceed 0.1 and the rate of emission of such gas does not exceed one cubic metre per tonne of coal produced;

(30)

"gassy seam of the second degree" means coal seam or part thereof lying within the precincts of a mine not being an open cast working in which the percentage of inflammable gas in the general body of air at any place in the working of the seam is more than 0.1 or the rate of emission of inflammable gas per tonne of coal produced exceeds one cubic metre but does not exceed ten cubic metres;

(31)

"gassy seam of the third degree" means coal seam or part thereof lying within the precincts of a mine not being an open cast working in which the rate of emission of inflammable gas per tonne of coal produced exceeds ten cubic metres;

(32)

"general body of air" means the general atmosphere in a seam and includes the atmosphere in the roof cavities, but does not include general atmosphere in the sealed off area or in any borehole drilled in coal or in the adjacent strata;

(33)

"goaf" means any part of working below ground wherefrom a pillar or part thereof, or in The case of longwall working, coal has been extracted but which is not a working place;

(34)

"haul road" means any passage of road, which is maintained and used in connection with the working of opencast mines for plying of machinery;

(35)

"Heavy Earth Moving Machinery (HEMM)" means automotive or electrical machinery used in mechanised opencast mines for digging, drilling (excluding hand held drills and drill machines capable of drilling hole of a diameter up to 50 mm), dredging, hydraulicking, ripping, dozing, grading, excavating, loading, transporting minerals or overburden;

(36)

"incline" means an inclined passage or road either on the surface or below ground;

(37)

"inset" means a landing or platform in a shaft, and includes an excavation therefrom between the top and the bottom of the shaft;

(38)

"machinery" means -

(i)

any locomotive or any stationary or portable engine, air compressor, boiler or steam apparatus or any such apparatus, apparatus or combination of appliances which is intended for developing, storing, transmitting, converting or utilizing energy; or

(ii)

any such apparatus, appliance or combination of appliances if any power developed, stored, transmitted, converted or utilised thereby is, used or intended for use in connection with mining operations; or

(iii)

any such equipment used for cutting, drilling loading and transport of material;

(39)

"manager" means a manager appointed under regulation 30;

(40)

"manufacturer" means an individual, association of individuals, company or institution who manufactures machinery, plant, equipment, appliance or substances for use in coal mines;

(41)

"material" includes coal, stone, debris, or any other material;

(42)

"misfire" means the failure to explode of an entire charge of explosives in a shot hole or blast hole;

(43)

"month" means a calendar month;

(44)

"official" means a person appointed in writing by the owner, agent or manager to perform duties or supervision in a mine or part thereof and includes an assistant manager, a ventilation officer, a safety officer, a sampling incharge, dust

incharge, an overman, a sirdar, an engineer and a surveyor;

(45)

"onsetter" means a person appointed to superintend the raising and lowering of persons, tools and materials and to transmit signals at any inset or shaft bottom;

(46)

"overman" means a person possessing a Manager's or Overman's Certificate, appointed by the manager in writing, under any designation whatsoever, to perform the duties of supervision of control in a mine or part thereof, and is as such superior to a sirdar;

(47)

"permitted explosive" means an explosive manufactured by such firm and of such type as the Chief Inspector may, from time to time specify by notification in the Official Gazette;

(48)

"pipe line" means Coal Bed Methane pipe lines;

(49)

"public road" means a road maintained for public use and under the jurisdiction of any government or local authority;

(50)

"quarter" means a period of three months ending on the 31st March, 30th June, 30th September or 31st December;

(51)

"railway" means a railway as defined in the Indian Railways Act, 1890;

(52)

"Regional Inspector" means the Inspector of Mines having jurisdiction over a geographical area in which the mine is situated and over which he exercises his powers under the Act;

(53)

"river" means any stream or current of Water, whether seasonal or perennial, and includes its banks extending up to the highest known flood level;

(54)

"risk" means combination of the likelihood of a specific unwanted event and the potential consequences, if it should occur;

(55)

"roadway" means any part of a passage or gallery below ground which is maintained in Connection with the working of a mine;

(56)

"Schedule" means a Schedule appended to these regulations;

(57)

"shaft" means a way or opening leading from the surface to working below ground or from one part of the working below ground to another, in which a cage or other means of conveyance can travel freely suspended, with or without the use of guides;

(58)

"shotfirer" means a person appointed under regulation 194;

(59)

"socket" means a shot hole or blast hole or part thereof remaining after being charged with explosive and blasted, and which is not known to be a misfired hole;

(60)

"supplier" means an individual, association (of individuals, company or institution who supplies a technology, machinery, plant, equipment, appliance or substances for use in coal mines;

(61)

"tub" includes a wagon, car, truck or other vehicle moving on rails for conveying material;

(62)

"ventilation district" means such part of a mine below ground as has an independent intake airway commencing from a main intake airway, and an independent return airway terminating at a main return airway, and in the case of a mine or part thereof which is ventilated by natural means, the whole mine or part;

(63)

"working" means any excavation made or being made in a mine for search of or obtaining coal;

(64)

"working place" means any place in a mine to which any person has lawful access.

(B)

Words and expressions used in these regulations and not defined herein but defined in the Act shall have the meanings respectively assigned to them under the Act.

Chapter-II Returns, Notices and Records

3. Notice of opening.

(1)

The notice for commencement of mining operations under section 16 of the Act shall be submitted in Form I accompanied by a plan showing the boundaries of the mine and the shafts or openings of the mine, tri-junction or revenue pillars and other prominent and permanent surface features to the Chief Inspector and a copy thereof to the Regional Inspector:

Provided that in case of change in the boundary of the mine under regulations 124 and 125, a plan showing the new boundary shall be submitted within seven days of the said change.

(2)

The notice referred to in sub-regulation (1) shall be accompanied by-

(a)

a copy of the surface plan prepared under sub-regulation (1) of regulation 68;

(b)

a copy of Safety Management Plan prepared under regulation 107;

(c)

a copy of Progressive Mine Closure Plan showing steps-proposed for reclamation while mining activities are continuing, at five yearly interval; and

(d)

a copy of Mine Closure Plan showing detailed steps proposed for final reclamation of the mining affected area:

Provided that in respect of a mine which has already been opened, the plan referred to in clauses (a) to (d) shall be submitted within one year of coming into force of these regulations.

(3)

The Mine Closure Plan referred to in clause (4) of sub-regulation (2) shall be in such form as may be specified by the Chief Inspector of Mines by a general or special order in writing and the plan duly prepared shall be submitted to the Chief Inspector of Mines and a copy thereof to the Regional Inspector:

Provided that, in respect of a mine, which has already been opened, such a plan shall be submitted within six months of coming into force of these regulations.

4. Quarterly returns.

- On or before the 20th day of January, April, July, and October every year, the owner, agent or manager shall submit to the Chief Inspector and the Regional Inspector correct returns in respect of the preceding quarter in Form II.

5. Annual returns.

(1)

On or before 20th day of February in every year, the owner, agent or manager shall submit to the District Magistrate and to the Chief Inspector annual returns in respect of the preceding year in Form III.

(2)

If a mine is abandoned or working thereof is discontinued over a period exceeding sixty days or if a change occurs in the ownership of a mine, the returns required under sub regulation (1) shall be submitted within thirty days of abandonment or change of ownership or within ninety days of discontinuance, as the case may be.

6. Notice of abandonment or discontinuance.

(1)

Where it is intended to abandon a mine or seam or to discontinue working thereof for a period exceeding 60 days, the owner, agent or manager shall give a notice to the Chief Inspector and the Regional Inspector stating the reasons for such abandonment or discontinuance and the number of persons likely to be affected thereby, not less than 40 days

before such abandonment or discontinuance:

Provided that where on account of unforeseen circumstances or without previous intention a mine is abandoned or discontinued beyond a period of 60 days before the said notice has been given, the notice shall be given forthwith in Form I to the Chief Inspector, the Regional Inspector and the District Magistrate.

(2)

Notwithstanding anything contained in sub-regulation (1), where it is intended to abandon or discontinue for more than 60 days any working below ground over which is situated any property vested in the Government or any local authority or any railway or any building or permanent structure not belonging to the owner, the owner, agent or manager thereof shall give notice to the Chief Inspector and the Regional Inspector of his intention of such abandonment or discontinuance, not less than 40 days before the date of abandonment or discontinuance.

7. Notice of reopening.

(1)

Where it is intended to reopen a mine or seam after abandonment or discontinuance for a period exceeding 60 days, the owner, agent or manager shall, not less than 30 days before resumption of mining operations, give to the Chief Inspector, the Regional Inspector and the District Magistrate notice in Form I.

(2)

Where a mine has been reopened, the owner, agent or manager shall forthwith communicate the actual date of reopening to the Chief Inspector and the Regional Inspector.

8. Change in ownership and appointment of agent, manager, etc.

(1)

(a)

Where a change occurs in the name of ownership of a mine or in the address of the owner, the owner, agent or manager thereof shall, within seven days from the date of the change, give to the Chief Inspector and the Regional Inspector a notice in Form I :

Provided that where the owner of a mine is a firm or other association of individuals, a change-

(i)

of any partner in the case of a firm;

(ii)

of any member in the case of an association;

(iii)

of any director in the case of a public company; or

(iv)

of any shareholder in the case of a private company,

shall also be intimated to the Chief Inspector and the Regional Inspector, within seven days from the date of the change.

(b)

Where the ownership of a mine is transferred, the previous owner or his agent shall make over to the new owner or his agent, within a period of seven days of the transfer of ownership, all plans, sections, reports, registers and other records maintained in pursuance of the Act and of these regulations, or orders made thereunder, and all correspondence relating to the working of the mine relevant thereto, and when the requirements, of this clause have been duly complied with, both the previous and the new owners or their respective agents shall forthwith inform the Chief Inspector and the Regional Inspector in writing.

(2)

Where any appointment is made of an agent, manager, engineer, surveyor, ventilation officer, safety officer, or assistant manager or when the employment of any such person is terminated or any such person leaves the said employment, or when any change occurs in the address of any agent or manager, the owner, agent or manager shall, within seven days from the date of such appointment, termination or change, give to the Chief Inspector and the Regional Inspector a notice in Form I.

(3)

The owner of a mine shall submit in writing to the Chief Inspector and the Regional Inspector, a statement showing names and designation of every person authorised to act on behalf of the owner in respect of management, control, supervision or direction of the mine.

(4)

The statement referred to in sub-regulation (3) shall state the responsibilities of every such person and the matters in respect of which he is authorised to act on behalf of the owner.

(5)

Every person referred to in sub-regulation (3) shall be deemed to be an agent for the mine or group of mines, as the case may be, in respect of the responsibilities as specified in the statement referred therein.

(6)

Any change, addition or alteration in the names - or other particulars of the statement referred to in sub-regulation (3) shall be reported in writing to the Chief Inspector and Regional Inspector within seven days from the date of such change, addition or alteration.

9. Notice of dangerous occurrence or accident.

(1)

(a)

Where there occurs a readily identifiable event such as a dangerous occurrence with potential to cause an injury or disease to persons at work or to the public around the workplace, or an accident in or about a mine such as -

(i)

an accident causing loss of life or serious bodily injury in connection with mining operations;

(ii)

an explosion or ignition;

(iii)

a spontaneous heating or outbreak of fire, or appearance of smoke or other indication of heating or outbreak of fire including cases in which there exists an underground working which is in close proximity to surface fire in coal heaps, OB dumps and coal ribs creating potential risk of transference of such hazard from surface to underground working but not cases of surface fires in coal heaps, OB dumps and coal ribs;

(iv)

fire in any part of working or in any machinery;

(v)

fall from height of any excavation, loading or transport machinery;

(vi)

bursting of equipment under pressure;

(vii)

an influx of inflammable or noxious gases;

(viii)

an eruption or inrush of water or other liquid matter;

(ix)

an instantaneous failure of a pillar, part of a pillar or several pillars of coal (i.e., a 'bump') in working below ground;

(x)

a premature collapse of any part of the working;

(xi)

any accident due to explosives;

(xii)

a breakage or fracture of rope, chain, headgear, pulley or axle or bearing thereof, or other gear by which persons or materials are lowered or raised in a shaft or an incline;

(xiii)

an over winding of cages or other means of conveyance while men or materials are being lowered or raised;

(xiv)

a breakage or fracture of any essential part of winding engine, crankshaft, coupling, bearing, gearing, clutch, drum or drum shaft, or failure of emergency brake;

(xv)

a bursting of any equipment containing steam, compressed air or other substance at high pressure; or

(xvi)

a breakage, fracture or failure of any essential part of any machine or apparatus whereby the safety of persons may be endangered;

(xvii)

any major slide causing injury to any person, damage to any machinery, and/or-interruption of normal mining operations;

(xviii)

failure of dump or side in opencast working,

the owner, agent or manager shall forthwith inform the Regional Inspector about the occurrence by telephone, express telegram, fax, e-mails or by special messenger; and shall also, within 24 hours of every such occurrence, give notice thereof in Form IV A to the District Magistrate, the Chief Inspector and the Regional Inspector and in the case of an accident mentioned in sub-clause (i), also to the Coal Mines Labour Welfare Commissioner, and shall simultaneously exhibit a copy of the notice on a special notice board outside the office of the mine for a period of not less than 14 days from the date of such exhibition:

Provided that in case such notice is sent by e-mails, it shall be immediately followed by fax or letter.

(b)

Where an accident causing loss of life or serious bodily injury occurs in or about a mine in connection with the generation, storage, transformation, transmission, supply or use of electrical energy, the owner, agent or manager shall forthwith inform the Electrical Inspector of Mines by telephone, express telegram or special messenger.

(2)

Where death results from any injury already reported as serious under sub regulation (1) or where any injury other than the serious injury become serious, the owner, agent or manager shall within 24 hours of his being informed of the same, give notice thereof to the District Magistrate, the Chief Inspector, the Regional Inspector and the Coal Mines Labour Welfare Commissioner.

(3)

In respect of every persons killed or injured, the owner, agent or manager shall, within seven days of the occurrence, send to the Chief Inspector, particulars in Form IV B.

10. Notice of disease.

- Where any person employed in a mine contract any disease notified by the Central Government in the Official Gazette under section 25 of the Act, the owner, agent or manager shall within three days of his being informed of the disease, give notice thereof in form V to the District Magistrate, the Coal Mines Labour Welfare Commissioner, the Chief Inspector, the Regional Inspector and Inspector of Mines (Medical).

Chapter-III Examination and Certificates of Competency and of Fitness

11. Board of Mining Examination.

(1)

For the purposes of these regulations, there shall be constituted a Board of Mining Examination, (hereinafter referred to as "the Board").

(2)

The Board shall consist of the Chief Inspector, who shall be its Chairman (ex-officio), and five members possessing degree in mining engineering; with

(a)

first class Manager's Certificate granted under regulation 12; or

(b)

practical experience in coal mines; or

(c)

serving in an institution imparting education in mining engineering at the degree or equivalent level; or

(d)

engaged in mining research or planning, to be appointed by the Central Government:

Provided that the Board shall be so constituted that it shall include at least three members possessing qualifications laid down in clause (a) and (b) and at least one member possessing qualifications laid down either in clause (c) or in clause (d).

(3)

Every member (other than the Chairman) of the Board shall hold office for a period of three years, or until his successor

is appointed, whichever is later:

Provided that -

(i)
a member may at any time resign his office by notice in writing addressed to the Chairman;

(ii)
a member appointed under clause (c) of sub-regulation (2) shall cease to hold office upon his ceasing to serve in any such institution, as is referred to in that clause;

(iii)
a person appointed to fill a vacancy caused by reason of the death, or by resignation, or otherwise, shall hold office for the remaining period for which such member would have, but for such reason, continued as member.

(4)
A person who holds, or who has held, office, as member of the Board shall, subject to the other provisions of this regulation, be eligible for re appointment to that office.

(5)
A member of the Board (other than the Chairman) Shall receive such remunerations as the Central Government may fix in this behalf.

(6)
An Inspector nominated in this behalf by the Chief Inspector shall act as Secretary to the Board (hereinafter referred to in these regulations as the Secretary).

(7)
Notwithstanding anything contained in this, regulation, the Central Government may, if satisfied that it is necessary so to do in the public interest, the constitute the Board even though the term of office of all or any of the members there of has not come to an end.

(8)
Meetings of the Board shall be held as and when the Chairman considers necessary, and unless otherwise decided by the Chairman, all meetings of the Board shall be held at Dhanbad.

(9)
(a)
For every meeting of the Board, not less than ten clear day, prior notice intimating the time and place of the proposed meeting and signed by the Chairman or the Secretary shall be given to each member who is not absent from India.

(b)
Such notice shall be delivered at, or posted to the usual place of residence of the member, and each such notice shall be accompanied by a list of items of business to be disposed of at that meeting.

(c)
Notwithstanding anything contained in clauses (a) and (b), in cases of urgency, an emergent meeting may be called for by the Chairman at any time by intimating the members only two days in advance, of the time and date of such meeting and the subject matter for discussion at such meeting.

(10)
(a)
The Chairman shall preside at every meeting of the Board.

(b)
If the Chairman is absent for any reason, the members present shall elect one from among themselves to preside over the meeting, and the members so elected shall, for the purposes of that meeting, have all powers of the Chairman.

(11)
No business shall be transacted at a meeting of the Board unless at least three members, including the Chairman, are present:

Provided that if at any meeting there is no quorum as aforesaid, the meeting shall automatically stand adjourned to a date which is seven days later or if that day is a public holiday to the next working day and the time, place and agenda for the adjourned meeting shall remain unchanged and shall thereupon be lawful to dispose of the business at such meeting irrespective of the number of members attending.

(12)

(a)

All matters which the Board is required to consider shall be considered at its meeting, or, if the Chairman so decides, by circulation of the papers, to every member who is not absent from India.

(b)

When any matter is referred to by circulation, a member may request that it should be considered at a meeting of the Board and the Chairman may direct that it shall be so Considered but when two or more members so request, the Chairman shall direct that the matter shall be considered at a meeting to be held.

(13)

(a)

The Secretary shall place before the Board a list of business to be transacted at the meeting.

(b)

No business which is not included in such list shall be considered unless the Chairman permits.

(14)

(a)

Every matter at a meeting shall be decided by the majority of votes of the members present at such meeting.

(b)

Every matter referred to the members by circulation under sub-regulation (12) shall be decided by the majority opinion of the members to whom the papers were circulated, unless the Chairman reserves it for consideration at a regular meeting to be held later.

(c)

In case of a tie, Chairman Shall have a casting vote.

(15)

(a)

The Secretary shall record the minutes of each meeting in a bound-paged book kept for the purpose and copies of such minutes of the meeting shall be circulated to all members present in India.

(b)

The minutes so recorded shall be confirmed at the next meeting of the Board and signed by the Chairman in token thereof.

(16)

(a)

The Chairman in addition to any other powers and duties conferred upon him under these regulations shall-

(i)

present all important papers and matters to the Board as early as possible;

(ii)

issue orders for carrying out the decisions of the Board;

(iii)

have power to refer, in his discretion, any matter to the Central Government for their orders; and

(iv)

have powers generally to take such action or pass such orders necessary to implement the decisions of the Board.

(b)

The Chairman may, during his temporary absence by an of leave or otherwise, authorise any member of the Board to perform all or any of the duties of the Chairman during such absence.

(c)

Unless the Chairman otherwise directs, all proceedings of the Board shall be conducted in camera and be regarded as confidential.

12. Certificate granted by the Board.

(1)

Certificate under these regulations shall be granted by the Board.

(2)

Certificate granted by the Board shall be valid throughout the territories to which these regulations extend, and shall be of the following kinds:-

(a)

Manager's first class certificate of competency to manage a coal mine (in these regulations referred to as a First Class Manager's Certificate);

(b)

Manager's second class certificate of Competency to manage a coal mine (in these regulations referred to as a Second Class Manager's Certificate);

(c)

Surveyor's certificate of competency to survey the working of a mine (in these regulations referred to as a Surveyor's Certificate);

(d)

Overman's certificate of competency to carry out inspections and duties as required under these regulations (in these regulations referred to as an Overman's Certificate);

(e)

Sirdar's Certificate of competency to carry out inspection and duties as required under these regulations (in these regulations referred to as a Sirdar's Certificate);

(f)

Winding engineman's certificate (in these regulations referred to as Engine Driver's Certificate) to drive a winding engine of any type or class; and

(g)

Certificate of competency to test for the presence of inflammable gas (in these regulations referred to as a Gas-testing Certificate):

Provided that any of the certificate aforesaid, other than Engine Driver's Certificate and Gas testing Certificate may be restricted to mines having opencast working only, and this fact shall be endorsed on the certificate.

13. Examinations and examiners.

(1)

Certificates shall be granted to candidates after such examination and in such form as the Board may specify:

Provided that the Board may, subject to the specified conditions, exempt any person from appearing at the examination or part thereof for the grant of a certificate referred to in regulation 12.

(2)

The examination shall be held at such time and at such centre as may be fixed by the Board, and shall be conducted by examiners appointed by the said Board.

(3)

The examiners so appointed shall be subject to the orders of the Board in respect of all matters relating to the conduct of the examinations, and shall receive such remuneration as the Board, with the sanction of the Central Government, may fix.

(4)

The Board may specify the procedure for the conduct of the examinations and to the granting of certificates of competency and of fitness as required under these regulations, and shall, so far as may be practicable, provide that the standard of knowledge required for the grant of certificate of any particular class and the standard of medical fitness shall be uniform throughout the territories to which these regulations extend.

14. Submission of applications.

(1)

Application for an examination shall be made to the Board not less than 60 days prior to the date fixed for the examination and on a form specified for the purpose.

(2)

Notice regarding the date and place of the examination for the Manager's, Surveyor's and Overman's certificate shall be published under the order of the Board, in such periodicals as the said Board may direct, not less than 60 days prior to the date fixed for receiving applications.

(3)

The certificate of experience etc., required for appearing in the examinations under these regulations shall be in the form specified by the Board.

15. Age and general qualification of candidate.

(1)

No persons shall be admitted as a candidate at any examination held by the Board unless he is 20 years of age.

(2)

No persons shall be admitted as a candidate at any examination for a Manager's, Surveyor's, Overman's or Sirdar's Certificate unless he holds a valid first-aid certificate of the standard of the St. John Ambulance Association (India).

(3)

Every application for any examination as aforesaid shall be accompanied by-

(i)

a certificate of age verified by a Gazetted Officer of the Government or by the headmaster of a recognised school of a Higher Secondary or equivalent standard:

Provided that in the case of a person holding a matriculation or equivalent certificate, such certificate shall be submitted as evidence of age;

(ii)

a medical certificate obtained not more than one year prior to the date of his application, from a qualified medical practitioner not below the rank of a Civil Assistant Surgeon or from a Certifying Surgeon, certifying the candidate to be free from deafness, defective vision or any other infirmity, mental or physical, likely to interfere with the efficiency of his work; and

(iii)

a certificate from some person of good repute as to the general good conduct and sobriety of the candidate.

(4)

No, person shall be admitted as a candidate to any examination for:-

(a)

manager's, surveyor's, overman's or sirdar's Certificate, unless he has passed the Senior Secondary School Examination or Intermediate Examination or its equivalent from a recognised Board or University or passed a Diploma or Degree in Engineering or other equivalent Qualifications approved in that behalf by the Central Government; and

(b)

An Engine Driver's Certificate unless he has passed the Secondary School Examination recognised by board or it's equivalent.

(5)

No person shall be admitted as a candidate at an examination for a Manager's or Overman's Certificate, which is not restricted to mines having opencast working only, unless he has obtained at least a Sirdar's Certificate (not so restricted) and a Gas testing Certificate, and no person shall be admitted, as a candidate at an examination for Manager's or Overman's Certificate restricted to mines having opencast working only unless he has obtained at least a Sirdar's Certificate:

Provided that this sub-regulation shall not apply -

(i)

to a candidate for a Manager's Certificate, if he already holds a Overman's Certificate granted under regulation 22; and

(ii)

to a candidate for a Manager's First Class Certificate if he already holds a Manager's Second Class Certificate granted under regulation 22.

16. Practical experience of candidate for Manager's Certificate examination.

(1)

No person shall, be admitted as a candidate at any examination for First or Second Class Manager's Certificate other than Exchange Certificate under regulation 22, unless the Board is satisfied that he has had practical experience in coal mine for a period of not less than, six and five years respectively:

Provided that in the case of a candidate -

(a)

who has received a diploma in mining engineering or other equivalent qualification approved in that behalf by the Central Government, such period of experience shall be five and four years respectively, and

(b)

who has passed a degree in mining engineering or other equivalent qualification approved in that behalf by the Central Government,-

(i)
such period of experience shall be two years for First Class Manager's Certificate; and

(ii)
shall be exempted from appearing at the Second Class Manager's examination on acquiring one year of practical experience and a Gas testing Certificate:

Explanation. - The experience referred to in this sub-regulation shall be the experience obtained after acquiring the relevant academic qualification.

(2)
The nature of the practical experience required for Manager's Certificate shall be in one of the following capacities in a coal mine having an average monthly output of not less than 1000 tonnes,-

(a)
as a workman, or mining apprentice having direct practical experience of getting coal and of drifting, shaft sinking development, depillaring, supporting and blasting; or

(b)
as an official in respect of mining operations:

Provided that out of the total period of experience required under sub-regulation (1), three fourth of the experience shall be of below ground coal mines, in respect of certificate not restricted to mines having opencast working only:

Provided further that the Board may approve a part of the period of the experience gained while engaged in inspections, rescue, research, planning, teaching or any other work, connected with mining operations, so however that, the aforesaid period inclusive of the period of experience in a mine other than a coal mine approved under sub-regulation (2) of regulation 18, shall not exceed one fourth of the total experience required for the purpose.

17. Practical experience of candidates for Surveyor's Certificate examination.

(1)
No person shall be admitted as candidate at any examination for a Surveyor's Certificate unless he has satisfied the Board that he has had not less than two years practical experience of surveying of a type specified by the Board:

Provided that in case of a certificate which is not restricted to mines having opencast working only, the experience referred to in this sub-regulation shall include practical experience for a period of not less than one year of surveying the working of a below ground mine having an average monthly output of not less than 1000 tonnes or such other mines as the Board may specify in this behalf.

(2)
In case of a candidate having diploma in mining or mine surveying or degree in civil engineering, the period of practical experience shall be one year and he shall be exempted from appearing at the examination.

(3)
In case of a candidate having degree in mining engineering or other equivalent qualification approved in that behalf by the Central Government, the period of practical experience shall be six months and he shall be exempted from appearing at the examination:

Provided that out of the total period of experience required under sub-regulations (2) and (3), at least three fourth of the experience shall be of below ground coal mines, in respect of certificate not restricted to mines having opencast working only.

18. Practical experience of candidate for Sirdar's Certificate Examination.

(1)
No person shall be admitted as a candidate at any examination for a sirdar's Certificate unless the Board is satisfied that he has had practical experience and training in a coal mine for a period of not less than three years:

Provided that in the case of a candidate who has received a diploma or certificate in scientific and mining training after a course of at least two years at an educational institution approved in that behalf by the Central Government or who has taken a degree in scientific and mining subject at a university approved in this behalf by the Central Government, the period of practical experience required shall be one year.

(2)
The nature of the practical experience required under this regulation shall be as specified by the Board and

notwithstanding anything contained in sub-regulation (1), the Board may approve a part of the period of practical experience which has been obtained in a mine other than a coal mine up to a period not exceeding one year.

19. Practical experience of candidate for Engine Driver's Certificate.

- No person shall be admitted as a candidate at any examination for an Engine Driver's Certificate unless the Board is satisfied that he has had practical experience of driving a winding engine or as an assistant to a qualified winding engine driver for a period of at least one year.

20. Number of attempts at examination.

- No person shall be admitted for examination for a particular certificate beyond seven attempts from the date of coming into force of these regulations.

21. Fees for grant of Certificates.

(1)

Fees on the following scale shall be paid in respect of every application for the grant of a certificate:

(a)

in the case of a First class Manager's Certificate.

1000 Rs.

(b)

in the case of a Second class Manager's Certificate.

750

(c)

in the case of a Surveyor's Certificate

500

(d)

in the case of an Overman's Certificate

500

(e)

in the case of a Sirdar's Certificate

300

(f)

in the case of a Winding Engine Driver's, Certificate

200

(g)

in the case of Gas-testing Certificate

200

(2)

The fee once paid shall not be refundable except where the candidate has died before the examination or where fee has been erroneously paid.

22. Exchange Certificate.

(1)

The Board may grant to any person, holding a Manager's, Surveyor's, Engine driver's, Foreman's, Mate's or Blaster's Certificate granted under any law for the regulation of mines in force in any other country or under the Metalliferous Mines Regulations made under the Mines Act, 1952, a corresponding certificate of a similar class under these regulations, if he possesses such qualification, experience and passes such examination as the Board may stipulate:

Provided that the Board may, subject to such conditions as it may specify, exempt any person from appearing at the examination or part thereof for the grant of an Exchange Certificate.

(2)

Every application for the grant of an Exchange Certificate under sub-regulation (1) shall be accompanied by -

(i)

a medical certificate obtained not more than six months prior to the date of his application from a qualified medical practitioner not below the rank of a Civil Assistant Surgeon or from a Certifying Surgeon certifying the candidate to be free from deafness, defective, on or any other infirmity, mental or physical, likely to interfere with the efficient discharge of his duties; and

(ii)

a certificate from some person of good repute as to the general good conduct and sobriety of the candidate:

Provided that in the case of a Manager's Certificate, the candidate shall possess practical training in India in the mines, for a period of not less than six months in such manner as may be specified by the Board or not less than six months practical experience of the nature specified in sub-regulation (2) of regulation 16, in a coal mine in India.

(3)

Fees on the scale laid down in regulation 21 shall be paid in respect of every examination under this regulation.

23. Duplicate Certificate.

- If any person proves to the satisfaction of the Board that he has, without any fault on his part, lost or been deprived of a certificate granted to him under these regulations, the said Committee may cause a copy of the certificate to be delivered to him subject to such terms and conditions as it thinks fit, and upon realization of the following fees:-

(a) in the case of Manager's or Surveyor's Certificate

Rs. 1000.00

(b) in the case of any other Certificate

Rs. 400.00

and the 'word "Duplicate" shall be stamped across every such copy.

24. Certificate to be delivered to the manager.

- The holder of an Overman's, Sirdar's, Engine Driver's and Gas testing Certificate shall require to deliver such certificate to the manager of the mine where he is for the time being employed in the capacity which requires the possession of the said certificate, and the manager shall deliver to such person a receipt for the same on retaining the certificate in the office at the mine so long as the holder thereof is so employed, and shall return it to the holder on his ceasing to be so employed.

25. Suspension or cancellation of Manager's, Surveyor's, Overman's, Sirdar's, Engine Driver's or Gas testing Certificate.

(1)

If on the basis of a report of the Inspector, the Regional Inspector is of the opinion that the holder of a Manager's, Surveyor's, Overman's, Sirdar's, Engine Driver's, or Gas testing Certificate is incompetent or is guilty of negligence or misconduct in the performance of his duties under the Act or under these regulations, he shall bring the matter to the notice of the Board.

(2)

The Board may, on the report of the Regional Inspector under sub-regulation (1), authorise an Inspector, not below the rank of an Inspector whose report formed the basis of the said opinion, to hold an enquiry to determine whether or not such a person (hereinafter referred to as the delinquent) is fit to continue to hold such certificate:

Provided that the Board shall, before the beginning of the enquiry, furnish to the delinquent a statement of the case on which the enquiry is instituted.

(3)

During the enquiry, the Inspector authorised to conduct the enquiry, shall be provided with all relevant documents and he shall record:

(a)

the evidence of any witness that formed the basis of the said opinion;

(b)

any evidence that the delinquent may like to give;

(c)

the evidence of any witness that the delinquent may like to produce;

(d)

the evidence of manager of the mine; and

(e)

any other evidence that may be considered necessary or relevant by the Inspector conducting the enquiry.

(4)

Unless the delinquent fails to be present in spite of sufficient notice, the evidence under sub-regulation (3) shall be recorded in the presence of the delinquent and he shall be given a reasonable opportunity to cross-examine the witness (other than those produced by him).

(5)
The Inspector conducting the enquiry also may cross-examine the delinquent and the witnesses.

(6)
The Inspector who conducted the enquiry Shall, within fifteen days from the date of conclusion of his enquiry, send a report to the Board together with his findings, the notes of evidence recorded during the enquiry and other relevant records.

(7)
Copies of the notes of evidence and the findings of the Inspector who conducted the enquiry shall also be sent to the delinquent who in any submit his written representation to the Board within thirty days from the date of despatch of such copies;

(8)
The Board may, after considering the evidence and other records and the written representation, if any, submitted by the delinquent, either cause further enquiry to be made in the case and thereupon, or otherwise, either exonerate the delinquent of the charges against him or suspend or cancel the certificate, as it deems fit.

(9)
Against any order of the Board under this regulation, an appeal shall lie before Central Government within 30 days of the order.

(10)
Where a certificate is suspended or cancelled under this regulation suitable endorsement may be made on such certificate or a duplicate thereof issued under regulation 23.

26. Validity of an Overman's, Sirdar's, Engine Driver's and Gas testing Certificate.

(1)
No overman's, Sirdar's, Engine Driver's and Gas testing certificate shall remain valid for a period of more than five years unless the certificate bears an endorsement by the Regional Inspector to the effect that the holder has within the preceding five years, been examined and certified by a qualified medical practitioner appointed by the Chief Inspector to be free from deafness, defective vision or any other infirmity, mental or physical, likely to interfere with the efficient discharge of his duties.

(2)
An application for endorsement on a certificate in respect of the medical examination undertaken under sub-regulation (1) shall be made to the Chief Inspector, accompanied by a fee of fifteen rupees,

(3)
A medical examination undertaken in accordance with rule 29B of the Mines Rules 1955 shall also be deemed to be a medical examination for the purposes of sub-regulation (1).

(4)
An application for endorsement on a certificate by the Regional inspector in respect of the medical examination undertaken under sub-regulation (3) shall be accompanied by the certificate of fitness granted under that sub-regulation along with a fee of five rupees.

27. Retirement age for managers and officials, etc.

(1)
No person shall act as a manager or an official or a winding engineman in a mine after attaining the age of 60 years unless he has obtained, within the preceding one year, a medical certificate of fitness certifying him fit to carry out the duties prescribed for him in the Act and in these regulations and orders made thereunder:

Provided that if the Chief Inspector or the Regional Inspector is of the opinion that a person as aforesaid, though less than 60 years of age, is medically unfit to carry on the duties prescribed for him in the Act and in these regulations and orders made thereunder, the Chief Inspector or the Regional Inspector may, by an order in writing, require such persons to obtain a medical certificate of fitness within such period, not exceeding three months, as he may specify therein; and no such persons shall continue to act in any capacity as aforesaid after the period so specified unless he has obtained a medical certificate of fitness.

(2)
The medical certificate of fitness as aforesaid shall be obtained from such authority and in such manner as the Board may specify.

(3)

Without prejudice to the provisions of sub-regulation (1), no person shall act as manager or an official after attaining the age of 65 years.

Chapter-IV Inspectors and Mine Officials

28. Qualification Inspection.

(1)

No person shall be appointed as Chief Inspector unless he holds a degree in mining engineering of an educational institution approved by the Central Government and also a First Class Manager's Certificate granted under regulation 12.

(2)

No person shall be appointed as an Inspector unless he holds a degree in mining engineering of an educational institution approved by the Central Government and also a First Class Manager's Certificate granted under regulation 12:

Provided that -

(i)

in relation to electrical machinery installed in mines, a person holding a degree in electrical engineering of an educational institution approved by the Central Government may be so appointed;

(ii)

in relation to other machinery or mechanical appliances installed in mines, a person holding a degree in mechanical engineering of an educational institution approved by the Central Government may be so appointed; and

(iii)

in relation to the provisions of the Act, rules and of the regulations which relate to matters concerning the health and welfare of persons, a person holding an MBBS degree and diploma in occupational health and hygiene, social science or labour welfare, as the case may be, of an educational institution approved by the Central Government or a person holding such other qualifications as the Central Government may approve in this behalf, may be so appointed.

29. Right of the workers representative to inspect the register maintained under the Mines Rules, 1955.

- The register maintained in Form "B" under the Mines Rules, 1955 shall be available for inspection to a worker's representative authorised by the persons employed in the mine on an application made by him in this behalf.

30. Qualification and appointment of manager.

(1)

No mine shall be opened, worked or re opened unless there is a manager of the mine, being a person duly appointed and having such qualifications under these regulations.

(2)

No person shall act or be employed as a manager unless he has attained 23 years of age and is paid by, and is directly answerable to the owner or agent of the mine.

(3)

Except as hereinafter provided in sub-regulation (6), and subject to the provisions of sub-regulation (4), no person shall act or continue to act, or be appointed, as manager of a mine or mines the average output of which corresponds to the figures given in column (i) of the table below unless he holds the corresponding qualifications given in column (ii) thereof:

(i)

(ii)

For below ground

mines:

(a) In

excess of 2,500 tonnes per month

(b) not exceeding 2,500 tonnes per

month Certificate not restricted to opencast mines only

A

First Class Manager's Certificate not restricted to opencast

mines only:

A First or Second Class Manager's Certificate not restricted to opencast mines only

For

opencast mines:

(c) In

excess 20,000 cum per months of material handled

(d) Not exceeding 20,000 cum material

handled

A First

Class Manager's Certificate.

A First or Second Class Manager's

Certificate:

Provided that in respect of a mine having both opencast and underground working, a person holding First Class Manager's Certificate not restricted to opencast mines shall only be appointed as the manager of the mine irrespective of production:

Provided further that where special conditions exist, the Chief Inspector may, by an order in writing, direct that in the case of any such mine as is referred to in clause (b) or (d) of the table, the manager thereof shall be the holder of a First Class Manager's Certificate.

(4)

Where under the provisions of sub-regulation (3) a person holding a First or Second Class Manager's Certificate has been appointed as manager, a person holding lower qualifications shall not, except with the previous permission in writing of the Chief Inspector and subject to such conditions as he may specify therein, be so appointed during the succeeding twelve months, notwithstanding any reduction in the average output of the mine.

(5)

No person shall act, or be appointed, as manager of more than one mine except with the previous permission in writing of the Regional Inspector and subject to such conditions as he may specify therein:

Provided that no such permission shall have effect for a period exceeding 12 months unless renewed:

Provided further that the Regional Inspector may at any time, by an order in writing, vary or revoke any such permission if the circumstances under which the permission was granted, have altered or the Regional Inspector finds that the manager has not been able to exercise effective supervision in the mines under his charge.

(6)

The Regional Inspector may, by an order in writing, and subject to such conditions as he may specify therein; authorise any person whom he considers competent, to be appointed or act as manager of any mine or mines for a specified period, notwithstanding that such person does not possess the qualifications specified under sub-regulation (3); and may by a like order revoke any such authority at any time.

(7)

Where by reason of absence or for any other reason the manager is unable to exercise daily personal supervision or is unable to perform his duties under the act or these regulations, or orders made thereunder the over agent or manager shall authorise in writing a person whom he considers competent, to act as manager of the time.

Provided that

(i)

such person holds a Manager's Certificate.

(ii)

no such authorisation shall have effect for a period in excess of 30 days except with the previous consent in writing of the Regional Inspector and subject to such conditions as he may specify therein.

(iii)

the owner, agent or manager as the case may be, shall forthwith send by registered post, Speed post or Fax to the Chief Inspector and the Regional Inspector a written notice intimating that such an authorisation has been made, and stating the reason for the authorisation the qualifications and experience of the person authorised, and the date of the commencement and ending of the authorisation;

(iv)

the Chief Inspector or the Regional Inspector may, except in the case of a person possessing the qualifications specified

in sub-regulation (3), by an order in writing, revoke any authority so granted.

(8)

The persons so authorised to act as manager under sub-regulation (7) shall, during the period of such authorisation, have the same responsibilities, discharge the same duties, and be subject to the same liabilities as the manager.

(9)

No manager shall vacate his office without giving due notice in writing to the owner or agent at least 30 days before the day on which he wishes to vacate his office.

Provided that the owner or agent may permit manager to his office after giving a shorter notice.

(10)

No owner or agent shall transfer, discharge or dismiss a manager unless the manager has been relieved by a duly qualified person.

(11)

Nothing in sub-regulation (7) shall confer on the owner, agent or manager the right to authorise any person not possessing the qualifications specified under sub-regulation (3) to act as the manager except in case of illness or other causes over which the manager has no control, or except with the previous written permission of the Regional Inspector and subject to such conditions as he may specify therein.

Provided that the Regional Inspector shall not permit any such authorisation for a period exceeding 60 days from the date on which the mine is worked if the person so appointed to act as a manager does not possess the qualifications specified under sub-regulation (2).

(12)

The owner or agent shall provide suitable residential accommodation for the manager and the assistant manager within a distance of five kilometers from all mine openings; and every manager, and assistant manager shall reside in the accommodation so provided.

Provided that where special difficulties exist, which render compliance with these provisions not reasonably practicable, the Regional Inspector may, by an order in writing and subject to such conditions as he may specify therein, grant exemption from the same.

(13)

No manager shall be entrusted by the owner or agent with any work, nor shall he himself perform any work, which will necessitate his frequent or prolonged absence from the mine.

(14)

If any doubt arises as to any matter under regulation (12) or regulation (13), it shall be referred to the Chief Inspector for decision.

(15)

No manager shall assistant manager or in any other supervisory capacity in another mine.

31. Charge report of managers.

- When there is a change of manager of any mine, the outgoing manager shall hand over to the incoming manager, a charge report in form VII, giving general information about the mine, sources of dangers, important safety works under execution requiring attention of manager and other important matters from the point of view of safety required to be attended by the incoming manager and the charge report shall be signed by both the outgoing and incoming managers and a copy of the charge report duly signed shall be sent to the Regional Inspector of the mine forthwith.

32. Qualification appointment of safety officer.

- In every below ground mine the average monthly output of which exceeds 5,000 tonnes or in every opencast mine the average monthly material handled of which exceeds 20,000 cum, the manager shall be assisted in the work of promoting safe practices in the mine by a Safety Officer who shall be a person holding the following qualifications:-

(i)

in the case of a below ground mine having an average monthly output in excess of 15,000 tonnes, or in case of a mine having opencast working with an average monthly material handled in excess of 50,000 cum, a First Class Manager's Certificate;

(ii)

in case of a below ground mine having an average monthly output in excess of 10,000 tonnes, but not exceeding 15,000 tonnes or in case of a mine with opencast working having an average monthly material handled in excess of 20,000 cum

but not exceeding 50,000 cum, a First or Second Class Manager's Certificate;

(iii)

in the case of a below ground mine having an average monthly output in excess of 5,000 tonnes but not exceeding 10,000 tonnes, holder of a First or Second Class Manager's Certificate or a degree or diploma in Mining or Mining Engineering approved by the Central Government:

Provided that where special conditions exist, the Chief Inspector may by an order in writing and subject to such conditions as he may specify therein, permit or require the appointment of a safety officer in variation of these provisions:

Provided further that where the Chief Inspector is of the opinion that, due to the large size of a mine, or due to other conditions existing at a mine, it is not possible for the safety officer to attend to his duties by himself, he may by an order in writing and for reasons to be recorded therein, require the appointment of such number of persons holding such qualification as he may specify in the order, to assist the safety officer.

33. Appointment of assistant manager.

(1)

In every mine having below ground working, the average monthly output of which exceeds 7,000 tonnes, the manager shall be assisted by assistant managers on the following scale;

Average monthly output

Number of assistant managers

Up-to and including 10,000 tonnes

One

Above 10,000 tonnes

One additional assistant manager for every 5,000 tonnes Output

or part thereof in excess of 10,000 tonnes

Provided that in a mine where the average output exceeds 20,000 tonnes, at least one of the assistant managers as aforesaid shall hold a First Class Manager's Certificate.

(2)

In every mine having opencast working, the manager shall be assisted by at least one assistant manager in each working shift on the following scale:

Average monthly material handled

Number of assistant managers.

More than 1.5 lakh cum and less than 7.5 lakh cum

One assistant manager for every 50,000 cum and at least one

Assistant Manager shall be holding First Class Certificate in each shift.

Above 7.5 lakhs cum

One assistant manager for every 1,00,000 cum of material

handled and at least one assistant manager shall be holding First Class Certificate in each shift.

(3)

The Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit or require the appointment of assistant managers in variation with these provisions.

34. Qualification and appointment of ventilation officer.

- In every below ground mine consisting of gassy seams of first degree, the average monthly output of which exceeds 5,000 tonnes or of second or third degree the average monthly output of which exceeds 2,500 tonnes, the manager shall be assisted in the work of supervising the maintenance of the ventilation system of the mine in accordance with the provisions of these regulations by a ventilation officer who shall be a person holding the following qualifications:-

(a)

in the case of a mine consisting of gassy seams of first degree and having an average monthly output in excess of 15,000 tonnes or a mine consisting of gassy seams of second or third degree and having an average output in excess of 10,000 tonnes, a Manager's Certificate; and

(b)

in every other case, a Manager's Certificate or a Degree, Diploma or Certificate in Mining or Mining Engineering approved by Central Government:

Provided that where special conditions exist, the Regional Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit or require the appointment of a ventilation officer in variance of these provisions or require the appointment of such number of persons to assist the ventilation officer, as may be specified in the order:

Provided further that in the case of a mine consisting of gassy seams of first degree and having an average output less than 15,000 tonnes, the Regional Inspector may, considering the nature and extent of working therein, permit, by an order in writing and subject to such conditions as he may specify therein, combine the post of ventilation officer with that of safety officer appointed under regulation 32.

Explanation 1: For the purposes of this regulation the expression "average output" means the average per month of the total output during the preceding financial year from the below ground working of all seams.

Explanation 2: Where the mine consists of gassy seams of different degrees, the average output shall be deemed to be from the seam or seams of the highest degree of gassiness.

35. Appointment of engineers.

(1)

At every mine where machinery is used, an engineer holding a degree or diploma in mechanical engineering, electrical engineering, mining machinery or equivalent qualification shall be appointed to hold general charge of such machinery, and to be responsible for its installation, maintenance and safe working:

Provided that where electrical energy at medium voltage and above is used and the installed capacity of all electrical equipment is 5 mega watts and above, an engineer holding a degree in electrical engineering or equivalent qualification shall be appointed to hold charge of the electrical equipment installed at the mine in addition to that specified above:

Provided further that nothing in this sub-regulation shall be deemed to prohibit the employment of two or more engineers at one mine so long as the jurisdiction and sphere of responsibility of every such engineer is defined by the manager in writing.

(2)

Notwithstanding anything contained in sub-regulation (1), the Chief Inspector may, by an order in writing, specify any qualification in addition to those referred to in that sub-regulation in respect of a mine or class of mines, if having regard to the conditions existing in such mine or class of mines, he is satisfied that it is necessary so to do in the interests of safety.

(3)

No person shall undertake the duties of an engineer of more than one mine without the previous permission in writing of the Regional Inspector and subject to such conditions as he may specify therein:

Provided that the Regional Inspector may at any time, by an order in writing, vary or revoke such permission.

(4)

Where by reason of temporary absence by any cause, the engineer, appointed under sub regulation(1) is unable to perform his duties, the manager shall authorise in writing a person whom he considers competent to act in his place:

Provided that -

(a)

notice of every such authorisation shall be sent to the Regional Inspector forthwith;

(b)

no such authorisation shall have effect for a period in excess of 30 days except with the previous written consent of the Regional Inspector and subject to such conditions as he may specify therein; and

(c)

the Regional Inspector may by an order in writing (revoke) authority so granted

36. Appointment and qualifications of senior officials.

(1)

At every one or more overman shall be appointed to hold charge of the different districts of the mine on each working shift unless, otherwise specified by the Regional Inspector.

(2)

The district assigned to an overman under sub-regulation (1) shall not be of such a size nor shall any additional duties

other than his duties under these regulation be such as are likely to prevent him from carrying out in a thorough manner, the duties prescribed for him under these regulations.

(3)

For the purposes of this regulation, every person employed as an official subordinate to the manager and superior to the Sirdar shall hold either a Manager's Certificate or an Overman's Certificate.

37. Appointment of Surveyor.

(1)

At every mine, one or more persons holding a Surveyor's Certificate shall be appointed to be the Surveyor for carrying out the surveys and levellings and for preparing the plans and sections required under the Act or the regulations, or orders made thereunder.

(2)

No person shall be appointed as a surveyor or more than one mine or in any other capacity in the same mine, without the previous permission in writing of the Regional Inspector and subject to such conditions as may be specified therein.

(3)

(a)

The number of surveyors required to be appointed shall be on the following scale, namely:-

The average monthly output in tonnes

No. of Surveyors

For below ground mines

8,000 tonnes and below Above 8,000 tonnes

One One for every additional 15,000 tonnes or part thereof

For opencast mines

The average monthly material handled in cum

No. of Surveyors

50,000 cum upto 9.00 lakhs cum

One for every 3.0 lakhs cum

Above 9.0 lakhs cum

One every 5.00 lakhs Cum

(b)

Notwithstanding anything contained in this regulation, the Regional Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit or require the appointment of surveyors in variation of these provisions depending upon the conditions prevailing in the mines.

(4)

If a mine has more than one surveyor, each shall carry the duties and the responsibilities of the surveyor for the part or section of the mine to be assigned in writing by the owner, agent or manager:

Provided that the owner, agent or manager shall appoint one of the surveyors to be responsible for the preparation and maintenance of the plans required to be prepared and maintained under these regulations Who shall also be responsible for co-ordination and overall supervision of survey work in the mine.

38. Appointment of officials and competent persons.

(1)

(a)

The owner, agent or manager shall appoint such number of competent persons, including officials and technicians as is sufficient to secure, during each of the working shifts -

(i)

adequate inspection of the mine and the equipment thereof

(ii)

a thorough supervision of all operations in the mine;

(iii)

the installation, running and maintenance, in safe working order, of all machinery in the mine; and

(iv)

the enforcement of the requirements of the Act and rules and regulations framed thereunder.

(b)

Without prejudice to the requirement of clause (a), where the mine is worked on more than one shift, the owner, agent or the manager shall arrange that during the afternoon shift and the night shift, the mine is under the general supervision of at least an assistant manager, if any, and of an experienced overman in other cases.

(2)

It shall be the responsibility of the manager to see that the persons so appointed are competent to perform the duties assigned to them:

Provided that no person shall be so appointed unless he is paid by the owner or agent and is answerable to the manager:

Provided further that the Chief inspector under special circumstances may vary the requirements of this sub-regulation by an order in writing.

(3)

Copies of all appointments made under sub-regulation (1) shall be entered in a bound-paged book kept for the purpose where a list of all such competent persons shall be maintained.

(4)

Without prejudice to the requirements of sub-regulation (2), every manager on taking over charge of a mine, shall satisfy himself that all persons already appointed under sub-regulation (1) are competent to perform the duties assigned to them; and if he finds them competent, he shall either countersign their authorisations or issue fresh ones.

39. General management.

(1)

The owner, agent and manager shall provide for the safety and proper discipline of persons employed in the mine.

(2)

Except in a case of emergency, no person who is not an official or a competent persons shall give, otherwise than through the manager, instructions to a person employed in a mine, who is responsible to the manager.

Chapter-V Duties and Responsibilities of Mine Management, Contractors, Manufacturers, Officials, Competent Persons and Workmen

40. Duties and responsibilities of owner.

(1)

In taking preventive and protective measures, the owner shall arrange for regular assessment of the risk and dealing with it in the following order of priority:-

(a)

eliminate the risk;

(b)

control the risk at source;

(c)

minimize the risk that include the design of safe work systems; and

(d)

in so far as the risk remains, provide for the use of personal protective equipment, having regard to what is reasonable, practicable and feasible and to good practice and the exercise of due diligence.

(2)

Owners shall take all necessary measures to eliminate or minimize the risks to safety and health in mines under their control and in particular

(a)

ensure that the mine is designed, constructed and provided with electrical, mechanical and other equipment, including a communication system to provide conditions for safe operation and a healthy working environment;

(b)

ensure that the machine is commissioned, operated, maintained and de-commissioned in such a way that workers can perform the work assigned to them without endangering their safety and health or that of other persons;

(c)

take steps to maintain the stability of the ground in which persons have access in the context of their work;

(d)

where practicable, provide, from every underground workplace, two exits each of which is connected to separate means of egress to the surface;

(e)

ensure the monitoring, assessment and regular inspection of the working environment to identify the various hazards to which the workers may be exposed, and to assess their level of exposure;

(f)

ensure adequate ventilation for all underground working to which access is permitted;

(g)

in respect of zones susceptible to particular hazards, draw up and implement an operating plan and procedures to ensure a safe system of work and the protection of workers;

(h)

take measures and precautions appropriate to the nature of a mine operation to prevent, detect and combat the start and spread of fires, explosion and inundations;

(i)

ensure that, when there is serious danger to the safety and health of workers, operations are stopped and workers are evacuated to a safe location; and

(j)

when managers or supervisors observe with safety and health regulations or codes of practice by any person they should; ensure that corrective actions are taken immediately and where necessary, the problem should be referred to a higher level of management immediately.

(3)

The owner shall prepare an emergency response plan specific to each mine for reasonably foreseeable industrial and natural disasters.

(4)

where workers are exposed to physical, chemical or biological hazards, the owner shall-

(a)

inform the workers, in a comprehensible manner, of the hazards associated with their work, the health risks involved and relevant preventive and protective measures;

(b)

take appropriate measures to eliminate or minimize the risks resulting from exposure to those hazards;

(c)

where adequate protection against risks of accident or injury to health including exposure to adverse conditions cannot be ensured by other means, provide and maintain at no cost to the worker, suitable protective equipment, clothing as necessary and other facilities as defined by these regulations; and,

(d)

provide workers who have suffered from an injury or illness at the workplace with first aid, appropriate transportation from the workplace and access to appropriate medical facilities.

(5)

The owner shall ensure that-

(a)

adequate training and re-training programs and comprehensible instructions are provided for workers, at no cost to them, on safety and health matters as well as on the work assigned;

(b)

adequate supervision and control are provided in each shift to secure the safe operation of the mine;

(c)

a system is established so that the names of all persons who are underground can be accurately known at any time, as well as their probable location;

(d)

all accidents and dangerous occurrences are investigated and appropriate remedial actions are taken; and

(e)

a report as specified by these regulations is made to the Regional Inspector of Mines and to the Chief Inspector of Mines

on accidents and dangerous occurrences

(6)

The owner shall ensure regular health surveillance of workers exposed to occupation health hazards specific to mining operations.

41. General responsibilities of supplier, manufacturer and designer.

- A person who design, manufacture, import, provide or transfer machinery, equipment or substances for use in coal mines, shall -

(a)

ensure that the machinery, equipment or substances do not entail dangers for the safety and health of those using them correctly;

(b)

make available:

(i)

information concerning their requirement for the correct installation, maintenance and use of machinery and equipment and the correct storage and use of substances;

(ii)

information concerning the hazards of machinery and equipment, the dangerous properties of hazardous substances and physical agents or products; and

(iii)

information on how to eliminate or control risks arising from the identified hazards associated with the products.

42. Responsibilities of contractor.

(1)

A contractor deployed in a mine for any work shall-

(a)

establish effective ongoing communication and coordination between appropriate levels of supervisors, officials and senior officials of the mine prior to commencing work, which shall include provisions for identifying hazards and the measures to eliminate and control risks;

(b)

ensure arrangements for reporting work related injuries and diseases, ill health and incidents among the contractors workers while performing work for the mine;

(c)

provide relevant workplace safety and health hazards awareness, and training to their workers prior to commencing and as work progresses as necessary;

(d)

ensure compliance of the provisions of the Act and the rules and regulations framed thereunder.

(2)

When using contractors, the owner, agent and manager shall ensure that:

(a)

the same safety and training requirements apply to the contractors and their workers as to the workers of the establishment;.

(b)

where required, only such contractors are used that have been duly registered or hold licenses;

(c)

the contract specify safety and health requirements as well as sanctions and penalties in case of non-compliance and such contract shall include the right for mine officials to stop the work whenever a risk of serious injury is apparent and to suspend operations until the necessary remedies have been put in place.

43. Duties of person employed in mine.

(1)

Every person shall strictly adhere to the provisions of the Act and of the rules and regulations made thereunder and to any order or direction issued by the manager or an official with a view to the safety or convenience of persons not, being inconsistent with the Act and these regulations; nor shall he neglect or refuse to obey such order or directions.

(2)

No person shall interfere with, impede or obstruct any person in the discharge of his duties, nor shall he offer or render any service, or use any threat to any other person with a view to preventing him from complying with the provisions of the Act and of the rules and regulations made thereunder or from performing his duties faithfully.

(3)

Every person shall, immediately before proceeding to work and immediately after terminating work at the end of his shift, have his name recorded in the register maintained under section 48(4) of the Act:

Provided that in case of working below ground, the person shall get his name recorded every time he proceeds below ground or returns to the surface:

Provided further that electronic punching or registry system as approved by the Chief Inspector may also be provided and used for the purpose of identification, marking attendance and recording the name of the person and a hard (printed) copy of such record shall be kept forthwith for the purpose of record in the aforesaid register or in any other format specified by the Chief Inspector.

(4)

Every person employed in a mine shall-

(a)

take reasonable care for their own safety and health and that of other persons who may be affected by their acts and omissions at work including the proper care and use of protective clothing, facilities and equipments placed at their disposal;

(b)

report forthwith to their immediate supervisor, any situation which they believe could pose a risk to their safety or health or that of other persons, and which they cannot properly deal with themselves; and

(c)

co-operate with the employer to permit compliance with the duties and responsibilities placed on the employer.

(5)

No person shall, except with the authority of an official, remove or pass through any fence, barrier or gate, or remove or pass any danger signal.

(6)

Subject to any directions that may be given by an official, no person shall, except for some justifiable purpose, go into any part of the mine other than that part in which he works, or travels to or from his working place by any roadway other than the proper travelling roadway.

(7)

No person shall sleep whilst on duty.

(8)

Subject to the provisions of the Act and of these regulations and orders made thereunder, no person shall remain in a mine beyond the period over which his shift extends.

44. Duties of competent person.

- Every competent person shall be subject to orders of superior officials, and shall not -

(a)

depute another person to perform his work without the sanction of his superior official;

(b)

absent himself without having previously obtained permission from such official, for the term of his absence or without having been relieved by a duly competent person; and

(c)

without permission from such official, perform during his shift any duties other than those for which he has been appointed.

45. Duties of officials.

(1)

Every official shall carry out the duties assigned to him by the manager or assistant manager in accordance with the provisions of the Act and of these regulations and orders made thereunder.

(2)

Every official shall ensure that persons under his charge understand and carry out their respective duties properly.

46. Duties and responsibilities of manager.

(1)

(a)

In every mine, daily personal supervision shall be exercised by the manager:

Provided that in case of working below ground, he shall visit and examine the working on at least four days in every week to ensure safety in every respect:

Provided further that at least one visit in every fortnight shall be made during the night shift:

Provided also that where owing to any unavoidable cause he is unable to carry out the aforesaid duties or inspections, he shall record the reason for the same in the book kept under clause (b).

(b)

The manager shall maintain, in a bound paged book kept for the purpose, a diary; and shall record therein the result of each of his inspections and also the action taken by him to rectify the defects noticed, if any.

(2)

The manager shall make arrangements for all overmen and other officials to meet him or the assistant manager once in every working day for the purpose of conferring on them matters connected with their duties.

(3)

(a)

The manager shall ensure sufficient supply of proper materials and appliances for the safety of the mine and the persons employed therein; and if he be not the owner or agent of the mine, shall report in writing to the owner or agent, where anything is required for the aforesaid purpose that is not within the scope of his authority to order, and a copy of every such report shall be recorded in a bound-paged book kept for the purpose.

(b)

On receipt of a requisition under clause (a), the owner or agent shall promptly arrange to supply the said materials and appliances, and shall within three days of receipt of the requisition, intimate to the manager in writing the action taken to meet the requisition.

(4)

The manager shall assign to every competent person his particular duties and take all possible steps to ensure that every such person understands and carries out the provisions contained in the Act or the rules or regulations made thereunder in a proper manner.

(5)

The manager shall provide every overman with a tracing, up-to the date of the last survey, showing the working of the district below ground assigned to him and such tracing shall, where any work of reduction or extraction of pillars is being carried out, show clearly the reference of the permission and the manner in which such reduction or extraction is to be carried out.

(6)

The manager shall examine all reports, registers and other records and shall countersign the same with date:

Provided that the manager may, by an order in writing, delegate this duty to an assistant manager except in cases where a specific provision is made requiring the manager to countersign a report or register.

(7)

The manager shall give attention to, and cause to be carefully investigated any specific representation or complaint that may be made to him in writing by an employee of the mine as to any matter affecting the safety or health of persons in or about the mine.

(8)

When any accident resulting in serious bodily injury to any person or loss of life occurs in a mine, the manager shall inspect the site of accident as soon as possible and shall either himself or through an assistant manager, have an inquiry made into the cause and circumstances attending the accident and the result of every such enquiry along with a plan and sections and, wherever practicable a photograph or photographs of the site of the accident showing details shall be recorded in a bound paged book kept for the purpose and copy thereof furnished to the Chief Inspector and Regional Inspector within fifteen days of the accident.

(9)

The manager shall perform such other duties as the Chief Inspector may assign to him.

(10)

The manager may suspend or take such disciplinary action against any employee for contravention of any of the provisions of the Act or the regulations and orders made thereunder.

47. Duties of safety officer.

(1)

The duties of the safety officer shall be -

(a)

to visit surface and underground parts of the mine with a view, to meet the workers on the spot, to talk to them on matters of safety and invite suggestions thereon;

(b)

to take charge of the newly recruited staff and show them around the mine pointing out the safe and unsafe acts during the course of their work in the mine.

(c)

to investigate all types of accidents and incidents in the mine including mine accidents and analyse the same with a view to pinpoint the nature and common cause of accidents in the mine;

(d)

to maintain detailed statistics about mine accidents and to analyse the same with a view to pinpoint the nature and common causes of the accidents in the mine;

(e)

to study and apprise the manager of all possible sources of danger such as information and, fire coal dust and others;

(f)

to hold safety classes and give safety talks and lectures to work members of the supervisory staff;

(g)

to organise safety weeks and other safety education and propaganda in mine;

(h)

to ensure that all concerned mine employees are fully conversant with various standing orders, codes of practices (such as those relating to stoppage of mine mechanical ventilators and to the occurrence of a fire or other emergency in the mine) and support plan;

(i)

to provide assistance in the formulation of programme for training at the mine level; including vocational training, training in gas testing, and training in First Aid;

(j)

to report to the manager any violations in the mine of the provision, of the Act, the rules, or the regulations made thereunder;

(k)

to promote safety practices generally and to lend active support to all measures intended for furthering the cause of safety in the mine and ensure reasonable compliance to the recommendations of the Safety Committee and Workman's Inspector and

(l)

to assist the manager in any other matter relating to safety in the mine -

(2)

The safety officer shall ensure that an appropriate emergency plan as required under these regulations is put in place and the requirements of the same are strictly implemented.

(3)

Except in an emergency no duties other than the of specified above shall be assigned to the safety officer.

(4)

The safety officer shall maintain in a bound paged book a detailed record of the work performed by him every day.

48. Duties and responsibilities of assistant manager.

(1)

The assistant manager shall carry out the duties assigned to him by the manager and shall see that in the part of the

mine assigned to him by the manager, all work is carried out in accordance with the provisions of the Act and of the regulations and orders made thereunder.

(2)

The assistant manager shall, subject to the orders of the manager, visit and examine the working under his charge, or part thereof, on every working day.

(3)

The assistant manager shall, from time to time, carefully examine every travelable part of the mine or part thereof placed under his charge, whether frequented by work persons or not.

(4)

In the absence of the manager, the assistant manager shall have the same responsibility, discharge the same duties and be subject to the Same liabilities as the manager, but not so as to exempt the manager therefrom.

(5)

The assistant manager shall maintain, in a bound paged book kept for the purpose, a diary; and shall record therein the result of each of his inspections and also the action taken by him to rectify the defects noticed, if any.

49. Duties of ventilation officer.

(1)

The ventilation officer shall -

(a)

ensure the observance of all regulations and orders concerning ventilation, spontaneous heating, fire, gas and coal dust including dust suppression and shall advise the manager, if any alteration is required in the ventilation system to ensure adequacy of ventilation in compliance with the regulations or orders;

(b)

advise the manager on day to day problems of ventilation, gas and coal dust;

(c)

maintain close liaison with the assistant managers and other supervisory staff, and assist them in their day-to-day ventilation problems;

(d)

carry out ventilation surveys of the mine and undertake any other special work relating thereto as may be directed by the manager from time to time;

(e)

take such steps as are necessary to ensure compliance with the ventilation standards required in terms of these regulations or otherwise;

(f)

check the speed of main mechanical ventilator, amperage by its electric motor, and fan drift water gauge at least once in a day and he shall investigate any unusual change in the water gauge and report to the manager;

(g)

determine the efficiency of the main mechanical ventilator once at least in every three months and get the fan blades and the fan drift cleaned when necessary;

(h)

ensure that copies of standing orders in the event of stoppage of the main mechanical ventilator are posted at conspicuous places at the mine, and ensure that the persons concerned understand the instructions contained therein;

(i)

ensure the correct sitting and installation of auxiliary and booster fans underground;

(j)

examine at frequent intervals all ventilation appliances like doors, brattices, air crossings, regulators, stoppings, booster and auxiliary fans, ventilation ductings and other devices of ventilation control in the mine and report any defect in the same to the manager;

(k)

take necessary steps to stop any leakage in any of the devices and ensure that the ventilation appliances are maintained in good order;

(l)

ensure that sufficient quantity of good air is coursed into all working places and reaches all other working belowground and shall -

- (i) ensure that the ventilation stoppings and brattices etc., are constructed as per specifications and are kept extended sufficiently;
- (ii) ensure that measurements of air quantity, temperature and humidity are regularly taken as specified and bring up-to-date the entries on the check boards provided at each air measurement station;
- (iii) determine the Ventilation Efficiency Quotient (VEQ);
- (iv) ensure that mine air samples are properly collected at the appointed time and place, and analysed immediately; and
- (v) make observations for inflammable and any other harmful gases;
- (m) maintain separate tracing of the ventilation, rescue, stone dusting and the dust sampling plans and bring them up-to-date;
- (n) bring to the notice of surveyor any changes in the ventilation system or ventilation appliances and shall see that all old markings on the ventilation and rescue plans are corrected and new ventilation circuits are shown forthwith;
- (o) regularly check the barometer provided at the mine and report any unusual change in barometric pressure to the manager for appropriate actions;
- (p) take care of the instruments and apparatus used in the mines for environmental monitoring and ensure that all such instruments are maintained in good order and calibrated at specified intervals;
- (q) regularly visit returns of working districts and old working including fire stoppings, if necessary, for symptoms of spontaneous heating and fire and report to the manager forthwith any such symptoms observed by him and shall himself take such steps as may be immediately necessary for the safety of the workers;
- (r) check the fire fighting measures and take necessary steps by regular rehearsals to ensure that all fire fighting equipments are maintained in working order and the concerned staff are fit and conversant with their duties in the event of a fire in the mine;
- (s) take necessary steps for proper cleaning, treatment and suppression of coal dust in the mine and see that the arrangements for wet-cutting at the faces and water spraying at and within 90 metres of the working places are properly installed and function satisfactorily;
- (t) ensure that the stone dust barriers are correctly sited, properly constructed and maintained in accordance with the statutory requirements or otherwise; and bring the entries on the check boards up-to-date from time to time;
- (u) ensure that samples of mine roadway dust and of airborne dust samples (if required by the manager) are regularly taken in the specified manner;
- (v) collect air samples from sealed off areas, exhaust gases from diesel Vehicles and from such other places as may be required by the manager;
- (w) ensure that all records and reports relating to ventilation and coal dust are kept up-to-date and entries are made regularly in the check boards for ventilation and stone dust barriers.

Provided that nothing contained above shall exempt the manager, assistant manager, surveyor, overman, sirdar or any

other competent person concerned, from any corresponding duties and responsibilities specified for them in these regulations or any orders made thereunder; and

(x)

assist the manager in any matter relating to the ventilation of the mine.

(2)

Except in an emergency no duties other than those specified above shall be assigned to the ventilation officer.

(3)

The ventilation officer shall maintain, in a bound paged book, a detailed record of the work performed by him every day.

50. Duties and responsibilities of overman.

(1)

The overman shall subject to the orders of superior officials, have responsibility, charge and control of such part of the mine, and shall carry out such duties, as may be assigned to him by the manager.

(2)

The overman shall -

(a)

while on duty, carry a tracing of the working of such district and shall keep the tracing up-to-date;

(b)

in his district, make the inspections and reports required by these regulations;

(c)

ensure that the subordinate officials and competent persons in his district carry out their respective duties in a proper manner.

(3)

Subject to the control of the manager and the assistant manager, if any, the overman shall give such directions to the workers under his charge as may be necessary, to, ensure compliance with the provisions of the Act, or the rules or regulations made thereunder, and to secure the safety of the district and the safety and proper discipline of the persons employed therein.

(4)

The overman shall ensure that sufficient supplies of timber, support materials, brattice and other tools and tackles, appliances, necessities required for the safe working of his district are kept at convenient places therein.

(5)

The overman shall -

(a)

ensure that every air-crossing, stopping, door, brattice and other ventilation device is maintained in good order;

(b)

ensure that the ventilation is effective in his district, and when brattices or air pipes or ducts are required to be used for the ventilation of the working places, he shall see that they are kept sufficiently advanced to ensure that an adequate amount of air reaches every such working place;

(c)

have power to send out of the mine any person under his charge who infringes or attempts to infringe any provision of the Act or of the regulations or orders made thereunder, or fails to carry out any direction given with regard to safety, and shall report such matter in writing to the manager;

(d)

ensure that all tracks and tramlines are properly laid, graded, ballasted or otherwise packed;

(e)

ensure that the manholes on the haulage roadways are kept safe, clear of any obstruction, and properly white-washed;

(f)

ensure see that the stop-blocks, runway switches and other safety devices are fixed and used as required under the regulations, drag or back-stays are provided and regularly used behind tubs ascending inclines/declines and that a sufficient supply of suitable sprags is provided where tubs are loaded on a gradient or lowered down a gradient by hand.

(g)

if he finds any of the ropes, chains, signals, brakes, jig wheels and post or other apparatus in use in his district to be in

an unsafe condition, stop the use of the same forthwith;

(h)

ensure that except for the purposes of inspection, examination and repairs every person other than an official or a haulage attendant travels by the travelling roadway;

(i)

give prompt attention to the removal of any danger observed or reported to him, and shall ensure that dangerous places are adequately fenced off;

(j)

under the direction of the manager, ensure that approved safety lamps are used and naked lights excluded wherever and whenever danger from inflammable gas is apprehended.

(6)

In case of opencast working, the overman shall ensure that-

(a)

sides of benches are kept dressed;

(b)

stability of benches is not endangered;

(c)

haul roads are kept maintained;

(d)

stability of overburden dumps is not endangered;

(e)

there is no over crowding of men and machinery at the working faces;

(f)

that adequate lighting is provided at area under his control; and

(g)

that adequate precautions as laid down in these regulations are taken before blasting operations is conducted.

(7)

The overman shall-

(a)

devote the whole of his time to his duties and shall visit each working place in his district as often as may be necessary or possible;

(b)

not, except for justifiable cause, leave the district in his charge until he had finished the inspections required under these regulations and any other duties that he is required to perform, or until relieved by a duly appointed substitute;

(c)

if the mine is working in a continuous succession of shifts, confer with the official succeeding him and give him such information as may be necessary for the safety of his district and of persons employed therein.

(d)

at the end of his shift, record in a bound paged book kept for the purpose a general report in the specified format on the performance of all his duties during the shift, including anything concerning the proper working of the mine and the safety and discipline of persons employed in his district.

51. Duties and responsibilities of Sirdar.

(1)

The Sirdar or other competent person appointed under regulation 133 shall, subject to orders of superior officials, he shall have responsibility, charge and control of the district of the mine assigned to him by the manager or assistant manager.

(2)

The sirdar shall -

(a)

ensure proper observance of the requirements of the Act and of the regulations, and orders made thereunder by persons under his charge and shall, as soon as practicable, report any contravention thereof to his superior official;

(b)

make such inspection and reports as are required by these regulations, and in making such examination, he shall pay particular attention to edges of the goaf, if any, for checking supports and for presence of gas;

(c)

except in the case of a mine working in a continuous succession of shifts, on completion of the first inspection of the district, proceed to the station specified under regulation 133 and instruct all persons as to their places of work and as to any special precautions necessary to be observed by them

(d)

if he finds any person in a place other than the one assigned to him, he may order such person out of the mine, and shall forthwith report the matter to his superior official;

(e)

ensure that no inexperienced person is employed on any work except under the supervision of an experienced person.

(f)

ensure that the roof and sides of all travelling roadways to, and working places in his district are made and kept secure;

(g)

where the height of any working place in the district in his charge measured from floor to roof exceeds three metres, ensure that a suitable wooden bunt or pole by which all parts of the roof may be effectively tested by a person standing on the floor and a ladder of suitable length are kept at convenient places in the district;

(h)

report to his superior official any deficiency in timber, support materials, appliances and other necessities required for the safe working of the district.

(i)

Where either of the two ways affording means of egress from the district to the surface is not ordinarily used for travelling, travel, once at least in every seven days, the whole of such roadway in order to make himself thoroughly acquainted with the same.

(j)

ensure that no timber or support is withdrawn except by means of a safety prop-with drawer.

(3)

If he observes any dangerous place during the course of his, inspections or if any danger at a place where work persons are employed is reported to him the Sirdar shall, if the danger cannot be removed forthwith withdraw all persons from such place and shall not leave the place until the danger has been removed in his presence or all approaches to the place have been fenced off so as to prevent persons from inadvertently entering such place.

(4)

The sirdar shall -

(a)

ensure that any dangerous operation is carried out with due precaution by being present throughout whenever any work of clearing falls of ground and setting of supports therein is being carried out;

(b)

cause the entrance to every place which is not in actual use or in course of working or extension, to be fenced across the whole width, so as to prevent persons from inadvertently entering such place;

(c)

if he finds any accumulation of inflammable or noxious gases, take such precautions specified in regulation 170 and shall not remove such accumulation until he has received instructions in that behalf from his superior official;

(d)

on receipt of information of an accident to any person in his district, proceed at once to the place of accident, inspect the place and, if required, supervise the rescue operations, and report or send notice of the accident to the manager or assistant manager;

(e)

devote the whole of his time to his duties, and not leave the mine until the end of the shift or until relieved by a duly appointed substitute;

(f)

if the mine is worked by a continuous succession of shifts, before leaving his district, confer with the or competent person succeeding him, and acquaint him with all matters requiring his personal attention and give him such other information as may be necessary for the safety of his district and of the persons employed therein.

52. Duties and responsibilities of shotfirer.

- The shotfirer shall -

(a)

carry out his duties in accordance with the provisions of these regulations and of any orders made thereunder with respect to the transport and use of explosives;

(b)

be responsible for the observance by his assistants, if any, of such provisions and of any direction with a view to safety which may be given to them by a superior official;

(c)

not hand over any explosives to any unauthorised person;

(d)

see that clay, sand or other suitable stemming material is available in sufficient quantities at convenient places;

(e)

be present when shots are being charged and stemmed and shall himself fire the shots; and

(f)

be responsible, when a shot has misfired, for ensuring that the place is adequately fenced, and that the provisions of regulation 208 are strictly observed.

53. Duties of support man.

- The support man shall -

(a)

carry out the orders of the manager or assistant manager, overman, sirdar or other competent person with respect to securing of roof and sides and the other working places;

(b)

ensure placement of supports strictly in accordance to the support plan;

(c)

at once report to the sirdar or other competent person any shortage of support materials in his district;

(d)

in case of use of timber, be responsible to see that wood cuttings are not left in any working belowground.

54. Duties of attendant of main mechanical ventilator.

- The person in charge of the main mechanical ventilator shall -

(a)

keep the ventilator running at the speed fixed by the manager;

(b)

examine the machinery and observe the pressure-recorded or water gauge and the speed indicator at intervals of not more than one hour, and shall enter the readings of the indicator in a bound paged book kept for the purpose at the fan-house;

(c)

immediately report to his superior official any stoppage of, damage to, or defect or derangement in the machinery, or any unusual variation in the water-gauge or other indicators and shall immediately report to him any unusual circumstances in regard to mine ventilation which may come to his notice; and

(d)

where the ventilator is continuously operated, not leave his place until relieved by a duly appointed substitute.

55. Duties of lamp-cabin in-charge.

- The competent person in charge of a lamp-cabin shall -

(a)

be responsible for ensuring that all lamps in the lamp-cabin including safety lamps are properly maintained as per manufacturers specification and in accordance with the provisions of these regulations;

(b)

ensure that the lamp-cabin and safety lamp-room is kept in a neat and tidy condition, and that all damaged and defective gauges, glasses and other parts of safety lamps are not kept or stored in such room;

(c)

ensure that fire extinguishers or other means of dealing with fires provided in the lamp cabin and safety lamps room are in good condition and readily available for use;

(d)

ensure that all records required by the regulations for the issue, return and maintenance of lamps including safety lamps are properly maintained;

(e)

ensure that every person going below ground is provided with a lamp having adequate charge to sustain at least whole of the shift; and

(f)

carry out such other duties relating to the maintenance, issue and return of lamps including safety lamps as may be specified by the manager or the assistant manager.

56. Duties and responsibilities of surveyor.

(1)

The surveyor shall -

(a)

make such accurate surveys and leveling, and prepare such plans and sections and tracings thereof, as the manager may direct or as may be required by the Act or by the regulations or orders made thereunder, and shall sign the plans, sections and tracings and date his signature; and

(b)

be responsible for the accuracy of any plan and section, or tracings thereof that has been prepared and signed by him.

(2)

The surveyor shall record in a bound paged book kept for the purpose-

(a)

the full facts when working of the mine have approached to about 120 metres from the mine boundary, or from disused or waterlogged working;

(b)

any doubts which may arise or exist concerning the accuracy of the plans and sections prepared under these regulations; and

(c)

any other matter relating to the preparation of the plans and sections that he may like to bring to the notice of the manager.

(3)

Every entry in the book shall be signed and dated by the surveyor and countersigned and dated by the manager:

Provided that where in any mine two or more surveyors are employed, each of the surveyors shall make the entries aforesaid in respect of the working in his jurisdiction or of the plans and sections in his charge.

(4)

Nothing in sub-regulation (3) shall absolve the owner agent or manager of his responsibility under the Act and under these regulations or orders made thereunder.

57. Duties and responsibilities of engineer.

- The engineer or other competent persons appointed for the purpose shall -

(a)

subject to the orders of the manager and other superior official, hold general charge of all machinery at the mine; and be responsible for the proper installation, maintenance and safe working of sub machinery;

(b)

when any machinery is shifted , newly installed, ensure that it is given a trial run before it is put into use, and be present during every such trial run;

(c)

be present throughout whenever any work, of installing, changing or recapping of any winding rope, or of installing,

changing or annealing any suspension gear, is being carried on;

(d)

ensure that the provisions of the Act and of these regulations and orders made thereunder relating to the installation, maintenance, operation or examination of machinery are properly carried out by himself or by Subordinate officials, competent persons or work-persons as the case may be, appointed for the purpose; and

(e)

if foreman, electrical supervisor, mechanics, electricians or other subordinate officials or competent persons are appointed for the purpose, examine all reports, registers and other records relating to the installation, maintenance, operation or examination of machinery required to be made or kept in pursuance, of the Act, these regulations or orders made thereunder, and countersign the same and date his signature.

58. Duties of winding engineman.

- A winding engineman shall -

(a)

at the beginning of his shift, examine the engine, brakes and all appliances in his charge and satisfy himself that they are in good working order;

(b)

during his shift keep the winding engine and apparatus connected therewith properly cleaned and old and ensure that the engine room is clean and free of inflammable material;

(c)

immediately report in writing to the engineer or other competent person appointed for the purpose any defect which he has noticed in the engine, brake, indicator, drum, rope or other appliances under his charge;

(d)

not allow any unauthorised person to enter the engine room or in any way to interfere with the engine;

(e)

thoroughly acquaint himself with, and carefully attend to, the code of signals and not start the engine until he has received the proper signal to do so and if the signal is indistinct, he shall not start the engine until it has been repeated and he clearly understands it.

(f)

avoid jerk in starting, running and stopping, the engine, and cause the cage or other means of conveyance to be brought gently to rest at any stopping place;

(g)

while persons are being lowered or raised in the shaft, not drive the engine at a speed higher than that fixed by the manager for man-riding purposes and approved by the Regional Inspector;

(h)

not unclutch the drum of his engine until he has assured himself immediately beforehand by testing the brake of the drum against the full power of the engine and ensure that the brake is in proper condition to hold the load suspended from the said drum or when the drum is unclutched, shall use the brake only for the purpose of maintaining such drum stationary, and not lower men or materials from an unclutched drum;

(i)

on no pretext leave the handle or brake whilst the engine is in motion, or while persons are riding a cage or other means of conveyance in the shaft;

(j)

not leave the engine whilst persons are at work in the shaft and whenever he has occasion to leave the engine, cut off the power and secure the drums with the brake:

(k)

not leave the engine at the end of his shift unless all the persons have come out of shaft or unless relieved by a duly appointed substitute.

59. Duties of banksman and onsetter.

(1)

Every banksman or onsetter shall -

(a)

subject to the orders of a superior official, have full control of the top or bottom of shaft and the inset, as the case may be, and report to such official any person who, without authority, gives a signal or disobeys instructions;

(b)

thoroughly acquaint himself with, and carefully attend to, the code of signals, and properly transmit the signals by the means provided and not act on any signal, the correctness of which he is in doubt, except a signal which he believes to be "to stop", and not allow any unauthorised person to give signals;

(c)

immediately report to his superior official any defect in the signalling installation;

(d)

devote the whole of his time to his duties, and not leave his post during the period of his duty;

(e)

where persons are raised or lowered in the shaft, not leave his post at the end of his shift unless all the persons have come out of the shaft or unless relieved by a duly appointed substitute;

(f)

not allow more than the authorised number of persons to enter the cage or other means of conveyance at anyone time;

(g)

not, unless specially authorised in writing by the manager in that behalf, allow any person when riding in a cage or other means of conveyance, to take with him any bulky material other than tools and instruments:

Provided that nothing in this clause shall be deemed to prohibit the carrying, in a cage or other means of conveyance, the explosives by a shotfirer or other competent person;

(h)

after any stoppage of winding for repairs or for any other cause for a period exceeding two hours, not allow any person to ride in the cage or other means of conveyance unless it has been run at least one complete trip up and down the working portion of the shaft;

(i)

not allow any person to ride on the top or edge of any cage or other means of conveyance except when engaged in examination, repair or any other work in the shaft

(j)

after persons have entered the cage, see that the cage gates on both sides are in position and closed, before signaling for the cage to be lowered or raised:

(k)

not allow any unauthorised person to handle tubs in or out of the cage:

(l)

while tubs are being lowered or raised ensure that the catches are holding the tubs properly before signaling the cage or other means of conveyance and if he notices any defect in the tub-catches, immediately inform his superior official;

(m)

at any entrance into a shaft or inset which is provided with gates or fences not worked by the cage or other, means of conveyance, not begin to remove the gate or fence until the cage or other means of conveyance has stopped at the entrance, and close the gate before he has signaled the cage or other means of conveyance away and not permit any unauthorised person to open or interfere with the gate;

(n)

ensure that all fences and gates provided at the top of the shaft or at any inset are in position;

(o)

not permit any unauthorised person to remove a fence or gate and if he notices any defect in such fence or gate, immediately inform his superior official;

(p)

keep the top of the shaft or the inset and the floor of every cage free from loose material;

(q)

when long timber, pipes, rails or other material projecting over the top of a cage or other means of conveyance are lowered or raised, ensure that the projecting ends are securely fastened to the rope, chains or bow;

(r)

when he suspects that the cages are not working smoothly in the shaft or when he hears anything unusual happening in the shaft while the winding engine is working, immediately give signal to the winding engineman to stop the engine.

(2)

The banksman shall-

(a)

at the beginning of his shift, ensure that the keeps are in proper working order;

(b)

when he is informed of any danger in the shaft, not allow any person to descend except for the purpose of examination or repair and during the time that such examination or repair is going on, be on duty and listen for signals;

(c)

not permit any person descending the shaft to carry any intoxicating drink or drug, or allow any intoxicated person to descend.

(3)

The banksman shall not leave his place of work unless duly relieved by his successor.

60. Duties of haulage engineman, attendant etc.

(1)

At the beginning of his shift, the haulage engineman shall examine the engine, its brake and all appliances in his charge, and shall satisfy himself that they are in good working order.

(2)

The haulage engineman shall, during his shift keep the haulage engine and apparatus connected therewith properly cleaned and oiled and the engine, room clean and free of inflammable material.

(3)

The haulage engineman shall report immediately to the engineer or other competent person appointed for the purpose any defect which he has noticed in the engine, brake, drum, rope or other appliances under his charge.

(4)

Whenever the haulage engineman has occasion to leave the engine, he shall cut off the power and secure the engine with the brake;

(5)

The haulage engineman and signaler shall not allow any unauthorised person to enter the engine-room or in any way to interfere with the engine or signal, as the case may be.

(6)

Every haulage engineman and signaler shall thoroughly acquaint himself with, and carefully attend to, the code of signals.

(7)

The haulage engineman shall not start the engine until he has received the proper signal to do so and if the signal is indistinct, shall not start the engine until it has been repeated and he clearly understands it.

(8)

The person in charge at the top of any haulage plane or incline shall ensure that the stop-blocks are put in position, before allowing any tub to be brought on to the top landing and shall cause the tubs to be securely coupled up to each other and to the rope or chain, before the stop block is put off in position and in case any alternative safety appliance is provided, he shall cause the same to be brought into use on every such occasion.

(9)

The person who is responsible for the attachment to the haulage rope, of any tub or set of tubs at any stopping place on any haulage plane or incline, shall ensure that no person remains in a position of danger at or near such stopping place while the rope is in motion.

(10)

The person in charge of any tubs or set of tubs, which it is intended to send up any haulage plane or incline on which drags or back-stays are required to be used, shall securely fix the drag or back-stay or cause it to be so fixed, before such tub or set of tubs is set in motion.

(11)

The person in charge at the top or bottom of the incline shall ensure that no unauthorised person rides on any tub.

(12)

Before a train of side-tipping tubs is set in motion, the person in charge shall ensure that the safety catches of all such tubs are properly secured.

61. Duties of locomotive driver.

- The locomotive driver shall-

(a)

before commencing work in his shifts, ensure that the audible signal, lights and the brakes of the locomotive are in proper working order;

(b)

not work the locomotive unless the locomotive is fitted with sufficient headlights;

(c)

immediately report to the engineer or other competent person appointed for the purpose any defect which he has noticed in the locomotive or any part or fitting thereof;

(d)

not set the locomotive in motion until audible warning has been given by him to persons whose safety may be endangered and give the audible warning when the locomotive is approaching a level crossing or any place where any person is at work or where the driver's sight is intercepted;

(e)

not leave a locomotive unattended away from the place where it is housed, unless he has ensured that it cannot be set in motion by any unauthorised person;

(f)

ensure that no unauthorised person drives, handles or rides on a locomotive;

(g)

ensure that when tubs or wagons are being pushed in front of the locomotive, the shunter shall accompany the leading wagon.

62. Duties of cutting and loading machine driver and mechanic fitter.

- The cutting and loading machine driver and mechanic/fitter shall-

(a)

where a machine is required, to work on a gradient exceeding 1 in 5, provide and use an effective contrivance to prevent the machine running back.

(b)

not fit any cutting or loading machine or otherwise move with the cutting or loading tool in motion, except in the actual process of cutting or loading and if the cutting or loading tool, as the case may be, cannot be locked out of gear securely, it shall be removed before fitting is started.

(c)

not make any repair or, adjustment to a cutting or loading machine or put in or take out a pick, until he has made such arrangements as will prevent the mechanism being inadvertently put into motion while such operation is being performed.

(d)

not open or replace the cover of any electrical part of a cutting or loading machine, except under the supervision and in the presence of an engineer, electrician or other competent person appointed for the purpose;

(e)

not leave the machine unless he has completely, cut off the power and has assured himself that the moving parts of the machine shall not be inadvertently set in motion.

63. Duties of magazine in-charge.

- The magazine in-charge shall -

(a)

subject to the orders of super officials, be responsible for the proper receipt, storage and issue of explosives in and from the magazine.

(b)

maintain such records of explosives so received, stored and issued, as are required by the rules made under Indian,

Explosives Act, 1884 and under the act and the regulations, rules, byelaws and orders made thereunder

(c)

not issue explosives to any person other than a competent person and when explosives are returned to the magazine, shall re-issue such explosives before issuing fresh stock;

(d)

record in bound paged book kept for the purpose, the names of various competent persons and the quantity and nature of explosives to each of them; and similarly record the quantity and nature of explosives returned to magazine by each such person;

(e)

securely lock each canister before issuing it to the competent persons and also check whether the canister is returned to the magazine in locked condition and shall not issue explosives in any canister which is not in proper repair or which cannot be securely locked;

(f)

not allow any unauthorised person to enter the magazine;

(g)

not issue any explosive for which the prescribed shelf life has expired; and

(h)

if he discovers any shortage of explosives in the magazine, forthwith inform the manager in writing.

64. Duties of register keeper and attendance clerk, etc.

(1)

Every person appointed to keep register or other record required to be kept by or under the Act or under these regulations, or orders made thereunder, or to make entries therein, shall make the necessary entries in ink and with reasonable despatch.

(2)

During the whole time that persons are at work, the attendance clerk shall remain on duty at attendance cabin which shall be provided near the workplaces, or in case of working below ground, near the outlet used by the work persons to enter and leave such working.

(3)

It shall be the duty of the attendance clerk to ensure that no person who is not an employee of the mine or is not entitled to enter the mine under the act or under the regulations, or orders made thereunder, or is not so authorised by the manager, shall enter the mine and if any such person forcibly enters the mine, the attendance clerk shall immediately report the matter in writing to the manger.

(4)

If after the commencement of a shift, any official or a competent person has not got his attendance recorded in the register maintained under sub-section (4) of section 48 of the Act, the attendance clerk concerned shall within two hours after the Commencement of the shift, report the fact in writing to the manager or the assistant manager or other official in charge of the shift.

65.

Duties of dragline, shovel, excavator, surface miner, spreader, mobile transfer conveyor operator, bucket wheel excavator, etc.

- Every person authorised to operate a dragline, shovel or excavator, etc., shall -

(a)

personally inspect the machine assigned to him in the beginning of his shift and test the various system, sub-systems and protective devices, as stipulated in this respect by the engineer in consultation with the manufacturer or supplier;

(b)

not take out the machine for work nor shall he work the machine unless he is satisfied that it is mechanically sound and in efficient working order;

(c)

maintain a record of every inspection made under clause (a) in a bound paged book kept for the purpose and shall sign every entry made therein;

(d)

keep the cabin or engine room and all window glasses clean;

(e)

keep the walkways, hand-rails, ladder-ways free of loose tools, lubricants or other material that might fall or cause a tripping hazard;

(f)

not allow any unauthorised person to ride on the machine;

(g)

not move or operate the machine when persons are in such proximity as to be endangered;

(h)

not swing the bucket over passing haulage units or over the cabin of units being loaded;

(i)

lower the bucket to ground, switch off the power supply to the machine or stop the prime mover and lock the cabin door before leaving the machine.

66.

Duties of truck and dumper operator.

(1)

Every person authorised to operate trucks and dumpers in a mine shall-

(a)

personally inspect the machine assigned to him in the beginning of his shift and test the various systems, sub-systems and protective devices;

(b)

not take out the machine for work nor shall he work the machine unless he is satisfied that it is mechanically sound and in efficient working order;

(c)

not drive the machine under his charge too fast, shall avoid distractions, and drive defensively, not attempt to overtake another vehicle unless he can see clearly far enough head to be sure that he can pass it safely and sound the audible warning signal before overtaking;

(d)

when approaching a stripping or loading equipment, sound the audible warning signal and not attempt to pass the stripping equipment until he has received proper audible signal in reply;

(e)

before crossing a road or railway line, reduce his speed, look in both directions along the road or railway line and proceed across the road or line only if it is safe to do so;

(f)

sound the audible warning signal while approaching 'blind' corner or any other points from where persons may walk in front unexpectedly;

(g)

not operate the truck or dumper in reverse unless he has a clear view of the area behind and give an audible warning signal before reversing a truck or dumper;

(h)

be sure of clearance before driving through tunnels, archway, plant structures etc.

(2)

The driver shall ensure that the vehicle is not overloaded and that the material is not loaded in a truck or dumper so as to project horizontally beyond the sides of its body and that any material projecting beyond the front or rear is indicated by a red flag during day and by red light after day light hours.

(3)

The driver shall not allow any unauthorised person(s) to ride on the vehicle.

Chapter-VI Plans and Sections

67.

General requirements about mine plans.

(1)

Every plan or section prepared or submitted in accordance with the provisions of these regulations shall -

(a)
show the name of the mine and of the owner and the purpose for which the plan is prepared;.

(b)
show the true north, or the magnetic meridian and the date of the later;

(c)
show a scale of the plan at least 25 centimeters long and suitably subdivided;

(d)
unless otherwise provided, be on a scale having a representative factor of 2000:1 or 1000:1.

Provided that the Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit or require the plans to be prepared on any other suitable scale; and

(e)
be properly inked in on durable paper, tracing cloth or on polyester film and be kept in good condition;

(f)
have an abstract of all statutory restrictions in respect of any specified working with a reference to the order imposing the same.

(2)
The conventions shown in the Second Schedule shall be used in preparing all plans and sections required by these regulations.

(3)
The plans and sections required by these regulations shall be accurate within such limits of error as the Chief Inspector may specify by a general or special order.

(4)
The plans and sections required under these regulations shall be maintained up-to-date which is not earlier than three months:

Provided that where any mine or seam or section is proposed to be abandoned or the working thereof to be discontinued or rendered inaccessible, the plan and section shall be brought up-to-date before such abandonment or at the time of discontinuance, as the case may be, unless such abandonment or discontinuance has been caused by circumstances beyond the control of the owner, agent or manager, in which case the fact that the plan or section is not up-to-date shall be recorded on it.

(5)
All the reference stations at surface and the reference points of underground surveys shall be shown in their correct position relative to the survey of India national grid within the limits of error of survey and plotting.

(6)
Plans and sections required to be maintained under these regulations shall be kept available for inspection in the office at the mine, and shall not be removed therefrom except by or with the approval in Writing of the Regional Inspector, unless a true copy thereof has been kept therein.

68.

Type of plans.

(1)
the owner, agent or manager of every mine shall keep the following plans and sections:-

(a)
a surface plan showing every surface feature within the boundaries, such as telephone, telegraph or power transmission line, watermain, tram-line, railway, road, river, watercourse, reservoir, tank, bore-hole, shaft and incline opening opencast working, subsidence and building on the surface;

(b)
an underground plan showing-

(i)
the position of the working of the mine below ground;

(ii)
every bore-hole and shaft with depth, incline opening, cross measure drift, goaf, fire-stopping or seal, water-dam (with

dimensions and other particulars of construction), pumping station and haulage roadway;

(iii)

every important surface feature within the boundaries, such as railway, road, river, stream, watercourse, tank, reservoir, opencast working and building which is within 200 metres of any part of the working measured on the horizontal plane;

(iv)

the general direction and rate of dip of the strata;

(v)

such sections of the seam as may be necessary to show any substantial variation in the thickness or character thereof and showing the working section, and such sections of the strata sunk or driven through in the mine or proved by boring as may be available;

(vi)

the position of every roll, washout, dyke and every fault with the amount and direction of its throw and hade;

(vii)

an abstract of all statutory restrictions in respect of any specified working with reference to, the order imposing the same, and

Whenever this plan is brought up-to-date, the then position of the working shall be shown by dotted line drawn through the ends of the working and such dotted line shall be marked with the date of the last survey;

(c)

where a seam has an average inclination of more than 30 degrees from the horizontal, prepare one or more vertical mine section or sections, as may be required by the Regional Inspector, showing a vertical projection of the mine working;

(d)

a ventilation plan, and section were necessary, showing the system of ventilation in the mine, and in particular -

(i)

the general direction of air-current;

(ii)

every point where the quantity of air is measured;

(iii)

every air-crossing, ventilation door, stopping and every other principle device for the regulation and distribution of air;

(iv)

every fire-stopping and its serial number;

(v)

every room used for storing inflammable material;

(vi)

the position of fire-fighting equipment;

(vii)

every water-dam with dimensions and other particulars of construction;

(viii)

every pumping, telephone and ambulance Station; and

(ix)

every haulage and travelling roadway;

(e)

a joint survey plan showing the details required under clause (b) of this regulation 68 and sub-regulation (6) & (7) signed by the surveyor and the manager of the adjoining mine(s) having working within 60 metres of the common boundary or where the boundary is in dispute, within 60 metres of the boundary claimed by the owner of the mine concerned signifying the correctness of the common boundary or the disputed boundaries, as the case may be, and of the position of the working in relation to one another;

(f)

a geological plan of the area of leasehold, on a suitable scale;

(g)

a water-danger plan and section showing-

- (i)
nullah, river, lake, water pond, water course, drainage or any other water bodies on surface or belowground existing upto 200 metres of the boundary of the mine;
- (ii)
the position of the working belowground and every borehole and shaft (with depth), drive, crosscut, staple pit, excavation and air passage connected therewith;
- (iii)
the position of every dyke, fault and other geological disturbance with the amount and direction of its throw as well as hade;
- (iv)
levels taken in working belowground at easily identifiable points sufficient in number to allow the construction of sections along all drives, main headings and haulage roadways;
- (v)
every source of water such as river, stream, water-course, reservoir, water-logged opencast working on the surface, and also, the outline of all water-logged working belowground lying within 60 metres of any Part of the working measured in any direction;
- (vi)
every reservoir, dam or other structure, either above or belowground, constructed to withstand a pressure of water or to control inrush of water, along with reference to its design and other details of construction; and
- (vii)
the highest flood level of the area.

(2)
Separate plans and sections for the working of every seam or of every separate section of every seam shall be kept in respect of clause (b),(c)(d) and (e) of sub-regulation (1):

Provided that in respect of plans maintained under clause (b) of sub-regulation (1), combined plans of all seams, which are lying within nine metres of each other and which are worked at the mine shall also be kept and in the combined plans, working in different seams or sections shall be shown in different colours.

(3)
(a)
The plans maintained under clauses (a), (b), (d), (e), (f) and (g) of sub regulation (1) shall also show surface contour lines drawn at vertical intervals not exceeding five metres (or ten metres in the case of a mine where there are no working belowground or in cases of mines situated in hilly terrain, such other larger interval as the Regional Inspector may permit by an order in writing-and subject to such conditions as he may specify) over the whole area lying within 200 metres of any part of the working.

(4)
The plans maintained under clauses (b) of sub-regulation (1) shall also show spot levels on the floor of the working-
(i)
along all haulage roadways, at every roadway junction, except in roadways where tramming is done by manual means where the spot levels may be shown at points not more than 150 metres apart; and

(ii)
in the case of headings which have been discontinued either temporarily or permanently, also at the end of such headings:
Provided where two drifts in stone or two galleries in coal pass over one another, this shall be clearly indicated on the plans, with appropriate noting, if necessary;

(5)
A permanent bench-mark shall be established on the surface, and all levels taken above and belowground shall be referred to a plane in relation to such bench-mark and particulars of the bench-mark, together with its height above Mean Sea Level, shall be shown on the plans required to be maintained under these regulations.

(6)
The plans maintained under clauses (a) and (b) of sub-regulation (1) shall also show the settled boundary of the mine, or where the boundary is in dispute, the boundaries claimed by the owner of the mine and by the owners of the mines

adjacent to the disputed boundary:

Provided that where it is not possible to show the complete boundary of leasehold on the same plan, an additional key plan or any other suitable scale showing such boundaries and the outline of the working shall also be maintained.

(7)

The plans maintained under clause (b) of sub-regulation (1) shall also show the working, and all features as specified in that clause, both above and belowground of all adjacent mines as are situated within 60 metres, measured on any plane, of the boundary claimed by the owners of the mines.

(8)

The owner, agent and manager of every mine shall as soon as its working extend to within 60 metres of the settled boundary with an adjustment (or where the boundary is in dispute within 60 metres of the boundary claimed by the owner of the adjacent mine) inform the owner, agent or manager of such mine of the fact of such extension and shall also give all reasonable facilities to the surveyors of its adjacent mines to carry out the surveys and levellings required to be made under this sub-regulation.

(9)

The Regional Inspector may, by an order in writing,-

(i)

require such additional details to be shown on the plans and sections under these, regulations, or the preparation and maintenance of such other plans and sections showing such details and on such scale and within such time as he may specify in the order;

(ii)

require the owner, agent or manager to submit to him within such time such plans and sections, or tracings thereof, as he may specify in the order;

(10)

The owner, agent or manager shall, at any time if required by the Regional Inspector, show on any plan or section the then position of the working of the mine.

69.

Plans and sections to be submitted after abandonment after discontinuance.

(1)

Where any mine or seam or section thereof is abandoned or the working thereof has been discontinued over a period exceeding 60 days, the person who was the owner of the mine at the time of abandonment or discontinuance shall, within 30 days after the abandonment or within 90 days after the discontinuance of the Working, as the case may be, submit to the Chief Inspector two true copies of the up-to-date plan and section of the working of the mine or seam or section maintained under clauses(a), (b), (c) and (e) of regulation 68 (1)and every such copy shall show the bearing and distance of at least one of the shafts or openings of the mine from a tri-junction or revenue pillar or from any other prominent and permanent surface feature, the position of all water-dams built belowground (with their dimensions and other particulars of construction) and also the spot levels at the ends of the working:

Provided that if a change of ownership occurs after the abandonment or discontinuance and before the expiry of the 30 days or the 90 days aforesaid, as the case may be; such plans and sections shall be submitted forthwith.

(2)

The original or a certified true copy of the plan and section submitted under sub-regulation (1) shall be kept in the office at the mine.

(3)

The Chief Inspector may, on such conditions as he thinks fit to impose, and on payment of the cost of preparing copies as determined by him, supply copies of a plan or section submitted to him under sub-regulation (1) or such, parts thereof as he thinks fit -

(a)

to any person having a bonafide interest in the mine, seam or section;

(b)

to the owner, agent or manager of an adjacent mine.

70.

Survey instruments and materials.

(1)

The owner or agent shall provide adequate number of accurate and reliable survey instruments and materials for the proper carrying out of all survey and levelling work and for the preparation of plans and sections required under these regulations and no other instruments shall be used in connection with any such survey or levelling work.

(2)

The survey instruments so provided under sub-regulation (1) shall be checked, maintained and calibrated at regular intervals to maintain their accuracy level.

71.

List of plans, sections and instruments and their storage.

(1)

All plans and sections, and tracings or copies thereof, kept at the mine shall be serially numbered.

(2)

Suitable arrangements shall be made at every mine for the proper storage and maintenance of every plan and section and of all instruments and materials and such arrangements shall provide for flat storage of every plan and section maintained under these regulations:

Provided that where special conditions exist the Chief Inspector may by an order in writing, permit storage and maintenance of plans and section in any other form subject to such conditions as he may specify in the order.

(3)

Every field book and other notes used in the preparation of plans and sections required under these regulations shall be duly indexed and kept in the office at the mine.

(4)

A list of all plans and sections maintained under these regulations, or any orders made thereunder, and tracings or copies thereof; of all survey instruments provided under regulation 70 with their respective types specifications and identification numbers; and of all field books and other notes kept under sub-regulation (3) shall be kept in a bound paged book kept for the purpose, and shall be brought up-to-date whenever necessary.

(5)

Every entry in the book shall be signed and dated by the surveyor, and countersigned and dated by the manager.

72.

Preparation of plans by surveyors.

(1)

Every plan and section, and tracing thereof, prepared under these regulations shall be prepared by or under the personal supervision of the surveyor.

(2)

Every plan or section, or any part thereof, prepared by or under the supervision of a surveyor shall carry thereon a certificate by him to the effect that the plan or section or part thereof is correct and shall be signed and dated by the surveyor and countersigned and dated by the manager on every occasion that the plan or section is brought up-to-date.

(3)

Every tracing of a plan or section or of any part thereof shall bear a reference to the original plan or section from which it was copied and shall be certified thereon by the surveyor to be a true copy of the original plan or section and the certificate shall be signed and dated by him.

(4)

If the surveyor fails or omits to show any part of the working or allow the plans or sections to be inaccurate, he shall be guilty of a breach of these regulations:

Provided that nothing in this sub-regulation shall, exempt the owner, agent or manager of their responsibility to ensure that every plan or section prepared, kept or submitted under these regulations or by any order made thereunder is correct and maintained up-to-date as required thereunder.

(5)

The owner, agent, or manager of every mine shall ensure that the plans and sections required to be kept and maintained under these regulations shall be surveyed and prepared using the latest survey instruments having facilities of automatic surveying, leveling, data storage and transfer to computers and plotters of compatible configurations incorporating all the required provisions of regulations in this regard:

Provided that in case of any difficulty arising during the working or compliance of this regulation, it shall referred to the Regional Inspector and the Chief Inspector who may change or specify the changes if so required by special order in writing.

73.

Plans to be checked on change of ownership or on re opening, etc.

(1)

When there is a change in ownership of a mine, or where a mine or part thereof is reopened, or where in any mine or part thereof it is intended to start any extraction or reduction of pillars, the owner, agent and manager shall ensure that the plans and sections of the mine or part are accurate and if any doubt arise as to the accuracy of the plans and sections in any respect, he shall have accurate plans and sections prepared afresh before any drifage or other work of development or of extraction or reduction of pillars is commenced.

(2)

If the Regional Inspector is of the opinion that any plan or section prepared, kept or submitted under this regulation is inaccurate, he may, by an order in writing, require a fresh survey made and a new plan or section prepared within such time as he may specify therein.

(3)

If the plan or section required to be prepared under sub-regulation (2) is not prepared within the time specified in the order, or to the satisfaction of the Regional Inspector, or the plan or section is not prepared or brought up-to-date as required under these regulations, he may get the plan or section prepared by any other agency and the cost thereof, as certified by the Chief Inspector, shall be defrayed by the owner of the mine and recoverable from him as an arrear of land revenue.

Chapter-VII Means of Access and Egress

74.

Outlets from a mine.

(1)

Except for the duration of shaft sinking or the drifage of other means of access and egress from the mine together with the necessary development work, no person shall be employed, or be permitted to enter or remain for purpose of employment, in any working below ground, unless the working is provided with at least two separate means of access to the mine and egress to surface from the mine:

(a)

with which every seam or section for the time being at work has a communication so as to afford separate means of ingress and egress to the persons employed therein;

(b)

which do not have their surface openings in the same building; and

(c)

which are under the sole control of the manager:

Provided that the Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit the employment belowground of persons even in a case where the two shafts, inclines or outlets are not under the control of the same manager.

(2)

The Regional Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit the employment belowground of restricted number of persons even in a case where places in the working of the mine do not contain two means of egress from each working place.

(3)

Suitable arrangements shall be made for persons to descend and ascend by each of such means of access and egress; such as

(a)

where the shaft is more than 30 metres in depth, such arrangements shall be by mechanical means so installed and maintained as to be constantly available for use; and

(b)

where the incline is more than one and half kilo metres in length (one way) or the travel by persons is arduous, a

suitable man-riding system approved in writing by the Chief Inspector shall be provided for persons to access and egress from the working of the mine:

Provided that in case of any doubt as to whether any such arrangement referred to clause(a) or clause (b) is suitable or not, it shall be referred to the Chief Inspector for decision.

(4)

Both the means of egress and the equipment used therein for the transport of persons shall be kept constantly in a safe condition to facilitate ease of traveling, including clearances for hand carried stretchers.

(5)

Where in any shaft, ladders are used as a means of ingress or egress of persons employed in a mine, every such ladder shall -

(i)

be of strong construction;

(ii)

be securely fixed in the shaft at an inclination of not more than 80 degrees from the horizontal;

(iii)

be made continuous or without perceptible overlapping or break except at platforms which shall be provided at intervals of not more than nine metres;

(iv)

project at least one metre above the mouth of the shaft, and above every platform, except where strong holdfasts or handrails are provided;

(v)

have rungs equally spaced and at a sufficient distance from the wall or any timber to ensure proper foothold; and

(vi)

be maintained in good repair.

(6)

Such shafts, inclines or outlets shall not be less than 13.5 metres distant from one another at any point, and each shall be connected with the other by means of a walkable passage, not less than 1.8 metres high and 1.5 metres wide, through the working belowground that are being served by such shafts, inclines or outlets.

(7)

Whenever the connection between two outlets which are required to be maintained under sub-regulation (1) has been obstructed or found dangerous, only such persons as are necessary to clear the obstruction or to repair the dangerous part of the connection or to make a new second outlet, as the case may be, shall be employed belowground until such time as the connection has been re-established or a new second outlet has been provided.

(8)

The foregoing provisions of this regulation with respect to shafts, inclines and outlets shall not apply -

(a)

to any working for the purpose of making a connection between two or more shafts, inclines or outlets; and

(b)

to any working for the sole purpose of searching for or proving minerals;

Provided that nothing in this sub-regulation shall be deemed to authorise the driving of roadways for the development of a seam before a second outlet has been made in accordance with the said provisions.

(9)

In both the means of access and egress in the mine, an effective two way communication facilities with broadcasting system of telecommunication shall be provided to ensure that persons from belowground may be able to directly communicate to surface without any obstruction or loss of message and communication link.

(10)

The system of communication and telecommunication so provided under sub-regulation (9) shall be of wired, wireless or any other type, as approved by the Chief Inspector.

(11)

The System, configuration and layout of the communication and telecommunication network to be provided in the mine either on surface or belowground as well as from surface to belowground shall be approved in writing by the Chief

Inspector.

75.

Working shafts.

(1)

All entrances to the mine shaft shall be adequately illuminated throughout working hours.

(2)

Every shaft in use or in course of being sunk and every incline or other outlet shall be made and kept secure.

(3)

Every shaft in the course of being sunk shall be provided with a permanent lining of metal, concrete or masonry, which shall at no time, be more than six metres from the bottom of the shaft:

Provided that where iron or steel rings with a substantial lagging are used below the permanent lagging and are kept close to the bottom of the shaft, this distance may be increased to not more than 20 metres:

Provided further that in the case of a shaft where special conditions exist which make compliance with the provisions of this sub-regulation not necessary, the Regional Inspector may, by an order in writing and subject to such conditions as he may specify therein, grant exemption from the compliance thereof.

(4)

All surface and seepage water shall be channeled in such a way as to prevent it from falling freely into the shaft.

(5)

The top, all insets and bottom of every working shaft and the sump thereof shall be kept clear and free from loose materials, tools and debris.

76.

Fencings and gates at outlets.

(1)

Every entrance to mine from the surface, and the top and all entrances between the top and bottom, including the sump, if any, of every working, ventilating or pumping shaft, shall be kept securely fenced so designed and constructed as to prevent any person accidentally falling down the shaft or coming into contact with a moving part of the hoisting equipment provided in the shaft.

(2)

Every walkable entrance from the surface to the working belowground shall be provided with a substantial gate which shall be kept closed and locked when there are no persons belowground:

Provided that where such entrance is not used as a means of ingress or egress in or out of the mine, it shall be permanently closed so as effectively to prevent persons from entering therein.

77.

Outlets from mine parts.

- Every part of a mine shall, where practicable, be provided with at least two ways of affording means of egress to the surface:

Provided that if any doubt arises as to whether the provision of two such ways is practicable or not, it shall be referred to the Chief Inspector for decision.

78.

Periodic examination, etc. of outlets.

(1)

Every shaft, incline and other outlets provided as required by regulation 74 shall be examined, once at least in every seven days, by an overman or other competent person and a report of every such examination shall immediately thereafter be recorded in a bound paged book kept for the purpose and shall be signed and dated by the person making the examination.

(2)

The mechanised outlet shall be examined at least once in every 7 days by an engineer or electrical supervisor or foreman or any other competent person only authorised in writing by the engineer and a report of every such examination shall immediately be recorded in a bound paged book kept for the purpose and signed and dated by the person making the examination.

(3)

The bound paged book so maintained under sub-regulation (2) by the electrical supervisor or foreman or any other competent person shall be checked and countersigned by the engineer and manager.

(4)

If at the time of such examination or at any other time, the shaft, incline or other outlet is found to be not safe, it shall not be used for any purpose, except as a natural airway, until it has been made safe in all respects and a report of every such action taken shall be recorded in the book kept under sub-regulation (1).

Chapter-VIII Winding in Shafts

79.

Appointment of winding enginemmen.

(1)

No person shall be appointed as a winding engineman unless he holds an Engine Driver's Certificate:

Provided that this sub-regulation shall not apply to the driver of an electrical winding engine upto 30 HP or of a steam or compressed air winding engine which has cylinders not exceeding 18 centimetres in diameter and which is not used for raising or lowering of persons.

(2)

Where special difficulties exist which make compliance with the provisions of sub-regulation (1) not reasonably practicable, the Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, relax the said provisions.

(3)

No person, other than a winding engineman appointed under sub-regulation (1) shall operate any winding engine:

Provided that in an emergency any other competent person may be permitted to operate the winding engine.

(4)

The name of the winding engineman on duty, together with the period of his shift shall be posted in the winding engine room:

Provided that where the Regional Inspector is of the opinion that the duties of any winding engineman are unduly arduous, he may by an order in writing require the period of his shift to be reduced to such period, not less than five hours, as he may specify.

80.

New winding installations.

(1)

Where it is intended to bring into use of any new winding installation for lowering and raising of persons, the owner, agent or manager shall, not less than 60 days before such use, give notice of his intension to the Chief Inspector in such form as may be specified by the Chief Inspector which shall contain detailed specifications of the installation.

(2)

A winding engine which is shifted from one site to the other within the same mine or from one mine to other shall be considered as a new installation for the purpose of this regulation.

(3)

If the Chief Inspector, by an order in writing requires, such additions or alterations shall be made to the installation, as he may specify in the order.

81.

Construction and installation of winding equipment.

(1)

Every part of a winding installation, including headgear shall be of sound construction and adequate strength, and shall be maintained in safe working order:

Provided that in case of any doubt as to the construction and strength of the installation, it shall be referred to the Chief Inspector for decision.

(2)

The engine shall be firmly connected to a rigid foundation and shall be so designed, constructed and maintained that with the power provided, the raising and lowering of persons or materials can be carried out with ease regularity and safety.

(3)

Unless otherwise permitted in writing by the Chief Inspector and subject to such conditions as he may specify therein, every engine for winding shall be so situate in relation to the headgear that the winding rope shall not, in the extreme position, subtend in either direction an angle more than one and a half degrees with the plane of the sheave or pulley used in connection with the rope.

(4)

The diameter of the drums or sheaves of the winding engine, and of the pulleys and sheaves used in connection with the winding shall, unless otherwise permitted in writing by the Chief Inspector and subject to such conditions as he may specify therein, be not less than 80 times the diameter of the rope in the case of winding installations installed before the 25th day of October, 1955 and not less than 100 times the diameter of the rope in other cases:

Provided that the Chief Inspector may, by an order in writing, require that in the case of any specified winding installation installed before the date aforesaid, the diameter of the said drums, pulley or sheaves shall not be less than such size, upto 100 times the diameter of the rope, as he may specify in the order.

(5)

The grooves of sheaves or pulleys used in winding installation shall be suited to the diameter of the rope.

(6)

Every pulley or sheave used in connection with winding shall, while in motion, rotate in a vertical plane, and shall be maintained in such a condition that slipping is reduced to a minimum.

82.

Fittings of winding engines.

(1)

At every Shaft, including a shaft in the course of being sunk, where winding is effected by means of an engine, the following provisions regarding winding engines shall have effect, namely:-

(a)

there shall be on the drum such flanges, and also if the drum is conical or spiral such other appliances, as will be sufficient to prevent the rope from slipping or coiling unevenly;

(b)

except in friction winder, the end of the rope shall be securely fixed in such a manner that the rope is not unduly strained;

(c)

there shall be at least two turns of the rope on the drum when the cage or other means of conveyance is at its lowest working point in the shaft.

(2)

There shall be provided one or more brakes on the drum or the drum-shaft, which -

(i)

if there are two cages or other means of conveyance, will hold such cages or other means of conveyance when the maximum torque is applied in either direction; or

(ii)

if there is only one cage or other means of conveyance, will hold the loaded cage or other means of conveyance in midshaft when the maximum torque is applied downwards:

Provided that at least one of the brakes shall be so designed that the brake remains at the 'on' position except when operated.

(3)

Where the brake or brakes are power-operated, at least one of them shall be arranged to be applied automatically at all times if the power supply fails.

(4)

The brake on the drum shall be used only for the purpose of keeping such drum stationary and not for lowering the cage or other means of conveyance, except in cases where the engine is to be worked at a very low speed as when examining the winding rope or the shaft.

(5)

Where the winding engine is worked by steam or compressed air, a screw stop-valve shall not be used as controlling valve of the engine.

(6)

Every engine shall be equipped with a reliable depth-indicator (in addition to any mark on the rope) showing to the winding engineman the position of the cage or other means of conveyance in the shaft, and an automatic device that will ring a bell in the engine room when the ascending cage or other means of conveyance is at a distance of not less than two revolutions of the drum from the top of the shaft.

(7)

The depth-indicator referred to sub-regulation (6) shall be tested after every adjustment or replacement of the winding rope.

83.

Shaft fittings.

(1)

At every winding shaft, other than a shaft in the course of being sunk to which the provisions of regulation 87 shall apply, the following provisions shall have effect, namely:-

(a)

two independent and efficient means shall be provided and maintained for interchanging separate, distinct and definite signals between the top of the shaft and -

(i)

the bottom or other permanent landing of the shaft; and

(ii)

every inset for the time being in use,
and one of such system shall be by electrical means.

(b)

there shall be provided and maintained efficient means for transmitting signals from the top of the shaft to the winding engineman which shall be transmitted by mechanical or electrical means.

(c)

the following code of signals shall be used and strictly observed:-

ONE RAP

:

STOP

when engine in motion

ONE RAP

:

RAISE

when engine at rest.

TWO RAPS

:

LOWER

when engine is at rest

THREE RAPS

:

MEN

ready to ascend or descend

THREE RAPS

:

In Reply

men may enter the cage or other means of Conveyance:

Provided that any other signals shall be in addition to, and shall not interfere with the foregoing.

(d)

a printed copy of the code of signals, including additional signals, if any, shall be posted prominently at the top of the shaft and at every such inset and landing and also in the winding engine room.

(e)

no person other than the banksman or onsetter shall give any signal unless he is an official of the mine or is authorised in writing by the manager to give signals.

(f)

any defect in the signaling system shall be immediately reported to a supervisory official who shall take action to have the defects rectified.

(g)

in addition to the system of signaling provided in the shaft and the winding system, another means of two way communication/telecommunication system shall also be provided so that the winding engine driver, banksman, onsetter, persons travelling inside the cage or any other person can contact and communicate with each other without any difficulties and hindrance.

(2)

The shaft shall be provided with sufficient number of guides to ensure smooth and safe travel of the cage or other means of conveyance:

Provided that where rope guides are used, the cheese-weights or bottom clamps shall be kept so exposed as to permit regular examination.

(3)

Above the topmost landing, 'dogs' or other devices for holding the cage or other means of conveyance in the event of an overwind shall be provided, or the guides, runners or receivers shall be sprung.

(4)

Except in friction winders, at the top of every shaft where cages are used, suitable keps shall be provided and so arranged as to fall into the 'on' position when the operating lever is released.

(5)

At every inset where keps or folding platforms are provided, arrangements shall be made to lock the keps or platforms securely in the off position; and when in the 'off' position such keps or platforms shall leave the shaft clear for the passage of the cage and in every such case, a proper automatic indicator to show the position of the keps or platforms, as the case may be, shall be provided in such a position as to be easily seen by the banksman.

(6)

Protective roofing, sufficient to prevent danger from anything falling in the shaft, shall be provided and maintained at the bottom of the shaft and the gap, both vertical and horizontal, between the protective roofing and the top of cage, when the cage is at the bottom of the shaft, shall not exceed 15 centimeters.

84.

Man Winding.

- At every shaft, other than a shaft in the course of being sunk, where a winding engine is used for the purpose of lowering or raising persons, the following provisions shall have effect, namely: -

(a)

In respect of every part of the winding installation, including pulleys or sheaves, cages, chains, distribution plates and detaching hooks, the following particulars shall be recorded in a bound paged book kept for the purpose: -

(i)

name of the manufacturer and the year of manufacture;

(ii)

specifications and dimensions;

(iii)

reference to every certificate supplied with the part;

(iv)

date of installation of each part;

(v)

any other detail that may be necessary or required by the Regional Inspector; and

(vi)

all entries in the book shall be made and signed by the engineer or other competent person, and shall be countersigned and dated by the manager.

(b)

Whenever any part or article is replaced or any repaired part or article is used in any winding installation, the fact of such replacement or repair shall be recorded in the book kept under clause (a).

(c)

A single linked chain shall not be used, except for the short coupling chain attached to a cage or other means of conveyance which shall be attached to the safety hook through a distribution plate or other approved appliance.

(d)

Where safety-chains are used, their length shall be such that if the king-bolt breaks, the shock to the cage or other means of conveyance is as minimum as possible.

(e)

Where drum-clutches are provided, the following provisions shall have effect, namely: -

(i)

the operating gear of the clutch of the drum shall be provided with locking gear to prevent inadvertent withdrawal of the clutch; and

(ii)

every engine used for the lowering or raising of persons shall have a suitable interlocking device so fitted that it is not possible to unclutch any drum unless the brakes of such drum are applied or to release the brakes until the drum clutch is fully engaged and securely locked.

(f)

Except when the cage or other means of conveyance attached to the drum is resting at the bottom of the shaft, the drum shall not be unclutched unless the winding engineman has assured himself immediately beforehand that the brake is fully applied.

(g)

Except in friction winders, there shall be provided between the rope and the cage or other means of conveyance, a detaching hook.

(h)

The space between the detaching hook referred to under clause (g), measured from the centre of the hole for attaching it to the rope shackle, and the detaching-bell or plate when the cage or other means of conveyance is at its normal position at the top of the shaft, shall be not less than 1.8 metres where a geared engine is used, and not less than 3.6 metres where a direct acting engine is used.

(i)

In every shaft, the engine shall be fitted with an automatically recording speed indicator.

(j)

In every shaft, there shall be provided an effective automatic contrivance to prevent overspeeding and overwinding, hereinafter called the 'Automatic Contrivance' which shall prevent the descending cage from being landing at the pit bottom or other permanent landing at a speed exceeding 1.5 metres per second and shall also control the movement of the ascending cage in such a manner as to prevent danger to persons riding therein.

(k)

The Chief Inspector may, by an order in writing, specify the maximum speed of winding in any shaft.

(l)

Tests of every Automatic Contrivance and every brake shall be made by the engineer or other competent persons appointed for the purpose, in the following manner:-

(i)

once at least in every seven days, by raising each cage or other means of conveyance, in turn, to pass the last control point above the topmost landing;

(ii)

once at least in every three months, by attempting to land the descending cage at excessive speed for which, the setting of the Automatic Contrivance may be altered so that pre-determined point in the shaft is regarded as the landing; and

(iii)

the results of every such tests shall be recorded in a bound paged book kept for the purpose, and shall be signed and dated by the person making the test.

(m)

Unless the Automatic Contrivance shall be fully engaged, either automatically or by the winding engineman, whenever persons are to be lowered or raised and an automatic indicator shall be provided in such a position to indicate the banksman that that Automatic Contrivance is fully engaged.

(n)

The banksman shall not allow any person to enter a cage or other means of conveyance until the indicator shows That the Automatic Contrivance has been fully engaged.

(o)

Except where an Automatic Contrivance is provided to prevent overwinding, a point shall be fixed and marked on the indicator of the engine in such a way as to show when the cage or other means of conveyance is at a distance of not less than twice the circumference of the drum from the completion of the wind; and if such cage or other means of conveyance contains persons, the winding engineman shall not, as soon as it has reached the point aforesaid, raise it for the remaining distance at a speed exceeding 1.2 metres per second.

(p)

Where the only means of egress in a mine is by apparatus worked by steam or electricity, precautions shall be taken to ensure that the two winding engines do not fail simultaneously, and in particular, in the case of electric winding engines, the engines shall be capable of being connected to two separate power supplies and unless otherwise directed by the Chief Inspector by an order in writing, the provisions of this clause shall be deemed to have been complied if an emergency winding gear is maintained.

(q)

Every cage or other means of conveyance in which persons ride, shall be -

(i)

covered completely at the top;

(ii)

closed in at the two sides in a manner sufficient to prevent persons or things from projecting beyond the sides;

(iii)

provided with a rigid hand-bar fixed in a position where it can be easily reached by all persons in the cage or other means of conveyance;

(iv)

provided with suitable gates or other rigid fences such that the gap between the floor of cage or other means of conveyance and the lowest part of the gate or fence does not exceed 15 centimetres and that between any two members of the gate or fence does not exceed 25 centimetres and gates or fences shall not open outwards and shall be so fitted and maintained that they cannot be accidentally opened; and

(v)

provided with an effective means of communication or telecommunication system having provisions of audio and visual, data and digital display including broadcasting system approved in writing by the Chief Inspector.

(r)

The floor of every cage or other means of conveyance shall be strongly constructed and so maintained as to prevent any part of the body of a person riding in the cage or other means of conveyance from projecting beyond the floor.

(s)

Not more than such number of persons as may be authorised by the manager shall be allowed to ride in the same cage or same deck of a cage or other means of conveyance at one time providing approximately 0.20 square metre of floor area per person and a notice specifying the number shall be posted at the top and bottom of every shaft and at every inset.

(t)

No person shall, when ascending or descending a shaft, take with him any bulky material other than tools and instruments, except when engaged in repairing the shaft or with the written authority of the manager.

(u)

Except as provided in clause (t), no person shall ride in a cage while materials or tubs are being raised or lowered in any of the cages or other means of conveyance.

(v)

The Chief Inspector may, subject to such conditions as he may specify, relax the requirements of this regulation if the

circumstances in any mine or part thereof are such as to tender compliance with such requirements not reasonably practicable.

85.

Multi-decks.

- Where a cage has two or more decks which are used simultaneously, each floor at a landing shall be connected by an effective signalling device with the main floor of the landing; and only the banksman or the onsetter or an official, as the case may be, at such main floor shall give action signal after he has satisfied himself that all cage gates are closed.

86.

Winding of material.

(1)

Every cage used for the raising or lowering of tubs shall be provided with catches or other effective contrivances to prevent the tubs falling out and the cage shall not be set in motion unless the catches or other effective contrivances are in position.

(2)

The floor of every cage shall be kept clean; and no skip, bucket or tub shall be filled up to such height that any of the contents can fall out.

(3)

Before long, timber, pipes, rails, or other material projecting over the top of the cage or other means of conveyance are lowered or raised, the projecting ends shall be securely fastened to the rope, chain or bow.

87.

Winding in sinking shafts.

- At every shaft in the Course of being sunk, where a winding engine is used, the following provisions shall have effect, namely:-

(a)

If the shaft exceeds 45 metres in depth, there shall be provided for each bucket or other means of conveyance a detaching-hook between the centre of the hole for attaching the detaching-hook to the rope shackle and the detaching bell or plate and when the bucket or other means of Conveyance is at the top landing, there shall be a clear over-run space of not less than 3.6 metres.

(b)

Where the shaft exceeds 150 metres in depth -

(i)

the bucket or other means of conveyance, when used for lowering or raising persons, shall be provided with sufficient cover overhead for protection from things falling down the shaft; and

(ii)

there shall be provided for each bucket or other means of conveyance, a sufficient number of guides which shall be kept extended to within 22.5 metres of the shaft bottom at all times when sinking is in progress:

Provided that the Regional Inspector may by an order in writing, require the guides to be provided in a shaft less than 150 metres in depth.

(c)

There shall be provided and maintained-

(i)

two separate means of interchanging distinct and definite signals between the bottom and the top of the shaft; and

(ii)

efficient means for transmitting such signals from the top of the shaft to the winding engineman.

(d)

The signaling appliances shall be examined by a competent person once at least in every 24 hours and the result of every such examination shall be recorded in a bound paged book kept for the purpose and shall be signed and dated by the person making the examination.

(e)

Except with the permission in writing of the Chief Inspector and subject to such conditions as he may specify therein, the following code of signals shall be used and observed in signaling:-

ONE RAP

STOP

when engine in motion

ONE RAP

TAKE UP SLACK

when engine at rest.

ONE RAP

RAISE SLOWLY

TWO RAPS

LOWER

THREE RAPS

TAKE UP SLACK

men are riding:

Provided any other signals shall be in addition to, and shall not interfere with, the foregoing.

(f)

A printed copy of the code of signals, including additional signals, if any, shall be posted prominently at the top of the shaft and also in the winding engine room.

(g)

Except while riding in a bucket or other means of conveyance, no person other than the person authorised in writing by the manager, shall give any signal.

(h)

Every bucket or other means of conveyance in which persons or materials are conveyed, shall be of strong construction and so maintained as to prevent persons or materials from falling.

(i)

At the top of the shaft or at the landing where the bucket or other means of conveyance is normally landed, suitable covering with door shall be provided and except as may be required for the passage of the bucket or other means of conveyance, the covering and doors shall always be kept closed.

(j)

Where the shaft exceeds 45 metres in depth, the persons working at the bottom of the shaft shall also be protected by an adequate protective covering extending over the whole area of the shaft which shall be provided with a door for the passage of the bucket or other means of conveyance, to be kept lowered to within 22.5 metres of the bottom of the shaft at all times when sinking is in progress:

Provided that where special circumstances exist, the Chief Inspector may, by an order in writing and subject to such conditions, as he may specify therein, grant an exemption from the provisions of this clause:

(k)

Not more than such number of persons as may be authorised by the manager shall be allowed to ride in the bucket or other means of conveyance at one time; and a notice specifying such number shall be posted prominently at the top of the shaft.

(l)

When tools, implements or other materials are lowered or raised, the banksman or an authorised persons, as the case may be, shall ensure that -

(i)

the bucket is properly loaded;

(ii)

materials are not loaded above the rim;

(iii)

long timber, pipes, rails, tools or other material with ends projecting over the rim are securely fastened to the rope, chains or bow; and

(iv)

the bucket, before being sent away, is steadied, and the bottom and sides thereof are free from adhering material.

(m)

Where guides are provided, the bucket or other means of conveyance shall be raised slowly from the bottom of the shaft, until the rider is picked up.

(n)

While persons are at work on any scaffold or platform in the shaft, the following precautions shall be strictly observed:-

(i)

the scaffold or platform shall be secured to the sides of the shaft in order to prevent it from swinging;

(ii)

the opening for the passage of the bucket or other means of conveyance, shall be so protected as effectively to prevent anything falling through it;

(iii)

the scaffold or platform shall not be lowered or raised except under the order of the authorised persons or pother competent person.

88.

Winding ropes, etc.

(1)

No rope, bar, link, chain or other attachment to a cage or other means of conveyance shall be used unless it is of good quality and manufacture, is free from any visible defect and is of adequate calculated strength:

Provided that the Chief Inspector may, by an order in writing prohibit the use of any rope or type of ropes where, in his opinion such use is unsafe.

(2)

The attachment between the rope and the cage or other means of conveyance shall be of such type and be maintained in such manner as to obviate accidental disconnection.

Provided that in case of a doubt, as to the fitness of any rope, bar, link, chain or other attachment used or intended for use, it shall be referred to the Chief Inspector for decision.

(3)

Except in a sinking shaft less than 30 metres in depth, every winding rope shall be made of cold drawn steel wire, and the gauge of the wires used in the construction of such rope shall be suited to the diameter of the drums, pulleys and sheaves of the winding installation.

(4)

In any shaft, including a shaft in course of being sunk, where persons are lowered or raised and where guides are not provided, no rope other than a rope of non-spinning type shall be used.

(5)

No rope which has been spliced shall be used for winding purposes.

(6)

Subject to such exemption as may be granted by the Chief Inspector in writing and any conditions as he may specify therein, no rope the braking load of which at anyone point therein is less than 10 times the maximum static load on it when the cage or other means of conveyance attached to the end of the rope is at the lowest working point, shall be used.

(7)

At every mine where a shaft is used for lowering or raising persons, at least one spare winding rope suitable for use in such shaft, shall, be kept in store.

(8)

For every rope in use or intended for use, a certificate showing its breaking load, quality, construction and diameter (obtained from the manufacture or supplier) and a history of its use, including a record of diameters of the drums, sheaves and pulleys used in conjunction with the rope, shall be kept in a bound paged book kept for the purpose and all entries therein shall be made and signed by the engineer or other competent person, and shall be countersigned and dated by the manager.

(9)

If in the case of a rope a test certificate as to the amount of its breaking load is not available, it shall not be used unless a portion thereof, not less than three metres in length, has been cut off from the end of the rope attached to the cappel and tested in a laboratory, institution or test house approved by the Central Government for the purpose.

(10)

No winding rope which has been in use for more than three and half years shall be used for winding purposes:

Provided that where the Regional Inspector is satisfied that due to sparing use any such rope is in good condition even after the expiry of the said period, he may, by an order in writing and subject to such conditions as he may specify therein, allow the use of such rope for a longer period:

Provided further that where the Regional Inspector is of the opinion that any rope has become unsafe for use in a shaft before the expiry of the period of three and a half years aforesaid, he may, by an order in writing, prohibit the use of such rope for winding purposes:

Provided also that an appeal against any order of prohibition of the Regional Inspector for use of the rope before expiry of three and half years may be preferred to the Chief Inspector.

(11)

Every application for permission to use a rope after the period of three and a half years shall be accompanied by a copy of the entries, in respect of the rope, in the book kept for the purpose under sub-regulation (8), and also by a certificate obtained in a manner laid down in the sub-regulation (9) as to the strength of the rope not more than three months prior to the date of the application.

(12)

No mode or type of capping shall be used, which fails to withstand a load of at least 10 times the maximum static load thereon.

(13)

The cappel of a round rope shall not be attached to the rope by rivets passing through the rope.

(14)

No bent back wire type cappel shall be used with winding rope.

(15)

Where white metal is used in the capping of ropes, the tapered portion of the socket shall not be less than six and a half times the diameter of the rope for the size up to 26 millimeter and seven and one third times in case of ropes of diameters higher than 26 millimeter.

(16)

If white metal is used in the capping of ropes, -

(i)

its melting point shall not exceed 300 degrees centigrade, and its temperature When poured into the socket shall not exceed 363 degree centigrade;

(ii)

in the length of rope which is to lie within the tapered part of the socket, the fibre core, if any, shall be cut and the wires shall be untwisted and thoroughly cleaned; and

(iii)

the socket shall be heated to a temperature of about 100 degrees centigrade before the white metal is poured into it.

(17)

Except in friction winder -

(i)

every rope shall be recapped once at least in every six months, or if necessary, at shorter intervals and also after every overwind; and

(ii)

before every recapping, a length, including the capping, of at least two metres shall be cut off the rope and every piece of rope so cut-off shall be opened and its internal condition examined.

(18)

The recapping of rope shall be carried out under the supervision of the engineer or other competent person, who shall record the date and other particulars thereof (including the length of the rope remaining after recapping) in a bound paged book kept for the purpose and shall sign and date the same.

89.

Suspension gear.

(1)

All parts of the suspension gears shall be of a type as approved by the Chief Inspector.

(2)

All parts of suspension gear in regular use shall unless otherwise permitted by the Chief Inspector, be renewed after a period of service of not more than six years, and at shorter intervals if necessary.

(3)

The factor of safety shall not be less than 10 for components of suspension gears and threaded joints shall be avoided: Provided that wherever it is not practicable to avoid a threaded joint, a factor of safety not less than 15 shall be ensured.

(4)

All cage chains in general use and all other parts of suspension gear between the rope and the cage or other means of conveyance, including the detaching-hook, shall be taken apart, cleaned and carefully examined as to wear and tear (where necessary by gauging) and for rust and cracks, once at least in every six months, or if necessary, at shorter intervals.

(5)

The various parts shall be annealed or given other proper heat treatment in a proper furnace where the temperature can be controlled before being refitted:

Provided that in the case of such chains or gear manufactured from a steel which is not liable to deterioration necessitating annealing or heat treatment, the Chief Inspector may by an order in writing and subject to such conditions as he may specify therein, grant exemption from the carrying out of this operation:

Provided further that detaching hooks used in sinking shafts shall be taken apart, cleaned and carefully examined once at least in every week and the shear pin replaced by a new one every time such examination is carried out

(6)

Non-destructive testing shall also be adopted for testing of vital components of machinery, such as drum shafts, brake tie rods and suspension gears at regular intervals and any harmful crack or flaw detected as a result of such tests shall be immediately reported to the Regional Inspector and use of such machinery or its components shall be discontinued forthwith.

(7)

Every detaching bell or plate used in connection with a safety-hook shall be examined, and the opening therein checked by calipers or gauges, once at least in every 30 days.

(8)

The operation and examination required under this regulation shall be carried out by or under the supervision of an engineer or other competent person, who shall record the date and other particulars thereof in a bound paged book kept for the purpose, and shall sign and date the same.

90.

Precautions after recapping, etc.

- After every installation or recapping of a rope and after every renewal or refitting of any suspension gear, the engineer or other competent person shall, after the cages or other means of conveyance fully loaded with materials have made five trips up and down the working portion of the shaft, examine the cappel and other parts of the suspension gear to see that they are in proper working order and a report of every such examination shall be recorded in the bound paged book and shall be signed and dated by the persons making the examination.

91.

Examination of winding equipment.

(1)

It shall be the duty of the engineer or other competent person to examine-

(a)

once at least in every 24 hours, -

(i)

the attachment of the winding rope to the drum, the depth indicator, every part of the suspension gear in the shaft, including cages or other means of conveyance and their gates, and every external part of the winding apparatus, upon the proper working of which the safety of persons depends; and

(ii)

the brakes of the winding engines;

(b)
once at least in every seven days, -
(i)
each winding rope, by passing the rope at a speed not exceeding one meter per second; and

(ii)
the external parts of the winding engine, the guides and the signalling arrangements fitted in a shaft;

(c)
once at least in every 30 days, every winding rope, by passing the rope at a speed not exceeding 0.5 metre per second. For the purpose of this examination, by cleaning the rope of any encrusted dirt and grease at all places particularly liable to deterioration and at other places, not more than 30 metres apart throughout the length, and any reduction in the circumference of the rope and the superficial condition of the wires as to wear, corrosion, brittleness and fracture at every such place shall be noted; and

(d)
once at least in every 12 months, the winding engine as to the condition of its internal parts.

(2)
A report of every such examination under sub-regulation (1) shall be recorded in a bound paged book kept for the purpose, and shall be signed and dated by the person who made the examination.

(3)
If on any examination made, there is discovered any weakness or defect by which the safety of persons may be endangered, such weakness or defect shall be immediately reported in writing to the engineer or other competent person and to the manager and the winding installation shall not be used until such weakness or defect is remedied.

92.

Gates and fences.

(1)
At the top of every shaft and at every inset which is in use, there shall be provided suitable gates or fences which shall effectively close the openings into the shaft at all times when a cage or other means of conveyance is not at the top or the inset and every such gate at the top of a shaft shall be self-operating.

(2)
At every landing where it is necessary for persons to pass from one side of the shaft to the other, an adequate bypass shall be provided for enabling them to do so without entering or crossing the shaft and every bypass so provided shall be not less than 1.8 metres high and 1.2 metres wide, and shall be kept clear of all obstructions.

(3)
No person shall enter or cross, or be permitted to enter or cross the exposed space at the bottom of any working shaft except for the purpose of entering or leaving a cage or other means of conveyance or for undertaking an examination, repair or any other work therein; and no person shall be allowed to work in any such space unless the cages or other means of conveyance, if any, have been stopped and adequate precautions have been taken for the protection of such person.

93.

Duties of persons riding or working in shafts.

(1)
No person shall get on or off a cage or other means of conveyance after the same has been signaled to be set in motion or leave it until it has reached the appointed stopping place; nor shall any person ride on the top or edge of any cage or other means of conveyance except when engaged in an examination, repair or any other work in the shaft.

(2)
Every person, when at or about the top or bottom of a shaft or any inset, shall obey the lawful orders and directions of the banksman or onsetter, as the case may be.

(3)
No person shall carry out any examination, repair or other work in any shaft while winding operations are being carried on; and no winding shall be carried on or permitted while persons are engaged in such examination, repair or work, except where winding is necessary for the same.

(4)

The person in immediate charge of any examination, repair or work in any shaft shall warn the banksman and the winding engineman that such examination, repair or work is about to be undertaken.

(5)

Every person while engaged in any examination, repair or other work in a shaft shall be accompanied by at least one other person; and all such persons shall be provided with safety belts of a type approved by Chief Inspectors and efficiently protected against the risk of falling.

(6)

Every person engaged in carrying out an examination, repair or other work in a shaft shall be protected by a suitable covering from objects falling from above and every such person shall also be provided with a protective hat; and shall wear the same when so engaged.

94.

General precautions.

(1)

No unauthorised person shall enter or be allowed, in a winding engine room.

(2)

No adolescent shall descend or ascend a shaft in a cage or other means of conveyance unless accompanied by one or more adult male.

Chapter-IX Haulage

95.

Haulage roadways.

(1)

Every length of roadway in a mine where materials are transported in tubs by means of gravity or mechanical power shall-

(a)

be of adequate dimensions and, as far as practicable, shall be straight and of regular gradient; and

(b)

have tracks properly laid with rails having at least 15 kg/metre section for general purpose and suitable heavier section where the gradient steeper than 1 in 6 or where mine cars of capacity higher than 1.5 cum are in use.

(2)

Pulleys, sheaves and rollers that alter the direction of a rope shall be securely fixed.

(3)

No person shall guide or adjust a moving rope on to a drum, pulley, sheave or roller except with a lever or other proper appliance.

(4)

Material for use in mine haulage tracks and standard in respect of construction, layout inspection of haulage tracks shall be in accordance with the guidelines notified by the Chief Inspector from time to time.

(5)

where haulage is effected by one or more ropes, there shall be provided and maintained-

(a)

at the top of every inclined plane, at least one stop-block or other effective contrivance to arrest tubs from running or moving out of control;

(b)

at least one run away switch or other effective contrivance below the first stop-block or other effective contrivance at a distance greater than the length of a set or train of tubs:

Provided that such distance shall not exceed the length of a set or train of tubs by more than 10 metres:

Provided further that where the Regional Inspector, by an order in writing so requires, the stop-block and the run away switch or other effective contrivance shall be so inter-coupled that they do not remain simultaneously ineffective.

(c)

an attachment, behind an ascending tub or set or train of tubs, a back-stay, drag or other suitable contrivance for preventing the tub, set or train of tubs running back:

Provided that where an endless rope or chain is used, the provisions of this clause shall be deemed to be satisfied if

suitable automatic catches or other effective contrivance are provided at suitable intervals along the track to prevent the ascending tubs running back:

Provided further that the Regional Inspector may, by an order in writing and subject to such conditions as he may specify therein, grant exemption from the operation of this clause on grounds that compliance with the provisions thereof are not reasonably practicable;

(d)
safety hooks, jazz-rails or other suitable contrivances to prevent runaway in the forward direction;

(e)
tub re-railers at intervals of not more than 250 metres:

Provided that where a tub is re-railed manually, it shall either be detached from the rope or ropes or the haulage engine shall be stopped; and

(f)
on every haulage roadway exceeding 30 metres in length, effective means of transmitting signal by mechanical or electrical means, from every stopping place on the roadway to the place at which the machinery working the rope is operated:

Provided that the Regional Inspector may, by an order in writing, require means of transmitting signals in the reverse direction also.

(g)
if any doubt arises as to whether any means of transmitting signal under clause (f) of sub-regulation (5) is effective or not, it shall be referred to the Chief Inspector for decision.

(6)
The following code of signals shall be used and strictly observed: -

ONE RAP

STOP

when in motion

TWO RAPS

LOWER

or haul in slowly

THREE RAPS

START

when at rest

FOUR RAPS

RAISE

or haul out slowly,

and any other signals shall be in addition to, and shall not interfere with, the foregoing.

(7)
A printed copy of the code of signals under sub-regulation (6) including additional signals, if any, shall be posted prominently at the place in which the machinery that works rope is operated and at all regular stopping places along the roadway.

(8)
No person other than a competent person or an official shall give any signal.

(9)
Where in any mine belowground, a system of haulage roadway (and conveyors, if any) extends to a distance of more than 300 metres from the shaft or the entrance to the mine, efficient telephonic communication shall be provided and maintained between the end of every such system and the bottom and top of the shaft or the entrance to the mine as the case may be: -

Provided that where travelling is unguly arduous, the telephonic communication between the end of every such system and the bottom and top of the shaft or the entrance to the mine or in any other, case as may be specified by the Regional Inspector shall be provided and maintained.

(10)
Where telephones or electrical signals are provided,

(a)

adequate precautions shall be taken to prevent signal and telephone wires coming into contact with other cables and electrical apparatus;

(b)

signal wires shall be supported on insulators, and shall not be energised at more than 30 volts;

(c)

contact makers shall be so constructed as to prevent accidental closing of the circuit; and

(d)

in every gassy seam of the second or third degree, all signaling or telephonic communication circuit shall be constructed, installed, protected, operated and maintained in such a manner as to be intrinsically safe.

(11)

At places where telephone receivers are installed or where signals and safety contrivances are regularly operated, every person using the telephone or operating any such signal or safety contrivance shall be afforded adequate protection against tubs moving out of control.

(12)

Where any person is allowed to work or pass while the haulage is in motion, manholes for refuge shall be provided at intervals of not more than 10 metres:

Provided that where the gradient is less than 1 in 6, such manholes may be provided at intervals of not more than 20 metres.

(13)

Manholes shall be not less than 1.8 metres in height and 1.2 metres in depth, and not less than 0.75 metres but not more than one metre in width:

Provided that where the roadway is less than 1.8 metres in height, the manholes may be made to the full height of the roadway:

Provided further that the Regional Inspector may, by an order in writing and subject to such condition as he may specify therein, permit the use as manholes or cross-roadways other than haulage roadways of dimensions larger than those aforesaid.

(14)

Where there are serious practical difficulties in providing manholes of the interval and the dimension specified in sub-regulations (12) and (13), the Regional Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit manholes to be at greater intervals or of other dimensions.

(15)

Every manhole shall be kept clean and clear of obstruction, and white washed both inside and upto a distance of not less than 0.3 metres around the aperture.

(16)

As far as practicable, all manholes shall be provided on one side of the haulage roadway.

(17)

Except where haulage is effected by means of an endless rope or chain, whenever the haulage rope is in motion, every person on the haulage roadway shall take shelter in a manhole.

(18)

The manager shall, by an order in writing, in respect of every haulage road or roadway fix the maximum number of loaded or empty tubs to be coupled together to run as a set or train, and a notice specifying the number of tubs so fixed shall be posted prominently at the top and at all regular stopping places of the haulage road or roadway.

(19)

At all places where tubs are coupled or uncoupled, there shall be a clear space of not less than one metre-

(a)

between, the tubs and one side of the roadway; and

(b)

where there are two or more tracks also between the adjacent tracks.

(20)

When any roadway or face is in direct line with a haulage track and persons may be exposed to danger from runaway

tubs, a strong buffer or other effective contrivance to prevent such danger shall be provided and maintained.

(21)

A stop-block or other effective contrivance shall be provided near the entrance of every tramming roadway branching off the main haulage road or roadway, and on every track which slopes towards a shaft.

96.

Travelling roadways.

(1)

Separate travelling roadways in the intake airways for persons to travel to and from their working places shall be provided.

Provided that where a separate traveling roadway cannot be provided and persons have to travel along a part of haulage roadway, the following precautions shall be taken:

(a)

there shall be provided and maintained effective two way means of transmitting signals between the two ends of the part of the roadway where persons may travel;

(b)

the manager shall fix specific periods for traveling; and notices specifying these periods shall be posted prominently at the top and bottom of and also at all regular stopping places along the haulage roadway;

(c)

any person who of necessity has to travel along the roadway outside the period fixed for traveling shall not be allowed to do so unless and until a special signal has been given to the haulage engineman; and

(d)

inexperienced persons shall not be permitted to travel the roadway or part unless accompanied by an experienced person:

Provided further that where the travelling roadways cannot be made in the intake air way, the matter shall be brought to the notice of the Regional Inspector who may permit in writing the use of travelling roadways other than in intake airways subject to the such conditions as he may specify therein.

(2)

Every travelling roadway shall -

(a)

be not less than 1.8 metres high throughout;

(b)

where the inclination exceeds 30 degrees from the horizontal, be provided with suitable steps or ladders;

(c)

where the inclination exceeds 45 degrees from the horizontal, be provided, in addition to steps or ladders, with handrails or ropes so as to ensure safe travel;

(d)

where the inclination exceeds 60 degrees from the horizontal, be provided, in addition to the steps or ladders and rails or ropes, with suitable platforms at intervals not exceeding 10 metres measured along the slope;

(e)

be provided with effective means of telecommunication facilities at suitable places; and

(f)

be provided with adequate number of digital display boards and communication ports through which any important message or information can be easily transmitted or broadcasted to persons working belowground or passing thereby.

(3)

Except for purposes of inspection, examination or repair, every person other than an official or a haulage attendant shall travel by the travelling roadway.

(4)

Where persons using a travelling roadway have to cross a conveyor or a haulage worked by mechanical means or gravity, a suitable cross-over or cross-under bridge or other suitable device approved in writing by the Regional Inspector shall be provided.

(5)

No haulage shall be used for the general conveyance of persons except with the permission in writing of the Chief Inspector and subject to such conditions as he may specify therein.

(6)

In case the traveling distance from the incline mouth or pit bottom exceeds one kilometer or the traveling is arduous, the owner, agent and manager shall provide suitable man-riding arrangement as approved by the Chief Inspector.

97.

Tubs and their movement.

(1)

On every tub there shall be provided and maintained at each coupling end a strong buffer projecting beyond the end and so arranged that when two such tubs are in tandem, the gap between the innermost ends shall not be less than 20 centimeters.

(2)

On every side-tipping tub in use, safety-catches shall be provided to prevent accidental tipping and no tub or set or train of tubs shall be set in motion unless all the safety catches are properly secured.

(3)

The attachment between a rope or locomotive and a tub or set or train of tubs and the attachment between any two tubs in a set or train, shall be of a type approved in writing by the Chief Inspector by a general or special order and so maintained as to obviate accidental disconnection.

(4)

The state of every buffer and drawbar of every tub in use and of every safety-catch, coupling-chain and other attachment shall be examined once at least in every 14 days, by a competent person appointed for the purpose and a report of every such examination shall be recorded in a bound paged book kept for the purpose, and shall be signed and dated by the person who made the examination.

(5)

Each component of coupling shall have factor of safety of not less than 7 in relation to maximum static load which shall be ensured by testing at an interval not exceeding 3 years and record thereof shall be maintained.

(6)

When tubs are about to be moved, persons likely to be endangered shall be warned.

(7)

Two or more tubs shall not be moved by hand in close succession but shall be coupled and moved together:

Provided that two tubs shall be deemed to be in close succession when the distance between them at any time is less than 10 metres.

(8)

No person shall cause or permit a tub to run uncontrolled except with the written permission of the manager:

Provided that the Regional Inspector may, by an order in writing, prohibit the uncontrolled movement of tubs at any place if he is of the opinion that such movement is likely to cause danger.

(9)

No person while taking a tub down a gradient exceeding 1 in 20, shall go in front of the tub; and in every case where conditions are such that a person cannot control the tub from behind, he shall not take the tub down unless sprags or other suitable contrivances are used to control it.

(10)

Where required for use, a sufficient number of sprags of suitable material and dimensions shall be provided.

(11)

Every tub while standing on a track having a gradient of more than 1 in 20 shall unless held effectively by brakes or securely coupled to a haulage rope or locomotive, be effectively blocked, chained or otherwise secured.

(12)

Except where haulage is effected by means of an endless rope, the coupling and uncoupling to tubs shall as far as practicable, be done only when the tub or set of the tubs, and the rope if connected to the set is not in motion.

(13)

Tubs shall not be coupled or uncoupled on a gradient.

(14)

No person shall ride on any tub or haulage rope.

98.

Brakes of haulage engines.

- Every haulage engine shall be provided with an effective brake.

99.

Haulage ropes.

(1)

No rope shall be used for purposes of haulage if it has any serious visible defect over any length and its factor of safety is less than 8.

(2)

Every rope which is capped shall be recapped once at least in every six months, and if necessary, at shorter intervals, under the supervision of a competent person.

(3)

For every haulage rope in use, a record of size, construction, quality, name of supplier and dates of installation and of recapping shall be kept in a bound paged book kept for the purpose, and all entries therein shall be made by the competent person who shall sign the same and date his signature.

100.

Roadway conveyors.

(1)

Every roadway conveyor shall be so installed that-

(a)

between the conveyor and one side of the roadway, there is a travelling space from obstruction not less than one metre wide;

(b)

the conveyor or any part thereof does not scrape against wooden props or supports;

(c)

the anchoring of the return station of the conveyor is independent of the face or roadway support;

(d)

in case a number of belt conveyors are used in series, safety fittings such as sequential control and sequential interlock shall be provided; and

(e)

it can be stopped from any place along the entire length of the conveyor by providing a pull cord switch or any other system, approved in writing by the Chief Inspector.

(2)

Where the inclination of the conveyor is such as to give rise to danger from sliding objects or material, suitable devices shall be used to provide adequate protection against such danger.

(3)

On every length of roadway in which a conveyor is installed for transporting loads over a distance exceeding 30 metres, there shall be provided and maintained effective means of transmitting signals from every point on the length of the road to the place at which the machinery working the conveyor is operated:

Provided that the Regional Inspector may, by an order in writing, require means of transmitting signals in the reverse direction also.

(4)

The conveyor operator and the cabin or place from where the conveyor is operated shall be provided with an effective means of telecommunication along with broadcasting facilities through which the operator can communicate to any person present in the conveyor roadway at any place of the installation and such system of telecommunication shall have facilities of both way communications.

(5)

Audio-visual Pre-start warning alarm of the type and standard approved by the Chief Inspector shall also be provided in the entire length of the roadway conveyor so as to warn persons of imminent dangers due to starting of the belt conveyor.

(6)

The manager shall formulate a code of practice for safe installation, operation, maintenance and use of belt conveyor belowground including extension of belt conveyor and shifting it from one place in the mine to another and submit the same to the Regional Inspector at least 30 days before the commencement of the installation of the belt conveyor belowground, and the Regional Inspector may at any time by an order in writing, require such modification in the Code of Practice as he may think fit in the interest of safety.

(7)

In case the belt conveyor is intended for the purpose of man-riding, the manager shall formulate a separate code of practice for safe installation, operation, maintenance and use of the belt conveyor for the said purpose in a specified location in the mine and shall submit the same to the Regional Inspector at least 90 days before the commencement of the installation of the said belt conveyor seeking permission for the said use and the Regional Inspector may at any time by an order in writing modify or revoke such permission as he may think fit in the interest of safety of persons using the same.

(8)

The manager and engineer shall both be responsible for implementation of the code of practice.

(9)

Adequate arrangement shall be provided in the belt conveyor to ensure that it gets automatically stopped,

(a)

in case of excessive friction between the belt and the drum, roller, scrapper, deflectors, guides or any other obstruction caused either due to spillage or otherwise;

(b)

in case of breakage of the belt; and

(c)

in case of fire or heating in the belt conveyor or in the vicinity thereof.

101.

Examination of haulage engines.

(1)

It shall be the duty of a competent person to examine carefully -

(a)

once at least in every 24 hours, every haulage engine, brake-wheel, rope and other appliance in use; and

(b)

once at least in every seven days, every track where the haulage is effected by means of mechanical power or gravity, and every safety contrivance fitted thereon.

(2)

A report of every such examination under sub-regulation (1) shall be recorded in a bound paged book kept for the purpose, and shall be signed and dated by the person who made the examination.

102.

Examination of haulage and travelling roadways.

- It shall be the duty of the overman or other competent person to examine carefully, once at least in every seven days, the state of all haulage and travelling roads and roadways, including roadways leading to all the outlets of the mine which are in use and a report of every such examination shall be recorded in a bound paged book kept for the purpose, and shall be signed and dated by the person who made the examination.

103.

Locomotives.

(1)

No locomotive shall be used belowground otherwise than in accordance with the permission in writing of the Chief Inspector and subject to such conditions as he may specify therein.

(2)

No locomotive shall be used where the gradient of the track exceeds 1 in 15.

(3)

No person other than the driver shall ride on any locomotive unless authorised in writing to do so by the manager.

(4)

Except during shunting operations, the locomotive shall lead the tubs or set or train of tubs.

104.

Roads for trucks and dumpers.

(1)

The design, construction, materials and layout of the haul roads shall be made in accordance with the standards and parameters required for the load and speed of the vehicles plying thereon which shall be approved in writing by the Chief Inspector.

(2)

All roads for trucks, dumpers or other mobile machinery shall be designed, constructed and maintained in such condition as to be fit for their use.

(3)

Except with the express permission of the Chief Inspector in writing and subject to such conditions as he may specify therein, no road shall have a gradient steeper than 1 in 16 at any place.

(4)

Where practicable, all roads from the opencast working shall be arranged to provide for one-way traffic and where not practicable, no road shall be of a width less than three times the width of the largest vehicle plying on that road, unless, definite turnouts and waiting points are designated.

(5)

All corners and bends in road shall be made in such a way that the operators and drivers of vehicles have a clear view upto a distance not less than 30 metres along the road.

Provided that where it is not possible to ensure visibility upto a distance of not less than 30 metres, there shall be provided two roads for the incoming and outgoing traffic or an alternative system:

Provided further that adoption of an alternative system shall be intimated to the Regional Inspector in writing.

(6)

Standard traffic signs shall be displayed at conspicuous places along the haul road.

(7)

Road surface shall be smooth and of sufficient strength.

(8)

All roads shall be provided with side drains to prevent water logging and damage to the road.

(9)

In case of ramps over small stretches, a gradient not steeper than 1 in 10 shall be provided.

(10)

Where any road exists above the level of the surrounding area, it shall be provided with strong parapet wall or embankment with a height of not less than half the diameter of the wheel of the largest dumper or other vehicle plying or one meter, whichever is greater and of adequate width to prevent any vehicle from getting off the road.

(11)

The portion of roads where there is heavy traffic of men and machine shall have separate lane properly fenced off from haul road for use by pedestrians and two wheelers.

(12)

Loaded trucks, dumpers or other vehicles shall not be reversed on gradient.

(13)

A berm of height at least half the diameter of the wheel of the largest dumper plying or one meter whichever is greater and of adequate width shall constantly be provided and maintained at the edge of the spoil or coal dump.

(14)

Sufficient number of stop-blocks shall be provided at every discharge point and it shall be used every time material is dumped from the truck, dumper or other such vehicle.

(15)

The manager shall formulate traffic rules for movement of trucks, dumpers or other vehicles which shall be prominently displayed at the relevant places in the opencast working and truck/dumper roads.

(16)

No person shall be permitted to work on the chassis of truck, dumper or other vehicles with the body in a rest position unless the truck or dumper body has been securely blocked in position and the mechanical hoist mechanism alone shall not be dependent upon to hold the body of the truck, dumper or other vehicles, in raised position.

(17)

Separate roads shall be provided for small vehicles.

(18)

Only authorised small vehicles shall be allowed in opencast working which have a raised red flag in day time and a red light in the night which shall be visible to a dumper operator from a distance of at least 30 metres.

(19)

All heavy earth moving machinery including light motor vehicle permitted to ply on the haul road shall be provided with flasher lights on top, indicator lights in front and rear sides of the vehicle as approved by the Chief Inspector.

(20)

Anti-Collision and Proximity Detection Device of a standard and type approved by the Chief Inspector shall be installed on every Heavy Earth Moving Machinery.

105.

Movement of wagons.

(1)

Only an adult shall be employed in moving railway wagons.

(2)

The movement of railway wagons shall be carried on under the supervision of a competent male person who shall himself control the brake.

(4)

Before wagons are moved, persons likely to be endangered shall be warned by the competent persons appointed under sub-regulation (2).

(5)

No person shall move or attempt to move a wagon by pushing at the buffer, or by pulling from in front.

(6)

Where two or more wagons are moved simultaneously, the wagons shall be coupled together which shall be moved only by pushing from the sides or from behind the last wagon and the number of such wagons shall not exceed beyond which can be effectively controlled.

(7)

No locomotive or wagon shall be moved when the natural light is insufficient, unless the approaching end is distinguished by a suitable light or is accompanied by a person carrying a lamp.

(8)

No person, other than the competent person referred to in-sub-regulation (2), shall pass immediately in front of wagons moving under bins or screens, nor between moving wagons and the under-structure of the bins or screens.

(9)

No person shall be upon the buffer of a locomotive or wagon in motion unless there is a secure hand-hold, or stand thereon or a secure footplate.

(10)

No person shall pass over the coupling between any two wagons while the wagons are moving.

(11)

No person shall cross a line of rails by crawling or passing underneath a train or wagon, nor shall a person sit or sleep underneath a wagon.

(12)

Wherever railway wagons are specially placed so as to afford a thoroughfare, such thoroughfare shall be not less than five metres in width.

(13)

No material shall be placed or dumped within 1.2 meters from either side of track of rails.

(14)

All space between the rails at switches and crossings in which the foot of a person is liable to be caught shall be kept

filled with concrete, tar, asphalt, or wooden blocks.

106.

Fencings and gates.

(1)

Where any haulage road, tramline, rail line or haul road passes over a public road, suitable gates shall be Provided to prevent danger to public from a moving tub set or train of tubs, locomotive or machinery and every such gate shall be fitted with a danger signal, and when the natural light is insufficient, also with warning lamps.

(2)

Where occupied buildings are situated within 15 metres of any haulage road, tramline, rail line or haul road, a substantial fence shall be provided and maintained between such buildings and the haulage road, tramline, rail line or haul road.

Chapter-X Mine Working

107.

Safety management plan.

(1)

The owner, agent and manager of every mine shall-

(a)

identify the hazards to health and safety of the persons employed at the mine to which they may be exposed while at work;

(b)

assess the risks to health and safety to which all employees may be exposed while they are at work;

(c)

record the significant hazards identified and risks assessed;

(d)

make those records available for inspection by the employees; and

(e)

follow an appropriate process for identification of the hazards and assessment of risks.

(2)

The owner, agent and manager of every mine, after consulting the safety committee of the mine and Internal Safety Organisation, shall determine all measures necessary to-

(a)

eliminate any recorded risk;

(b)

control the risk at source;

(c)

minimize the risk; and

(d)

in so far as the risk remains,

(i)

provide for personal protective equipment; and

(ii)

institute a program to monitor the risk to which employees may be exposed.

(3)

Based on the identified hazards and risks, owner, agent and manager of every mine shall prepare an auditable document called "Safety Management Plan" that forms part of the overall management and includes organisational structure, planning, activities, responsibilities, practices, procedures, processes and resources for developing, implementing, achieving, reviewing and maintaining a safety and health policy of a company.

(4)

It shall be the duty of the owner, agent and manager to implement the measures determined necessary and contained in the Safety Management Plan for achieving the objectives set out in sub-regulation (2) in the order in which the measures are listed in the said sub-regulation.

(5)

The Safety Management plan shall contain-

- (a) defined mine safety and health policy of the company;
- (b) a plan to implement the policy;
- (c) how the mine or mines intend to develop capabilities to achieve the policy;
- (d) principal hazard management plans;
- (e) standard operating procedures;
- (f) ways to measure, monitor and evaluate performance of the safety management plan and correct matters that do not conform with the safety management plan;
- (g) a plan to regularly review and continually improve the safety management plan;
- (h) a plan to review the safety management plan if significant changes occur;
- (i) details of involvement of mine workers in their development and application.

(6)

The owner, agent and manager of every mine shall periodically review the hazards identified and risks assessed, to determine whether further elimination, control and minimization of risk is possible and consult with the safety committee on the review.

(7)

The owner, agent or manager of every mine shall submit a copy of the 'Safety Management Plan' to the Regional Inspector who may, at any time by an order in writing, require such modifications in the plan as he may specify therein.

(8)

The owner, agent and manager of every mine shall be responsible for effective implementation of the 'Safety Management Plan'.

108.

Manual opencast working.

- In manual opencast working, the following precautions shall be observed, namely: - (1) In alluvial soil, morum, gravel, clay, debris or other similar ground, -

(a)

the sides shall be sloped at an angle of safety not exceeding 45 degrees from the horizontal or such other angle as the Regional Inspector may permit by an order in writing and subject to such conditions as he may specify therein; or

(b)

the sides shall be kept benched and the height of any bench shall not exceed 1.5 metres and the breadth thereof shall not be less than the height:

Provided that the Regional Inspector may, by an order in writing and subject to such conditions as he may specify therein, exempt from the operation of this clause in any working in the case of which special difficulties exist, which in his opinion make compliance with the provisions thereof not reasonably practicable; and

(2)

Where any pillar is left 'in situ' for the purpose of measurement, its height shall not exceed 2.5 metres; and where the height of such pillar exceeds 1.25 metres, the base of the pillar shall not be less than 1.5 metres in diameter.

(3)

In coal, the sides shall either be kept sloped at an angle of safety not exceeding 45 degree from the horizontal, or the sides shall be kept benched and the height of any bench shall not exceed three metres and the width thereof shall not be less than the height:

Provided that the Regional Inspector may, by an order in writing and subject to such conditions as he may specify

therein, exempt from the operations of this sub-regulation any working, in the case of which special difficulties exist which in his opinion make compliance with the provisions thereof not reasonably practicable.

(4)

In an excavation in any hard and compact ground or in prospecting trenches or pits, the sides shall be adequately benched, sloped or secured so as to prevent danger from fall of sides.

Provided that the height of the bench shall not exceed 6 metres.

(5)

No tree, loose stone or debris shall unless otherwise permitted in writing by the Chief Inspector, be allowed to remain within a distance of three metres from the edge or side of the excavation.

(6)

No person shall undercut any face or side or cause or permit such undercutting as to cause any overhanging.

109.

Mechanised opencast working.

- In all mechanised opencast working, the following precautions shall be observed, namely - (1) Before starting a mechanised opencast working, design of the pit, including method of working, ultimate pit and dump slope shall be planned and designed as determined by a scientific study and a copy of the report of such study shall be kept available in the office of the mine:

Provided that in case of mines where such a study has not been made, it shall be the responsibility of the owner or agent to get the aforesaid study made within one year from the date of coming into force of these regulations.

(2)

The height of the benches in overburden consisting of alluvium or other soft soil shall not exceed 3 metres and the width thereof shall not be less than three times the height of the bench and the height of the benches in overburden of other rock formation shall not be more than the designed reach of the excavation machine in use for digging, excavation or removal.

(3)

The width of any bench shall not be less than-

(a)

the width of the widest machine plying on the bench plus two metres, or

(b)

if dumpers ply on the bench, three times the width of the dumper, or

(c)

the height of the bench, whichever is more.

(4)

Notwithstanding anything contained in sub-regulations (1), (2) and (3), the Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, require or permit the height and width of benches to be less than or more than aforesaid.

110.

Reclamation.

- All excavations made by opencast mining shall be suitably reclaimed by back filling or by any other means.

111.

Spoil-banks and dumps.

(1)

While removing overburden, the top soil shall be stacked at a separate place, so that, the same is used to cover the reclaimed area.

(2)

The slope of a spoil bank shall be determined by the natural angle of repose of the material being deposited but shall not exceed 37.5 degree from the horizontal:

Provided that if scientific studies indicate possibility of information of steeper slope of a spoil bank, approval of such slope angle subject to the conditions specified shall be obtained from the Regional Inspectors.

(3)

Loose overburden and other such material from opencast working or other rejects from washeries or from other source

shall be dumped in such a manner that there is no possibility of dumped material sliding.

(4)

Any spoil bank exceeding 30m in height shall be benched so that no bench exceeds 30m in height and the overall slope shall not exceed 1 vertical to 1.5 horizontal.

(5)

The toe of a spoil-bank shall not be extended to any point within 100m of a mine opening, railway or other public works, public road or building or other permanent structure not belonging to the owner.

(6)

A suitable fence shall be erected between any railway or public works or road or building or structure not belonging to the owner and the toe of an active spoil bank so as to prevent unauthorised persons from approaching the spoil-bank.

(7)

No person shall approach or be permitted to approach the toe of an active spoil bank where he may be endangered from material sliding or rolling down the face.

(8)

Adequate precautions shall be taken to prevent failure of the slopes of the spoil banks or dumps.

112.

Transport Rules.

(1)

The manager of every mine shall frame and enforce a code of transport rules with due regard to the size and capacity of the transportation machinery in use and prevailing local conditions and a copy of the same shall be submitted to the Regional Inspector, who may at any time, by an order in writing require such modifications in the transport rules as he may specify therein:

Provided that in mines where such machinery are already in use, the aforesaid transport rules shall be framed and enforced within 90 days from the date of coming into force of these regulations.

(2)

The manager shall hand over copies of the Transport rules to all supervisory officials concerned including the assistant managers and shall also post such copies at all conspicuous places in the mine in language(s) comprehensible by the workers.

(3)

The Manager and such supervising officials shall each be responsible for securing effective compliances with the provisions of the Transport rules, and no mine or part of a mine shall be worked in contravention thereof.

113.

Codes of Practice.

(1)

The manager of every mine shall, before commissioning of any excavator or drilling machinery, other than hand held drills, frame and enforce a code of practice for every type of machinery to be used in the mine.

(2)

The codes of practice shall be framed with due regard to the type, size and capacity of the machinery in use and prevailing local conditions and a copy of the same shall be submitted to the Regional Inspector, who may at any time, by an order in writing require such modifications in the codes as he may specify therein:

Provided that in mines where such machinery are already in use, the aforesaid codes of practice shall be framed and enforced within 90 days from the date of coming into force of these regulations.

(3)

The codes of practices shall cover -

(a)

examination and testing of the machinery before first use after installation, reinstallation, modification, alteration or repair;

(b)

schedule and nature of examination and testing of the machine, including its subassemblies, so as to ensure its safe operation;

(c)

the manner in which the records of examination shall be kept.

(4)

A copy of the codes of practice framed under sub-regulation (3) shall always be kept in the office of the mine.

114.

Development work.

(1)

The dimensions of pillars and galleries, and the shape of pillars, formed in any seam or section shall be such as to ensure stability during the formation and extraction of pillars and during the period between such formation and extraction.

(2)

Save with the previous permission in writing of the Regional Inspector and subject to such conditions as, he may specify therein no gallery in a seam or section shall exceed three metres in height or 4.8 metres in width at any place,

(3)

The pillars formed in any seam or section shall normally be rectangular in shape.

(4)

The distance between the centres of any two adjacent pillars left in a seam or section shall not be less than that specified in the table below as corresponding to the depth of the seam or section from the surface at that point and the width of the galleries in the working in question.

Table

Depth of seam from surface

Where the width of galleries does

not exceed 3.0 meters

Where the width of galleries does

not exceed 3.6 meters

Where the width of the galleries

does not exceed 4.2 meters

Where the width of the galleries

does not exceed 4.8 meters

The distance between centres of adjacent pillars

shall not be less than

1

2

3

4

5

Metres

Metres

Metres

Metres

Not exceeding 60 metres

12.0

15.0

18.0

19.5

Exceeding 60 but notExceeding 90 metres

13.5

16.5

19.5

21.0

Exceeding 90 but notexceeding 150 metres

16.5

19.5

22.5

25.5

Exceeding 150 but not exceeding 240 metres

22.5

25.5

30.0

34.5

Exceeding 240 but not exceeding 360 metres

28.5

34.5

39.0

45.0

Exceeding 360 metres

39.0

42.0

45.0

48.0

(5)

The Chief Inspector may by an order in writing, subject to such conditions as he may specify therein, exempt a mine or part thereof to form pillars in variance to that specified in the table under sub-regulation (4).

(6)

Nothing in sub-regulation (2),(3),(4) and (5) shall apply to working in a mine made before 7th September, 1926. The following provisions shall apply to all working made prior to 7th September, 1926 the except during the extraction or reduction of pillars :-

(a)

If the distance between the centres of adjacent pillars is smaller than that specified in the table under sub-regulation (4), the pillars shall not be further reduced; or

(b)

If the distance between the centres of adjacent pillar is not smaller than that specified in the table under sub-regulation (4), the pillars shall not be so reduced as to render such distance smaller than -

(i)

the distance so specified; or

(ii)

any distance required in this behalf by the Regional Inspector; and

(c)

The height and width of the galleries shall not be further increased without the permission in writing of the Regional Inspector and subject to such conditions as he may specify therein.

(8)

In the case of all working, where in the opinion of the Regional Inspector the dimensions of pillars or galleries are such as to render it likely that crushing of pillars or the premature collapse of any part of the working will occur either before or during the extraction of pillars, he may, by an order in writing require such modification of the dimensions aforesaid in respect of any future working as he may specify.

115.

Depillaring operations.

(1)

No extraction or reduction of pillars shall be commenced, conducted or carried out except with the permission in writing of the Regional Inspector and in accordance with such conditions as he may specify therein.

(2)

An application in prescribed proforma as given in schedule-III (form) for permission under sub-regulation (1) shall be accompanied by two copies of an up-to-date plan of the area where pillars are proposed to be reduced or extracted

showing the proposed extent of extraction or reduction of pillars, the manner in which such extraction or reduction is to be carried out.

(3)

The extraction or reduction of pillars shall be conducted in such a way as to prevent, as far as possible the extension of a collapse or subsidence of the goaf over pillars which have not been extracted.

(4)

Save as otherwise provided under sub-regulation (5), no pillars shall be reduced or split in such a manner as to reduce the dimensions of the resultant pillars below those required by regulation 114 or by any order made thereunder nor shall any gallery be so heightened as to exceed three metres.

(5)

During the extraction of pillars, no splitting or reduction of pillars or heightening of galleries shall be affected for a distance greater than the length of two pillars ahead of the pillar that is being extracted or reduced:

Provided that where pillar extraction is about to begin in a district, such splitting or reduction of pillars or the heightening of galleries shall be restricted to a maximum of four pillars and the width of the split-galleries shall not exceed the width prescribed for galleries under regulation 114(4).

(6)

The Regional Inspector may, by an order in writing and stating the reasons therefor, relax or restrict the provisions of this sub-regulation (4) or sub-regulation (5) in respect of any specified working to such extent and on such conditions as he may specify therein.

(7)

Whether the method of extraction is to remove all the coal or as much of the coal as practicable, and to allow the roof to cave in, the operations shall be conducted in such a way as to leave as small an area of un-collapsed roof as possible with due regard to danger from an air blast or weighting on pillars and where possible, suitable means shall be adopted to bring down the goaf at regular intervals.

(8)

Where the voids formed as a result of extraction are stowed with sand or other materials, the owner, agent or manager shall, on or before the 10th day of every month submit to the Regional Inspector a statement giving the quantity of coal raised and the quantity of sand or other material stowed in every district during the preceding month.

116.

Extraction of Coal by method other than bord & pillar system.

(1)

No development or extraction of coal by a system other than bord and pillar system shall be commenced, conducted and carried out except with permission in writing of the Chief Inspector and in accordance with such conditions as he may specify therein.

(2)

An application for permission under this sub-regulation (1), shall be made in Performa given in form-I of the Third Schedule accompanied by two copies of an upto date plan of the area where such extraction is proposed, showing the proposed extent of extraction and the manner of extraction.

117.

Saving clause.

(1)

Nothing in regulation 114 or regulation 115 shall prevent the driving of any gallery through any pillar or the enlargement of any gallery beyond the limits specified by or under those regulations, wherein the opinion of the manager such work is necessary for haulage, ventilation, drainage or any other purpose necessary for the proper working of the mine, if 14 days previous notice in writing of the intention to commence such work has been given to the Regional Inspector.

(2)

Every notice under sub-regulation (1) shall be accompanied by an offset plan showing details of the operation.

(3)

If in the opinion of the Regional Inspector such work under sub-regulation (1) is likely to endanger the stability of the working, he may, by an order in writing, require the completion, before commencing such drivage or enlargement, of such protective works as he may specify therein.

118.

Roads and working places.

(1)

The roof and sides of all working places and travelling roadways, including airways and travelling road-ways to second outlets, shall be made and kept secure.

(2)

Proper provision shall be made to prevent premature collapse of working; and adequate steps shall be taken to isolate, control or remedy any such collapse which may occur.

(3)

Whenever crush of pillars or any symptom of an impending collapse other than ordinarily caused by pillar extraction is detected, the manager shall inform the Regional Inspector forthwith.

119.

Powers of Inspectors.

(1)

If in any mine or part thereof, it appears to the Regional Inspector that the provisions of regulations 108, 109, 114, 115 and 118 or of any order issued under any of those regulations have not been complied with, he may give notice in writing to the owner, agent or manager requiring him to take such, protective measures, within such time as he may specify in the notice.

(2)

In case of non-compliance with the requirements of the notice issued under sub-regulation (1), the Regional Inspector may, by an order in writing, prohibit the extraction of coal in the part or parts of the mine in which protective measures are required to be taken, until the requirements specified in the notice are complied with.

120.

Pointing out of contraventions during inspections.

(1)

If the Chief Inspector or an Inspector, during his inspection of any mine, finds or comes to know of any contravention of any provisions of the Act or the regulations, rules, bylaws or orders made thereunder, he shall enter such contravention in an interleaved paged and bound register kept for the purpose at the mine, in Form VI and shall also point out such contravention to the owner, agent or manager, if present on the spot.

(2)

The Chief Inspector or the Inspector making the entry in the register of the contravention under sub-regulation (1) shall duly sign such entries with date, and take a carbon copy of the entries for his record:

Provided that the Chief Inspector or the Inspector need not enter such contravention which require confirmation after a surveyor other further examination and he may subsequently intimate the owner, agent or manager, specifying the contravention, if confirmed, and also any other contraventions which were, by inadvertence, not entered in the register aforesaid:

Provided further that an entry made in the register or the absence of an entry therein as also a communication in pursuance with the aforesaid proviso or absence thereof, shall not in any way limit the duties or obligations of a person under the Act or the regulations, rules, bye-laws or orders made thereunder.

(3)

When an entry is made in the register,-

(a)

the owner, agent and manager shall each be deemed to know what is contained in that entry; and

(b)

a copy thereof Shall be displayed within one day of the date of such entry on the notice board of the mine for not less than fifteen days.

(4)

The owner, agent or manager of the mine shall return one copy, within a period not exceeding fifteen days from the date of the entry, to the Chief Inspector or the Inspector who made the entry with remarks thereon showing the action taken to remedy the contravention and the date on which such action was taken.

(5)

The register-

(a)

shall be kept available for inspection in the office of the mine for a period of at least three years after the date of making of the last entry in it; and

(b)

shall not be removed therefrom before the expiry of the aforesaid period, except by or with approval in writing of the Regional Inspector.

121.

Multi-section and contiguous working.

(1)

No work in a higher seam or section shall be done over an area in a lower seam or section which may collapse.

(2)

No working shall be made in more than one section in any seam, nor shall working be made in any two seams lying within nine metres of each other, without the prior permission in writing of the Chief Inspector and subject to such conditions as he may specify therein.

(3)

Every application for permission under sub-regulation (2) shall be accompanied by two copies of a plan showing the proposed layout of the working, a section of the seam or seams, the depth of the seam(s) from the surface, the rate and direction of dip, the proposed dimensions of pillars and galleries in each seam or section, and the thickness of the parting between the seams or sections.

(4)

Where two or more seams or sections are worked in a mine, the pillars in one seam or section shall as far as practicable, be vertically above or below the pillars in the other seam or section unless the strata are inclined at an angle of more than 30 degrees from the horizontal.

(5)

The parting left between any two seams or section shall not be less than three metres in thickness at any place: Provided that the Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit or require a smaller or greater thickness of parting, as the case may be.

122.

Working under railways and roads, etc.

(1)

No working shall be made and no work of extraction or reduction of pillars shall be conducted at, or extended to, any point within 45 metres of any railway, or of any public works in respect of which this regulation is applicable by reason of any general or special order of the Central Government, or of any public road or building, or of other permanent structure not belonging to the owner of the mine, without the prior permission in writing of the Chief Inspector and subject to such conditions as he may specify therein.

(2)

Every application for permission under sub-regulation (1) shall specify the position of the working of the mine in relation to the railway or public road or works or building or structure concerned, the manner in which it is proposed to carry out the intended operations, and the limits to which it is proposed to carry out the said operations; and shall be accompanied by two copies of a plan showing the existing and the intended mining operations in so far as they affect the railway or public road or works or building or structure concerned.

(3)

A copy of the application for permission under sub-regulation (1), shall also be sent in the case of a railway, to the railway administration concerned; and in the case of any public works as aforesaid, to such authority as the Central Government may by general or special order direct.

(4)

Notwithstanding anything contained in these regulations, the stability of such railway, road, works, building or structure shall not be endangered until it has been dismantled, diverted or vacated, as the case may be.

(5)

Where the stability of such railway, road, works, buildings or structure has been endangered due to any mining

operations, the Chief Inspector may, by an order in writing, require the owner to construct in the mine belowground or on the surface such protective works within such time as he may specify in the order.

123.

Protective work before a mine is closed.

(1)

The Chief Inspector may, by an order in Writing, require the owner of any mine to which regulation 6 applies, to construct in the mine belowground or on the surface such protective works within such time as he may specify therein.

(2)

If the owner fails to construct such protective work within the time specified in the order, the Chief Inspector may get the work executed by other agency, and the cost thereof, as certified by the Chief Inspector, shall be defrayed by the owner of the mine and recoverable from him as an arrear of land revenue.

(3)

Until the protective work have been constructed to the satisfaction of the Chief Inspector, the means of entering the mine at not less than two entrances shall be kept intact and in working order.

124.

Working near mine boundaries in belowground mines.

(1)

The owner, agent or manager of every belowground mine shall have fixed boundaries of the mine and notwithstanding anything contained in sub-regulation (2), these shall not be changed except with the express permission of the Chief Inspector in writing and subject to such conditions as he may specify therein.

(2)

No working shall be made within a distance equal to half the distance as specified in column (5) of table under sub-regulation (4) of regulation 114, corresponding to the depth of the seam being worked, of the boundary of any mine and in case of a disputed boundary no working shall be made within the aforesaid distance of the boundary claimed by the owner of an adjacent mine until such time as a binding agreement has been reached as to the correct boundary or the question has been finally determined by a court of law:

Provided that, where work is done in more than one seam, the barrier kept at the boundary shall, as far as practicable, be vertically coincident and of the same dimensions:

Provided further that, where the working of any seam, for any reason, are extended or get extended within any shorter distance than is laid down herein above, the Chief Inspector may, by an order in writing, require the owner to construct such protective work within such time as he may specify in the order.

(3)

Notwithstanding anything contained in sub-regulation (2), the Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit the working of any mine or part thereof to extend within any shorter distance than what is laid down in sub-regulation (2) or may require that the said working shall not extend further than a specified distance, not exceeding 60m of such boundary.

125.

Working near mine boundaries in opencast mines.

(1)

The owner, agent or manager of every open cast mine shall have fixed boundaries of the mine and notwithstanding anything contained in sub-regulation (2), the boundary shall not be changed except with the permission of the Chief Inspector in writing and subject to such conditions as he may specify therein:

Provided that if any mine consists of two or more separate excavations and if, in the opinion of the Chief Inspector, they are not sufficiently near to one another to permit daily personal supervision being exercised by one manager, the Chief Inspector may, by an order in writing, require the mine to be split into two separate mines.

(2)

No working shall be made within a distance of 7.5 metres of the boundary of any mine and, in case of a disputed boundary, no working shall be made within a distance of 7.5 metres of the boundary claimed by the owner of an adjacent mine until such time a binding agreement has been reached as to the correct boundary or the question has been finally determined by a court of law:

Provided further that, where the working of any mine, for any reason, are extended or get extended within any shorter

distance than what is laid down herein above, the Chief Inspector may, by an order in writing, require the owner to construct such protective works within such time as he may specify in the order;

(3)

Notwithstanding anything contained in sub-regulation (1), the Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, Permit the working of any mine or part thereof to extend within any shorter distance than 7.5 metres aforesaid, or may require that the said working shall not extend further than a specified distance, not exceeding 60 metres, of such boundary.

126.

Strata Control and Management Plan (SCAMP).

(1)

The owner, agent and manager of every mine, in consultation with a scientific and research organisation having experience and expertise in Strata Control, Rock Mechanics and Instrumentation, shall prepare, formulate and implement a Strata Control and Monitoring Plan (SCAMP) based on the geotechnical data, information and the method of development and extraction of coal or the excavation required therefor, which shall be approved in writing by the Chief Inspector and subject to such conditions as he may specify therein.

(2)

The owner, agent or manager of every mine shall formulate a support plan to secure the roof and sides of belowground workplaces, which shall be subject to revision with change in condition, for all working belowground.

(3)

The owner, agent or manager of every mine having working below ground shall, before commencing any operation frame, with due regard to the engineering classification of strata, local geological conditions, system of work, mechanisation, and past experience, and enforce the support plan specifying in relation to each working place the type and specifications of supports and the intervals between:

(i)

supports on roadways including places where machinery is used for cutting, conveying or loading;

(ii)

each row of props, roof bolts or other supports;

(iii)

adjacent props, roof bolts or other supports in the same row;

(iv)

last row of supports and the face;

(v)

powered supports;

(vi)

fore-poles or sprags;

(vii)

shields; and

(viii)

the pack and the face:

Provided that in respect of a mine where development operations are already in progress, the support plan shall be framed and enforced within 30 days of the date of coming into force of these regulations.

(4)

The manager shall, at least 30 days before the commencement of any operation other than in respect of a mine where development operations are already in progress, submit a copy of the support plan to the Regional Inspector who may at any time, by an order in writing, require such modification in the plan as he may specify therein.

(5)

The Manager shall hand over copies of the support plan in English as well as in a local language understood by majority of the persons employed in the mine together with illustrative sketches, to all supervisory officials concerned including the assistant manager and shall also post such copies at all conspicuous places in the mine.

(6)

The manager and such supervising officials shall be responsible for securing effective compliance with the provisions of

the support plan, and no mine or part of a mine shall be worked in contravention thereof.

(7)

The support plan shall include interalia, system of, monitoring of the support performance, measurement of strata behaviour, re-setting of supports, provision of temporary support, replacement of old supports, withdrawal of supports and clearing of falls of ground.

(8)

The support plan shall also include the implementation strategy of the plan, training and inspection and supervision policies.

(9)

The owner, agent or manager shall formulate and implement a code of standing orders specifying -

(a)

the system and the organisation for procurement and supply of supports of suitable material, adequate strength and in sufficient quantity where these are required to be readily available for use;

(b)

the method of handling including dismissing and assembling where necessary and transportation of the supports from the surface to the face and from the face line to their new site;

(c)

the system and the organisation for maintenance and checking of supports, dressing the roof and side, erecting, examining and re-tightening of supports and re-erecting dislodged supports, including the use of appropriate tools;

(d)

the panel of competent persons for engagement as substitutes in the event of a regular supportsmen or dresser absenting from duty; and

(e)

the manner of making all concerned persons such as loaders, dressers, supportsmen, shortfirers, sirdars, overmen and assistant managers including persons empanelled for engagement as substitute supportsmen or dresser fully conversant with the support plan and the Codes of Standing Orders under this sub-regulation and under regulation 128 and the nature of work to be performed by each in that behalf.

127.

Use of supports or support materials.

(1)

All types of steel supports, powered supports, chock shields, roof bolts, cement, resin grouts or any other material used as grout along with the rock bolt to be used for the purpose of supporting roof and sides of working in a mine, shall be of such type, standard and make as approved by the Chief Inspector.

(2)

The approval standard, test procedures and protocol for such supports shall be framed and notified by the Chief Inspector from time to time.

128.

Setting of support.

(1)

Every prop shall be set securely on a sound foundation and shall be kept tight against the roof.

(2)

Where a prop is set on sand or, other loose material, a flat base-piece not less than 5 centimeter in thickness, 25 centimeters in width and 75 centimeters in length shall be used.

(3)

The lid used over a prop shall have a width not less than the diameter of the prop, a thickness not less than 8 centimeters and a length not less than 50 centimeters.

(4)

Every bar set for supporting the roof of a roadway shall be set securely on props or on cogs or shall be securely fixed on the sides of the roadway in holes at least 50 centimeter deep made in the sides of the roadway and shall be made and kept tight against roof and where lagging is necessary the number of laggings shall not be less than one for every metre length of the bar and the laggings shall be made and kept tight against the roof.

(5)

Every cog used as a support shall be well built and set on the natural floor or on a secure foundation, and shall be made and kept tight maintaining maximum possible contact against the roof.

(6)

In case of timber for erection of a cog, it shall be sufficient to joggle two opposite sides, provided that only four sides sawn sleepers shall be used for goaf edge supports in depillaring panels.

(7)

The cogging members shall be not less than 1.2 metres in length.

(8)

Before erecting cogs in a depillaring area, props shall be erected at the corners of each cog.

(9)

In inclined seams, the supporting props and cogs shall be so set as to ensure maximum support having regard to the inclination of the seam or roadway and probable strata movement and where necessary such supports shall be reinforced to prevent displacement.

(10)

Every ledge and every prominent crack or slip in the roof shall be kept supported with at least a pair of cogs and or cross-bars suitably lagged.

(11)

Overhanging sides shall be dressed down:

Provided that where this is not practicable, stay props or other suitable means of support shall be erected at intervals not exceeding one metre.

(12)

Where sand or other material is stowed or a pack is formed for the purpose of support, it shall be packed or made as tight against the roof as practicable over its whole area.

(13)

Roof and sides and support shall be tested as often as necessary; and except where it is no longer necessary for purpose of support, any support loosened, broken or dislodged by or removed in any operation shall be tightened, replaced or reset with the least possible delay and particularly before persons are allowed to pass or resume work after an interruption.

(14)

Where floor coal or roof coal is taken, shorter props shall be replaced with longer props.

(15)

In every place wherein roof coal is taken or a fall of roof or sides has occurred, no work of cleaning the dislodged coal or the fall or any part thereof shall be undertaken nor shall any person be allowed to pass, until the newly exposed roof and sides in the vicinity thereof have been examined and made safe, if necessary, by temporary supports.

(16)

Notwithstanding anything contained in sub-regulation (9), (13), (14), or (15) only such minimum number of persons may be engaged under the supervision of a sirdar or overman as may be necessary for securing the roof and sides thereat.

(17)

Where roof bolts are used for support,

(a)

only full column grouted roof bolts with adequate bearing plates shall be used,

(b)

holes shall be drilled to correct diameter and length with a suitable drill maintaining verticality/ correct inclination in accordance to the support plan,

(c)

hole diameter shall not be more than 8 to 12 mm larger than the bolt diameter,

(d)

roof bolts shall be installed as soon as the roof has been exposed, under the direct supervision of an assistant manager or overman or mining sirdar.

(18)

Powered supports, hydraulic chocks or link-bars shall be advanced as soon as practicable after a web of coal has been taken off the face so as to ensure that the area of unsupported newly exposed roof is kept to a minimum.

(19)

Powered supports, hydraulic chocks, props and friction props shall be set securely and checked from time to time.

(20)

When any defect is detected in any powered support or hydraulic chock, the same shall be attended to as soon as possible and any defective hydraulic or friction prop shall be replaced immediately.

(21)

Where, by reason of any irregularity in the roof, floor or sides or due to any other reasons, any powered support or hydraulic chock becomes ineffective, conventional supports in sufficient number shall be used.

(22)

A code of safe practices shall be framed and implemented by the manager of the mine for correct installation, maintenance and monitoring of all types of supports.

129.

Withdrawal of supports.

(1)

Whenever supports are to be withdrawn, the withdrawal shall be done in such maner as the manager may by orders specify.

(2)

The orders referred to in sub-regulation (1) shall cover -

(a)

the supply and use of appropriate tools and safety contrivances;

(b)

the setting of extra supports to control the collapse of roof from which supports are being withdrawn;

(c)

the sequence of withdrawal of supports; withdrawal of a cog to precede withdrawal of its corner props;

(d)

safe positioning of the persons engaged in the operation and all other persons present nearby;

(e)

training of competent persons who are entrusted with the operations; and

(f)

supervision during withdrawal of supports.

(3)

In every mine where powered supports are used, it shall be the duty of the owner, agent and manager to prepare a plan for the installation of powered supports and a plan for their withdrawal and transport.

(4)

The plan referred under sub-regulation (3), shall be reviewed and approved with or without any necessary amendment by the Regional Inspector before implementation.

130.

Provision of roof canopies or cabs.

(1)

Electrical, battery, or diesel-powered, self-propelled machinery including side discharge loaders, load haul dumpers, coal haulers, shuttle cars etc., used below ground in coal mines shall be provided with substantially constructed roof canopies or cabs which will give adequate protection against falling of ground from the roof or sides.

(2)

The cabin or seat of the operator provided is such machine shall be ergonomically designed and shall be such that the operator has clear line of sight in front as well as at rear of the machine without involving any constraint or strain.

131.

Steep working.

(1)

In working having an inclination of 30 degrees or more from the horizontal, adequate precaution shall be taken to

prevent danger to persons from falling or rolling of timber, tools or other appliances or material.

(2)

No person shall work or be permitted to work at any place having an inclination of 45 degrees or more from the horizontal, where he is likely to slip or unbalance, unless he is secured by a safety belt or life line or is otherwise safeguarded.

132.

Fencings and gates.

(1)

The top of every opencast working shall be kept securely fenced.

(2)

Where an excavation which has been formed as a result of any mining operation, extends within a distance of 15 metres from a public road or any building, substantial fencing shall be erected and maintained around the excavation.

(3)

Where as a result of mining operations, a subsidence of the surface has taken place or is likely to take place and persons are likely to be endangered thereby, the owner, agent or manager shall keep the entire surface area securely and effectively fenced.

(4)

Every entrance to a shaft, staple pit, sump, goaf or other dangerous place shall be provided with an efficient fence, barrier or gate, so designed and constructed as to effectively prevent any person from entering or falling therein.

(5)

Where a shaft, staple pit or a gallery having an inclination of more than 30 degrees from the horizontal leads directly into a working place or travelling roadway, such place or roadway and any working place situated on its dip side, shall be securely guarded or otherwise protected as to prevent danger to persons from falling materials.

(6)

Every entrance from a roadway into a mine or a part thereof which, for the time being, is neither being worked nor being used for any purpose, by reason of any cause whatsoever, shall be provided with a fence, barrier or gate so designed and constructed as to prevent any person from inadvertently entering that part of the mine.

(7)

Shafts and opencast working temporarily or permanently out of use and any place in or about an excavation which is dangerous shall be completely filled in or kept securely fenced:

Provided that if in the opinion of the Regional Inspector, any disused trench, pit or other excavation is dangerous, he may, by an order in writing, require the same to be filled in to the level of the adjacent ground.

(8)

Before a mine is abandoned or the working thereof discontinued, the owner, agent or manager shall cause the top or entrance of every shaft, incline or other opening into the mine to be fenced by a structure of a permanent character sufficient effectively to prevent persons falling into or entering the same.

133.

Examination by sirdars.

(1)

Every place in a mine, whether belowground or in manual opencast working including travelling roadways and landings, where work is carried on or where persons are stationed or required to pass shall be placed under the charge of a sirdar or other competent person.

(2)

The size of the mine or district, or the nature of any additional duties assigned to a sirdar or other competent person shall not be such as to incapacitate the sirdars or the competent person from efficient discharge of his duties under the regulations.

(3)

Any complaint as to the size of the mine or district or any additional duties referred to under sub-regulation (2) shall be referred to the Chief Inspector for decision.

(4)

At the entrance to every mine or district, one or more stations shall be fixed by the manager; and except in the case of a

mine working in a continuous succession of shifts no person other the persons making the inspection under sub-regulation (6) or an official shall pass beyond any such station until all the roadways and working places to which such persons are required to have access, have been examined by the competent person in charge of the mine or district and found to be satisfactorily ventilated and in safe condition.

(5)

Every such station shall be legibly marked 'STATION' and shall be of such a size as to accommodate all the persons employed in the district in any one shift.

(6)

The sirdar or other competent person accompanied by such assistants as may be required shall, within two hours before the commencement of work in a shift, inspect every part of the mine or district assigned to him, in which persons have to work or pass during the shift and all roadways and working places where work is temporarily stopped and shall ascertain the condition thereof as regards ventilation, sanitation, the presence of gases, the state of the roof and sides, presence of spontaneous heating and other fire risks, and generally so far as the safety of the persons is concerned.

(7)

Inspections shall be made once at least in every four hours during which the shift continues, of all the roadways and other working places to which persons engaged in the mine or district are required to have access;

(8)

The inspection under sub-regulation (6) shall be made with an approved flame safety lamp or any other approved apparatus made for the purpose and in case of a fiery seam, also with an apparatus for detecting carbon monoxide gas approved by the Chief Inspector;

(9)

In case of manual opencast working, the sirdar shall pay attention to overhangs, undercuts, presence of loose stone, materials or trees, etc., within 3 metres of the edges, foot paths for traveling and carrying loads, fencing on top of the quarry and disused part or abandoned quarries, fencing along footpaths and benches where required, use of personal protective equipment and condition of dumps; and

(10)

The sirdar or other competent person shall, at the completion of his shift, record without delay the result of his inspections in a bound paged book in the format specified for the purpose and every such report shall include the following :-

(a)

the details referred to in sub-regulation (6) and (9);

(b)

the number of persons working under his charge;

(c)

such instructions for the purposes of securing the safety of the persons as he may have given during his shift; and

(d)

the date and time of the inspections, the signature of the sirdar or other competent person and the date and time when the report was written.

(11)

In the case of a shaft in the course of being sunk, the competent person or chargeman shall...

(a)

have entire charge of the shaft bottom and shall, in his shift, remain in the shaft while persons are at work at the bottom of the shaft.

(b)

be the last person to ascend the shaft at the end of the shift and if his shift is succeeded immediately by another shift, he shall not leave the bottom of the shaft until after the descent of the chargeman of the succeeding shift; and

(c)

after each round of shots, and at the beginning of every shift, and after every cessation of work in the shaft for a period exceeding two hours, examine the sides of the shaft and remove all loose pieces before persons are allowed to descend.

134.

Avoidance of dangers.

(1)

If at any time it is found by a competent person in charge of a mine or district, that by reason of any cause whatsoever, the mine or district is dangerous, he shall immediately withdraw all persons from the mine or district; and the mine or district shall be fenced off so as to prevent persons inadvertently entering therein.

(2)

The competent person shall also immediately inform the manager or assistant manager about the danger, and shall record the fact in the bound paged book kept for the purpose.

(3)

The manager shall make, or cause to be made by a competent person, a careful examination of the mine or district; and no person shall, except in so far as is necessary for enquiring into the cause of the danger or for the removal thereof or for exploration, be readmitted into the mine or district until the mine or district is reported to be safe.

(4)

A report of every such examination under sub-regulation (3) shall be recorded without delay in a bound paged book kept for the purpose and shall be signed and dated by the person who made the examination.

(5)

If the work of removing the danger is suspended before the danger is removed, the mine or district shall be securely fenced off effectively to prevent persons entering therein during the period of suspension.

(6)

Notwithstanding anything contained in these regulations-

(a)

where the danger arises from the presence of inflammable or noxious gas, the provisions of regulation 170 shall apply; and

(b)

where in any part of a mine the appearance of smoke or other sign indicates that a fire or spontaneous heating has broken out, the provisions of regulation 141 shall apply.

135.

General precautions.

(1)

No person shall cut or remove coal from or in the vicinity of any place unless it is his authorised working place.

(2)

Every person shall-

(a)

carefully examine his working place before commencing work and also at intervals during the shift.

(b)

if any dangerous condition is observed, cease all work at that place and shall either take immediate steps to remove such danger or inform an official or the competent person in charge of the mine or district.

(3)

Where several persons are working together and one of them is in-charge, the examination required under sub-regulation (2) shall be made by the person in-charge.

(4)

Every person cutting coal and every person operating a coal-cutting machine or any other cutting or loading machine in any place shall ensure that the dimensions of that place do not exceed the dimensions specified in that behalf by these regulations.

(5)

No person shall work or travel on any ledge or footpath less than 1.5 metres wide, from where he is likely to fall more than 1.8 metres, unless he is protected by guard rails, fence or safety belt or rope suitably fixed and sufficiently strong to prevent him from falling.

(6)

When a plant, machinery, workshop, or any other shed or structure is under construction, repair or renovation, and persons are allowed to work at heights, proper walkway, stairs or ladderway With hand rails, guards or stages and

platforms with fencing shall be provided to avoid danger or risk of persons falling from height.

(7)

Every person permitted to work at height shall be provided with safety belt, of the type and standard suitable for the nature of work to be performed by him and approved by the Chief Inspector.

(8)

Where it is required to work at heights involving danger or risk of falling, proper stage or working platform of adequate and suitable design and strength shall be provided.

(9)

The stage or working platform referred to in sub-regulation (7) shall have proper approachway to and from the stage or platform for the persons to ascend or descend, as well as the fencing to prevent such falls.

(10)

A safety net of adequate strength and design shall also be provided immediately below every working place so that the danger or risk of injuries to persons falling from working at heights is completely eliminated.

(11)

A code of safe practices shall be framed by the manager and implemented under the supervision and control of a competent person and official especially authorised for the purpose by the engineer and manager, to ensure such safety precautions while executing such work in the mine.

(12)

No untrained and inexperienced person(s) and contractor(s) shall be engaged for execution of any such work in the mine which involves working at heights and the dangers associated therewith.

(13)

A system of Check List and Work Permit shall be maintained by the agent and manager where any such work in the mine which involves working at heights and the dangers associated therewith is undertaken.

(14)

No person shall carry or be permitted to carry any load along a road or footpath having an inclination of 30 degrees or more from the horizontal.

(15)

Every road or footpath, along which loads are carried by human agency, shall comply with the following requirements: -

(i)

its breadth shall not be less than one metre; and

(ii)

at every place where the inclination exceeds 15 degrees from the horizontal, level steps shall be provided such that the vertical height of every step does not exceed 0.18 metre and the distance from the edge to the back is not less than 0.35 metre.

Explanation - gang-planks used for loading purposes shall not be deemed to be part of a footpath for the purposes of this sub-regulation, provided that every gang-plank shall be so inclined or constructed as to give a secure foot-hold.

(16)

No person shall be employed to lift, carry or move a load so heavy as is likely to cause bodily injury or injury to health of that person. In case of any doubt as to whether risk of bodily injury or injury to health is involved, it shall be referred to the Chief Inspector for decision.

(17)

Every person shall ensure that tools, wood, stones, or other articles are not put down or allowed to remain, in or near a shaft or dip gallery where work is going on, in such position as may result in their falling into the shaft or gallery.

(18)

No person shall work or be permitted to work alone in any remote part of a mine where, if any accident occurred he would not soon be discovered or assisted.

(19)

No inexperienced person shall be employed in the mine for any work whereby he or other persons can be seriously endangered except under the supervision and guidance of an experienced person.

Chapter-XI Precautions against Dangers from Fire, dust, gas and Water

136.

Classification of coal seams according to their degree of gassiness.

(1)

All coal seams shall be classified into different degrees of gassiness by the Regional Inspector or an Inspector assisted by such assistants and after such investigation as he may consider necessary.

(2)

If in a gassy seam the percentage of inflammable gas in the general body of air or the rate of emission of such gas increases so as to bring that seam into a higher degree of gassiness, the owner, agent or manager shall within 24 hours from his knowledge of such increase, inform the Regional Inspector and also observe all the precautions required to be taken under these regulations in respect of a gassy seam of that degree and the Regional Inspector, shall within 30 days of the receipt of such information, verify and investigate the degree of gassiness and classify the seam into the appropriate degree of gassiness.

Provided that if it is not practicable to observe all the precautions required to be taken under this regulation within 24 hours, the Regional Inspector, on a request in writing by the owner, agent or manager, may defer the observance of the required precautions, subject to such conditions as he may specify, for a period not exceeding sixty days.

(3)

If in a gassy seam the percentage of inflammable gas in the general body of air or the rate of emission of inflammable gas decreases so as to bring that seam to a lower degree of gassiness, the owner, agent or manager may inform the Regional Inspector of the same and the Regional Inspector shall within 30 days from the receipt of such information verify and investigate the degree of gassiness and classify the same into appropriate degree of gassiness.

(4)

Notwithstanding anything contained in sub-regulation (2) or sub-regulation (3), the Regional Inspector may at any time make investigation and reclassify a gassy seam into the appropriate degree of gassiness.

(5)

The owner, agent or manager shall at least once in every 30 days examine the rate of emission of inflammable gas in cubic metres per tonne of coal raised and the percentage of inflammable gas in the general body of air as laid down in regulation 173 and the result of every such examination shall be recorded in a bound paged book kept for the purpose.

(6)

Every instrument, apparatus and the system used for detection and measuring of the inflammable gas shall be of a type and standard approved in writing by the Chief Inspector.

(7)

Every instrument, apparatus and the system used for detection and measuring of the inflammable gas and analysis of gas samples, shall be calibrated at specified interval by an approved or accredited agency or laboratory notified by the Chief Inspector by a special or general order in writing from time to time.

137.

General precautions against fire.

(1)

No oil, grease, canvas or other inflammable material shall be stored in any mine except in a fire-proof receptacle.

(2)

Greasy or oily waste in working belowground shall be regularly removed to the surface.

(3)

In case of opencast working or workshops, greasy and oily wastes shall be disposed off regularly in a safe manner.

(4)

No person shall place or throw, or cause or permit to be placed or thrown, any naked light or lamp on or near any timber, wooden structure or other combustible material.

(5)

Adequate and sufficient arrangements shall be made in every mine for early detection, control and extinguishing any fire.

(6)

The owner, agent and manager of every mine shall take measure and precaution appropriate to the nature of a mine operation to prevent, detect and combat the start and spread of mine fires.

(7)

The owner, agent and manager of every mine shall ensure that operations are stopped and workers are evacuated to a safe location, when there is serious danger due to fire, threatening the safety and health of workers.

138.

Surface precautions against fire.

(1)

All surface structures and supports within a horizontal distance of 10 metres from all entrances to a mine shall be of incombustible material:

Provided that this sub-regulation shall not apply to temporary structures, supports and coverings at the top of a shaft which is in the course of being sunk and to the small lid of a shaft-covering operated by the rope cappel.

(2)

Shale or other carbonaceous material shall not be heaped or dumped, dead leaves or dry vegetation shall not be allowed to accumulate, combustible materials (other than materials required for use within a period of 24 hours) and inflammable materials shall not be stored within a distance of 15 metres from any entrance to a mine which is not effectively sealed off from the working below ground:

Provided that nothing in this sub-regulation shall prevent the dumping of coal raised from the mine, near the entrance to the mine.

(3)

In opencast working and in any ground broken by extraction of coal, all wild or herbaceous plants shall be removed and all dead leaves and dry vegetation cleared as often as may be necessary, to prevent an outbreak of fire.

(4)

No person shall deposit any heated material or ashes on any outcrop of coal seam or in any opencast working or any ground broken by extraction of coal.

(5)

No person shall light a fire or permit a fire to be lighted in any opencast working within a distance of 15 metres from any entrance to a mine, except by the permission in writing of the manager and only for a special purpose specified in such order and all such orders shall be recorded in a bound paged book kept for the purpose:

Provided that this sub-regulation shall not apply to boilers other than vertical boilers.

(6)

A competent person shall, once at least in every seven days, inspect the top of all entrances to a mine, all opencast working and any ground broken by extraction of coal in order to ascertain whether the precaution laid down in this regulation have been complied with, and for the presence of any fire that may have broken out or any indications thereof and a record of every such inspection shall be maintained in a bound paged book kept for the purpose, duly signed and dated by the person making the inspection.

139.

Underground precautions against fire.

(1)

No timber or other combustible material shall be used in the construction of, or in connection with, any shaft lining or any room housing any machinery or apparatus belowground.

(2)

Wood cuttings Shall not be left in any working belowground, but shall be removed to the surface at the end of every shift.

(3)

No person shall light a fire or permit a fire to be lighted in any working belowground :

Provided that -

(a)

in the case of a gassy seam of the first or second degree, flame or electric welding or repairing apparatus may be used belowground if permitted by an order in writing of the manager and every such order shall specify the person who shall be in-charge of the apparatus; and it Shall be the duty of such person to bring the apparatus back to the surface when no longer required belowground; and

(b)

in the case of a gassy seam of third degree a flame or electric welding or repairing apparatus may be used belowground

if prior permission in writing has been obtained from the Regional Inspector and subject to such conditions as he may specify therein.

(4)

No person shall leave a portable light or lamp belowground unless he has placed it in-charge of some other person remaining therein.

(5)

At the end of a shift, unless the mine is worked by a continuous succession of shifts, after all persons have left the mine, all lights shall be extinguished and all power cut off.

(6)

Provision shall be made to prevent an outbreak of the fire belowground or the spread of fire from any part of the same mine or from any adjoining mine, and adequate steps shall be taken to control or isolate any such fire or heating that may occur.

(7)

All unused working connected to the surface though a walkable entrance which is not permanently closed, Shall once at least in every 30 days be inspected by a competent person for signs of illicit distillation of liquor and a report of every such inspection Shall be recorded in a bound paged book kept for the purpose, duly signed and dated by the person making the inspection.

140.

Precaution against spontaneous heating.

- The following precautions shall be taken against the danger of spontaneous heating:-

(1)

The seam or section shall be worked in panels having independent ventilation in such a manner that it is possible to isolate one from another easily when necessary.

(2)

Where the seam or section has already been developed without complying with the provisions of sub-regulation (1), artificial panels shall be created by construction of stoppings.

(3)

In determining the size of the panel under sub-regulations (1) and (2), due consideration shall be given to enabling complete extraction of the pillars therein, within the incubation period of the coal seam.

(4)

No coal, shale or other carbonaceous material shall be left or stacked belowground.

(5)

Where removal of fallen coal from the mine is not practicable, the area shall be effectively sealed off.

(6)

Except where otherwise permitted by the Chief Inspector by an order in writing and subject to such conditions as he may specify therein, no extraction of pillars in any seam or section shall be commenced until fire dams or stoppings have been provided in all entrance to the pane.

(7)

In the fire dams or stoppings built in entrances which are to be kept open for ventilation or haulage, suitable doors or openings may be left and bricks and other suitable materials shall be kept readily available there in the vicinity.

(8)

Shale or other carbonaceous material shall not be used in the construction of fire dams or stoppings.

(9)

A panel shall be isolated by adequate stoppings as soon as it has been goaved out.

(10)

All the isolation stoppings erected to seal off the goaves or to isolate old, abandoned or disused working or to isolate area affected by fire or spontaneous heating shall be plastered or painted with fire resistant sealant materials and chemicals which shall be of a type and standard approved by the Chief Inspector.

(11)

Sufficient material for dealing with fire shall be kept ready at suitable places below ground for transport and use and sufficient number of persons shall be trained in the use of such material.

(12)

In order to detect spontaneous heating in early stages, the air in the return airway of every depillaring district and of every goaf which has not been isolated, shall be -

(a)

tested for percentage of carbon monoxide once at least in every seven days with an automatic detector of a type approved by the Chief Inspector; and

(b)

completely analysed once at least in every 30 days with a view to determining the ratio CO-formed/O₂-absorbed.

and the result of every such tests shall be recorded in a bound paged book kept for the purpose duly signed and dated by the person carrying out the test:

Provided that if successive tests show any steady increase in the CO-formed/O₂ absorbed ratio, suitable measures shall be taken to determine the site of the heating and to deal with it.

(13)

Every depillaring district shall be inspected on every idle day and a report of every such inspection shall be recorded in a bound paged book kept for the purpose, duly signed and dated by the person making the inspection.

(14)

All unused working including unused working which have not been sealed off and isolation stoppings built around goaved out area shall be inspected once at least in every seven days, by a competent person for any fire risks, and a report of every such inspection shall be recorded in a bound paged book kept for the purpose, duly signed and dated by the person making the inspection.

(15)

Where at any mine or part, special conditions exist which make compliance with any of the provisions of this regulation not necessary or reasonably practicable, the Regional Inspector may, by an order in writing and subject to such conditions as he may specify therein, grant a relaxation from the provision.

(16)

Where coal is stacked on surface, suitable arrangement shall be made for spraying of water or any other fire resistant or suppressant materials on the coal stack at regular interval so as to prevent spontaneous heating.

(17)

Hydraulic fluid which is not fire resistant shall not be used belowground.

(18)

Precaution shall be taken to minimize the possibility of hydraulic fluids or oils coming in contact with hot surface, electrical apparatus or cables.

(19)

Combustible oils or material shall not be used for filling in electrical equipment.

(20)

Notwithstanding any thing contained in the sub-regulations (17), (18) & (19), the Regional Inspector may by an order in writing, exempt from use of fire resistant hydraulic fluid in degree-I gassy mine subject to such conditions as he may specify therein.

141.

Precaution after a fire has broken out.

(1)

On the appearance in any part of a mine, of smoke or other signs indicating that a fire or spontaneous heating has or may have broken out, all persons other than those whose presence in the mine is deemed necessary for dealing with the fire or heating shall be immediately withdrawn from the mine.

(2)

No person, other than those required for dealing with or sealing off the fire or heating referred to in sub-regulation (1), shall be admitted in the mine until the fire or heating has been extinguished or effectively sealed off and an examination has been made by the manager or by the assistant manager and the mine has been declared to be safe and a report of every such examination shall be recorded in a bound paged book kept for the purpose duly signed and dated by the person making the examination:

Provided that the Regional Inspector may, by an order in writing and subject to such conditions as he may specify

therein, permit the employment in the mine, of persons other than those required to deal with the fire or heating.

(3)

The examination required under sub-regulation (2), shall be made with an approved flame safety lamp and other means of detecting carbon monoxide gas approved by the Chief Inspector.

(4)

During the whole time that any work of dealing with or sealing off a fire or heating belowground is in progress, -

(a)

a competent person shall be present on the spot throughout;

(b)

adequate precaution shall be taken to prevent danger to persons from any noxious, asphyxiating or inflammable gases, flame, steam and ejected or rolling down hot material, explosion of water gas, falling into crevices or pot holes, that may occur in the area on fire; and

(c)

there shall be kept available, at or near all places belowground:

(i)

adequate number of self-rescuers and at least two smoke helmets or other suitable apparatus for use in emergency;

(ii)

an apparatus for detecting carbon monoxide gas approved by the Chief Inspector; and

(iii)

a flame safety lamp or other means of detecting carbon dioxide gas and oxygen deficiency, approved by the Chief Inspector.

(5)

The manager of every mine shall prepare and establish a detailed scheme for-

(a)

the provision and maintenance of suitable fire-fighting arrangements;

(b)

the prevention, detection, dealing and control of any heating or fire;

(c)

the examination and maintenance of the protective measures taken to control or isolate a fire or heating;

(d)

ensuring safety of persons engaged in the said operations, and the scheme shall be suitably modified and kept updated as the situation warrants.

142.

Equipment for fire-fighting.

(1)

In every mine,-

(a)

unless expressly exempted in writing by the Regional Inspector, adequate quantity of water at sufficient pressure shall be provided to all working places belowground and all other places of fire risk such as coal stocks, spoil heaps containing carbonaceous material and exposed coal surfaces liable to heating, for the purpose of efficient fire fighting;

(b)

fire stations with suitable supply of fire-fighting equipment shall be established and kept maintained at convenient points, both on surface and belowground; and

(c)

sufficient supply of sand or incombustible material and suitable portable fire extinguishers in sufficient quantity or automatically operated fire suppression devices shall be Provided at-

(i)

every entrance to a mine or district and at every landing and shaft bottom in use;

(ii)

every place where timber, grease oil or other inflammable material is stored;

(iii)

every engine room, diesel engine maintenance workshop, filling station and storage battery charging station;

(iv)

on each track and off-track locomotive, self propelled manriding car and personnel carrier;

(v)

each permanent and temporary electrical installation;

(vi)

at locations where welding, cutting or soldering with arc or flame is being done;

(vii)

every HEMM used in opencast working;

(viii)

every machinery, plant and installations; and

(ix)

such other special places of fire risk as may be specified by the manager;

(d)

conveyor belt installation, specially designed water, foam spray, deluge systems or dry chemicals shall be installed above each belt drive, belt take up, electrical control, gear reducing unit and other strategic locations on the conveyor belt system.

(e)

adequate number of suitable fire extinguishers or automatically operated fire suppression devices shall be provided on continuous mining machines, other face cutting machines, loading machines, roof bolting machines and other hauling machines.

(2)

Soda-acid type extinguishers or water shall not be used for fighting oil or electrical fires.

(3)

Foam type extinguishers shall not be used for fighting electrical fires.

(4)

Fire-extinguishers containing chemicals which are liable, when operated, to give off poisonous or noxious gases shall not be provided or used belowground:

Provided that nothing in sub-regulations (2), (3) or (4) shall prohibit the use belowground of fire-extinguishers giving off carbon dioxide when operated.

(5)

All types of fire fighting and fire suppression systems including automatic fire detection and suppression systems to be used in machinery and plant including HEMMs, materials and chemicals to be used in fire sealing, fighting or suppression systems in mines both on surface as well as belowground shall be of such type, standard and make, as approved by the Chief Inspector by a general or special order in writing.

(6)

A competent person shall, once at least in every month, examine all the equipment, material and arrangements provided for fire-fighting and shall discharge and re-fill the fire extinguishers as often as may be necessary to ensure that these are in proper working order and any deficiency found during any such examination or otherwise shall be immediately remedied, and a report of every such examination shall be made in a bound-paged book kept for the purpose, duly signed and dated by the person making the examination.

143.

Organisation for fire fighting.

(1)

The owner, agent and manager of every mine shall establish a proper organisation for fire fighting in the mine by installing fire stations at surface and also on every main haulage roadway belowground at suitable places in the intake airway near the main shaft with adequate fire fighting equipment kept in every such fire station.

(2)

Sufficient number of plans shall be prepared showing the fire fighting equipment, the water mains, taps, fire-stations, pumping stations, ventilation system, escape route etc. and containing such other information as may be useful for the purpose of fighting fires, and up-to-date copies of these plans shall be kept available at suitable places both on the

surface and belowground.

(3)

Adequate number of persons shall be trained in the use of fire-extinguishers and in fire fighting and such persons shall be made familiar With the position of all fire fighting equipments provided in the mine in general and near their places of work in particular.

(4)

The manager of every mine shall, with the approval of the Regional Inspector, frame standing orders containing the procedures that may be adopted in giving warnings of fire, timely withdrawal of personnel from the mine and for the conduct of fire fighting operation.

144.

Apparatus for testing for carbon monoxide.

- In every belowground mine there shall be kept at the mine constantly available for use suitable apparatus approved by the Chief Inspector for detecting carbon monoxide gas.

145.

Precautions when a fire exists.

(1)

No person shall be employed in any seam, -

(a)

where a fire or spontaneous heating exists in a lower seam whether such fire has been sealed off by means of fire stoppings or not; or

(b)

where the seam has a common ventilation system with another seam on fire; or

(c)

where the outlets or openings of the seam are within 60 metres of an active fire or spontaneous heating in a higher seam or on the surface in any ash heap or spoil heap or in any other heap or place or any other fire or spontaneous heating which cannot be controlled immediately or where broken ground connected with the seam exists within 60 metres of such fire or spontaneous heating; or

(d)

where the parting, with an overlying seam on fire or in which spontaneous heating has taken place, or with surface containing an active fire or spontaneous heating in any spoil heap or ash heap or in any other heap or place, or with any other fire or spontaneous heating which cannot be controlled immediately, consist of less than 10 metres of hard rock, except with the permission in writing of the Chief Inspector and subject to such conditions as he may specify therein.

(2)

In any working mine, in which a fire is known or believed to exist, -

(a)

adequate precautions shall be taken to prevent the passage of air, from any part of the mine on from the surface, into the fire area through any broken strata; and

(b)

no work other than the operations required under clause (a) shall be done in any part of the mine which is not effectively sealed off from any such goaf or broken strata.

(3)

In every fiery seam or gassy seam of the second or third degree, arrangements shall be made Once at least every 30 days to ascertain the atmospheric condition behind the stoppings built to seal off the area of old working or such goaf or a fire or spontaneous heating, unless such stoppings are capable of resisting force of an explosion.

(4)

Every stopping erected to isolate or control a fire or spontaneous heating belowground or to seal off goaf or an area of old working shall be numbered, and shall be of adequate strength and so maintained as to prevent any leakage of air or gas through it:

Provided that where water is likely to accumulate behind any such stopping there shall be provided in the stopping a suitable pipe or other device to drain away the water without permitting any leakage of air or gas, etc.

(5)

Every stopping erected in accordance with the provision of sub-regulation (4) and the pillars containing such stoppings, shall be plastered with fire resistant or fire retardant sealant of adequate thickness using high pressure guns so as to completely fill up the cracks, cavities, crevices, joints, slips, fractures and cleats present in the working in order to completely seal off such area, which shall be kept maintained as leak proof.

(6)

Where in any mine or part thereof the provisions of sub-regulations (4) or (5) have not been complied with or where in the opinion of the Regional Inspector the steps so taken are inadequate, he may give notice in writing to the, owner, agent or manager requiring him to take such protective measures, within such time, as he may specify therein:

Provided that in case of non-compliance with the requirements of the notice under this sub-regulation, the Regional Inspector may, by an order in writing, prohibit until the requirements of the notice have been complied with to his satisfaction, the employment in the mine or part, of any person whose employment is not, in his opinion, necessary for the purpose of complying with the requirements aforesaid.

(7)

A competent person shall, once at least in every seven days, inspect all stoppings erected for isolation or control of fire or spontaneous heating belowground to ascertain the general condition of every stopping by checking it for leakage and presence of gas, and the temperature and humidity of the atmosphere outside the stopping.

(8)

The competent persons shall after carrying out the inspection under sub-regulation (7), place his signature, with date, on a check-board provided for the purpose at a suitable position on the stopping and which shall be maintained for a period of not less than three months, and a report of every such inspection shall also be recorded in a bound pagged book kept for the purpose duly signed and dated by the person making the inspection:

Provided that the Regional Inspector may, by an order in writing, require the inspection of stoppings to be made at such shorten intervals as he may specify therein.

146.

Precaution against dust.

(1)

The owner, agent or manager of every mine shall take such steps as are necessary for minimizing emissions of dust and for the suppression of dust which enters the air at any work place belowground or on surface and for ensuring that the exposure of workers to respirable dust is limited to an extent that is reasonably practicable but in any case not exceeding the limits that are harmful to the health of persons.

(2)

For the purposes of this regulation, a place shall not be deemed to be in a harmless state for person to work or pass or remain therein if the 8 hours time-weighted average concentration of airborne respirable dust in milligrams per cubic metre of air sampled by dust sampler of a type approved by and determined in accordance with the procedure as specified by the Chief Inspector by a general or special order, exceeds two, where working is being made wholly in seam or where free respirable silica present is less than 5 percent and the value is arrived at by dividing the figures of ten with the percentage of free respirable silica present in other cases; or

(3)

The owner, agent or manger of every mine shall, within three months of the coming into force of these regulations and once at least in every month thereafter or whenever the Regional Inspector so requires by an order in writing, cause the air at every work place where airborne dust is generated, to be sampled and the concentration of respirable dust therein determined:

Provided that, such measurements shall also be made immediately upon the commissioning of any plant, equipment or machinery or upon the introduction of any new work practice or upon any alteration therein that is likely to bring about any substantial change in the level of airborne respirable dust.

(4)

The samples drawn under sub-regulation (3) shall as far as practicable, be representative of the levels of dust exposure of work-persons and for this purpose, the sampler shall be positioned on the return side of the point of dust generation and within 1 metre of the normal working position of but not behind the operator or other worker whose exposure is deemed to be maximum in his working group.

(5)

Based on the results of static or personal sampling, the representative dust exposure profiles for different categories of workers shall be estimated by portal to portal monitoring of selected workers whose exposure is deemed to be representative of their working groups.

(6)

Samples shall be taken by a person who has been specially trained for the purpose in the sampling equipment and accessories that have been checked to ensure correct maintenance and efficient operation thereof and examined, treated and calibrated on a date which is not earlier than one year:

Provided that if the dust sampling is carried out by a laboratory other than belonging to the mine, such laboratory shall be approved in writing by the Chief Inspector in this behalf.

(7)

Respirable dust content of the samples and in case of samples collected from a working other than the working being made wholly in a coal seam, quartz content shall be determined at a laboratory approved in writing by the Chief Inspector in that behalf.

(8)

All result of measurements of airborne respirable dust and all other relevant particulars shall be systematically recorded within fourteen days of the date of collection of samples, in a bound paged book kept for the purpose and every entry in the book aforesaid shall be countersigned and dated by the manager within twenty four hours.

(9)

When the dust monitoring results have established that the permissible limit of dust concentrations are exceeded at any place, immediate steps shall be taken to minimize the emission of dust and to notify the Regional Inspector.

(10)

If the average concentration of respirable dust in a series of 5 samples taken in 7 successive normal working shifts during the subsequent month exceeds one and a half times the permissible limit, the relevant operation or operations causing excessive dust shall be stopped and the operation or operations shall not be resumed or allowed to be carried on until improvements have been made in the prevention and suppression of dust and fresh sampling carried out immediately on resumption of the said operation or operations has established that such improvements have reduced the dust concentration below the permissible limit:

Provided that if the dust prevention and suppression device of any machinery or equipment fails to operate efficiently, the operation of the said machinery or equipment shall likewise be stopped and shall not be resumed until the defect therein has been rectified:

Provided further that, purely as a contingency measure or as a secondary means of protection in a work situation wherein it is technically not feasible to reduce the respirable dust concentration below the permissible limit or during the time period necessary to install and commission any device or to institute any new, work practice for dust prevention or suppression, compliance with the permissible limit of dust may be achieved by remote operation or by job rotation and failing which by the Use of suitable dust respirator.

(11)

The owner, agent or manager of every mine where need of dust respirators might arise, shall provide dust respirators of approved type in sufficient number to the concerned workpersons for their use and the dust respirators shall be regularly cleaned, disinfected and maintained in efficient working order for the proper fitting of and for thorough training of the concerned workers in the need for correct use of respirators.

(12)

To prevent the liberation, accumulation or propagation of airborne dust, the following provisions shall have effect, namely:-

(a)

dust shall be suppressed as close as possible to its source of formation;

(b)

during any operation of drilling or boring on surface or belowground, -

(i)

the production of dust shall be reduced by using bits which are sharp and of proper shape, by keeping suitable pressure on the bits and by keeping the holes clear of the cuttings;

(ii)

except in naturally wet ground, a jet water shall be directed on to the cutting edge to wet the cuttings or other equally efficient device approved by the Chief Inspector which shall be provided and kept in operation throughout the drilling or boring operation to prevent the atmosphere being charged with dust;

(iii)

where pneumatic drilling is performed, water shall be turned on before turning on compressed air to the drill;

(iv)

when drilling is done by hand, it shall be sufficient if holes are kept constantly moist during such drilling:

(c)

every roadway on surface or belowground, where mobile mining machinery ply, shall be kept regularly wet or Shall be effectively treated with some other equally efficient dust suppressant agent approved by the Chief Inspector to reduce dust being raised in the atmosphere to a minimum;

(d)

no plant for the screening or sorting of coal and as far as practicable, no heap of cinder, cement, sand, mortar or other dry and fine material Shall be placed within 80 metre of the top of any down cast shaft or other intake airway nor shall any such material be so handled as to make it air-borne and drawn into such shaft or airway;

(e)

in every working belowground,-

(i)

no machinery or equipment which is likely to emit dust in excess of permissible limit shall be operated unless it is equipped with a suitable dust-prevention and suppression device which shall be properly interlocked with the operating lever or switch and unless such device is operating efficiently;

(ii)

the design, arrangement, material and condition of picks on every mechanical coal cutter Shall be such as to reduce the formation of dust to the minimum and no mechanical coal cutter Shall be operated unless suitable water sprays or jets of water are directed on the cutting edges thereof so as to damp the cuttings as they are formed;

(iii)

every working face and the floor, roof and sides of every roadway or airway within 50 metres thereof shall be, unless naturally wet throughout, regularly washed down to prevent accumulation of dust and shall be kept thoroughly wet during the work shifts;

(iv)

a current of air sufficient to clear away the dust emitted by any machinery or operation and to dilute the dust concentration below the permissible limit, shall be maintained by means of general ventilation and if necessary, by local ventilation, so however that, as far as practicable, the velocity of air in any roadway or workplace shall not be such as to raise dust in the atmosphere;

(v)

after blasting, no person shall enter working places unless sufficient time has elapsed for dust, smoke and fumes to be cleared by a current of air and the broken material shall not be moved unless it has been thoroughly made wet with water:

(vi)

vehicles, tubs and conveyors used for transport of coal shall be maintained in good condition so as to minimise spillage or leakage and chutes, spiral conveyors, bins, tipplers conveyor, discharge points and skip loading and unloading installations shall be so controlled as to reduce the formation of dust to the minimum and be kept thoroughly wet with water unless it is already wet or other effective means of dust suppression are used; and

(vii)

unless, owing to special difficulties, exempted in writing by Regional Inspector in that behalf and subject to such conditions as he might specify therein, water in pipes in sufficient quantity and under adequate pressure and independent of any pumping system, shall be provided and maintained so as to get maximum efficiency in allaying of dust:

(f)

no process of crushing, breaking, disintegrating, opening, grinding, screening or sieving of coal or any operation incidental thereto, shall be carried out at any mine unless efficient watering or other appropriate and effective dust

control measures, such as, but not limited to isolation, enclosure, exhaust ventilation and dust collection are designed, provided, maintained and used;

(g)

the exhausted air, belowground, which contains dust in excess of the permissible limit shall be efficiently diluted and if necessary filtered so as to reduce the concentration of respirable dust therein below ten percent of the permissible limit before being re-circulated into working places;

(h)

every device used for the prevention and suppression of dust produced by any machinery, equipment or process as also for the filtering of the exhausted air and every dust respirator shall be inspected once at least in every seven days and shall be thoroughly examined and tested at least once in every month and reports of the results of every such inspection, examination and test shall be recorded in the register kept for the purpose:

(13)

The manager of every mine where airborne dust is generated shall formulate and implement a scheme specifying -

(a)

the location, frequency, timing, duration and pattern of sampling;

(b)

the instruments and accessories to be used for sampling;

(c)

the laboratory at which respirable dust content of samples and quartz content shall be determined;

(d)

the format in which the results of measurements of dust concentration and other particulars are to be recorded;

(e)

the organisation for dust monitoring and for the examination and maintenance of dust prevention and suppression measures and dust respirators; and

(f)

the manner of making all persons concerned with the implementation of the dust control measures fully conversant with the nature of work to be performed by each in that behalf.

(14)

The Regional Inspector may, where special conditions exist, permit to require by an order in writing and subject to such condition as he may specify therein, any variation in the foregoing provisions or in the scheme made under sub-regulation (13).

(15)

If any doubt arises as to any matter referred to in this regulation, the same shall be referred to the Chief Inspector for decision.

147.

Execution of measures for dust control.

(1)

There shall be maintained at every mine having working belowground a dust plan on a scale having representative factor of not less than 2000 : 1, clearly showing by distinctive colours, code letter and or numbers, the separate areas: -

(i)

which are naturally wet;

(ii)

which require treatment with water indicating the system of water pipe lines laid down for the purpose; and

(iii)

which require treatment with incombustible dust at such intervals of 24 hours, 7 days, 14 days, 30 days, three months or other specified period as the case may be.

Explanation: The intervals referred to in clause (iii) shall be based on the results of analysis of routine mine dust samples collected from the areas concerned.

(2)

The areas referred to in sub-regulation (1) shall be clearly demarcated in the working belowground by means of suitable notice boards or by other suitable means.

(3)

Every part of mine which is not naturally wet throughout or which is not isolated by explosion-proof stoppings shall be treated -

(i)

with fine incombustible dust in such manner and at such intervals as will ensure that the dust on the floor, roof and sides and any support or structure shall always consist of a mixture containing not less than 75% of incombustible matter in case of coal seams containing less than 30% volatile matter (on dry ash free basis) and 85% of incombustible matter in case of coal seams containing more than 30% of such volatile matter; or

(ii)

with water in such manner and at such intervals as will ensure that the dust on the floor, roof and sides or any support or structure is always combined with not less than 30% by weight of water in intimate mixture; or

(iii)

in such manner as the Regional Inspector may approve by an order in writing.

(4)

The incombustible dust used for the purpose of sub-regulation (3) shall be -

(i)

such that it does not contain more than 5% of free silica;

(ii)

of such fineness and character, that it is readily dispersible into the air and that, when used in plates which are not directly made wet by water from the strata it does not cake but it is dispersed into the air, when blown upon with mouth or by a suitable appliance; and

(iii)

as far as practicable light in colour.

(5)

No incombustible dust shall continue to be used if it is found by tests which shall be carried out once at least in every three months, not to comply with the foregoing requirements:

Provided that when the supply of incombustible dust used in a mine is not from a regular source, these tests shall be carried out whenever a fresh supply of incombustible dust is received.

(6)

Where any place or part of the mine is to be treated with incombustible dust -

(i)

before treating with incombustible dust, all coal dust shall be cleaned, as far as practicable, from the roof, sides, floor, props, cogs, bars, brattice cloth or any other objects or structure or place on which coal dust may deposit, and all dust so collected shall be removed to the surface within 24 hours;

(ii)

incombustible dust shall be spread on the objects, structure and places aforesaid in adequate quantity and at such intervals as may be necessary to ensure compliance with the provisions of this sub-regulation;

(iii)

the cleaning of coal dust and spreading of incombustible dust shall be carried out in the direction of the flow of the air;

(iv)

a sufficient supply of incombustible dust shall be kept readily available at suitable places in the mine, and any deficiency in the supply of dust underground shall immediately be brought to the notice of the manager; and

(v)

incombustible dust stacked at different places and kept on pans or on dust barriers in the mine shall be changed whenever it is no longer readily dispersible or whenever it becomes coated with coal dust, such dust shall be removed.

(7)

A daily record of the areas cleaned of coal dust and of the areas treated with incombustible dust or with Water and the amount of incombustible dust used shall be maintained in a bound pagged book kept of the purpose and every entry in this book shall be signed and dated the dust in-charge, and countersigned by the manager or the ventilation officer.

(8)

The dust control measures shall be carried out under the supervision of a competent person holding a manager's or overman's certificate or a degree or diploma in mining or mining engineering from a university or institution approved by the Central Government and no duties with respect of sampling of dust under regulation 148 shall, be entrusted to this person who may be designated as the Dust In-Charge, nor any other duties shall be entrusted to such person except with the previous permission of the Regional Inspector.

Provided that in the case of a mine having an average monthly output of less than 5000 tonnes, the Dust Incharge shall act as the Sampling In-charge referred to in regulation 148.

(9)

The Dust Incharge shall ensure -

(a)

that every part of the mine which, under these regulations requires treatment with water, is thoroughly drenched or sprayed with water immediately before firing shots and also at intervals during the working hours so as to strictly comply with the provisions of regulation 147(3) (ii);

(b)

that every part of the mine which, under these regulations can be treated with incombustible dust, is so treated as to Strictly comply with the provisions of regulation 147(3) (i); and

(c)

that the arrangements for treating with water or incombustible dust as aforesaid are maintained in good order.

148.

Check on measures for dust control.

(1)

For the purposes of ensuring adequate treatment of coal dust as required under regulations 146 and 147, systematic samples of mine dust shall be collected, tested and analysed at intervals and in the manner specified in this regulation.

(2)

Every return airway as lies within two hundred metres of the last working face and every haulage, tramming or conveyor roadway which is not naturally wet throughout, shall be divided into zones not longer than one hundred and fifty metres each:

Provided that where in a mine some parts are being treated with water and others with incombustible dust, the zones shall be so formed that in each zone only one system of treating coal dust is being followed.

(3)

Every zone formed shall be divided into three equal sections, each not exceeding 50 metres in length.

(4)

Every zone shall be given a distinct number and every section, the code letters a, b or c in a systematic manner and such zones and sections, with their numbers and code-letters, shall be clearly marked on a plan prepared on a scale having a representative factor of not less than 2000: 1 hereinafter called the sampling plan, clearly showing the areas of the mine that are naturally wet throughout.

(5)

Every zone and section shall also be distinctly demarcated in the working below ground by means of suitable notice boards or by other suitable means.

(6)

Representative samples of dust shall be collected once in every 30 days from every zone, and for this purpose samples may be collected from different sections such as a, b or c in rotation such that; during every such period of 30 days, all samples are collected from the Section a or section b or section c.

(7)

Representative samples referred to in sub-regulation (6) shall be collected in a systematic manner irrespective of the cleaning and treating operations but shall in no case be collected within 24 hours of cleaning and treating of any zone, section or part thereof.

(8)

If the representative sample of mine dust from any particular zone shows that the provisions of regulation 147(1) have not been complied with, immediate steps shall be taken to clean and treat whole of the zone so as to comply with the provisions of the said regulation.

(9)

In every travelling roadway, and in every airway other than those specified in sub-regulation(2), samples shall be taken in such a systematic manner and at such intervals not exceeding three months, to maintain proper check on the efficiency of the treatment thereof in terms of regulation 146(2).

(10)

Samples of dust shall be collected from roof, sides, and floor, and shall comprise of dust collected to a depth not exceeding five millimeters on the roof and sides, and to a depth not exceeding 10 millimeters on the floor.

(11)

Where a zone is treated with incombustible dust, the samples shall be collected by a method of strip sampling, the strips being as near as possible of equal width of not less than 10 centimetres, and at uniform intervals not exceeding five metres.

(12)

Where a zone is treated with water, the samples shall be collected by a method of spot sampling such that a spot-collection of dust is made for every metre of the length sampled, as nearly as possible at regularly spaced intervals along a zigzag path.

(13)

In collecting the samples, the strips shall be extended into or spot collection made from any cross galleries upto the air-stoppings, if any.

(14)

Each sample shall be well-mixed and then reduced in bulk (by quartering) to a weight of not less than 30 grammes and be reduced and packed in a moisture-proof container which shall be suitably labeled or marked.

(15)

The sampling operations shall be carried out under the supervision of a competent person holding a manager's or overman's certificate or degree or diploma in mining or mining engineering from a university or institution approved for the purpose, who shall be designated as the Sampling Incharge and no other duties shall be entrusted to him except with the previous permission of the Regional Inspector.

(16)

Within seven days of taking of each sample, it shall be sent for analysis and the result of such analysis, immediately on its receipt, shall be recorded in a bound-paged book kept for the purpose and every entry in that book shall be signed and dated by the sampling incharge and be countersigned and dated by the manager.

Explanation - A place in a mine is considered naturally wet throughout if it is moist enough to keep the coal dust present, at any time, on the roof, sides and floor and other objects at that place so that it is always combined with not less than 30 percent by weight of water in intimate mixture.

149.

Stone dust barriers.

(1)

In every gassy seam of the second or third degree or in the development working in a gassy seam of the first degree in which there is likelihood of occurrence of inflammable gas in dangerous quantities, additional precautions shall be taken by providing stone dust barriers to prevent an ignition or explosion from extending from one part of the mine to the other.

(2)

Every stone dust barrier shall be of such type as may be approved by the Chief Inspector by a general or special order in writing and shall be maintained in such manner as may be specified in the said order:

Provided that the Chief Inspector may permit in any mine or part thereof alternative precautionary measures to be taken in lieu of stone dust barriers.

(3)

If any dispute arises as to whether stone dust barriers or other alternative precautionary measures are required to be provided in any part of a gassy seam of the first degree, under sub-regulation (1), the question shall be referred to the Chief Inspector who shall decide the same.

150.

Precautions against eruption of gas.

- Where any working is extended to within 30 metres of any goaf or disused working containing or likely to contain an

accumulation of inflammable or noxious gases, there shall be maintained at least one bore-hole not less than 1.5 metres in advance of the working and the operation of drilling the bore hole shall be carried out under the supervision of a competent person, and no lamp or light other than an approved safety lamp or torch shall be used in any such working.

151.

Recovery and exploratory work.

(1)

After an explosion of inflammable gas or coal dust has occurred in a mine, only such persons as are authorised by the manager or by the principal official present at the surface, shall be allowed to enter the mine.

(2)

Where it is intended or proposed to reopen a mine or part thereof, which has been isolated, sealed off or flooded with water to deal with a fire or spontaneous heating, the owner, agent or manager shall, not less than 30 days before the commencement of such work, give notice in writing of such intention or proposal to the Regional Inspector and the Chief Inspector.

(3)

Where it is intended to carry out any exploratory work in a mine or part belowground likely to contain irrespirable atmosphere, -

(a)

all work shall be done under rescue cover only;

(b)

no party of less than three persons shall be allowed to proceed to carry out such work; and

(c)

every such party shall carry a cage containing suitable birds or other means of detecting carbon monoxide gas approved by the Chief Inspector and also an approved flame safety lamp.

152.

Danger from surface water.

(1)

Where any mine or part thereof is so situated that there is any danger of inrush of surface water into the mine or part, adequate protection against such inrush shall be provided and maintained and whether such protection is adequate or not shall be determined by the Chief Inspector whose decision shall be final.

(2)

Except with the permission of the Chief Inspector in writing and subject to such conditions as he may specify therein and subject to the provisions of sub-regulation (1), every entrance into a mine shall be so designed, constructed, and maintained that its lowest point (which means the point at which a body of rising water on surface can enter the mine) shall be not less than 1.5 metres above the highest flood level at that point.

(3)

Every year, during the rains constant watch shall be kept on the flood levels on the surface of the mine and if at any time the levels cross the highest levels earlier recorded, such levels shall be marked by permanent posts along the edges of water and the new highest levels thus observed shall be recorded with the date of the highest flood level on the plans by an actual survey:

Provide that the highest flood level shall not be plotted on plans by interpolations.

(4)

If there are water dams or reservoirs built across rivers and water courses on the upstream side, arrangements shall be made for communication between appropriate authorities for the purpose of ascertaining the quantity and timing of water released from the dams which is like to endanger safety of the mine and arrangement for similar communication shall be made when water level rises on the upstream side which is likely to endanger any mine.

(5)

In every mine which is likely to be endangered by surface water, the highest flood levels and danger levels at least 1.2 metres or as required by the Regional Inspector, below the highest flood level, shall be permanently marked at appropriate places on the surface and whenever water rises towards the danger level at any place, all persons shall be withdrawn from the mine sufficiently in advance and for this purpose adequate arrangements of quick communication to an parts of the mine by effective systems shall be provided and maintained.

(6)

No working shall be made in any mine vertically below-

(a)

any part of any river, canal, lake, tank or other surface reservoir; or

(b)

any spot lying within a horizontal distance of 15 metres from either bank of a river or canal or from the boundary of a lake, tank or other surface reservoir, except with the permission in writing of the Chief Inspector and subject to such condition, as he may specify therein.

(7)

Every application for permission under sub-regulation (6) shall be accompanied by two copies of a plan and section showing the existing position of the working of the mine, the proposed layout of working, the depth of the surface, the position and depth of any goaves in every seam in the neighborhood, all faults, dikes and other geological disturbances and such other particulars as may affect the safety of the mine or of the persons employed therein.

Explanation. - Where sand or alluvium are lying in the course of a river, canal, lake, tank or reservoir, the depth from the surface shall be reckoned from the surface of hard ground underlying such sand or alluvium.

(8)

An working made under sub-regulation (6) shall be clearly demarcated belowground.

(9)

A competent person shall, once at least in every 14 days during the rainy season and once at least in every 30 days during other periods of the year, examine every protective measure provided under sub-regulations (1), (2), (3), (4) and (5), whether in use or not, for their stability and a report of every such examination shall be recorded in a bound paged book kept for the purpose, duly signed and dated by the person making the examination and countersigned by the manager.

(10)

The protective measure and working shall also be inspected, once at least in every quarter by the manager personally.

153.

Danger from underground inundation.

(1)

Proper provision shall be made in every mine to prevent eruption of water or other liquid matter or any material that is likely to flow when wet from the working of the same mine or of an adjoining mine and to prevent accidents while drilling bore-holes for probe or release of a body of water or other liquid matter or any material that is likely to flow when wet.

(2)

Where work is being done in-

(i)

any seam, a section below another seam or section, or

(ii)

any place in a seam or section, which is at a lower level than any other place in a lower seam or section, or

(iii)

any place in a seam approaching a fault passing through an upper seam or section, which contains or may contain an accumulation of water or other liquid matter or any material that is likely to flow when wet, or

(iv)

any water-bearing strata,

all useful information including the position, extent and depth of the above mentioned features shall be acquired and kept recorded and a scheme of working designed to prevent eruption of water or other liquid matter or any material that is likely to flow when wet shall be prepared and put into operation.

(3)

Without prejudice to the requirement of sub-regulation (1) and sub regulation (2), no working which has approached within a distance of 60 metres of any other working (not being the working which has been physically examined and found to be free from accumulation of water or other liquid matter or any material that is likely to flow when wet) and whether in the same mine or in an adjoining mine, shall be extended further except with the prior permission in writing of the Chief Inspector and subject to such conditions as he may specify therein.

Explanation. - For the purposes of this sub-regulation, the distance between the said working shall mean the shortest distance between the working of the same seam or between any two seams or sections, as the case may be, measured in any direction whether horizontal vertical or inclined.

(4)

Every application for permission to extend any working referred to in sub-regulation (3) shall be accompanied by two copies of the plan and section showing-

(a)

the outlines of all such disused or abandoned working in relation to the working approaching them and also the depth of such disused or abandoned working from the surface;

(b)

the outlines, the layout and the method of the proposed working for which permission is sought;

(c)

the faults, dykes and other geological disturbances in relation to working specified in clause (a) or (b); and

(d)

any other information that is available with the management and any other particulars or information that may be required by the Chief Inspector.

(5)

When permission is granted to extend any working referred to in sub-regulation (3) or sub-regulation (6), it shall be extended strictly in accordance with the plan and the method approved under, and the conditions specified in such permission and there shall be no variation therefrom unless such variation is again approved by the Chief Inspector.

(6)

Whenever seepage of water which is not normal to the seam is noticed at any place or working or if there be any such suspicion or doubt, such working shall immediately be stopped and the Chief Inspector and the Regional Inspector shall forthwith be informed of such seepage:

Provided that such working shall not be extended further except with the prior permission in writing of the Chief Inspector and subject to such conditions as he may specify therein.

(7)

The height or width of any working referred to in sub-regulation (3) or sub-regulation (6) shall not exceed 2.4 metres and there shall be maintained at least one bore-hole near the centre of the working face, and sufficient flank bore-holes on each side; and where necessary, bore-holes above and below the working at intervals of not more than five metres.

(8)

All bore-holes shall be drilled sufficiently close to each other to ensure that the advancing face will not accidentally hole through into a working containing water or liquid matter or any material that is likely to flow when wet and shall be constantly maintained at sufficient distance in advance of the working and such distance shall in no case be less than three metres.

(9)

The precautions under this regulation shall be carried out under the direct supervision of an official, having manager's or overman's certificate specially authorised for the purpose.

(10)

A record showing the exact height and width of such working, the number of boreholes driven, the length of each bore-hole, the places at which and the direction in which each bore-hole was driven, shall be maintained by the official referred to in sub-regulation (9) in a bound paged book kept for the purpose and the entries made therein shall be signed and dated by such competent person and shall be countersigned by the manager every day.

(11)

A plan and section of working referred to in sub-regulation (10), showing the particulars referred therein shall be prepared and maintained and they shall be brought up to date at least once in every fifteen days.

(12)

Unless specific relaxation is granted by the Chief Inspector in writing, the provisions of sub-regulations (7), (8) and (9), shall be strictly complied with while extending any working referred to in sub-regulation (3) or sub-regulation (6), whether or not the permission granted to extend such working requires compliance with all or any of the provisions of sub-regulations (7), (8) or (9).

(13)

If the Chief Inspector is satisfied that the conditions in any mine or part thereof are such as to render compliance with all or any of the provisions specified in sub-regulations (7), (8) or (9), unnecessary or impracticable, he may, by an order in writing and subject to such conditions as may be specified therein, relax, vary or dispense with all or any of the conditions and requirements contained in those sub-regulations, and, if he is of the opinion that the conditions at any mine or part thereof are such as to require additional precautions to be taken, he may by an order in writing require that such additional precautions besides those specified in those sub-regulations shall be taken.

154.

Intentional flooding.

(1)

Where the owner, agent or manager intends or propose, by introducing water from the surface or from any other part of the mine or from an adjacent mine, to flood any part of the working of his mine, he shall give, in writing, not less than 14 days notice of such intention or proposal to the Chief Inspector and the Regional Inspector and to the management of all adjoining mines and of such other mines as might be affected by such flooding;

Provided that the Regional Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit such operations to be commenced on any day prior to the expiry of the said 14 days;

Provided further that the Regional Inspector may, by an order in writing, either prohibit any such operation or require that such operation shall not be commenced until such precautions as he may specify therein, have been taken to his satisfaction.

(2)

If the operations in respect of which notice is given under sub-regulation (1) are not commenced within 60 days from the expiry of the said 14 days, the notice shall be deemed to have lapsed and the provision of the sub-regulation (1) shall apply as if no such notice had been given.

155.

Construction of a water dam, etc.

(1)

Where in any mine, it is intended to construct a reservoir, dam or other structure to withstand a pressure of water or other material which will flow when wet, or to control an inrush of water, the owner, agent or manager shall give in writing not less than 14 days notice of such intention to the Regional Inspector and every such notice shall be accompanied by two copies of plans and sections showing the design and other details of the proposed construction:

Provided that where the safety of the mine or of the persons employed therein is seriously threatened, the provisions of this regulation shall be deemed to have been complied with if the said notice is given to the Regional Inspector as soon as the work of construction is commenced:

Provided further that where such a reservoir, dam or other structure was constructed before the coming into force of these regulations, the said copies of the plans and sections shall be submitted to the Regional Inspector within three months of the coming into force of these regulations and where these details are not available, the Regional Inspector shall be informed of the fact within the aforesaid period.

(2)

The Regional Inspector may, by an order in writing, require such modification or alternations to be made by the design of any such reservoir, dam or structure, as he may specify therein.

Chapter-XII Ventilation

156.

Standard of ventilation.

(1)

It shall be the duty of the owner, agent or manager of every mine to take such steps as are necessary to provide in all parts of the mine belowground which are not sealed off, adequate ventilation to clear away smoke, steam and dust, to dilute gases that are inflammable or noxious so as to render them harmless, to provide air containing sufficient oxygen and to prevent such excessive rise of temperature or humidity which may be harmful to the health of persons.

(2)

For the purposes of securing adequate ventilation, the owner, agent and manager shall ensure that -

(a)

in every ventilating district, not less than six cubic metre per minute of air per person employed in the district on the largest shift or not less than 2.5 cubic metres per minute of air per tonne output daily, whichever is larger, is passed along the last ventilation connection in the district i.e., the inbye-most gallery in the district along which the air passes;

(b)

at every place in the mine where person are required to work or pass, the air does not contain less than 19 percent of oxygen or more than 0.5 percent of carbondioxide or any noxious gas in quantity likely to affect the health of any person;

(c)

the percentage of inflammable gas does not exceed 0.75 in the general body of the return air of any ventilating district and 1.25 in any place in the mine;

(d)

the wet bulb temperature in any working place does not exceed 33.5 degrees centigrade and where the wet bulb temperature exceeds 30.5 degrees centigrade, arrangements are made to ventilate the same with a current of air moving at a speed of not less than one metre per second,

and for ensuring compliance with the provisions of clauses (b), (c) and (d) of this sub-regulation; air samples and temperature readings shall be taken at least once in 30 days and the result shall be recorded in a bound paged book kept for the purpose:

Provided that where at any mine or part, special conditions exist which make compliance with any of the above provisions not necessary or reasonably not practicable the Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, grant a relaxation from the provisions.

(3)

If with respect to any mine or part thereof the Regional Inspector is of the opinion that the ventilation is not adequate, he may, by an order in writing, require the installation and maintenance of such a mechanical ventilator as is capable of producing adequate ventilation in the mine or part.

157.

Main mechanical ventilator, its drive and fittings etc.

(1)

Every main mechanical ventilator in a mine shall be capable of producing adequate ventilation in the mine or part and shall be installed on the surface at a distance of not less than 10 metres from the opening of the shaft or incline at any point.

(2)

If electricity is used for driving the mechanical ventilator, electrical energy shall be supplied to the drive motor of the ventilator through a separate circuit from the main distribution point of the mine.

(3)

In every gassy seam of second or third degree, two different sources of power to the mechanical ventilator shall be provided.

(4)

There shall be provided and maintained at every main mechanical ventilator, a recording instrument to continuously register the pressure developed.

(5)

Every main mechanical ventilator shall be so designed, installed and maintained that the current of air can be reversed when necessary.

(6)

At every shaft or incline ordinarily used for lowering or raising of persons or materials where a mechanical ventilator is installed, there shall be provided a properly constructed air lock.

158.

Restriction of installation of mechanical ventilator belowground.

(1)

In every gassy seam of the second or third degree, the following provisions shall have effect in relation to the installation belowground of booster fans:-

(a)

No booster fan shall be installed belowground in the mine unless the manager is satisfied, as a result of a survey of the

ventilation of every part of the mine liable to be affected, that such installation is necessary or expedient for the proper ventilation of the mine and that it should be installed.

(b)

Seven days' prior notice of every such installation under clause (a), together with particulars of the survey aforesaid, shall be sent to the Regional Inspector.

Explanation. - The shifting of a booster fan from one place to another shall be deemed to be an installation of a booster fan for the purposes of this regulation.

(2)

The Regional Inspector may at any time, by an order in writing, require the use of any booster fan installed belowground in the mine to be discontinued.

159.

Installation and maintenance of mechanical ventilators.

(1)

In every below ground coal mine where a booster or auxiliary fan is electrically driven, the drive motor, unless it is so constructed, installed, operated and maintained as to prevent the risk of open sparking, shall not be placed in a return airway.

(2)

The installation of every mechanical ventilator and booster fan shall be supervised and controlled by a competent person appointed for the purpose.

(3)

A competent person appointed for maintenance of mechanical ventilator and booster fan shall once at least in every seven days, examine every mechanical ventilator and booster fan in use and shall record the results of such inspection in a bound paged book kept for the purpose.

(4)

Except in an emergency, no person shall start, stop, restart, remove or in any way alter, repair or interfere with any ventilator or booster fan, except on the written authority of the manager or other official authorised in writing in this behalf.

(5)

The written authority referred to under sub-regulation (4) shall clearly specify the conditions under which a ventilator or booster fan shall be started, stopped or restarted, the period for which it can be stopped and the procedure for removal, repair, alterations or interference with such fans.

(6)

Particulars of every alteration and every stoppage including any stoppage beyond control together with the duration thereof, shall be recorded by the competent person appointed under this regulations, in a bound paged book kept for the purpose:

Provided that whenever a mechanical ventilator or a booster fan has been stopped for any reason whatsoever, the competent person shall immediately record the time of stoppage in the aforesaid book and send it to the manager or the person authorised in this behalf for his appraisal and such the manager, or the person authorised, or the official, as the case may be, shall sign the entry in the bound paged book in respect of the stoppage, duly signed and dated and return the book to the competent person:

Provided further that when the stoppage of mechanical ventilator or a booster fan has been brought to the notice of the official other than the manager or person authorised in this behalf, the said official shall immediately inform the fact in writing to the manager or in his absence to any competent person appointed under this regulation or the official superior to the manager, about such stoppage and the manager or such person or superior official, as the case may be, shall acknowledge in writing the receipt of such information sent to him and shall, without prejudice to the standing order under regulation 160 take such action as may be expedited for the safety of persons employed in the mine and the details of every action thus taken by him shall be recorded in the aforesaid book.

(7)

The manager or the person authorised or the senior official, referred to in the sub-regulation (6), on being informed or coming to know of the stoppage of a mechanical ventilator or a booster fan, shall decide about disconnecting electric supply to the mine and the extent of such disconnection and the details of such disconnection shall be entered in the

bound paged book-kept for the purpose:

Provided that in the case of stoppage of the mechanical ventilator or the booster fan, the electrical supply or the other source of drive to any auxiliary fan, if, installed, shall be disconnected immediately by an arrangement of sequence control, or other effective arrangement.

(8)

In every mine in which a mechanical ventilator is in use, the quantity of air shall, once at least in every 14 days be measured -

(i)

In every main intake and return airway of every seam or section, as near as practicable to the entrance to the mine;

(ii)

in every split, as near as practicable to the point at which the split commences;

(iii)

in every ventilating district, as near as practicable to the point where the air is subdivided at the end of a split and also where it enters the first working place; and

(iv)

any other point the Regional Inspector may, by an order in writing, specify,

and the measurements shall be entered in a bound paged book kept for the purpose:

Provided that in a non-fiery seam or a gassy seam of the first degree it shall be sufficient to take the air measurement once in every 30 days.

(9)

The measurements referred to in sub-regulation (8) shall also be taken and recorded whenever the system of ventilation is so altered as to substantially affect or likely to affect the ventilation of the mine:

(10)

Every such ventilator or fan shall be in charge of a competent person appointed for the purpose, who shall not be entrusted with any other additional duties which may interfere with his duties as in charge of the ventilator or fan, as the case may be.

160.

Standing orders.

(1)

The manager of every mine in which a mechanical ventilator other than an auxiliary fan is installed shall submit to the Regional Inspector in the case of a new installation, within 30 days of the installation, standing orders specifying the action that shall be taken with respect to the withdrawal of persons from the mine or part thereof in the event of a stoppage of the ventilator.

(2)

The Regional Inspector may, by an order in writing, approve the standing orders referred to in sub-regulation (1), either in the form submitted to him or with such additions and alterations as he may think fit and the standing orders so approved shall be enforced at the mine.

(3)

A copy of the Standing Orders in English and other local language understood by majority of the persons employed in the mine shall be pasted at conspicuous places in the mine, both above and belowground.

161.

Splits and airways.

(1)

For the purposes of ventilation, every mine shall be divided into such number of districts or splits as to ensure that separate current of fresh air is made available in every such district or split.

(2)

The intake air shall be so arranged as to travel away from all stagnant water.

(3)

In every ventilating district there shall be provided two main intake airways one of which shall be used as a travelling roadway:

Provided that if the Regional Inspector is satisfied that compliance with this regulation is not reasonably practicable, he

may, by an order in writing and subject to such conditions as he may specify therein, grant exemption from the provisions thereof.

(4)

Every connection between a main intake airway and a main return airway shall until it is no more required and has been sealed off, be provided with at least two doors so spaced that whenever one door is opened, the other door can be kept closed and steps shall be taken to ensure that at least one of the doors is always closed.

Provided that any such connection which is no more required shall be effectively sealed;

162.

Brattices, doors, stoppings and air-crossings.

(1)

There shall be provided and maintained in every mine, such number of air-crossings, stoppings, doors and other devices as may be adequate to ensure compliance with the provision of regulation 156 and in case of any doubt as to the adequacy of such ventilation devices, it shall be referred to the Chief Inspector for decision.

(2)

The space between the frame of every ventilation door and the roof and sides of the roadway shall be built up with masonry or concrete; not less than 25 centimetres in thickness.

(3)

Every ventilation door shall be self-closing and whenever opened, it shall be closed as soon as possible, and shall not be propped or fixed so as to remain open.

(4)

If the ventilation door is required to be frequently kept open for the passage of men or material, there shall be, throughout every working shift, a door attendant at the door.

(5)

If a door is not in use, it shall be taken off its hinges and placed in such a position that it shall not cause any obstruction to the air current.

(6)

Every stopping between the main intake and main return airways shall be constructed of masonry or brickwork or of concrete without reinforcement, not less than 25 centimetres in thickness, and if constructed of properly reinforced concrete, of 15 centimeter thickness or such greater thickness as may be required by the Regional Inspector, and shall be faced with a sufficient thickness of lime or cement plaster to prevent leakage of air.

(7)

In case of mine having problems of fires or spontaneous heating along with the working of coal seam(s) of degree two or degree three gassiness, the Chief Inspector may by an order in writing and subject to such conditions as he may specify therein, require construction of explosion proof ventilation stoppings between the main intake and the main return airways and at such other places which he may specify.

(8)

Every stopping in use shall be kept accessible for inspection.

(9)

The partitions and walls of every air-crossing shall be not less than 25 centimetres in thickness if constructed of masonry or of concrete not properly reinforced, and not less than 15 centimeters in thickness if constructed of properly reinforced concrete.

(10)

Every air-crossing, ventilation stopping, poor or brattice shall be maintained in efficient working order and good repair.

(11)

A competent person shall, once at least in every 14 days, examine every airway, air crossing, ventilation stopping and door in use, and shall record the result thereof in a bound paged book kept for the purpose, and shall sign the same and date his signature.

163.

Velocity of air current.

- The velocity of air current measured in metres per minute at the place shown in column (2) shall be not less than that shown in column (3) for the different seams shown in column (1) of the table given below:-

Table

Degree of gassiness

Place where velocity of air is to be measured

Velocity of air

1

2

3

First, Second or Third degree

Immediate outbye ventilation connection from the face

30

First or Second degree

(i) 4.5 metres from any face

whether working or discontinued on the intake side of the brattice or partition;

(ii) 7.5 metres outbye of the

discharge end of an air pipe;

(iii) at the maximum span of a longwall face.

30

15

60

Third degree

(i) 4.5 metres from any face

whether working or discontinued on the intake side of the brattice or partition;

(ii) 7.5 metres outbye of the

discharge end of an air pipe;

(iii) at the maximum span of a longwall face:

45

25

75

Provided that if in the opinion of the Chief Inspector or the Regional Inspector the compliance with the above provision is not practicable or not necessary, he may, by an order in writing and subject to the conditions specified therein exempt partially or totally any mine from the provisions of this regulation.

164.

Auxiliary fans.

(1)

Every auxiliary fan,-

(a)

shall be installed, located and worked in such a manner that-

(i)

a sufficient quantity of air shall, at all times, reach it so as to ensure that it does not re-circulate air;

(ii)

there is no risk of the air which it circulates being contaminated by any substantial quantity of inflammable or noxious gases or dust; and

(b)

shall, whether driven electrically or otherwise, be efficiently connected with earth so as to avoid the accumulation of an electrostatic charge;

(c)

shall have an approved type of air-duct for conducting the air to or from the face or blind end; and

(d)

such air-duct shall be so maintained as to minimize any leakage of air and to ensure an adequate supply of air to within

3.0 metres of the face or blind end.

(2)

No auxiliary fan shall be started, stopped, removed, replaced or in any way altered or interfered with, except by or on the authority of an official.

(3)

No person shall enter or remain in any place which is dependent for its ventilation on an auxiliary fan, unless such fan is operating efficiently:

Provided that whenever the ventilation of any such place has been interrupted, whether by the stoppage of the fan or otherwise, no person shall so enter or remain therein, except for the purpose of restoring the ventilation, unless the place has been examined by a competent person and declared safe.

(4)

In every gassy seam of the second or third degree,-

(a)

two or more auxiliary fans shall not be installed in the same ventilating district or split, except with the permission in writing of the Regional Inspector and subject to such conditions as he may specify therein;

(b)

the application made under clause (a) for any such installation shall be accompanied by a plan showing the general system of ventilation, the quantity of air flowing in the split and the proposed position of the fans; and

(c)

the Regional Inspector may at any time, by an order in writing require that the Use of any auxiliary fan shall be discontinued.

165.

Conditions for installation and operation of two or more auxiliary fans.

(1)

Before any auxiliary fan is installed, and also, at least once every week thereafter, the quantity of air flowing in the airway at the point of its installation shall be measured by the ventilation officer:

Provided that the fan shall not be installed or continued to be operated unless the measurement shows that sufficient quantity of air is available so that -

(i)

not more than 50% of total air passing in the roadway is taken by the auxiliary fan; and

(ii)

ventilation of the inbye faces when the fan is running is adequate.

(2)

Particulars of the quantities measured as referred to in sub-regulation (1), shall be duly recorded in a bound paged book kept for the purpose.

(3)

There shall also be kept at the mine an accurate plan showing the general system of ventilation in the district and quantity of air in each air current therein.

(4)

The auxiliary fans shall be installed, as forcing fan, on the intake side of the place to be ventilated by it:

Provided that no auxiliary fan shall be installed at a point within or less than 4.5 m from the nearer side of the entrance to the place to be ventilated by it.

(5)

Air entering any of the auxiliary fans in the district shall not contain-

(i)

more than 0.3% CH₄; and

(ii)

excessive quantity of dust.

(6)

The percentage of CH₄ in the return air in any heading ventilated by an auxiliary fan shall not exceed 0.5.

(7)

The auxiliary fans in the district shall be electrically interlocked by a sequence control which ensures that unless the first fan is running the second fan cannot be started and so on.

(8)

No person shall start an auxiliary fan on any occasion unless the overman or a competent person authorised in this behalf considers that it is safe for the fan to be so started.

(9)

An auxiliary fan on the return side shall not be started or re-started after stoppage of the auxiliary fan on its intake side due to power failure or otherwise, unless an overman or a competent person specially authorised by the manager for the purpose, has also satisfied himself that the condition referred to in sub-regulation (5) is fulfilled.

(10)

No workman shall enter or be allowed to enter in any heading ventilated by an auxiliary fan unless the overman or competent person has inspected the place and found it safe and the requirements referred to in sub-regulation (5) are also satisfied.

(11)

In case of any interruption in the ventilation arrangements for working places, all persons shall be withdrawn from the area affected, with the exception of those required to rectify the situation.

(12)

No person other than an official of the mine, ventilation officer or a person authorised by the manager to do so, shall regulate the quantity of air passing through or delivered by any auxiliary fan:

Provided that if it is necessary to regulate an auxiliary fan, it shall be done in such a way as to prevent unauthorised or inadvertent alteration.

(13)

Fans delivering air to flexible ducting shall not be regulated by constricting the ducting.

(14)

Fans with rigid ducting shall not be regulated by placing loose obstruction such as bricks or stone in the ducting.

(15)

The supply of electricity to the working place shall only be effected when the auxiliary fan is running satisfactorily:

Provided that on restarting the fan after a prolonged shutdown, sufficient time shall be allowed to elapse to clear any accumulation of gas or foul air before re-establishing the supply of electricity to the working place.

166.

Precautions against fire in ventilation appliances.

(1)

Every mechanical ventilator on the surface shall be installed in a suitable fire proof housing.

(2)

In the case of every fan other than an auxiliary fan installed belowground, the coal or other carbonaceous material exposed in the sides, roof and floor shall be covered with masonry or other fire resistant sealant as protection against fire, for a distance of not less than 5 metres in every direction from the fan.

(3)

The covering of every shaft sealed off or covered for ventilation purposes, every fan drift, duct or casing and every part of a mechanical ventilator or fan within such drift, duct or casing, and every air-crossing and ventilation door shall be constructed of fire-proof material:

Provided that this regulation shall not apply to the small lid of a shaft-covering operated by the rope cappel.

167.

Ventilation plans to be brought up-to-date.

- The manager shall ensure that as soon as any alteration is made in the ventilation of a mine involving the erection or removal of an air-crossing or stopping or the alteration in the position of or installation of a ventilator or fan other than an auxiliary fan belowground, the erection, removal, alteration or installation, as the case may be, is notified to the surveyor who shall forthwith make necessary alterations on the ventilation plan maintained under regulation 68.

168.

Obstructions, interruptions and alterations.

(1)

No material or debris shall be allowed to accumulate in any level, drive, crosscut or any other part of the working below ground so as to impede the ventilation.

(2)

Every roadway in working below ground which is not adequately ventilated shall be fenced or barricaded so as to effectively prevent persons entering the same.

(3)

If any person becomes aware of any obstruction in, or interference with, or deficiency of, ventilation in any mine or part thereof, he shall -

(a)

if it falls within his power to remedy such obstruction, interference or deficiency, immediately take steps to do so; or

(b)

cease all work at that place, and shall forthwith inform his superior official of such obstruction, interference or deficiency.

(4)

Whenever there is any interruption of ventilation by the stoppage or any mechanical ventilator, including an auxiliary fan, installed below ground, the official in charge of the mine or part shall immediately take precautionary measures including withdrawal of men, if necessary against dangers that may arise out of non-compliance with the provisions of regulation 156, to restore the ventilation in the mine or part.

(5)

No person shall alter the general system of ventilation in any mine or part except with the authority of the manager:

Provided that, in an emergency, an official of the mine may carry out such alteration as he may deem necessary for the safety of persons, but he shall as soon as possible inform his superior official and the manager about the same in writing.

169.

Precautions against gas during de-watering and re-opening.

(1)

No disused mine or shaft shall be de-watered except under the constant supervision of a competent person and during such de-watering, approved safety lamps or torches shall be exclusively used, and there shall also be kept burning at every place where persons are at work, at least one approved flame safety lamp.

(2)

The first inspection of a mine or part which is re-opened after discontinuance of mining operations for a period exceeding seven days and of any part of a mine after being de-watered, shall be made by a competent person with an approved flame safety lamp or other approved apparatus for determining presence of inflammable or noxious gases and deficiency of Oxygen and during such inspection, no additional light or lamp other than an approved electric torch or lamp shall be used.

(3)

The result of every such inspection shall be recorded in a bound paged book kept for the purpose, and shall be signed and dated by the persons making the inspection, and countersigned and dated by the manager.

170.

Precautions against inflammable and noxious gases.

(1)

For the purposes of this regulation, inflammable gas shall be deemed to have been found or detected -

(i)

when it is indicated by a methane detector to be 0.1% or more in case of mine having only degree one gassy seams;

(ii)

when it is indicated by a methane detector to be 0.5% or more in case of mine having degree two seams; and

(iii)

when with the lowered flame of a flame safety lamp or, where methane indicators are used, they indicate one and a quarter percent, or more of inflammable gas in case of mine having only degree three gassy seams.

(2)

When any person detects the presence of inflammable gas, he shall not brush or waft it out, but shall immediately withdraw from the place and shall inform his superior official about the same.

(3)

Where in any place in a mine, inflammable or noxious gas is detected all persons shall be withdrawn from the place, and the place shall be immediately fenced off so as to prevent persons inadvertently entering the same and the competent person in charge shall, without delay, take steps to remove the gas by improving the ventilation.

(4)

During the removal of such gas under sub-regulation (3) all persons except those necessary for such removal shall be withdrawn from the return side of the ventilating district in which the gas has been detected unless the quantity of gas is, in the opinion of the competent persons, so small that such withdrawal is not necessary:

Provided that where the danger arises from the presence of inflammable gas, no naked light shall be used in the ventilating district in which the gas is detected.

(5)

No person shall be re-admitted into the place where the gas was detected until a competent person has examined the place and has reported that the place is free from gas and every such examination shall be made with an approved flame safety lamp and, in the case of noxious gas, also with other means of detecting carbon monoxide gas approved by the Chief Inspector.

(6)

Particulars of every occurrence referred to in sub-regulation (2) and of every examination made under sub-regulation (5) together with a statement as to where and when the gas was found and when it was removed, and in case of inflammable gas, the percentage thereof, shall be recorded in a bound paged book kept for the purpose and every such entry shall be signed and dated by the competent person making the report, and countersigned and dated by the manager.

171.

Inspection of unused working for gas.

- In any fiery seam or gassy seam of the second or third degree or where the Regional Inspector may require by an order in writing, all unused working which have not been sealed off, shall, once at least in every seven days, be inspected by a competent person for the presence of inflammable or noxious gas and a report of every such inspection shall be recorded in a bound paged book kept for the purpose, duly signed and dated by the person who made the inspection.

172.

Safety lamps to be used in belowground mines.

- No lamp or light other than an approved safety lamp or torch or other installation permitted under the Indian Electricity Rules, 1956 shall be used or permitted to be used below ground in any mine:

Provided that the Chief Inspector may, by an order in writing and subject to such condition as he may specify therein, exempt any mine or part thereof from the operation of this regulation on the ground that on account of its special character the use of safety lamps is not necessary therein.

173.

Determination of percentage of Inflammable gas and of environmental conditions.

(1)

Where electric energy is used in any ventilating district, determination shall be made of the percentage of inflammable gas present in the general body of air and the following provisions shall apply in respect of such determination, namely: -

(a)

The determination shall be made by a competent person either by means of an apparatus of a type approved for the purpose by the Chief Inspector, or by analysis of samples of air:

Provided that if determination is made by the analysis of air, the samples shall be analyzed within three days of the taking thereof.

(b)

The determination shall be made or samples of air taken at suitable point fixed by the manager, on the intake side of the first working place and on the return side of the last working place in the district:

Provided that where the Regional Inspector is of the opinion that the location of any such point is unsuitable, he may, by an order in writing, require the manager to fix some other point or points in substitution thereof.

(c)

The determination shall be made or samples of air taken, as the case may be, once at least in every seven days, so however that -

(i)

if any determination shows the percentage of inflammable gas exceeding 0.8, determination shall be made or samples of air taken at intervals not exceeding hours for so long as such content exceeds that percentage and for the seven next following days, unless the Regional Inspector otherwise permits by an order in writing and subject to such conditions as he may specify therein; and

(ii)

if the determination made during the 30 days immediately preceding any day have shown the percentage of inflammable gas to be below 0.6, it shall be sufficient to make such determination or take such samples, once in every 30 days for so long as such content does not exceed that percentage:

Provided that when any alteration is made in the system of ventilation so as to substantially affect or likely to affect the ventilation of the mine, such determination shall be made within 24 hours of such alteration.

(d)

Particulars of every such determination under this regulation shall be recorded in a bound paged book kept for the purpose.

(e)

If any determination in any ventilating district shows the percentage of inflammable gas to exceed one and a quarter, the supply of electric energy shall be cut off immediately from all cables and apparatus in the district, and a written report thereof submitted to the Regional Inspector forthwith.

(2)

If the Regional Inspector so requires by an order in writing in respect of any mine having working belowground, determination shall, once at least in every 30 days, be made of temperature, humidity and such other environmental conditions at such points as the Regional Inspector may specify in the order.

174.

Monitoring devices.

(1)

The Chief Inspector may, if he considers necessary for the safety of persons, require by an order in writing that in any mine or any class of mines belowground, approved environmental monitoring devices to continuously record information regarding environmental conditions, to be installed belowground within such time and subject to such conditions as he may specify therein.

(2)

The Chief Inspector may, if he considers necessary for the safety of persons, require by a general or special order in writing analysis of mine air samples by gas chromatographs.

175.

General precautions in gassy mines.

- In every mine, the following provisions shall have effect, namely: -

(a)

No working or gallery shall be extended to a distance of more than 4.5 metres from the nearest ventilation connection unless the current of air is coursed upto a point within 4.5 metres of the face by means of fire resistant pipes, tubes brattices or other material.

(b)

No narrow main or advance gallery shall be driven more than three meters ahead of the widened gallery.

(c)

Every stopping between the main intake and the main return airways shall be substantially built and every air-crossing shall be constructed and maintained as to withstand the force of an explosion.

(d)

The main air current shall be so split and coursed that an air current which ventilates a goaved out area, whether packed or unpacked, or any disused working shall not, except with the prior permission in writing of the Regional Inspector and subject to such conditions as he may specify therein, ventilate ally working where coal is being extracted.

(e)

No major alteration shall be made in the system of ventilation except with the prior permission in writing of the Regional Inspector and subject to such conditions as he may specify therein:

Provided that where the safety of the mine or of the persons employed therein is seriously threatened, the provision of this clause shall be deemed to have been complied with, if information of such alteration is sent to the Regional Inspector forthwith:

(f)

Except in an emergency, when a major alteration is made in the system of ventilation, only such persons as are engaged in making the alteration shall be present belowground.

176.

Contrabands.

- In every mine the following provisions shall have effect, namely:-

(a)

No person shall have in his possession belowground any cigar, cigarette, biri, mobile phone or other smoking apparatus, or any match or other apparatus of any kind capable of producing a light, name or spark:

Provided that nothing in this clause shall be deemed to prohibit the use belowground of any apparatus for the purpose of shot firing or of relighting safety lamps, of a type approved by the Chief Inspector.

(b)

For the purpose of ascertaining whether any person proceeding belowground has in his possession any article referred to in clause (a), a competent person other than the banksman, if any, shall be appointed to search every such person immediately before he enters the mine and the competent person shall be on duty throughout the shift, and no duties other than those under this regulation and regulation 183(b) shall be entrusted to him.

(c)

The competent person so appointed shall make a thorough search for the articles referred to in clause (a) and in particular shall -

(i)

search or turn out all pockets;

(ii)

pass his hand over all clothing; and

(iii)

examine any article in the possession of the person searched.

(d)

If the competent person suspects that the person searched is concealing any articles he shall detain him, and as soon as possible refer the matter to the manager or assistant manager and no such person shall be allowed to enter the mine until the manager or other superior official is satisfied that the person has no such article in his possession.

177.

Underground relighting stations.

(1)

In every mine, lamp stations for relighting safety lamp may be fixed by the manager at suitable places belowground legibly marked Relighting Station, which shall be situated in a main intake airway, and shall be placed in charge of a competent person.

(2)

No person shall be appointed as a competent person under this regulation unless he holds a Gas testing Certificate.

Chapter-XIII Lighting and Safety Lamps

178.

Whitewashing.

- The roof and sides of the following places belowground in a mine shall be kept effectively whitewashed, namely: -

(a)

every shaft inset and shaft bottom or siding and every by-pass which is in regular use;

(b)

the top and bottom of every haulage plane, every regular stopping place, siding, landing, by-pass, and junction, except within 100 metres of the face;

(c)

every travelling roadway;

(d)

every room and place containing any engine, motor or other apparatus; and

(e)

every first aid station belowground.

179.

General lighting.

(1)

Adequate general lighting arrangements shall be provided during working hours -

(a)

on the surface where the natural light is insufficient-

(i)

in every engine house;

(ii)

in the vicinity of every working shaft;

(iii)

at every open cast working;

(iv)

at every shunting or marshaling yard; and

(v)

at every place where persons have to work; and

(b)

belowground-

(i)

at every shaft landing and shaft bottom or siding which is in regular use;

(ii)

in every traveling roadway normally used by 50 or more persons during any shift:

Provided that the provisions of this sub-clause shall be deemed to have been complied with where electric lamps or lights are provided to every work person;

(iii)

at the top and bottom of every self-acting incline in regular use;

(iv)

at every place on haulage roadway, at which tubs are regularly coupled or uncoupled or attached to or detached from a haulage rope;

(v)

at every place at which tubs are regularly filled mechanically;

(vi)

at every room and place containing any engine, motor or other apparatus;

(vii)

at every place where any pillar is under extraction; and

(viii)

at every first aid station below ground :

Provided that the lighting fixtures installed in a gassy seam of the second or third degree and in the blind ends of a gassy seam of the first degree which are not ventilated by a mechanical ventilator shall comply with the provisions of the Indian Electricity Rules, 1956.

(2)

The lighting provided in a mine shall, as far as possible, be so arranged as to prevent glare or eye strain.

(3)

Where electric energy is available at the mine, the lighting arrangement made under sub-regulation (1) shall be by electrical means.

(4)

Where electric lighting is used, an addition light or lamp having no connection with electric supply shall be kept continuously burning -

(i)

belowground, in every shaft bottom or landing in regular, use and in every engine room;

(ii)

on the surface, after dark, at the top of every working shaft and in every engine room; and

(iii)

in travelling roadway and escape routes.

(5)

Every electrical lamp-fitting shall be so constructed as to protect it from accidental damage and adequate precaution shall be taken to prevent lamps being damaged from shot-firing.

(6)

In addition to the lights required to be provided under this regulation, fluorescent or luminescent path finder or indicator of a type approved by the chief Inspector shall be provided all along the travelling roadway and if the escape route in the mine.

180.

Electric lighting in gassy mines.

(1)

Subject to the provisions of Indian Electricity Rules, 1956 relating to the use of electric energy in parts of mines in which inflammable gas is likely to occur in quantity sufficient to be indicative of danger, electric lighting from a source of supply external to the lighting unit may be used -

(a)

on any roadway or place ventilated by intake air; and

(b)

on any other roadway or place not within 270 metres of the nearest face.

(2)

In every mine or part thereof to which regulation 172 applies, every electrical lighting apparatus shall be of a type approved by the Chief Inspector:

Provided that electric lighting from a source of electric power enclosed in the lighting unit, electric lights which are fittings or accessories to machinery or electrical plant including signaling apparatus, and any other means of lighting not specifically mentioned in this regulation, may be used in the mine if so permitted by the Chief Inspector by an order in writing and subject to such conditions as he may specify therein.

181.

Every person to carry a light.

(1)

The owner, agent or manager shall provide every person employed belowground with a light or lamp adequate to enable him to perform his duties in a proper and thorough manner and no such person shall proceed or remain belowground without such light or lamp.

Provided that on his return to the surface, every such person shall, unless otherwise directed by the manager by a general or special order in writing immediately return his lamp to the lamp room.

(2)

The number of safety lamps at every mine shall be adequate to permit thorough cleaning and checking before they are issued and in case of any doubt as to the sufficiency or otherwise of the safety lamps provided at a mine, it shall be referred to the Chief Inspector for decision.

182.

Standard of lighting.

(1)

If any doubt arises as to whether any lamp or light is of adequate lighting performance or not, it shall be referred to the Chief Inspector for decision.

(2)

The Chief Inspector may, from time to time, by notification in the Official Gazette, specify-

- (a)
the type of lamp to be provided to specified categories of persons employed in a mine; or
- (b)
the standard of lighting to be provided in specified areas or places in a mine.

(3)
All lights and lighting fixtures used to provide required lighting at specified places in the mine shall be of a type, make and standard approved by the Chief Inspector.

183.

Maintenance and examination of safety lamps.

- For ensuring proper maintenance of safety lamps in use in the mines the following provisions shall have effect, namely:

-

(a)
a competent person appointed for the purpose shall clean, trim, examine and lock securely all such lamps before they are issued for use, and no such lamp shall be issued for use unless it is in safe working order and securely locked.

(b)
a competent person appointed for the purpose shall examine every safety lamp at the surface immediately before it is taken belowground for use and shall assure himself, as far as practicable from external observation that such lamp is in safe working order and securely locked:

Provided that the person so appointed shall not perform any other duties, other than those prescribed under regulation 176(2).

(c)
a competent person appointed for the purpose shall examine every safety lamp at the on its being returned after use and if on such examination, any lamp is found to be damaged or misused, he shall record the nature of the damage or misuse in a bound paged book kept for the purpose, every such entry shall be countersigned and dated by the manager.

(d)
the manager, assistant manager, or a competent person appointed for the purpose shall, once at least In every seven days, examine thoroughly every safety lamp in use, and shall record the result of examination of every such lamp in a bound paged book kept for the purpose.

(e)
no person shall be appointed as a competent person under this regulation unless he holds a Manager's, Overman's or Gas testing Certificate.

184.

Use of safety lamps.

(1)
Every safety lamp shall be numbered and so long as the lamp is in use the number thereof shall be maintained in a legible condition.

(2)
A competent person appointed for the purpose shall maintain a correct record of the lamps issued from and returned to the lamp, and in the record so maintained, the number of the lamp issued to and person shall be entered against his name.

(3)
If any person returns to the lamp-room a lamp other than the one issued to him, he shall explain the cause and circumstances of the change.

(4)
No unauthorised person shall either himself take or give out any safety lamp from the lamp-room.

(5)
Every person who receives a lamp shall satisfy himself that it is complete and in good order and should he find any defect therein, he shall immediately return it to the lamp-room.

(6)
No person shall willfully damage or improperly use, or unlock or open, or attempt to unlock or open any safety lamp.

(7)

Should any person find that the safety lamp in his possession has become defective, he shall at once carefully extinguish the flame, if any, and report the defect to his superior official.

185.

Maintenance and repairs of safety lamps and electric torch or lamp.

(1)

Every approved safety lamp and electric torch or lamp shall be properly assembled and maintained in good order and if any such lamp or torch is found to be defective or damaged, it shall not be used or issued for use until the defect or damage has been remedied.

(2)

If the wires of any gauge of a flame safety lamp and electric torch or lamp are broken or burnt away, the gauge shall not be reconditioned for further use.

(3)

Damaged and defective gauges, glasses or other parts of a safety lamp shall not be kept or stored in the safety lamp-room.

(4)

No glass of a safety lamp and no bulb of an electric safety lamp shall be replaced except by a glass or bulb of such type as the Chief Inspector may, from time to time specify by notification in the Official Gazette, and no other part of a safety lamp, other than a wick or battery, as the case may be, shall be replaced except by a part manufactured by the manufacturers of the lamp to approved specifications.

(5)

No repaired part shall be used in a safety lamp:

Provided that in the case of an imported safety lamp, a part manufactured indigenously may be used if it is of such design and make as is approved by the Chief Inspector.

(6)

In every flame safety lamp kept for the purpose of inspection or of testing for or detecting the presence of inflammable gas, no oil other than an oil of a type approved by the Chief Inspector shall be used.

(7)

No electric safety lamp shall be issued for use unless the covers of the battery and of the headpiece are properly assembled, securely locked and sealed, and the battery is properly charged.

(8)

No electric safety lamp shall be hung or held by the cable.

186.

Precautions to be taken in safety lamp-rooms.

(1)

No unauthorised person shall enter the safety lamp room.

(2)

No person shall smoke in the safety lamp-room.

(3)

Where petrol, benzol, or any other highly volatile spirit is used in safety lamps, the following precautions, shall be observed:-

(a)

lamps shall be cleaned, refitted and refilled in a separate room;

(b)

only such quantity of volatile spirit as is required for one working day shall be kept in any such room;

(c)

internal relighters shall not be taken out of lamps and cleaned, repaired or refitted on the same table where lamps are cleaned, refitted or refilled; and

(d)

adequate number of suitable fire extinguishers shall be provided and kept ready for use in every such room.

Chapter-XIV Explosives and Shot-firing

187.

Type of Explosives to be used in mines.

(1)

No explosive shall be used in a mine except that which is provided by the owner, agent or manager which shall be of good quality and in good condition.

(2)

No explosive, other than a fuse, a detonator, site mixed slurry or emulsion shall be issued for use in mine, or taken into or used in any part of a mine, unless it is in the form of a cartridge.

(3)

In belowground mines, explosive cartridges shall be used only in the form in which they are received.

(4)

No liquid oxygen explosives shall be used in any mine.

188.

Storage of explosives.

(1)

No owner, agent or manager shall store, or knowingly allow any other person to store, within the premises of a mine any explosive otherwise than in accordance with the provisions of rules made under the Indian Explosives Act, 1884.

(2)

Explosives shall not be taken into or kept in any building except a magazine duly approved by the Licensing Authority under the Indian Explosives Act, 1884.

Provided that the Regional Inspector may by an order in writing and subject to such conditions as he may specify therein, permit the use of any store or premises specially constructed at or near the entrance to a mine for the temporary storage of explosives intended for use in the mine or of surplus explosives brought out of the mine at the end of a shift.

(3)

Explosives shall not be stored belowground in a mine except with the approval in writing of the Chief Inspector and subject to such conditions as he may specify therein and such storage shall be done only in a magazine or magazine or duly licensed in accordance with the provisions of rules made under the Indian Explosives Act, 1884.

(4)

Every license granted by the Licensing Authority under the Indian Explosives Act, 1884 for the storage of explosives, or a true copy thereof shall be kept at the office of the mine.

189.

Magazines, stores and premises to store explosives.

(1)

Every magazine, or store or premises, where explosives are stored shall be in charge of a competent person who shall be responsible for the proper receipt, storage and issue of explosives.

(2)

Explosives shall not be issued from the magazine unless they are required for immediate use:

Provided that if any explosive is returned to the magazine or store or premises, they shall be re-issued before fresh stock is used.

(3)

Explosives shall be issued only to competent persons upon written requisition signed by the shot-firer or by an official authorised for the purpose, and only against their signature or thumb impression and such requisition shall be preserved by the person in charge of the magazine or store or premises.

(4)

The person in charge of the magazine or store or premises shall maintain, in a bound paged book kept for the purpose, a clear and accurate record of explosives issued to each competent person and a similar record of explosives returned to the magazine or store or premises.

190.

Cases and containers for carrying explosives.

(1)

No explosive shall be issued from the magazine or taken into any mine except in a case or container of substantial

construction which is securely locked:

Provided that cases or containers made of iron or steel shall be heavily galvanised and no case or container provided for carrying detonators shall be constructed of metal or other conductive material.

(2)

No detonator shall be kept in a case or container which contains other explosives, materials or tools and two or more types of detonators shall not be kept in the same case or container:

Provided that nothing in this sub-regulation shall restrict the conveyance of primer cartridges fitted with detonators in the same case or container for use in a wet working or in a sinking shaft.

(3)

No detonator shall be taken out from a case or container unless it is required for immediate use.

(4)

Except as otherwise provided for in regulation 191, no case or container shall contain more than five kilogrammes of explosives, and no person shall have in his possession at one time in any place more than one such case or container:

Provided that the Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit the carrying of a larger quantity of explosives in a single case or container, or the use, at one time in one place, of more than one such case or container.

(5)

Every case or container shall be numbered and as far as practicable, the same case or container shall be issued to the same shot-firer or competent person, as the case may be, every day.

(6)

The key of every case or container shall be retained by the shot-firer in his own possession throughout his shift.

191.

Transport of explosives.

(1)

While explosives are being carried on a ladder, every case or container shall be securely fastened to the person carrying it.

(2)

No person other than a shotfirer shall carry any priming cartridge into a shaft which is in the course of being sunk and no such cartridge shall be so carried except in a thick felt bag or other container sufficient to protect it from shock.

192.

Transport of explosives in bulk.

(1)

Transport of explosives from the magazine to the priming station or the site of blasting shall not be done except in day light and in the original wooden or card board packing case.

(2)

The quantity of explosive transported at one time to the site of blasting shall not exceed the actual quantity required for use in one round of shots.

(3)

The explosives shall be transported to the site of blasting not more than 30 minute before the commencement of charging of the holes.

(4)

No mechanically propelled vehicle shall be used for the transport of explosives unless it is of a type approved in writing by the Chief Inspector:

Provided that a Jeep or Land Rover or Sports Utility Vehicle may be used for the transport of detonators from magazines or priming stations subject to the following conditions:

(a)

not more than 200 detonators are transported in a vehicle at a time;

(b)

the detonators are packed suitably in a wooden box;

(c)

the wooden box containing detonators is placed inside an outer metal case of a construction approved by the Chief

Inspector;

(d)

the outer metal case shall be suitably bolted to the floor of the vehicle or otherwise fixed in a wooden frame so that the container is not displaced while the vehicle is in motion; and

(e)

no person shall ride on the rear Portion of the vehicle.

(5)

Every vehicle used for the transport of explosives shall be marked or placarded, on both sides and ends, with the word 'Explosives' in red letters not less than 15 centimetres high on a white background and all such vehicles shall be provided with red flashing beacon light.

(6)

Every mechanically propelled vehicle transporting explosives shall be provided with not less than two fire extinguishers, one of dry powder type for petroleum fire and the other of Carbon Dioxide under pressure type for electrical fire, suitably placed for immediate use.

(7)

The vehicle used for transport of explosives shall not be overloaded, and in no case shall the explosive cases be piled higher than the sides of its body.

(8)

Explosives and detonators shall not be transported in the same vehicle.

(9)

No person other than the driver and his helper, not below 18 years of age, shall ride on a mechanically propelled vehicle used for the transport of explosives.

(10)

A vehicle loaded with explosives shall not be left unattended.

(11)

The engine of a vehicle transporting explosives shall be stopped and the brakes set securely before it is loaded or unloaded or left standing.

(12)

A vehicle transporting explosives shall not be driven at a speed exceeding 25 Kilometers per hour.

(13)

A vehicle loaded with explosives shall not be taken into garage or repair shop and shall not be parked in a congested place.

(14)

A vehicle transporting explosives shall not be refueled except in emergencies even when its engine shall be stopped and other precautions taken to prevent accidents.

(15)

No trailer shall be attached to a vehicle transporting explosives.

(16)

Every vehicle used for the transport of explosives shall be carefully inspected once in every 24 hours by a competent person, to ensure that-

(a)

fire extinguishers are filled and in place;

(b)

the electric wiring is well insulated and firmly secured;

(c)

the chassis, engine and body are clean and free from surplus oil and grease;

(d)

the fuel tank and feed lines are not leaking; and

(e)

lights, brakes and steering mechanism are in good working order.

(17)

A report of every inspection made under sub-regulation (16) shall be recorded in a bound paged book kept for the purpose duly signed and dated by the competent persons making the inspection.

(18)

All operations connected with the transport of explosives shall be conducted under the personal supervision of an overman solely placed in charge of blasting operations at the mine.

(19)

The shotfirer shall personally search every persons engaged in the transport and use of explosives and shall satisfy himself that no person so engaged has in his possession any mobile phone, cigar, cigarette, biri or other smoking materials or any match or any other apparatus of any kind capable of producing a light, flame or spark.

193.

Reserve Station.

- No case or container containing explosives shall be left or kept in a mine except in a place appointed by the manager or assistant manger for the purpose and so situated that it is not frequented by workpersons and every such place shall be kept clean, safe and adequately fenced and legibly marked 'Reserve Station'.

194.

Shotfirers.

(1)

The preparation of charges and the charging and stemming of holes shall be carried out by or under the personal Supervision of a competent person, in these regulations referred to as a shotfirer, who shall fire the shots himself.

(2)

No person shall be appointed to be a shotfirer unless he holds -

(a)

a manager's or overman's certificate or a sirdar's certificate together with a gas testing Certificate in case of belowground mines; and

(b)

a manager's, overman's or a sirdar's certificate in the case of open cast working.

(3)

The competent person appointed as shotfirer shall not be given any other duty nor any one performing any other duty shall be allowed to perform shotfiring.

(4)

No person whose wages depend on the amount of mineral, rock or debris obtained by firing shots, shall be appointed to perform the duties of a shot-firer.

(5)

The manger shall fix, from time to time, the maximum number of shots that a shotfirer may fire in anyone shift and such number shall be based on-

(a)

the time normally require to prepare and fire a shot in accordance with the provisions of these regulations;

(b)

the time required for that shotfirer to move between places where shots are fired;

(c)

the assistance, if any, available to him in the performance of his said duties; and

(d)

any other duties assigned to him, whether statutory or otherwise, and shall not in any case exceed-

(i)

in the case of a gassy seam of second or third degree or a fiery seam, forty, if a single shot exploder is used and eighty; if a multi-shot exploder is used;

(ii)

in the case of other seams fifty, if a single-shot exploder is used and hundred, if a multi-shot exploder is used; and

(iii)

in the case of open cast mines sixty, if a single shot exploder is used or if blasting is done with ordinary detonators and one hundred and twenty, if a multi-shot exploder is used:

Provided that if thirty or more persons are employed belowground at anyone time in any mine or district under the charge of any official who is qualified to perform the duties of the shotfirer, such official shall not fire or be permitted to fire more than half the maximum number of shots specified in sub-clauses (i), (ii) and (iii) depending on the category of the seam or mine specified therein:

Provided further that where special conditions exist, the Regional Inspector may by an order in writing and subject to such conditions as he may specify therein permit a larger maximum number to be fixed:

Provided also that where the Regional Inspector is of the opinion that, for the proper observance of the provisions of these regulations the number of shots so fired shall be reduced, and if he so required by an order in writing, the manager shall fix a lower maximum number of shots as specified by the Regional Inspector.

(6)

The number of detonators issued to, and in the possession of, a shotfirer during his shift shall not exceed the maximum number of shots that he is permitted to fire under sub-regulation(5).

195.

Shotfiring tools.

(1)

Every shotfirer on duty shall be provided with -

(a)

a suitable shotfiring apparatus of approved type;

(b)

suitable shotfiring cable of approved type;

(c)

a suitable electric lamp or torch, a whistle and a stop watch;

(d)

a tool, made entirely of wood, suitable for charging and stemming shotholes;

(e)

a scraper made of brass or wood suitable for cleaning out shotholes;

(f)

where fuses are used, a knife for cutting off fuses and, unless machine-capped fuses are provided, also a pair of suitable crimpers for crimping detonators;

(g)

where detonators are used, a pricker made of wood or a non-ferrous metal for priming cartridges;

(h)

a tool of a type approved by the Chief Inspector for detecting cracks;

(i)

an approved type of methanometer for detection of inflammable gas in case of solid blasting; and

(j)

a circuit tester for checking shotfiring circuits.

(2)

No tool or appliance other than that provided under sub-regulation (1) shall be used by a shotfirer.

196.

Drilling, charging, stemming and firing of shotholes.

(1)

No drill shall be used for drilling a shothole unless it allows a clearance of at least 0.3 centimetre over the diameter of the cartridge of explosive, which it is intended to use.

(2)

No shothole shall be charged before it is thoroughly cleaned.

(3)

Before any shothole is charged, the direction of the hole shall, where practicable, be distinctly marked on the roof or other convenient place.

(4)

No detonator shall be inserted into a priming cartridge until immediately before it is to be used, however that in case of

wet working, priming cartridges may be prepared at the nearest convenient dry place and such primed cartridges shall be carried to the working place in a securely closed case or container.

(5)

Detonators once inserted into a priming cartridge shall not be taken out.

(6)

In belowground working the explosive used in any shot-hole shall be of the same type.

(7)

In opencast mines, to use two types of explosives in any shot-hole, the manager shall frame and enforce standing orders for the safe use of explosives and a copy of the same shall be submitted to the Regional Inspector.

(8)

The shotfirer shall, to the best of his judgment, ensure that no shothole is over-charged or under-charged, having regard to the task to be performed.

(9)

Shots shall be fired electrically or by any other means or instruments or apparatus as approved by the Chief Inspector.

(10)

Every shothole shall be stemmed with sufficient and suitable non-inflammable stemming so as to prevent the shot from blowing out and only sand loosely filled in, or soft clay lightly pressed home, or a compact but not bard mixture of sand and clay or water shall be used as stemming.

(11)

In charging or stemming a shothole, no metallic tool, scraper or rod shall be used and no explosive shall be forcibly pressed into a hole of insufficient size.

(12)

No shot shall be fired except in a properly drilled, charged and stemmed shothole.

(13)

Blasting gelatin or other high explosives shall not be lighted in order to set fire to fuses.

(14)

All surplus explosives shall be removed from the vicinity of a shothole before a light is brought near it for the purpose of lighting the fuse.

(15)

As far as practicable a shot shall be fired by the same blaster who charged it.

(16)

Except in a stone drift or a sinking shaft, not more than 16 shots shall be fired in any one round:

Provided that where more than six shots ate to be fired in one round, they shall be fired electrically:

Provided further that in the case of opencast working any number of shots can be fired in one round if they are fired electrically by an approved type of shotfiring device of adequate capacity.

(17)

No shothole shall be charged except those which are to be fired in that round and all shotholes which have been charged shall be fired in one round.

(18)

Where a large number of shots have to be fired, shotfiring shall, as far as practicable, be carried out between shifts.

(19)

No person shall remove any stemming, or pull out any detonator lead, or remove any explosive, from a shothole either before firing or after a misfire, or bore out a hole that has once been charged, or deepen or tamper with empty holes or sockets.

197.

Use of Ammonium Nitrate-Fuel Oil.

- Where in any opencast working a mixture of Ammonium Nitrate-Fuel Oil is used as explosive, the following provisions shall apply: -

(a)

the Ammonium Nitrate-Fuel Oil mixture (hereinafter referred to ANFO) shall not be used for firing without possessing a valid license for storage and preparation of ANFO explosives and also for transport of such explosives on surface from

the licencing authority under the explosives Act, 1884;

(b)

the mixing or impregnating of ammonium nitrate or a non-explosive mixture of ammonium nitrate with other substances with fuel oil shall be done in a special shed in the vicinity of the mine working constructed from incombustible material and equipped with remote control water sprays to combat a fire in an emergency;

(c)

if mechanical mixer is used the machine in which the mixing takes place, shall be of a type approved by the licencing authority under the Explosives Act, 1884;

(d)

the design of mixer shall be such as to minimise the possibility of frictional heating, Compaction and confinement and bearings and gears shall be protected against accumulation of product dusts and shall preferably be of outboard type;

(e)

persons employed on mixing shall wear rubber gloves;

(f)

the mixer shall be thoroughly cleaned after use;

(g)

smoking, open flames or any other source of fire shall not be allowed when mixing, carrying or handling operations are being carried on;

(h)

when mixing, loading, charging or handling operations are being carried on, any naked light or fire shall not be allowed to be brought within 10 metres of the shot hole or the explosive;

(i)

the area surrounding the place of mixing for a distance of atleast 10 metres shall be kept free of rubbish, dry grass or other combustible materials;

(j)

the mixture shall be kept in water-proof bag until the time of use, separate from all other materials;

(k)

the mixture shall be carried only in containers of wooden or of similar materials or in polythene bags kept inside jute or rexin bags and not in a metallic container;

(l)

only the necessary minimum number of person shall be allowed to be present at the premises where the process of mixing is being done;

(m)

liquid fuels with a flash point lower than that of diesel fuel or more volatile fuels such as gasoline, kerosene or number one diesel fuel oil shall not be used for mixing with ammonium nitrate;

(n)

crude oil or crank case oil shall not be used for mixing with ammonium nitrate;

(o)

the cap sensitivity of a composition of ANFO shall be determined at a regular interval and after every change in proportion, and character of ingredients;

(p)

ANFO explosives shall be manufactured in quantities only immediately required for use:

Provided that if due to unforeseen circumstances, explosives in excess of immediate requirement is prepared, it shall be packed in an inner package of polythene and outer package of wood or fire board and removed to the magazine licenced for storage of such mixture;

(q)

persons engaged in the process of firing with ANFO explosive including the Shotfirer shall be previously trained in correct techniques of such blasting;

(r)

adequate priming shall be done to guard against misfires, increased toxic fumes and poor performance:

Provided that the kind and amount of primer used shall be governed by the sensitivity of the explosive, hole diameter

and other factors;

(s)

if a pneumatic loader (hereinafter referred to as ANFO loader) is to be used, the following precautions shall be taken:-

(i)

the ANFO loader shall be regularly examined by a competent person appointed by the manager to ensure proper operation of the injector and other parts;

(ii)

pneumatic equipment used for loading ANFO explosives shall be adequately grounded to dissipate any static charges that may be generated during the loading operation;

(iii)

water lines, air-lines, rail, or any permanent electrical grounding system shall not be used to ground pneumatic loading equipment; and

(iv)

the owner, agent or manager of the mine shall within 30 days of using ANFO explosives in the mine submit to the Regional Inspector a notice giving details of the method and the precautions to be taken during firing operations.

198.

Deep-hole blasting.

- The blast holes made more than 3 metre in depth may be called deep-holes and where deep-holes are blasted in an opencast mine, the following provisions shall apply:-

(a)

the position of every deep-hole to be drilled, shall be distinctly marked by the overman so as to be readily identified by the driller;

(b)

no drilling shall be commenced in an area where shots have been fired, until the Shot-firer has made a thorough examination at all places, including the remaining portion of old blast holes for unexploded charges;

(c)

no drill rod or pick shall be inserted in remaining portion of old blast holes;

(d)

after completion of drilling the holes shall be kept plugged or covered till loading is commenced;

(e)

notwithstanding anything contained in clause (b) of sub-regulation (2) of Regulation 194, the firing of deep-holes shall be done by a person holding a manager's or an overman's Certificate;

(f)

shots shall not be fired except during the hours of day light, as far as practicable, either between the shifts or during rest interval or at the end of the work for the day:

Provided that all holes charged on any one day, shall be fired on the same day:

Provided further that where special conditions exist, the Regional Inspector may, by an order in writing and subject to such conditions as he may, specify therein, relax from the operation of this clause any working in case of which special difficulties exist which in his opinion make compliance with the provisions thereof not reasonably practicable or unnecessary;

(g)

during the approach and progress of an electric storm,-

(i)

no explosive or detonators shall be handled;

(ii)

if loading or charging operations have begun, the work shall be discontinued until the storm has passed;

(iii)

if the shots are to be fired electrically, all exposed wires shall be coiled up and if practicable, placed in the mouths of the holes or kept covered with non-metallic objects;

(h)

when detonating fuse is used for blasting, such fuse shall not be cut after it has been inserted into a shot hole;

(i)
no cartridge of any explosive shall be dumped freely into a shot hole without using proper lowering device;

(j)
no person shall be allowed to possess or use any mobile phone or radio transmitters during transport and handling of explosives.

199.

Electric Shotfiring.

(1)
No shot shall be fired except by means of a suitable shotfiring apparatus of a type approved by the Chief Inspector and the number of shots fired at any one time by the apparatus shall not exceed the number for which it is designed.

(2)
Every electrical shotfiring apparatus shall be so constructed and used that -

(a)
it can only be operated by a removable handle or plug;

(b)
the handle or plug shall not be placed in position until a shot is about to be fired and shall be removed as soon as a shot has been fired; and

(c)
the firing circuit is made and broken either automatically or by means of a push-button switch.

(3)
No apparatus shall be used which is defective and every apparatus shall once at least in every three months, be cleaned and thoroughly overhauled by the manufacturer or his authorised representative.

(4)
If the apparatus fails to fire all the shots in a properly connected circuit, the shotfirer shall return the apparatus to the manager or assistant manager as soon as possible, and it shall not be used again unless it has been tested on the surface and found to be in safe working order.

(5)
The result of every overhaul, test or repair made under sub-regulation (4) shall be recorded in a bound paged book kept for the purpose and shall be signed and dated by the person making the overhaul, test or repair.

(6)
No current from a signaling, lighting or power circuit shall be used for firing shots.

(7)
The Shotfirer shall -

(a)
retain the key of the firing apparatus in his possession throughout his shift;

(b)
use a well-insulated cable of sufficient length to permit him to take proper shelter and in case of belowground working sufficient to take two right angle turns of pillar, and in no case, shall this cable be less than 50 meters in length;

(c)
before coupling the cable to the firing apparatus, couple up the cable itself to the detonator leads;

(d)
take care to prevent the cable from coming into contact with any power or lighting cable or other electrical apparatus;

(e)
take adequate precautions to protect electrical conductors and apparatus from injury;

(f)
himself couple the cable to the firing apparatus and before doing so, see that all persons in the vicinity have taken proper shelter as provided under regulation 200; and

(g)
after firing the shots and before entering the place of firing, disconnect the cable from the firing apparatus.

(8)
Where more than one shot are to be fired at the same time,-

- (a) care shall be taken that all connections are properly made;
- (b) all shots if fired belowground shall be connected in series;
- (c) the circuit shall be tested either for electrical resistance or for continuity before connecting it to the firing apparatus and such a test shall be made with an apparatus specifically designed for the purpose and only after all persons in the vicinity have taken proper shelter as provided under regulation 200;
- (d) the cable to the shotfiring apparatus shall be connected last; and
- (e) detonators of the same electrical resistance shall only be used.

200.

Taking shelter, etc.

- (1) The shotfirer shall, before a shot is charged, stemmed or fired, see that all persons other than his assistants, if any, in the vicinity, have taken proper shelter and he shall also take suitable steps to prevent any person approaching the shot and shall himself take adequate shelter, along with his assistants if any, before firing the shots.

- (2) In the case of an opencast working the shotfirer shall not charge or fire a shot-

- (a) unless he has taken the precautions laid down in sub-regulation (1);

- (b) unless sufficient warning by efficient signals or other means approved by the manager, is given over the entire area falling within a radius of 500 metres from the place of firing (hereinafter referred to as the danger zone) and that he has ensured that all persons within such area have taken proper shelter; and

- (c) where any part of a public road or railway lies Within the danger zone, unless two persons are posted, one in either direction at the two extreme points of such road or railway which fall within the danger zone who have, by an efficient system of telephonic communication or hooter or loudspeakers or other means approved by the Chief Inspector or Regional Inspector intimated clearance of traffic to the shotfirer and have also warned the passersby and whenever possible the vehicle also, if any, which have passed by such road or railway.

- (3) In the case of an opencast working, where any permanent building or structure of permanent nature, lies within the danger zone, the aggregate maximum charge per delay and per round shall be fixed by a scientific study and permission in writing for such blasting shall be obtained from the Chief Inspector, who may specify the conditions for such blasting.

- (4) Notwithstanding anything contained in sub-regulation (3), the Chief Inspector may, by an order in writing and subject to such conditions all he may specify, exempt any mine or part thereof from the operation of all or any of the provisions of sub-regulation (3), on the ground that the observance of its provisions is not necessary or reasonably practicable on account of the special conditions existing thereat.

- (5) Where the working, either above or belowgroup offer insufficient protection against flying fragments or missiles, adequate shelter or other protection shall be provided.

- (6) When two working places belowground have approached within 9 metres of each other, the shotfirer shall not fire any shot in anyone of the said working unless all persons have been withdrawn from the other working place and the same has been so fenced off as to prevent persons inadvertently coming in direct line of the shot.

201.

Precautions against dry coal dust.

- No shots shall be fired at any place belowground unless the place itself and all accessible places, including roof and

sides, within a distance of 18 metres have been treated in the manner specified in clause (ii) of sub-regulation (3) of regulation 147 unless such places are naturally wet as defined in regulation 147.

202.

Conditions requiring use of permitted explosives.

(1)

Notwithstanding anything contained in these regulations, no shots shall be charged or fired in the belowground working if the explosive used is not a permitted explosive, except in-

(a)

a stone-drift, if it does not contain dry coal dust; and

(b)

a shaft which is in the course of being sunk.

(2)

In a gassy seam of the second or third degree, no explosive other than the permitted sheathed explosives or other explosives equally safe or any device or apparatus for breaking coal approved by the Chief Inspector shall be used, while in a gassy seam of the first degree, permitted sheathed explosives or permitted explosives or any device for breaking coal approved by the Chief Inspector in writing may be used:

Provided that the Chief Inspector may by any order in writing and subject to such conditions as he may specify therein permit, in any gassy seam of the first degree, the use of any explosives other than the permitted explosives.

(3)

Notwithstanding anything contained in sub-regulation (1), if blasting is done in any stone drift or sinking shaft within five metres of any coal seam or in coal measure drifts or staple shaft from one seam to another only permitted explosives of such types as may be required by the Chief Inspector shall be used:

Provided that in case of special difficulties, the Chief Inspector may exempt any stone drift or sinking shaft from the provisions of this sub-regulation subject to such conditions as he may specify therein.

203.

Precautions in the use of permitted explosives.

(1)

No detonator shall be used, unless it is of a type approved by the Chief Inspector.

(2)

Where more than one shots are charged for firing, the shots shall be fired simultaneously.

(3)

The aggregate charge in any shot to be fired in coal shall not exceed such permissible maximum charge, as the Chief Inspector may, by a notification in the Official Gazette, specified for the kind of permitted explosives used.

204.

Approved shotfiring apparatus.

- No shot shall be fired in a mine except by means of a shotfiring apparatus of a type approved by the Chief Inspector and subject to such conditions as he may from time to time specified by notification in the Official Gazette:

Provided that where special conditions exist, the Chief Inspector may, by an order in writing and subject to such conditions as he may specify therein, permit the use of any other shotfiring apparatus.

205.

Additional Precautions.

(1)

If in a ventilating district, presence of inflammable gas is detected in any place, no shothole shall be charged, stemmed or fired in that place or in any other place situated on its return side till such place has been cleared of gas and declared safe.

(2)

Immediately before charging a shothole or a round of shotholes, and again before firing the shots the shotfirer shall carefully test for inflammable gas at all places within a radius of 18 metres of the place of firing.

(3)

No shothole shall be charged if any break is found therein, or if inflammable gas is found issuing therefrom.

(4)

If after charging a shothole, inflammable gas is found in any place within the prescribed radius no shot shall be fired until the place has been cleared of gas and declared safe.

(5)

No delay-action detonator shall be used, except with the previous permission in writing of the Chief Inspector and subject to such condition as he may specify therein.

206.

Blasting in fire area in opencast mines.

(1)

While blasting in hot strata, either in overburden or coal in opencast mines, the following precautionary measures shall be adopted:-

(a)

no explosive other than slurry and emulsion explosives shall be used; and

(b)

blasting shall be done with detonating fuse down the hole:

(2)

Temperature inside the blast holes shall be measured, before filling with water, and if the temperature exceeds 80°C in any hole, such hole shall not be charged and records of measurement of temperature in each hole shall be maintained in a bound pagged boots.

(3)

All blast holes shall be kept filled with water and where any hole is traversed by cracks or fissures, such holes shall not be charged unless it is lined with a suitable fire resistant casing and the hole filled with water and in addition, bentonite shall be used for sealing any cracks at the bottom of the holes.

(4)

Detonating fuse shall not be laid on hot ground without taking suitable precautions which will prevent it from coming in contact with hot strata.

(5)

The charging and firing of the holes in any one round shall be completed expeditiously and in any case within two hours.

(6)

Blasting operations shall be carried out under the direct supervision of an assistant manager having experience of blasting in hot Strata.

207.

Inspections after shotfiring.

(1)

After a shot has been fired, no person other than the shotfirer or any other person shall enter the place until the area is free from dust, smoke or fumes:

Provided that the shotfirer shall before any other person enters the place, make a careful examination and with his assistants, if any, make the place safe.

(2)

No other person shall enter the place, and where guards have been posted, they shall not be withdrawn, until the examination has been made and the place has been declared safe in all respects.

(3)

In the case of opencast working, after shots have been fired, an all-clear signal shall be given except in the case of a misfire.

208.

Misfires.

(1)

The number of shots which explode shall, unless shots are fired electrically, be counted by the shotfirer and any another competent person authorised for the purpose and unless it is certain that all the shots have been exploded, no person shall re-enter or be permitted to reenter the place until 30 minutes after the firing of shots:

Provided that where shots are fired electrically, this interval may be reduced to not less than five minutes after the source of electricity has been disconnected from the cable.

(2)

In the event of a misfire, the entrance or entrances to the working place shall be fenced so as to prevent inadvertent access and no work other than that of locating or relieving the misfire shall be done therein until the misfire has been located and relieved.

(3)

In opencast working, it shall be sufficient to mark the place of the misfire with a red flag.

(4)

In the event of a misfire, a second charge shall not be placed in the same hole.

(5)

If the misfire contains a detonator, the leads or fuse thereof shall be attached by a string to the shotfiring cable or some distinctive marker.

(6)

Except where the misfire is due to faulty cable or a faulty connection, and the shot is fired as soon as practicable after the defect is remedied, another shot shall be fired in a relieving hole which shall be so placed and drilled in such a direction that at no point shall it be nearer than 30 centimetres from the misfired hole:

Provided that the new hole shall be bored in the presence of a shotfirer, preferably the same person who fired the shot.

(7)

After a relieving shot has been fired, a careful search for cartridges and detonators, if any, shall be made in the presence of the shotfirer, amongst the material brought down by the shot:

Provided that in the case of working belowground if such cartridge or detonator is not recovered, the tubs into which the material is loaded shall be marked and a further search made on the surface and the loading of any coal, stone or debris which may contain a detonator, shall be carried out without the aid of tools.

(8)

If a misfired hole is not dislodged by a relieving shot, the procedure laid down in sub-regulation (6) and (7) shall be repeated and a misfired hole which cannot be dealt with in the manner so provided, shall be securely plugged with a wooden plug and no person other than a shotfirer an official or a person authorised for the purpose shall remove or attempt to remove such plug.

(9)

When a misfired shot is not found, or when a misfired shot is not relieved, the shotfirer shall, before leaving the mine, give information of the failure to such shotfirer or official as may relieve or take over charge from him and he shall also record, in a bound paged book kept for the purpose, a report on every misfire, whether suspected, and whether the shothole is relieved or not relieved.

Provided that the relieving shotfirer or official shall sign the report and record in the said book the action taken for relieving the misfired shothole.

(10)

The shotfirer of the next shifts shall locate and blast the misfired hole, but if after a thorough examination of the place where the misfire was reported to have occurred he is satisfied that no misfire has actually occurred, he may permit drilling in the place.

(11)

In case of opencast mines, the owner, agent and manager of a mine shall draw up a plan which shall instruct all shotfirers the detail procedure to be followed in the event of a misfired shot.

209.

Special precautions in stone drifts.

- In stone drifts, -

(a)

after shots have been fired, all loose rock shall be removed from the face, and the area lying within a distance of 1.2 metres from the face shall be thoroughly cleaned or washed down with water and carefully examined for the presence of misfires or sockets and unless such precautions have been taken, the next round of shots shall not be fired; and

(b)

if any socket is found, it shall be dealt with in the manner provided in regulation 208.

210.

Duties of shotfirer at the end of his shift.

- Immediately after the end of his shift, the shotfirer-

(a)

shall return all unused explosive to the magazine, or where a store or premises is provided under regulation 188, to such store or premises; and

(b)

shall record, in a bound paged book kept for the purpose, the quantity of explosive taken, used and returned, the places where shots were fired and the number of shots fired by him, and misfires, if any, and every such entry shall be signed and dated by him.

211.

General precautions regarding explosives.

(1)

No person, whilst handling explosives or engaged or assisting in the preparation of charges or in the charging of holes, shall smoke or carry or use a light other than an enclosed light, electric torch or lamp :

Provided that nothing in this sub-regulation shall be deemed to prohibit the use of an open light for lighting fuses.

(2)

No person shall take any light other than an electric torch or an enclosed electric lamp into any explosive magazine or store or premises.

(3)

The owner, agent, or manager shall take adequate steps to prevent pilferage of explosives during its storage, transport and use in the mine.

(4)

No person shall have explosives in his possession except as provided for in these regulations, or hide or keep explosives in a dwelling house.

(5)

Any person finding any explosives in or about a mine shall deposit the same in the magazine or store or premises and every such occurrence shall be reported to the manager in writing.

(6)

Shotfirers and their helpers shall-

(a)

not use battery operated watches, synthetic clothes and shocks;

(b)

use only conducting type of foot-wears; and

(c)

in case of leather shoes or boots, the sole shall also be of leather and without hobnails.

Chapter-XV Machinery, Plant and Equipment

212.

Use of certain machinery in coal mines.

(1)

No internal combustion engine or steam boiler shall be used belowground in a mine except with the permission in writing of the Chief Inspector and subject to such conditions as he may specify therein.

(2)

In every gassy seam of the second or third degree, only flame proof electrical apparatus and equipment shall be used belowground unless otherwise provided for under the Indian Electricity Rules, 1956:

Provided that the Central Government or the Chief Inspector or an Inspector authorised by the Central Government may, subject to such conditions as may be specified, permit the continuance of the use of non-flame proof apparatus or equipment for a specified period, not exceeding one year, in any mine of the first degree gassiness, which has subsequently been classified as second or third degree gassiness.

(3)

The Chief Inspector may, from time to time by notification in the Official Gazette, specify appliances, equipment, machinery, or other material, that are or may be used in a mine, which shall be of such type, standard and make as

approved by the Chief Inspector by a general or special order, and where any such appliance, equipment, machinery or other material had been specified by the Chief Inspector, any such appliance, equipment, machinery, or material, other than that approved by the Chief Inspector as aforesaid, shall not be used in any mine.

(4)

Where in respect of any appliance, equipment, machinery or other material, the Chief Inspector has not made any notification under sub-regulation (3) and any such appliance, equipment, machinery or material is used in any mine, the Chief Inspector or Regional Inspector may, if he is of the opinion that the use of such appliance, equipment, machinery, or material is likely to endanger safety in the mine, by an order in writing, prohibit the use thereof until the same is approved by the Chief Inspector.

213.

Approved machinery, equipment and devices.

(1)

The owner, agent or manager of a mine while acquiring any approved type of machinery, equipment, apparatus, device, lamp light, materials, etc. shall ensure that they conform to approved specifications in all respects and shall also, be responsible for maintaining them as per the approved standard.

(2)

A copy of approval of every approved machinery, equipment and device being used shall be kept at the office of the mine.

(3)

Where machinery is used for lifting, pulling, drilling, other than by hand held drill, dinting, ripping, cutting, loading, hauling, dumping; etc., safe code of practices Separately for each type of machinery with respect to the method of work, shall be framed by the person authorised for the purpose, containing codes for the control and guidance of persons employed for the erection, installation, operation, repairs, maintenance, dismantling and transportation of such machinery, and ancillary equipments as well as for the prevention of accident and to provide for the safety, health, convenience and discipline of the persons so employed and the engineer authorised for the purpose shall be responsible for the implementation of above safe code of practices.

(4)

Where surface transportation and handling machinery including coal handling plants, repair sheds or workshops are provided, safe code of practices for their erection, installation, operation, repairs, maintenance, dismantling and transportation of such machinery, plants and ancillary equipments as well as for the prevention of accident and to provide for the safety, health, convenience and discipline of the persons so employed shall be framed by the person authorised for the purpose, and the engineer authorised for the purpose shall be responsible for the implementation of above safe code of practices.

214.

General provisions for construction and maintenance of machinery.

- All parts and working gear whether fixed or movable, including the anchoring and fixing appliances, of all machinery and apparatus used as or forming part of the equipment of a mine, and all foundations in or to, which any such appliances are anchored or fixed shall be of good design, sound construction, suitable material, adequate strength and free from visible defect and shall be properly maintained.

215.

Apparatus under pressure.

(1)

All apparatus, used as forming part of the equipment in a mine, which contains or produces air, gas or steam at a pressure greater than atmospheric pressure shall be so designed, constructed, installed and maintained as to obviate any risk of fire, bursting, explosion or production of noxious gases.

(2)

Every air receiver forming part of a compressing plant shall be fitted with a safety valve and an air gauge which shows pressure in excess of the atmospheric pressure.

(3)

Before an air-receiver is case in or put in commission, the engineer or other competent person shall subject it to a hydraulic test at a pressure at least one-and-a-half times the maximum permissible working pressure and a similar test

shall be made after every renewal or repair and in any case at intervals of not more than three years.

(4)

The result of every such test under sub-regulation (3) shall be recorded in a bound paged book kept for the purpose duly signed and dated by the person carrying out the test.

(5)

The supply of air for air-compressors shall be drawn from a source free from dust and fumes.

(6)

All apparatus used as or forming part of the equipment in a mine which contains or produces hydraulic fluid or emulsion under pressure shall be so designed, constructed, installed and maintained as to obviate any risk of bursting and fire.

216.

Precautions regarding moving parts of machinery.

(1)

Every winch or windlass shall be provided with a stopper, pawl or other reliable holder.

(2)

Every drums, fly-wheel and every other dangerous exposed part of any machinery used as, or forming part of, the equipment in a mine shall be adequately fenced by suitable guards of substantial construction to prevent danger and such guards shall be kept in position while the parts of the machinery are in motion or in use, which may be removed for carrying out any examination, adjustment or repair if adequate precautions are taken.

(3)

It shall be the duty of the engineer in charge, the supervisory officials and other authorised persons to keep all guarding properly maintained in good condition and in the correct position.

(4)

No person shall, or shall be allowed to, repair, adjust, clean or lubricate machinery in motion where there is risk of injury.

(5)

No person shall, or shall be allowed to, shift or adjust a driving belt or rope while the machinery is in motion unless a proper mechanical appliance is provided and used for the purpose.

(6)

No person in close proximity to moving machinery shall wear, or be permitted to wear, loose outer clothing.

(7)

No unauthorised person shall enter any engine room or in any way interfere with the engine.

217.

Engine rooms and their exits.

- Every engine, motor, transformer and battery charging room, and every room in which highly inflammable materials are stored on the surface shall be kept clean, and be provided with at least two exits and every such exit shall be properly maintained and kept free from obstruction.

218.

Working and examination of machinery.

(1)

No machinery shall be operated otherwise than by or under the constant supervision of a competent person.

(2)

In a gassy seam of the second or third degree no person shall be appointed to supervise or operate any electrical machinery, apparatus or appliance other than a telephone or signaling device or an electric lamp or light, unless he holds a gas-testing certificate.

(3)

Every competent person shall whilst on duty be provided with and carry an approved flame safety lamp or any other apparatus for determination of inflammable gas as approved by the Chief Inspector and check for presence of inflammable gas in the atmosphere.

(4)

Every person in charge of any machinery, apparatus or appliance shall, before commencing work, ensure that it is in proper working order and if he observes any defect therein, he shall immediately report the fact to the manager, engineer or other competent person.

(5)

Every person in charge of an air-receiver shall ensure that no extra weight is added to the safety valves and that the permissible pressure of air is not exceeded.

(6)

A competent person or persons appointed for the purpose shall, once at least in every seven days, make a thorough inspection of all machinery and plant in use, and shall record the result thereof in a bound paged book kept for the purpose.

(7)

In respect of electrical machinery and plant, the competent person shall be an engineer or electrician holding qualifications specified in the Indian Electricity Rules, 1956.

219.

Belowground face equipment, continuous mining and coal cutting machines.

(1)

All self propelled face equipments, including shuttle cars, which are used in belowground coal mines shall be equipped with substantially constructed canopies or cabs, located and installed in such a manner that when the operator is at the operating control of such equipment he shall be protected from falls of roof, face or side.

(2)

Methane monitors shall be installed on all roof bolting machines and cutting machines, continuous miners, longwall face equipments, loading machines and other mechanised equipments used to extract or load coal.

(3)

Roof bolting machines, continuous miners and coal loading equipments shall be equipped with adequate lighting to illuminate the workplace.

(4)

All face equipments used to cut coal or drill holes for roof support shall be equipped with engineering controls such as water sprays, dust collector and air-scrubber system to control respirable coal mine dust.

(5)

Where remote controlled devices are used in belowground coal mines and on coal faces to operate continuous mining machines or other such equipment, the owner, agent and manager shall ensure-

(a)

testing of the remote controlled devices to ensure they are all on separate frequencies and will not accidentally Cause a machine not been intentionally operated to activate or move;

(b)

proper training of all operators on the use of the remote control device; and

(c)

designing of a mining plan for the use of remote control mining equipment which include the safe location for the machine operators and any other workers in the area to prevent crushing accidents while the machine is in movement, and from respirable dust and noise hazard.

220.

Cranes and lifting gears.

(1)

All machinery used to lift or transport equipment and materials, shall be designed, constructed and erected, inspected, maintained and operated as specified by the manufacturer.

(2)

The rated capacity or legible load chart, where appropriate, of a crane, hoist, grab or winch shall be permanently marked on the structure and clearly visible, without in any case, exceeding the rated capacity.

(3)

No person shall load any crane, grab or winch in excess of the safe working load except for the purpose of a test, which shall be carried out by an authorised person and only in a manner specified by the manufacturer.

(4)

The rated capacity of a hoist shall not exceed the capacity of the structure supporting the hoist.

(5)

Cranes and hoists shall be regularly inspected and maintained to ensure that every component is capable of carrying out its original design function, and records shall be kept thereof.

(6)
A crane or hoist shall not be used until any condition that could endanger workers is remedied.

(7)
All installations, modifications and repairs to load-bearing equipment shall be certified by a competent person or authorised organisation in accordance with the original design and safety standard.

(8)
All cranes or hoists with a boom that is movable in the vertical plan shall-

(a)
have a device that can be clearly read by the operator, to indicate the boom angle if the rated capacity is affected; and

(b)
be fitted with an automatic load indicator showing the safe working load.

(9)
All modifications that affect the rated capacity of a crane or hoist shall be assessed, and the rated capacity adjusted by the original equipment manufacturer, a competent person or authorised organisation.

(10)
There shall be a safe means of access and egress to the operator's position and to maintenance locations for all cranes and hoists.

(11)
If a normal means of egress is not always available to the operator, then an alternative safe means shall be provided to get from the operating position to a safe area in the event of a power failure or other emergency.

(12)
Effective audible and visual communications devices shall be installed on a crane or hoist.

(13)
The crane or hoist operator shall sound a warning signal when it is necessary to alert workers.

(14)
All controls on a crane or hoist shall be clearly identified and shall return to neutral when released, and an automatic braking system shall be activated.

(15)
The operator of a crane or hoist shall be protected against hazardous conditions such as airborne contaminants, falling or flying objects and excessive heat or cold.

(16)
The operator's seat on a crane or hoist shall be of an ergonomic design that allows the operator to operate the equipment safely.

(17)
All the hooks, hook guards or latches, wire ropes, chains and other attachments and fittings that may be safety critical shall be maintained and inspected on a regular basis.

(18)
The operator of a crane or hoist shall perform a check at the start of each shift and test the limit switch and record of the check shall be kept.

(19)
Transport routes for cranes shall be clear of obstructions and transport routes for wheeled or tracked cranes shall, as far as practicable be level.

(20)
Training shall be given for crane operators to ensure proper and safe operation of the crane and rigging loads.

221.
General precautions.

(1)
All machinery and plant used in connection with working of a mine shall be of good design, sound construction, suitable material, adequate strength, free from patent defect and properly maintained.

(2)

The owner, agent and manager shall provide adequate training facilities and training of an adequate number of persons.

(3)

No person except a qualified engineer or mechanic or a competent person under his supervision shall undertake any work on machinery and plant in which technical knowledge or experience is required.

Chapter-XVI Extraction of Methane from Coal Seam, Working or Abandoned Coal Mine

222.

Notice of Intention or proposal of exploration for methane.

(1)

Whenever there is any proposal or intention to explore for presence of methane, in any coal seam or coal measure strata or in any working or abandoned mines or part thereof, with an objective to exploit the methane or any other gases for captive, domestic or industrial purposes, the owner, agent and manager of every mine or coal block shall, not less than six months prior to the date of start of such exploration work, give notice in writing in Form I of the First Schedule, to the Chief Inspector and also to the Regional Inspector.

(2)

The notice so required under sub-regulation (1) shall be submitted along with-

(a)

a pre-feasibility report prepared by an authorised agency or accredited institution having knowledge, experience and expertise in the required fields and subjects;

(b)

a report on the manner of exploration work proposed to be Undertaken and the organisation, agency and any other contractor by which all or the parts of the jobs are proposed to be carried out within such time period as planned or proposed;

(c)

details of machinery, equipments, instruments and apparatuses along with their specifications, parameters and all relevant information, that are proposed to be used in the exploration work;

(d)

persons and the category of persons including officials, competent persons, officers and staffs with organisation structure and chart clearly specifying duties and responsibilities thereof;

(e)

a plan and section of the mine or of the coal block clearly showing the coal seams, coal measure strata, target seams and strata or the places from where the methane is proposed to be extracted, existing or old mine working showing their extent and all other details as required, present and future working from where coal is or shall be extracted; and

(f)

a plan and section of the area showing details of exploration work that is proposed to be undertaken indicating the proposed layout of the area over which drilling or any other exploration work are to be undertaken.

(3)

The Chief Inspector may by an order in writing and subject to such conditions as may be specified therein require any alteration, modification and an additional requirements to be incorporated in the exploration plan or the project.

(4)

No exploration plan or project as mentioned aforesaid shall be started or undertaken unless a written order or approval in that respect is obtained from the Chief Inspector.

(5)

Notice of starting of the execution of exploration plan or project shall also be submitted to the Chief Inspector and a copy thereof to the Regional Inspector in Form I of the First Schedule at least seven days prior to the actual date of starting of exploration work.

223.

Submission of the exploration or abandonment of the exploration report.

(1)

On completion of the exploration work in the mine or part thereof or in the coal block, a detailed exploration report complete in all respect shall be submitted to the Chief Inspector and a copy thereof to the Regional Inspector:

Provided that an interim report if so required by the Chief Inspector may also be submitted from time to time.

(2)

If during exploration, it is decided to abandon or discontinue further exploration in the area, mine or part thereof, a notice to that effect enclosing therewith the detailed exploration work carried thereon shall be submitted forthwith to the Chief Inspector and the Regional Inspector.

(3)

A plan and section showing the details of all the boreholes drilled, abandoned or incomplete with details such as diameter, depth and direction of every section of the borehole, extent of water, gas etc; and any other information, shall be submitted along with the report mentioned under sub-regulation (1).

(4)

All the boreholes drilled or excavation made therein during exploration shall be effectively sealed or plugged so as to prevent any leakage of gas therefrom or any other gas or liquid matter flowing thereinto unless otherwise is required to be used further.

(5)

A record of all such boreholes giving details as mentioned in this regulation shall be maintained in a bound paged book kept for the purpose duly signed and dated by the driller, drilling engineer, assistant manager and countersigned and dated by the manager.

224.

Notice of commencement of extraction of methane.

(1)

Whenever there is any proposal or intention to commence or start extraction of methane from any coal seam or coal measure strata or from any working or abandoned mines or part thereof, with an objective to exploit the methane or any other gases for captive, domestic or industrial purposes, the owner, agent and manager of every mine or coal block shall, not less than thirty days prior to the date of start of such extraction work, give notice in writing in Form I of the First Schedule, to the Chief Inspector and also to the Regional Inspector.

(2)

The notice so required under sub-regulation (1) shall be submitted along with a detailed project report prepared by an authorised agency or accredited institution having knowledge, experience and expertise in the required fields and subjects incorporating the following:-

(a)

the manner of extraction or exploitation work proposed to be undertaken and the organisation, agency and any other contractor by which all or the parts of the jobs are proposed to be carried out within such time period as planned or proposed;

(b)

details of machinery, equipments, instruments and apparatuses along with their specifications, parameters and all relevant information, that are proposed to be used;

(c)

persons and the category of persons including officials, competent persons, officers and staffs with organisation structure and chart clearly specifying duties and responsibilities thereof;

(d)

plan and section of the mine or of the coal block clearly showing the coal seams, coal measure strata, target seams and strata or the places from where the methane is proposed to be extracted, existing or old mine working showing their extent and all other details as required, present and future working from where methane as well as coal is or shall be extracted; and

(e)

plan and section of the area showing details of exploration work that is already completed and also the remaining area over which drilling or any other exploration work are yet to be undertaken.

(3)

The Chief Inspector may, by an order in writing and subject to such conditions as may be specified therein, require any alteration, modification and an additional requirements to be incorporated in the extraction plan or the project.

(4)

No extraction plan or project as mentioned in this regulation shall be started or undertaken unless a written order or approval in that respect is obtained from the Chief Inspector.

225.

Notice of closure, abandonment or discontinuance of extraction of methane.

(1)

Whenever there is any proposal or intention to close or abandon or discontinue extraction of methane from any coal seam or coal measure strata or from any working or abandoned mines or part thereof, the owner, agent and manager of every mine or of coal block shall, not less than sixty days prior to the date of start of such closure, abandonment or discontinuance of extraction work, give notice in writing in Form I of the First Schedule, to the Chief Inspector and also to the Regional Inspector.

(2)

The notice so required under sub-regulation (1) shall be submitted with the following information:-

(a)

the reasons of closure, abandonment or discontinuation;

(b)

plan and section of the mine or part thereof or of the coal block clearly showing the coal seams, coal measure strata, target seams and strata or the places from where the methane has already been extracted, existing or old mine working showing their extent and all other details as required, present and future working from where methane as well as coal are likely to be extracted;

(c)

plan and section of the area showing details of extraction work that is already completed and the conditions of all the boreholes and the part of working abandoned or discontinued; and

(d)

protective work like sealing or isolation of working or any other work already completed and all those work required to be done so as to make the area safe and secure in all respect.

(3)

The Chief Inspector may, by an order in writing and subject to such conditions as may be specified therein, require any other work as required to be completed before closure, abandonment, or discontinuation of such working, by the owner of the mine of the coal block.

(4)

If owner and agent fails to complete the remaining work required to be done before closure or abandonment or discontinuance of the working in a mine or part thereof or in a coal block, the Chief Inspector may get such protective work done by any other agency and the charges so incurred shall be defrayed from the owner as an arrear of land revenue.

226.

Quarterly returns for the coal bed methane (CBM), coal mine methane (CMM) or abandoned mine methane (ABM).

- Quarterly return in the Form prescribed by Chief Inspector shall be submitted at the end of every quarter to the Chief Inspector and the Regional Inspector.

227.

Annual returns for the coal bed methane (CBM), coal mine methane (CMM) or abandoned mine methane (ABM).

- Annual return in the Form prescribed by Chief Inspector shall be submitted at the end of every year to the Chief Inspector and the Regional Inspector.

228.

Manner of extraction of methane.

(1)

Owner, agent and manager of every mine from where CBM is extracted, shall prepare a standard manner of extraction of methane from the target seam or the coal measure strata and submit the same to the Chief Inspector for approval.

(2)

If the methane is proposed to be extracted from an abandoned mine or part thereof, or from a discontinued working of a mine or part thereof or from an existing working of a mine or part thereof, the owner, agent and manager of such mine shall also prepare a standard manner of extraction of methane from such place and submit the same to the chief

Inspector for approval.

(3)

No extraction of methane from coal seam and the coal measure strata or from closed or abandoned or discontinued working of a coal mine or from existing working or part thereof a mine shall be done unless a permission in writing and subject to conditions as he may specify therein is obtained from the Chief Inspector.

229.

Drill machine.

(1)

No drill machine shall be used for the purpose of Coal Bed Methane (CBM), Coal Mine Methane (CMM) or Abandoned Mine Methane (ABM) work either for exploration or extraction unless it is of a type approved by Chief Inspector.

(2)

The drill machine and each of its accessories shall as far as practicable, be of non-inflammable material and any inflammable material, if used, shall be shrouded with substantial metallic covering to render it non inflammable.

(3)

Drill machine shall be provided with-

(a)

an efficient head light capable of showing any obstruction in the working place ahead;

(b)

an efficient portable fire extinguisher so placed as to be within easy reach of operator and also with an automatic type of fire detection and suppression system; and

(c)

a seat for operator and a canopy over head shall be provided to protect the operator from falling objects.

(4)

No drill machine shall be operated otherwise than by a competent person appointed in writing by the manager to be the operator of the drill machine:

Provided that for repairs or tests, the drill machine can be operated by a competent person, other than operator, authorized in writing by the manager.

(5)

Drill machine shall not be used at any work place where there is after allowing for swing of the machine, clear space not less than the following:-

(a)

below the roof or its support 0.3 metr; and

(b)

on the sides 0.6 meter.

(6)

While drill machine is in operation. no person other than authorised in writing by the manger, shall be allowed to stay on the machine or in the vicinity thereof.

(7)

Every work place in which drill machine is used shall be placed under the charge of a competent person or persons who shall once at least in every shift, examine every such work place with particular regards to -

(i)

clearance and free from any obstruction;

(ii)

the state of roof and sides;

(iii)

ventilation;

(iv)

general safety,

and the competent person making such examination shall record the results thereof in the bound paged book kept for the purpose duly signed by the competent person and countersigned by the manger.

230.

Drilling operation.

(1)

The drill machine shall not be used at a place -

(i)

where ventilation is such that the velocity of air current is less than 60 meters per minute;

(ii)

where methane concentration in the general body of the air exceeds 0.5%;

(iii)

which is in the return airway of the ventilation,

unless provided with heavy type of stone dust barriers of a design as specified by the Chief Inspector.

(2)

The power unit and associated switch gears shall always be placed in the upstream side of the air-current (intake side) and of the boreholes being drilled or the boreholes from where gases are being extracted.

(3)

The power unit of the drill shall be equipped with an automatic type of methane monitoring and indicating system interlocked with the electric power switch so that all electrical components on the power unit of the drill machine gets automatically switched off in case the concentration of methane reaches 1.0%.

(4)

Permissibility checks and methane monitor calibration shall be conducted and recorded weekly.

(5)

Borehole survey shall be conducted at regular interval to maintain desired borehole trajectory.

(6)

Borehole projections and the actual co-ordinates should be plotted on the mine plan and section regularly to guide the drill operator in time.

(7)

In order to maintain the desired borehole trajectory, established procedures and techniques shall be adopted.

(8)

All instruments, apparatuses, tools and tackles used to conduct survey of the borehole and its deviations (trajectory) shall be of a type approved by the Chief Inspector.

(9)

Details regarding the trajectory of each borehole shall be maintained separately in a bound paged book kept for the purpose and the details about the boreholes with logging thereof as well as observations of the drill operators shall be recorded therein, duly signed and dated by competent person appointed for the purpose in each shift, and counter signed and dated by the drilling engineer, assistant manager and the manager.

(10)

In case any defect is noticed either during examination or while in use, the drill shall be stopped and shall not be put in to use before it is rectified.

(11)

Transportation of the drill machine in belowground from one working place to another or from surface to underground or underground to surface shall be done under the constant personal supervision of a competent person authorised in writing for the purpose by the manager.

(12)

On both sides of the drill machine two hydraulic jack shall be provided which shall be jammed against the roof and floor of gallery to provide the rigidity to the machine during drilling operations or grouting.

(13)

At the end or every shift, the drill operator shall write his report which may be called Driller's Report in which the operator shall enter all his observations and the work that has been performed by him during the shift and any period of obstruction or stoppage of drilling shall also be recorded with clearly specifying the reasons thereof.

231.

Production drilling and extraction of methane from boreholes.

(1)

While drilling for exploration or extraction of methane through boreholes, all precautions and arrangements shall be made to ensure that no frictional or open spark occurs inside the borehole-containing methane gas or likely to contain methane which may cause explosion or blow out in the borehole.

(2)

At the start of each degasification hole, necessary arrangements shall be made to control the release of the methane through the degasification hole.

(3)

Arrangement shall always be made and kept functional to ensure that there is no leakage of gas from the borehole.

(4)

Automatic gas and leakage detectors fitted With audio-visual alarm shall be provided at the collar of the borehole to ensure that the borehole is kept leakage proof and no inflammable gas or any other noxious gas leaks out of the borehole into the atmosphere.

(5)

In addition to the automatic detector mentioned in sub-regulation (4), hand held portable detector shall also be used to detect leakage and inflammable or noxious gases coming out of the borehole.

(6)

A competent person shall check for the leakage of inflammable or noxious gases using hand held gas detector at regular interval during the shift and a record thereof shall be maintained in a bound paged book kept for the purpose.

(7)

Gas and leakage detection book required to be kept under this regulation, shall be checked by the engineer in-charge of the borehole and shall be signed and dated by the assistant manager in each shift which shall be countersigned and dated by the manager.

232.

Separate ventilation plan for CBM, AMM or CMM.

(1)

Separate ventilation plan shall be maintained showing the position of each exploration and production borehole and gas transportation pipe lines.

(2)

Ventilation planning of the mine shall be done in consultation with the scientific body of repute, and quantity as well as quality of air reaching in each split and gallery through which transport gas transportation pipe line passes, shall be fixed.

(3)

Air measurement stations shall be fixed at each split through which gas transportation pipe line passes in belowground working and air measurement at all such stations shall be taken in each shift and the record thereof maintained in the bound paged book kept for the purpose, signed by the ventilation officer, official in-charge of the CBM or AMM or CMM operation and counter signed and dated by the manager of the mine.

(4)

Whenever the normal ventilation of the mine is disturbed, all CBM or AMM or CMM activities shall immediately be stopped and CBM recovery work shall not be resumed till the normal ventilation of the mine is restored.

(5)

A coal barrier of not less than 150 metre in thickness shall be maintained in the same seam from old boreholes and present working of the mine from where extraction of methane is in progress.

(6)

The ventilation plan and section required to be kept maintained under sub-regulation (1) shall be kept updated in every shift and shall be signed by the ventilation officer and countersigned and dated by the assistant manager in-charge of the CBM or AMM or CMM project or work.

233.

Water and gas separation.

- Water separation and removal equipment shall be installed at each degasification hole, at lower points in the main gas pipe line.

234.

Welding, cutting or fusion, etc.

(1)

No welding or cutting operations shall be done in AMM or CMM district in case main mechanical ventilator or auxiliary fan is not working.

(2)

A written work permit shall be obtained by a competent person from the ventilation officer to carry out any cutting or welding operation in the AMM, CMM or CBM district or area which shall be placed under the direct supervision of an assistant manager in-charge of the district.

(3)

The apparatus or machine to be used for welding or cutting shall only be used in fresh intake airway and at least 270 metres out-by of any working place or any goaf unless it is sealed off by means of explosion proof stopping.

(4)

The quantity of air passing at the place where it is intended to use the apparatus or machine for welding or cutting shall not be less than 284 cubic metre per minute.

(5)

The manager shall exercise overall control on the use of such apparatus or machine to be used for welding or cutting.

(6)

Assistant manager duly authorised in writing by the manager shall be placed in-charge of the entire work, including the operation of carrying the apparatus or machine to be used for welding or cutting from surface to belowground and bringing the same back to surface when no longer required belowground.

(7)

A person holding at least an overman's certificate and duly authorised in writing by the manager shall be present throughout the entire work and be responsible for general safety including checking and recording the percentage of inflammable gas.

(8)

During every welding or cutting operation, water hose with running water shall be kept ready at the site and at the insets below.

(9)

The presence of inflammable gas shall be constantly monitored by an automatic on line gas detection, display system provided with audio visual alarm before commencement and during the welding or cutting operations and at all places within 30 metres distance of the site of cutting or welding.

(10)

A flame safety lamp shall be kept burning at the site of welding or cutting.

(11)

The apparatus or machine to be used for welding or cutting shall not be used if 0.2 percent or more inflammable gas is detected at any time and intimation of such occurrence shall be given to the Chief Inspector and Regional Inspector in writing.

(12)

Sparks or particles from welding or cutting shall be quenched immediately with water.

(13)

The place where the apparatus or machine for welding or cutting is used and its vicinity shall be kept clear of grease, oil etc. and no inflammable material shall be kept at or within metres of such place.

(14)

The place where welding or cutting and the area lying within 90 meters thereof shall be adequately cleaned of loose coal and coal dust and treated with water or incombustible dust before commencement of use of apparatus or machine for welding or cutting and after completion of work.

(15)

All combustible material at and in the vicinity of the place shall be thoroughly kept wet with water or completely covered with sand or stone dust.

(16)

Where cutting or welding takes place in close proximity to any electrical apparatus, the latter shall be kept securely

screened.

(17)

Adequate number of suitable fire extinguishers in good condition Shall be provided and kept in readiness for immediate use therein.

(18)

As soon as cutting or welding operations are completed, the official authorised by the manager for supervision, shall examine the place and satisfy himself that it is safe and further inspection shall be made from time to time during the next four hours and if necessary thereafter until the assigned officials have satisfied themselves that place is safe and has so reported to the manager.

(19)

During the inspections, all fire precautions including those mentioned in the foregoing provisions shall be kept in commission.

(20)

A written record of every occasion when the apparatus or machine for welding or cutting was used belowground, the circumstances under which it was used, the duration for which it was used, the time the apparatus was taken belowground and removed to the surface, the result of tests for gas with methanometer and signs of heating or fire detected if any, together with the name and designation of persons who supervised the operation shall be maintained in a bound paged book kept for the purpose and the record shall be signed and dated by the officials referred to above and countersigned by the manager.

(21)

Only the minimum number of persons shall work belowground when the apparatus or machine for welding or cutting is in use.

(22)

Air sample shall also be collected and analysed by chromatograph where practicable, at frequent intervals at the place where the apparatus or machine for welding or cutting is in operation.

(23)

A copy of the entries made in the book referred under this regulation shall be sent to the Regional Inspector within 24 hours of making the entry.

235.

Methane gas transportation belowground.

(1)

Valves, pipes, bends and other fittings of the gas pipe lines shall be of BIS standard.

(2)

The compressed air distribution system shall be provided with suitable isolating valve at every branching off point.

(3)

Main air line shall be provided with suitable isolating valve in the underground so located that air supply to underground can be effectively and quickly stopped if required.

(4)

Suitable water tap or drain cock shall be provided before the isolating valve and the accumulated water in the tap shall be drained regularly before opening the isolating valve.

(5)

Suitable pressure gauge shall be provided at the outlet side of the isolating valves on the main air range on the surface as well underground to ensure proper functioning of the valves.

(6)

All the pipe conveying pressurised air shall be of standard make, type, robust and tested for quality and of antistatic type and the fitting of the pipes shall be properly secured to prevent accidental disconnection.

(7)

Underground pipe lines carrying methane shall be installed on the floor or hung from the roof in the return air way which shall be plotted on the mine plan.

(8)

The pipeline shall not be installed on the floor where haulage is operative or conveyor is in operation and be graded to

minimize sagging in order to avoid accumulation of water:

Provided that in case there exists a sagged part of the pipelines, water taps shall be installed at such places to drain out water regularly.

(9)

The underground main transport line shall be grounded using metal bands and grounding wires shall be connected to the general earthing system of the mine.

(10)

Testing of earthing system of the gas transportation pipe lines at a interval of not more than one month shall be done by a competent person authorised for the purpose and record of the same shall be kept in a bound paged book duly signed by the competent person and counter signed by the official in - charge of CBM or CMM or AMM project, colliery engineer and manager.

(11)

Suitable arrangements shall be made to dissipate static electric charge from the full length of gas pipeline.

(12)

The places in the mine where gas pipelines pass or cross through the intake airway, the pipeline should be installed within a larger pipe and routed to the return and suitable warning notice shall be displayed at such places.

(13)

The gas pipelines shall be accompanied by tracer tubing safety system.

(14)

Suitable arrangement shall be incorporated in the pipe lines to protect against the leakage or rapture of gas pipelines in underground.

236.

Sectionalisation of methane pipe lines.

(1)

The main gas pipelines shall be sectionalised so that in the event of rapture in the pipe line the methane gas in the general body of the air gets diluted within the permissible limit.

(2)

Sectionalisation shall be accomplished with automatic control valves which shall be of spring loaded type and pneumatic valves that fail close.

(3)

The tracer tubing shall be connected to pneumatic valve on each degasification hole. Which shall be activated (fail close) whenever pressure is lost in tracer tubing and also when the automatic control valves on the pipeline are actuated.

237.

Methane and other gas monitoring system (in case of positive pressure in the reservoir).

(1)

A proper automatic on line or continuous methane monitoring system fitted with audio visual alarm shall be installed along the gas pipelines.

(2)

Methane monitors will be placed at every 500 meters interval along the main pipeline in the return air way or at closer interval if so required by the Regional Inspector.

(3)

Methane monitors shall be inter-faced to electrically actuated valves attached to the tracer tubing.

(4)

Methane monitoring system shall be provided with uninterrupted power supply arrangement.

(5)

All air that cross or passes the pipe-line shall be monitored by methane monitors.

(6)

Calibration of each methane monitor and the system so installed shall be done quarterly and record of the same shall be maintained in a bound paged book kept for the purpose duly signed by the competent person and counter signed by official in-charge of CBM or AMM or CMM and the manager.

(7)

All the gas monitors and the automatic gas detecting system shall be of a type approved in writing by the Chief inspector and the repairs and calibrations of all such monitors and system shall be undertaken only at an approved laboratory.

238.

Communication.

- Proper means of efficient and effective telephonic communication system shall be installed and maintained in between surface and belowground working and at the strategic points all along gas transportation pipe lines in belowground ground as well as on surface, which shall be of type approved by the Chief Inspector.

239.

Inspection and examination of machinery and equipment.

(1)

All the installations, pipelines and safety system installed therein shall be examined daily by competent person authorised for the purpose and the results of all such examinations shall be recorded in bound paged book kept for the purpose duly signed by the competent person and countersigned by colliery engineer and manager.

(2)

Suitable log book shall be maintained wherein shall be recorded the repair or maintenance job in the pipeline and other installations and the record of the same shall be kept in a bound paged book kept for the purpose duly signed by the competent person and countersigned by colliery engineer and manager.

240.

Duties and responsibilities.

- The owner, agent, manager, colliery engineer and the person in-charge of the CBM or AMM or CMM recovery operations shall be responsible for securing effective compliance with the provisions relating to extraction of methane from coal seam, working or abandoned coal mine.

241.

General provisions.

(1)

Owner, agent and manager in consultation with the expert agencies shall frame suitable code of safe practices and standard operating procedures for development extraction, storage, transport and all other related operations, required for extraction of coal bed methane or abandoned mine methane or coal mine methane as the case may.

(2)

The code of safe practices and standard operating procedures shall be circulated to all the concerned persons and officials who shall ensure their compliance.

(3)

A copy of the code of safe practices and standard operating procedures so framed shall be sent to the Regional Inspector who may examine and modify the same if so require.

(4)

The work persons, staff and officials involved in coal bed methane or coal mine methane or abandoned mine methane project and activities shall be adequately trained.

(5)

Work persons, staff and officials involved in the coal bed methane or coal mine methane or abandoned mine methane project and activities shall not be changed normally unless replaced with equally trained and competent persons, staff or officials.

Chapter-XVII Miscellaneous

242.

Fences.

(1)

Every tank or reservoir or other dangerous place in or about a mine which has been formed as a result of, or is used in connection with, mining operation, shall be kept securely fenced.

(2)

Every fence erected on the surface shall, once at least in every seven days, be examined by a competent person and a report of every such inspection shall be recorded in a bound paged book kept for the purpose, duly signed and dated by

the person who made the examination.

(3)

Any fence, gate or barricade may be temporarily removed for the purpose of repairs or other operations, if adequate precautions are taken.

(4)

If any doubt arises as to whether any fence, guard, barrier or gate provided under this regulation is adequate, proper or secure, or as to whether the precautions taken under sub-regulation (3) are adequate, it shall be referred to the Chief Inspector for decision.

243.

Notices.

- Where at any place smoking or unauthorised entry is prohibited, notices to that effect shall be posted at conspicuous places at every entrance to the place.

244.

General safety.

- No person shall negligently or willfully do anything likely to endanger life or limb in the mine, or negligently or willfully omit to do anything necessary for the safety of the mine or the persons employed therein.

245.

Use, supply, maintenance of Protective footwear.

(1)

No person shall go into, or work in, or be allowed to go into, or work in, a mine, unless he wears a protective footwear of such type as may be approved by the Chief Inspector by a general or special order in writing.

(2)

The protective footwear referred to in sub-regulation (1) shall be supplied free of charge, at intervals not exceeding six months, by the owner, agent or manager of a mine who shall at all times maintain a sufficient stock of protective footwear in order to ensure immediate supply as and when need for the same arises.

(3)

Where a footwear is provided otherwise than as specified in this regulation, the supply shall be made on payment of full cost.

(4)

The owner, agent or manager of a mine shall provide at suitable places in the mine dubbing and revolving brushes or make other suitable alternative arrangements for the cleaning of protective footwear by the persons using them:

Provided that it shall however be the responsibility of the person supplied with the protective footwear to arrange the repair of the same at his own cost.

246.

Use and supply of helmet.

(1)

No person shall go into, or work in, or be allowed to go into or work in, a mine, other than the precincts of a mine occupied by an office building, canteen, creche, rest shelter, first aid room or any other building of a similar type, unless he wears a helmet of such type as may be approved by the Chief Inspector by a general or special order in writing:

Provided that where the Chief Inspector is of the opinion that due to special circumstances it is not necessary or reasonably practicable for any person or class of persons going into, or working in, a mine to wear a helmet, he may, by a general or special order in writing and subject to such conditions specify therein, exempt such person or class of persons, from the operation of the provisions of this sub-regulation.

(2)

The helmet referred to in sub-regulation (1) shall be supplied free of charge, at intervals not exceeding three years or such other interval as the Chief Inspector may specify by a general or special order in writing by the owner, agent or manager of a mine, who shall at all times maintain a sufficient stock thereof in order to ensure immediate supply as and when need for the same arises :

Provided that when a helmet is accidentally damaged during legitimate use, the owner, agent or manager shall immediately replace the damaged helmet free of cost.

(3)

Where a helmet is provided otherwise than as specified in this regulation, the supply shall be made on payment of full cost.

247.

Supply of other personal protective equipment.

(1)

Where it appears to the Regional Inspector or the Chief Inspector that any person or class of persons employed in a mine is exposed to undue hazard by reason of the nature of his employment, he may, by a general or special order in writing, require the owner, agent or manager of the mine to supply to such person or class of persons, free of charge, gloves, goggles, shin guards, respirator or such other protective equipment as may be specified in the order.

(2)

The protective equipment provided under sub-regulation (1) shall be replaced free of charge by the owner, agent or manager whenever it is rendered unserviceable by legitimate use and in any other event, the replacement shall be made on payment of full cost.

(3)

If any dispute arises as to the life of any protective equipment, it shall be referred to the Chief Inspector for decision.

248.

Use, supply and maintenance of self-rescuer.

(1)

No person shall go into, work or be permitted to go into or work belowground in any mine unless he is provided with and carries with him a self-rescuer of such type as may be approved by the Chief Inspector by a general or special order in writing.

(2)

If such a self-rescuer is accidentally damaged during use or goes out of order or becomes unserviceable on having gained weight in excess of specified limits or having exceeded its specified life or has been used, the owner, agent or manager shall immediately replace such self-rescuer.

(3)

The owner, agent or manager of every mine where self-rescuers are to be used, shall-

(a)

at all times keep sufficient stock of self-rescuers so that they are readily available whenever needed;

(b)

provide, at the mine, adequate arrangements for cleaning, maintenance and inspection of self-rescuers;

(c)

ensure that every person who may be required to use self-rescuer under sub-regulation (1) undergoes a course of training in the use of self rescuer, as may be specified by the Chief Inspector by a general or special order in writing.

249.

Obligation of persons provided with personal protective equipment.

- Whenever any person is supplied by the owner, agent or manager of a mine with any personal protective equipment, he shall use the same While doing the work for which he is supplied with such protective equipment.

250.

Information about sickness.

- Every official or competent person shall, in case of sickness or of absence, give early and sufficient notice thereof to his superior official or the assistant manager or the manager, as the case may be, so that a substitute may be arranged.

251.

Manpower distribution plan.

- During the first week of every month, a survey shall be made of the number of persons normally employed in every district and other places belowground in the mine and a sketch plan showing the results of such a manpower survey, signed and dated by the manager shall be kept in the office of the mine and a copy thereof shall be kept with the attendance clerk.

252.

Sirdan and overmen.

(1)

No person shall be appointed as a competent person under regulations 36, 78, 102, 140(13) and (14), 141(4)(a), 142(6), 145(7) and (8), 153(9), 162(11), 164(3), 169(2), 171, 173 and 198, unless he is the holder of either an overman's or a manager's certificate.

(2)

No person shall be appointed as a competent person under regulations 133, 134, 138(6), 139(7), 150, 169(1) and 170 and unless he is the holder of either a manager's or overman's certificate or a sirdar's certificate together with a gas testing certificate:

Provided that so much of this regulation as requires a person holding a sirdar's certificate to hold gas testing certificate also shall not apply to, persons employed aboveground, or in opencast working, or competent persons under regulation 138(6) or 152(9).

(3)

Notwithstanding anything contained, in sub-regulation (2), the Chief Inspector may, where special conditions exist, permit or require appointment of any person, not necessarily holding either a manager's or an overman's or a sirdar's certificate as a competent person under regulation 133, if such person possess otherwise a suitable qualification and experience for effective supervision of the working places;

253.

Officials to be literate.

- No person shall be appointed as an official of a mine unless he is literate and is conversant with the language of the district in which the mine is situated or with the language understood by a majority of the persons employed in the mine:

Provided that so much of this regulation as requires a person to be conversant with the language of the district or of the majority of persons, shall not apply to managers, assistant managers, engineers and surveyors.

254.

Writing of reports.

(1)

If any person required to make any report is unable to write, the competent person so authorised by the manager shall write the report on his behalf and in his presence, and he shall attach his thumb mark to it or sign on it after the report is read over to him by the competent person.

(2)

The competent person writing the report shall certify that it has been read over to the person for whom it was written, and shall sign the certificate and date his signature.

255.

Payment of fees.

- Any fees payable under these regulations shall be paid by means of a crossed Indian Postal Order or Bank Draft and any other means as specified from time to time by the Chief Inspector.

256.

Place of accident not to be disturbed.

(1)

Whenever an accident occurs in or about a mine causing loss of life or serious bodily injury to any person, the place of accident shall not be disturbed or altered before the arrival or without the consent of the Chief Inspector or the Inspector to whom notice of the accident is required to be given under sub-section (1) of section 23 of the Act unless such disturbance or alteration is necessary to prevent any further accident, to remove bodies of the deceased, or to rescue any person from danger, or unless discontinuance of work at the place of accident would seriously impede the working of the mine:

Provided that where the Chief Inspector or the said Inspector fail to inspect the place of accident within seventy-two hours of the time of the accident, work may be resumed at the place of accident.

(2)

Before the place of accident involving a fatal or serious accident is disturbed or altered due to any reason whatsoever, a sketch of the site illustrating the accident and all relevant details shall be prepared in duplicate and such sketch shall be duly signed by the manager or assistant manager, safety officers, surveyor and the workmen's inspector or, where there is no workmen's inspector by a work person present at the place of accident, supported by the photographs of the place of accident:

Provided that, if the place is disturbed or altered to prevent further accident or rescue persons from danger before the sketch could be prepared, the same shall be prepared immediately thereafter giving all relevant details as existed before the place was disturbed or altered.

(3)

One of the authenticated sketches shall be delivered or sent to the concerned Inspector.

257.

Emergency response and evacuation plan.

(1)

The owner, agent and manager of every mine shall have a comprehensive programme in place to respond to any injury, illness or mine emergency that may occur at each mine including foreseeable industrial and natural disasters which shall include immediate first-aid treatment, medical treatment, transportation and evacuation of injured persons procedures to respond to emergencies that arise at the mine and arrangements for the rescue of persons incapacitated or trapped in coal mines.

(2)

The plan referred to in sub-regulation (1) shall cover mine evacuations and include-

(a)

establishment of individual responsibilities for administering actions identified to implement an emergency response;

(b)

establishment of emergency communication systems, procedures and individual responsibilities for carrying out emergency communications;

(c)

a system in place to provide immediate notification to all persons affected by the emergency, including alarms in place which shall be capable of being seen and heard by everyone affected.

(d)

a procedure to allow for the safe, orderly immediate withdrawal of persons from the mine or area of danger, including training on emergency escape routes and procedures;

(e)

procedures in place to be followed by workers who remain to perform critical operations before they evacuate, which include-

(i)

the selection of only those personnel who have received special training to respond to critical operations and mine emergencies, including mine fires and explosions;

(ii)

having procedures in place to ensure the locations that are safe for persons who would not be immediately withdrawn; and

(iii)

having the personnel equipped with the necessary gas-detection, equipment and other equipment or tools necessary to respond to the critical operation at hand;

(f)

providing persons with particular risks with the equipment necessary for escape, such as self-contained self-breathing devices etc.;

(g)

a response team that is trained and equipped and immediately available to respond to fires or other hazards that create mine emergencies;

(h)

procedures to account for all workers after the emergency evacuation is complete; and

(i)

providing relevant information and training all personnel, at all levels, including regular exercises in emergency prevention, preparedness and response procedures and periodic emergency drills.

(2)

The owner, agent and manager shall submit a copy of the emergency plan prepared by him to the regional inspector

who may, by an order in writing approve such action plan, either in the Form submitted to him or with such additions and alterations as he may think fit, and the action plan so approved shall be enforced at the mine.

(3)

On receiving information of any emergency, the owner, agent and manager and in his absence the principal official present at the surface shall immediately put emergency action plan in operation.

258.

Taking samples from mine.

- Where for official purposes, an Inspector considers it necessary to take samples of any coal, rope or other material, the owner, agent or manager shall make over to him such samples in such quantities as he may require.

259.

Chief Inspector to exercise powers of the Regional Inspector.

- Any power granted under these regulations to the Regional Inspector may be exercised by the Chief Inspector or any other Inspector authorised in writing in his behalf by the Chief Inspector.

260.

Appeal to the Chief Inspector.

(1)

An appeal shall lie against an order made by the Regional Inspector under any of these regulations, to the Chief Inspector who may confirm, modify or cancel the order.

(2)

Every appeal under sub-regulation (1) shall be preferred within fifteen days of the receipt of the order by the appellant.

261.

Appeal to Committee.

(1)

An appeal against any order made by the Chief Inspector under any of these regulations or against any order passed under regulation 260 shall lie, within twenty days of the receipt of the order by the appellant, to the Committee constituted under section 12 of the Act.

(2)

Every order of the Chief Inspector, against which an appeal is preferred under sub-regulation (1) shall be complied with, pending the receipt at the mine of the decision of the Committee:

Provided that the Committee may, on an application by the appellant, suspend the operation of the order appealed against, pending the disposal of the appeal.

262.

Repeal and savings.

- The Indian Coal Mines Regulations, 1957 are hereby replaced:

Provided that all acts done or orders issued or certificates, authorisation or permits granted or renewed under the said regulations shall be deemed to have been done, issued, granted or renewed under the corresponding provisions of these regulations.

First Schedule

Form I

(See Regulation 3, 6, 7, 8)

(Notice of Opening, Closing or Change of name)

From

.....

.....

To

1. The Chief Inspector of Mines, DGMS, Dhanbad - 826001

2. Regional Inspector of Mines,

3.

4.

Sir,

I have to furnish the following particulars in respect of _____ at (mine of

* _____ owner)

1. In case of change of name of mine: Old name of mine

**Date of Change

2. (a) Situation of the mine District

Village Sub-Division State

Police Station (Taluq)

(b) In case of a new mine, particulars of situation of mine;

Post Office

Telegraph Office

Railway Station

Rest House

(Give distances therefore) Means of traveling

3. (a) Name and postal address of;

(i) Owner

(ii) Managing agent, if any

(iii) Agent, if any

(iv) Manager

(b) In case of change, date of change-

Present

Previous

4. (a) Name and qualification etc. of Manager/Under

Manager/Assistant Manager [Safety Officer /Ventilation Officer

/Engineer/] Surveyor whose appointment is terminated /who is

appointed. ****.

(b) Date of appointment/termination of appointment

5. Date on which it is intended to

open/reopen/abandon/discontinue the mine

6. Actual date of opening/re-opening/abandonment

/discontinuance on the mine.

Yours faithfully

Signature:

Designation: Owner/ Agent / Manager

Date:

Instructions

*Mention the matter to which the notice refers.

**Need not be filled in if the notice relates to item 4.

***To be filled in only when the notice refers to change, and only against the item in respect of which notice is given.

****Delete whatever is not applicable.

First Schedule

Form II

(See Regulation 4)

Quarterly Return for the ---Quarter (--- to ----) of

200

1.

Name of Mine

Postal address of Mine

2.

Situation of Mine

Place State

District

3.

Name of owner

Postal address of owner

4.

Name of managing agent, if any

Postal address of managing agent

5.

Name of agent, if any

Postal address of agent

6.

Name of manager

Postal address of manager

7.

Tables A to D duly filled in, are attached.

Certified that the information given above and in Tables A to D below is correct to the best of my knowledge.

Signature:

Designation: Owner / Agent / Manager

Date:

Table-A: Raising, Despatched and Stocks (in tones)

1. Name of Colliery siding or loading point

2. Grade of coal

Size of Coal

Stock at the beginning of quarter

Coal Rsize

Colliery consumption(boilers,

Domestic etc.

Coal Used in making coke in colliery

Coke produced

Coal Despatched

Stock at the end of quarter

Working below ground

Open cast working

Development district

Depillaring district

By rail

By road

By other means

1

2

3

4

5

6

7

8

9

10

11

12

Stem

Slack

Rubble

Total, Coal

Hard Coke

Soft Coke

Instructions

1. A separate return should be submitted in respect of each grade of coal produced.
2. Figures relating to dispatches of coal should be given separately in respect of each loading point from which coal was dispatched.
3. All figures should be rounded up to the nearest tonne.
4. Columns 3, 4 and 5 should include all coal brought to the surface, irrespective of the use to which it may be put.
5. Column 6 should include all coal consumed at or about the colliery in connection with colliery operations (including domestic use and coal taken home by the workers.)
6. Column 7 should include all coal used at colliery for the purpose of making coke irrespective of the method used.
7. Adjustment in stocks made, if any, should be clearly indicated by means of footnotes.

Table - B: Machinery

Coal-cutting machines

Mechanical Loaders

Conveyors

Number in use

Type

Square meters cut

Coal cut (tonne)

Number in use

Type

Coal loaded (tonnes)

Type

Length (meters)

Coal conveyed (tones)

1. Development

District (District wise)

2. Depillaring District (District wise)

Table - C: Number of Man-Days etc.

Maximum number of persons employed on anyone day during the month and Quarter.

(I)

In workings belowground on (a) :

(II)

In all In the mine on (a) :

Aggregate number of

Aggregate number of man-day lost on account of absence

Men

Women

Sickness-Accident Leave Other causes Total

Belowground

(I) Miners & loaders

(II) Others

(III) Sirdar

(IV) Overman

(V) Asst. manager

(VI) Engineer

(VII) Surveyor

Opencast Working

(I) Miners & loaders
(II) Others
(III) Sirdar
(IV) Overman
(V) Asst. manager
(VI) Engineer
(VII) Surveyor
Aboveground
Contractual Workers
Total

If there is any marked Increase or decrease in attendance of absence, please account for it.

Instructions

- a. Give day of the week and the date and month.
- b. The Information should cover all person "employed" In the mine as defined in clause (h) of section 2 of the Mines Act, 1952 Including clerical and subordinate supervisory staff.
- c. Total number of man-days worked should be obtained by adding the daily attendance for the whole month.
- d. Total number of man-days lost by absence should be obtained by adding the dally absences for the whole month.
- e. Absences should Include all cases in which a person is "scheduled to work" or is expected to turn up for work but does not. All permanent employees are to be treated as "scheduled" to work. So far as temporary or casual employees are concerned, a person Who attended work during the preceding week should be considered as scheduled to work during the week under consideration unless (I) he has reported his Intention to quit, or (II) his services have been terminated by the management, or (III) he does not turn up for work during the whole week. A person who has not worked during the preceding week, should be considered as "scheduled to work" only from the day on which he joins work during the week under consideration. Absence due to strike, lay off or maternity, lockout should not be Included as absence here.
- f. Persons employed in the removal of overburden should be included amongst 'Others' or among 'Miners or Loaders'.

Table - D: Hours of Work and Earnings

Information should be furnished in respect of one complete working week during the Quarter (a).

1. Attendances, man-hours worked and cash earnings.

Total cash payments for work done during the week(d)

Average dally attendance during the week (b)

Aggregate number of man-hours worked during the week

Basic wages

Dearness allowance

Other cash payments

Total

(c)

(d)

(e)

Rs.

Rs.

Rs.

Rs.

Belowground

(I) Overmen & Sirdars

(II) Miners and Loaders

(III) Others

Opencast Working

a) Overman & Sirdars

b) Miners & Loaders

(c) Other

MenWomen

Aboveground

(I) Clerical & Supervisory staff

(II) others

MenWomen

2. Total estimated value of concessions in kind (s) given during the week:

Rs.....

3. Normal hours of production shifts.

From

To

1stShift

2ndShift

3rdShift

4. If there is any major change in wages or hours of work as compared to the preceding month, please account for the change.

Instructions

(a)

The Information should cover all persons 'employed as in Table-C'. Particulars relating to payments etc. to monthly-paid staff should be included on a prorated basis.

(b)

Average daily attendance should be obtained by dividing the aggregate number of attendance on all the shifts on all days during the week by the number of working days. A day on which the mine did not work, for any cause whatsoever, should not be treated as working day.

(c)

Aggregate number of man-hours worked during week should be obtained by adding, for the whole week, the number of man-hours worked every day. The number of man-hours worked on a day is obtained by summing up the number of hours worked by each person attending work on each of the shifts during the day including overtime worked, if any.

(d)

Total of each payment should include all remuneration payable (and paid) for work during the work before making deductions if any, towards fines, provident fund contributions etc. Employer's contribution to the provident fund or on account of welfare provisions should not be included. Bonuses not payable for every pay-period should also not be included.

(e)

Including over-time payments.

(f)

Persons employed in the removal of overburden should be among "Others" and not among "Miners and Loaders."

(g)

Concessions in kind (such as supply of food-stuff, etc.) free or at subsidized prices should be estimated in terms of the difference between the monetary value of the food-stuff, etc. at cost price and the value realized by sale at concessional price.

(h)

Data should be tabulated month-wise and totaled to compile for the whole year. Soft copy should also be submitted.

First Schedule

Form III

(See regulation 5)

Annual Return for the year ending on the 31st December,.....

Mineral: Coal

Table A. Identification particulars of the Mine

Item no.

Item description

Detail information

(1)

(2)

(3)

1.

Name of Mine

2.

State

3.

District

4.

Name of Zonal office of DGMS

5.

Name of Regional office of DGMS

6.

Name of Sub-regional office of DGMS

7.

Postal Address of mine

8.

Date of opening of mine

9.

Date of closing, if any

10.

Date of re-opening of mine, if any

11.

Name of owner

12.

Postal address of owner

13.

Name of Agent

14.

Postal address of Agent

15.

Name of Manager

16.

Postal address of Manager

17.

Telephone no. of Manager

18.

E-mail ID of Manager

19.

FAX No. of Manager

20.

Nature of power used

21.

Type of mine (please write BG for below ground working, OC for opencast working, BO for both below ground & opencast working)

22.

Please specify which of the tables filled up Summary card, A1, A2, A3. B1. B2. C1, C2, C3, C4, C5, D & weekly wage table

W

23.

Whether all the tables are filled up (write Yes/No)

24.

If 'No' in item 23, please specify name of tables not filled up

25.

Please give reasons if entry is there in item 24

Certified that the tables, as mentioned in item 22 above, in prescribed format are duly filled in & enclosed, no additional sheet is attached and information and/figures given in all the tables are correct to the best of my knowledge.

Signature of Manager with seal

Place:

Dated:

Summary card

Item no.

Item description

Below ground (BG)

Opencast (OC)

Above ground(AG)

Total

(1)

(2)

(3)

(4)

(5)

(6)

1.

Average daily employment

2.

Output (raising) in tonnes

Production related tables

1. Table A1- Details of production

2. Table A2- Details of below ground (BG) output (raising)

3. Table A3- Details of opencast(OC) output (raising)

Employment related tables

1. Table B1- Details of manshift, no. of days worked and average daily employment

2. Table B2- Details of employment on rolls

Machinery related tables

1. Table C1 - Details of below ground(BG) machineries

2. Table C2 - Details of Opencast(OC) machineries

3. Table C3 - Details of above ground (AG) machineries

4. Table C4 - Details of mechanical ventilators and winding engines

5. Table C5 - Details of pumps, coal handing plant and workshop machineries

Explosives

Table D - Details of explosives and detonators used.

Safety Lamps

Table E - Details of Safety Lamps used

Wages

Table W - Wages during the last week of the month of December

Table A1. Details of production

Item no.

Item description

Quantity /Value (as indicated against each item)

- (1)
 - (2)
 - (3)
 - 1.
 - Opening stock (in tonnes)
 - 2.
 - Output (Raising) (in tonnes)
 - 3.
 - Value of output (raising)(in Rs.)
 - 4.
 - Surplus (in tonnes), if observed
 - 5.
 - Receipts (in tonnes), from other colliery
 - 6.
 - Used for coke making (in tonnes)
 - 7.
 - Colliery consumption (in tonnes)
 - 8.
 - Transfer (in tonnes)
 - 9.
 - Shortage (in tonnes)
 - 10
 - Total dispatch (in tonnes)
 - 11.
 - closing stock (in tonnes)
 - 12.
 - Sum of item 1, item 2, item 4 & item 5
 - 13.
 - Sum of item 6 to item 11
- Instructions for Table A1:
- i) Item 12 must be equal to item 13
 - ii) output (item 2 of Table A1) = BG output (item 1 of Table A2) + OC output (item 1 of Table A3)

Table A2. Details of below ground (BG) output (raising)

Item no.

Item description

Output (in tonnes)

- (1)
- (2)
- (3)
- 1.
- Below ground output
- 2.
- Output by Mechanisation
- 2.1
- Fully mechanized
- 2.2
- Semi mechanized
- 2.3
- Solid blasting

2.4

Others

3.

Output by method of work

3.1

Development

3.2

Depillaring

3.3

Others

3.4

Longwall

4.

Average daily employment and output for degree II

& III gassy mines

Item no.

Item description

Average daily employment

Output(in tonnes)

(1)

(2)

(3)

(4)

4.1

Degree II

4.2

Degree III

Instructions for Table A2:

Belowground output (item 1) = sum of item 2.1 to item 2.4

= sum of item 3.1 to item 3.5

item 2.1(Fully mechanized): if coal is cut and loaded by machines, output is to be reported in item 2.1

item 2.2(Semi mechanized): if coal is cut by CCM and blasted with explosives but handloaded output is to be reported in item 2.2

item 2.3(Solid blasting): if coal is obtained by Solid blasting, output is to be reported in item 2.3

item 2.4 (Others): if coal is cut and loaded by manual means only, output is to be reported in item 2.4

If BG output(raising)>0, entry must be there in corresponding items of Table B1 and in Table CI, C4, C5 & D

Table A3. Details of opencast (OC) output (raising)

Item no.

Item description

Output (as indicated against each item)

(1)

(2)

(3)

1.

Opencast output (in tonnes)

2.

Output by Mechanisation

2.1

Fully mechanized (in tonnes)

2.2

Semi mechanized (in tonnes)

2.3

Other (in tonnes)

3.

Over burden (in cubic metre)

Instructions for Table A3:

Opencast output (Item 1) = sum of item 2.1 to item 2.3

item 2.1(Fully mechanized): if coal is obtained by deep hole blasting and heavy earth machinery, output is to be reported in item 2.1

item 2.2(Semi mechanized): if coal is obtained by compressed air/power drills but hand loaded, output is to be reported in item 2.2

item 2.3(Others): if coal is cut and loaded by manual means only, output is to be reported in item 2.3

item 3(Over burden):

If OC output(raising)>0, entry must be there in corresponding items of Table B1 and in Table C2 & D

Table B1. Details of manshift, no. of days worked and average daily employment

Item no.

Item description

Manshift

No. of days worked

(1)

(2)

(3)

(4)

1

Manshifts and no. of days worked details

1.1

Below ground (BG)

1.2

Open cast (OC)

1.3

above ground

1.4

Total manshift

Item no.

Item description

Average daily employment

(1)

(2)

(3)

2.

Average daily employment details

2.1

Direct employment

2.2

Contract employment

2.3

Total employment

Average daily employment category wise

3.

Below ground employment (BG) details

3.1

Overman & Sirdar

- 3.2
- Miners & Loaders
- 3.3
- Other face
- 3.4
- Other
- 3.5
- Total BG employment
- Item no.
- Item description
- Average daily employment
- (1)
- (2)
- (3)
- 4.
- Opencast employment (OC) details
- 4.1
- Overman & Sirdar
- 4.2
- Miners & Loaders
- 4.3
- Other face workers
- 4.4
- Others
- 4.5
- Total OC employment
- 4.6
- OC Male employment
- 4.7
- OC Female
- 5.
- Above ground employment (AG) details
- 5.1
- Clerical & Supervisory
- 5.2
- Others
- 5.3
- Total AG employment
- 6.
- Average daily employment by sex
- 6.1
- Total Male employment
- 6.2
- Total Female Employment

Instructions for Table B1:

Contractual employment to be taken into account in reporting manshift and average daily employment.

Clerical & Supervisory: Senior Supervisory Officials like Agents, Managers, Under Managers, Engineers, Doctors to be excluded.

Please check if Table B1 is non blank, entry must be there in Table B2.

i) Total manshift (col. 3 item 1.4) = sum of col. 3 of item 1.1 to item 1.3

ii) Total employment (Item 2.3) = sum of Direct employment (item 2.1) & Contract employment (item 2.2)

= Total BG employment (item 3.5) + Total OC employment (item 4.5) + Total AG employment (item 5.3)

= sum of Total male employment (item 6.1) & Total female employment (item 6.2)

iii) Total BG employment (item 3.5) = sum of item 3.1 to item 3.4

= BG manshift (col. 3 item 1.1) / No. of days worked in BG working (col. 4 item 1.2)

iv) Total OC employment (item 4.5) = sum of item 4.1 to item 4.4

= OC Male employment (item 4.6) + OC Female employment (Item 4.7)

= OC manshift (col. 3 item 1.2)/ No. of days worked in OC working (col. 4 item 1.2)

v) Total AG employment (item 5.3) = Clerical & Supervisory (item 5.1) + Others (item 5.2)

= AG manshift (col. 3 item 1.3) /No. of days worked in AG (col. 4 item 1.3)

Table B2. Details of employment on rolls

Item no.

Item description

No. of persons

(1)

(2)

(3)

1.1

Below ground (BG) employment on rolls

1.2

Opencast (OC) employment on roll

1.3

Above ground (AG) employment on rolls

1.4

Total employment

Direct employment on rolls

1.51

Male direct employment on rolls

1.52

Female direct employment on rolls

1.53

Total direct employment on rolls

Contract employment on rolls

1.61

Male contract employment on roll

1.62

Female contract employment on rolls

1.63

Total contract employment on rolls

Instructions for Table B2:

i) Total employment on rolls (Item 1.4)- sum of item 1.1 to item 1.3

ii) Total direct employment on rolls (item 1.53) = sum of Male direct employment on rolls (item 1.51) & Female direct employment on rolls (item 1.52)

iii) Total Contract employment on rolls (Item 1.63) = sum of Male contract employment on rolls (item 1.61) & Female contract employment on rolls (Item 1.62)

iv) Total employment on rolls (Item 1.4) = sum of Total direct employment on rolls (item 1.53) & Total Contract employment on rolls (Item 1.63)

v) Total employment on rolls (item 1.4)> Total average daily employment (item 2.3 of Table B1)

Table C1. Details of below ground (BG) machineries

Item no.

Item description

Total number of units in use

Total HP

(1)

(2)

(3)

(4)

1.1

Coal Cutting Machines

1.2

Loaders

1.3

Cutter com Loaders

1.4

Haulages

1.5

Conveyors

1.6

Drills

1.7

Shuttle Cars

1.8

Locomotives

1.9

Others

Instruction for Table C1: If BG output (raising)>0, entry must be there in Table C1

Table C2. Details of Opencast (OC) machineries

Item no.

Item description

Total number of units in use

Total HP

(1)

(2)

(3)

(4)

1.1

Shovels

1.2

Dumpers

1.3

Drills Loaders

1.4

Dozers

1.5

Cranes

1.6

Scrapers

1.7

Draglines Cars

1.8

Loaders

1.9

Graders

2.0

Others

Instructions for Table C2: If OC output (raising)>0, entry must be there in Table C2

Table C3. Details of above ground (AG) machineries

Item no.

Item description

Total number of units in use

Total HP

(1)

(2)

(3)

(4)

1.1

Haulages

1.2

Dumpers

1.3

Conveyors

1.4

Locomotives

1.5

Loaders

1.6

Others

Instructions for Tables C1, C2 & C3: i) All machineries, as classified above, deployed in mines including those of hired from outside agencies to be reported.

ii) Total HP is the sum of HP of number of used of the same item

Total C4. Details of mechanical ventilators and winding engines

Item no.

Item description

Total number of units in use

Total HP

(1)

(2)

(3)

(4)

1.1

Main Ventilators (MVN)

1.2

Booster ventilators (BVN)

1.3

Winding engines electrical

1.4

Winding engines steam

1.5

Name and size of Mechanical Ventilator

Position where installed

Average total quantity of air delivered per minute

Water gauge obtained (in centimetres)

Instructions for Table C4: If BG output(raising)>0, entry must be ther in item 1.1m table C4

Table C5. Details of pumps, coal handling plants and workshop machineries

Item no.

Item description

Total number of units in use

Total HP

(1)

(2)

(3)

(4)

1.1

Below ground pumps (BPP)

1.2

Opencast pumps (OPP)

1.3

Above ground pumps (APP)

1.4

Coal handling plants (CHP)

1

1.5

Workshop machineries

Instructions for Table C5:

(i)

Coal handling plants in a mine to be treated as single unit

(ii)

welding machines to be excluded from working machinery

(iii)

If BG output (raising)>0, entry must be there in item 1.1, Table C5

Instructions for Tables C4 & C5:

i) All machineries, as classified above, deployed in mines including those of hired from outside agencies to be reported.

ii) Total HP is the sum of HP of number of units need of the same item

Table D. Details of explosive and detonators used

Item no.

Item description

Quantity as indicated against each item

(1)

(2)

(3)

1.1

Permitted explosive used (for BG mines) in K.g.

1.2

Non permitted explosive used (for OC mines) in kg.

1.3

Total explosive used in K.g

2

Detonators used in numbers

Instruction: If BG output (raising)>0, entry must be there in item 1.1 & 2, Table D If OC output (raising)>0, entry must be there in item 1.2 & 2 Table D

Table E. Details of Safety Lamps used

Name and type of safety lamps*

Number of safety lamps according to

method of locking

Lead rivet

Magnetic

other

• Mention type, such as flame type electric hand type, electric cap type etc.

Table W. Wages during the last week of the month of December

Item no.

Item description

Average daily attendances during the week

Average number of manhours worked during the week

Total cash payments for work done during the week (in Rs.)

(1)

(2)

(3)

(4)

1.

Below ground (BG)

1.1

Overman & Sirdar

1.2

Miners & Loaders

1.3

Other face workers

1.4

Others

1.5

Total BG

2

Opencast (OC)

2.1

Overman & Sirdar

2.2

Miners & Loaders

2.3

Other face workers

2.4

Others

2.5

Total OC

3.

Above ground (AG)

3.1

Clerical & Supervisory

3.2

other

3.3

Total AG

Instructions for Table W:

Contractual employment to be taken into account in reporting average daily attendances.

Clerical & Supervisory: Senior Supervisory Officials like Agents Managers, under Managers, Engineers, Doctors are excluded.

Form IV-A

(See Regulation 9)

Notice of Accident/Dangerous Occurrence

From:

To:

1. The Chief Inspector of Mines/Director General of Mines Safety, Dhanbad-826001
2. The Regional Inspector of Mines/Director of Mines Safety Region / Dy. Director in Charge of Sub-Region.
3. The District Magistrate / District Collector.
4. The Electrical Inspector of Mines/Director/Dy. Director of Mines Safety (Electrical) (Electrical accidents only)

Sir,

I have to furnish the following particulars of a fatal/serious accident/a dangerous occurrence(*) which occurred at(**) mine. of (owner)

1.0

Mine Code (nnnnnn):

2.1

Date of accident (dd/mm/yy)

2.2 Time of Accident (hhmm)

3.0

Particulars of the mine (if mine code is not known):

3.1 Mineral

:

3.2 Village

:

3.3 Post Office

:

3.4 Pin Code

:

3.5 Police Station

:

3.6 Sub Division/Taluq

:

3.7 District

:

3.8 State

:

3.9 Owner's address

:

4.0

Particulars of the Accident :

4.1

No/name of shift

4.2

Cause Code

:

(nnn)

4.3.1

Place Code (n)

:

4.3.2

Working code (n):

4.4

No. of persons killed

:

4.5

No. of persons seriously injured :

4.6

Location of accident

:

4.6.1

Pit/Incline No.

:

4.6.2

District/Panel/bench/outside working district :

4.6.3

Seam/vein

4.6.4

Level, dip no. :

4.6.5

Distance from the face (in mtrs)

4.6.6

Details not covered above :

(If within district)

5.0

Particulars of person(s) involved:

SI No.

Name of victim

Designation

Desig.code(nn)

If contractor worker(Y/N)

Sex(M/F)

Age (Years)(nn)

Hours at work(hhmm)

Nature of injury and cause of death

Killed :

1.2.

Seriously injured:

1.2.

6.0

Brief description of the cause of accident/occurrence:

7.0

Immediate last accident:[Fatal][Serious]

7.1

Type of accident :

7.2

Date of Immediate last accident : [dd][mm][yy]

Particulars in respect of every person killed or Injured in Form IV-B are enclosed/shall be forwarded within a week(*)

Your faithfully

Date:Place:

Signature:Name in block letters:Designation:

(Owner/agent/manager)

?.....

*delete whichever is not applicable.

**name of mine.

Codes to Be Used in Form IV - A

Cause Code

Code

Long Name

0111

Fall of Roof

0112

Fall of Sides

0199

Other falls of Ground

0221

Over winding of Cages Skips etc.

0299

Other accidents/Dangerous occurrences due to winding

0332

Rope Haulage

0334

Conveyors

0335

Dumpers

0339

Other Wheeled Trackless

0399

other non-winding Machinery

0448

HEMM other than Dumpers

0499

Other machinery

0500

Explosives

0600

Electricity

0775

Spontaneous Heating

0799

Other accidents/dangerous occurrence due to dust/gas/fires

0881

Fall of Persons

0883

Fall of Objects

0889

Others (falls other than ground movement)

0999

Other than above

Place Code

Code

Long Name

1

Below ground

2

Opencast

3

Above ground

Working Code

1

Development

2

Longwall

3

Depillaring/stopping

4

Coal/Ore bench

5

O/B/Bench

6

Waste Dump

9

Others

Site Code

1.

Haul Road

2

Rope haulage Road

3

Other Roadways

4

Miscellaneous

Designation

Code

CODE

LONG NAME

20

Management

30

Subordinate Supervision Officials

41

Loaders

42

Support Personal

43

Dresser

44

Shotfirer

49

Other B/G Face Worker

50

Other B/G worker

59
Trammer
60
Fitter
71
Dumper Operator
72
Truck etc. Operator
79
Other Operator
99
Any other
For office use only
Acc Class
Plc Code
Cos Code
RESP
Operation
Scrutiny
DE
Form IV-B
(see regulation 9)
Particulars of deceased/injured person (s) (to be submitted within 7 days of occurrence)
1.0
General
1.1
Name of Mine :
1.2
Mine Code (nnnnnn) :
1.3
Name of Owner :
1.4
Date of accident (dd/mm/yy) :
1.5
Time of accident (hhmm) :
2.0
Details of the injured/deceased workers
S.No.
Name of victim(s)
Injury code(n)
Body part Code (nn)
Ex-gratia payment made (Rs.)
Compensation paid (Rs.)
Other benefits if any
Total Experience (months)
What was the
Experience of victim in
If vocationally trained in
Initial vocational training (Yes/No)
Date of refresher training
Nature of job (regular/temporary)

Cause of death

Job being done at the time of
accident

Months

(yes/No)

Killed

1.

2.

3.

Seriously injured

1.

2.

3.

3.0

Responsibility for the accident:

Sr. No.

Name(s) of person (s) held
responsible

Desig. Code(nn)

Designation

Rules/Reg. Contravened

Nature of contravention

Action taken

4.0

As a Sequel to this accident :

4.1 Loss of production (in
tonnes)

4.2 Loss of manshifts(indirect loss)

:

:

5.0

Could the accident have been averted, if so, how?

Date:

Place:

Signature :

Name in block letters :

Designation : Owner/Agent/Manager

**see overleaf for codes.

Codes to be used in Form IV-B

Injury Codes(N)

CODE

LONG NAME

1

2

3

4

Amputation

Fracture

Internal injury

Others

Parts of Body Codes (NN)

CODE

LONG NAME

10

Head

12

Eye

31

Back

32

Chest

34

Pelvis

38

Spinal column

40

Upper limb

45

Fingers

49

Other upper limbs

50

Lower limbs

99

Others

Form IV-C

Particulars of injured person returning to duty

(to be given in respect of every person within 15 days of his return to duty)

1.0

General

1.1

Name of Mine :

1.2

Mine code (nnnnnn)

1.3

Name of Owner :

1.4

Date of accident (dd/mm/yy) :

1.5

Time of accident (hhmm) :

S.No.

Name(s) of injured workers(s)

Date of return to duty (dd/mm/yy)

Whether returned to regular or other job

Permanent Disablement

Exgratia payment made Rs.

Statutory Compensation paid (Rs.)

Benefits to dependants or other benefits

Part(s) of body

Loss

Gone out of use

1.

2.

3.

4.

Date:

Place:

Signature:

Name in Block letters:

Designation: Owner / Agent /Manager

First Schedule

Form V

(see Regulation 10)

(Notice of Disease notified under section 25 of the Mines Act)

From.....

.....

To

1. The Chief Inspector of Mines! Director General of Mines Safety, DGMS, Dhanbad-826001.

2. The Regional Inspector of Mines/ Director of mines Safety

3. The Inspector of Mines (Medical)/Dy. Director of Mines Safety(OH) Dhanbad-826001.

4. The District Magistrate/District Collector

5. Coal Mines Welfare Commissioner, Dhanbad.

Sir,

I have to furnish the following particulars with respect to an occupational disease contracted by a person employed in the mine of (owner).

1. Particulars of Mine, etc.

(i)

Situation of Mine:

Village

Pot Office

Police Station

Sub-Division (Talug)

District

State

(ii)

Mineral Worked

(iii)

Name and postal address of owner

2. Particulars of person Affected

(i)

Name (in Block Capitals)

(ii)

Caste or surname

(iii)

Permanent address:

Village

Police Station

Post Office

District

Sub-Division (Talug)

State

(iv)

Sex

(v)

Date of birth (or age)

(vi)

Occupation

How long engaged?

(vii)

Date of commencement of employment

(a) in your mine

(b) in mining

|

3. Particulars of diseases etc.

(i)

Nature of disease from which the person is suffering (state stage)

(ii)

Date of detection of disease

(iii)

Name, registration number and address of Medical

Practitioner suspecting disease:

Signature :

Designation : Owner / Agent / Manager

Date :

First Schedule

[Form VI]

Name of Mine Seam/vein etc. Inspected by on

200--.

Owner Section/Area etc. Accompanied by Shri

Manager

Places Inspected

Mines Act/Coal Mines Regulations/mines Rules

Section/Clause number etc.

Contravention observed

Action taken by Management to remedy the contravention

Date of rectification of the contravention

Remark any

The contraventions mentioned above are not exhaustive. A letter giving the details of other contraventions observed may follow in due course,

Signature of Inspection Officer (IO) Date Designation

Signature of Mine Official Accompanying I.O Date Designation

First Schedule

Form-VII

Manager's Charge Report

Name of Mine.....

Name of Owner

Name of outgoing Manager

Name of incoming Manager

1.0

General information about the mine-

2.0

Details of Workings

2.1

Present Workings

2.2

Past Workings

2.3

Future Workings

3.0

Information about Gases:

3.1

Problems of Gases

4.0

Information about Fires and Spontaneous Heating

4.1

Problems of Fires or Spontaneous Heating

5.0

Information on Strata and Geological disturbances

5.1

Problems of Strata and Dumps

6.0

Sources of Water from Surface

6.1

Precautions against Danger of inundation from Surface Water

7.0

Sources of Water from belowground or adjoining mines

7.1

Precautions against Danger of Inundation from Underground Sources of Water

8.0

Ventilation and Problems if any

9.0

Workings near important Surface Features

10. Other important matters from the point of view of safety required to be attended by the incoming manager

Signature of incoming

Signature of the outgoing

Manager

Manager

Date.....

Date.....

Second Schedule

(See Regulation 67) Conventions for Preparing Plans and Sections

NAME

SYMBOL

REMARK

BOUNDARY OF LEASE HOLD

UNDERGROUND COALBARRIER

SHAFT

ABANDONED SHAFT

INCLINE

ABANDONED INCLINE

PILLARS AND

GALLERIES

DRIFT
QUARTERLY SURVEY LINE
STAPLE SHAFT
ABANDONED STAPIE
SHAFT
FAULT
DYKE OR OTHERINTRUSION
GOAF
SUBSIDENCE
BENCH MARK
SURFACE CONTOUR
UNDERGROUND SPOTLEVEL
WATER DAM
DIRECTION OF AIRCURRENT
BRATTICE
DOORS
BRICK/STONE OR CONCRETEVENTILATION STOPPING
In
RedIn Green
Workings shown by
Dotted lines are not surveyed and their extent is not correctly known
In Burnt Sienna showing gradient in Black
In Red should state distance up and down to all insets
In Red
In Red showing the amount and direction of throw
In Green
In Red
In Burnt Sienna
In Blue
In Red
Intake in Blue ReturnRed
In Red
In Red
In Red
Third Schedule
NAME
SYMBOL
REMARK
FIRE DAM, SEAL OR
STOPPING
EXPLOSION PROOF
STOPPING
AIR CROSSING
EXPLOSION PROOF
AIR CROSSING
REGULATOR
AUXILIARY FAN
TELEPHONE
UNDERGROUND FIRST AIDSTATION
ENGINE HOUSE ORROOM
BORE HOLE

SURVEY LINES &

STATIONS

SECTION OF SEAM

In Red

In Red

In Red

In Red

In Red

Thick cross in Red

Should show serialnumber & diameter

In Red

Form III

Extraction of coal by a system of Board and Pillar system

1. General

(i)

Name of Mine

(ii)

Owner

(iii)

District

(iv)

State

2. Particulars of Seam to be worked

(i)

Name/Number

(ii)

Total thickness (give section)

(iii)

Working thickness

(iv)

Nature of roof upto a thickness of at least 1 metre

(v)

Nature of floor upto a thickness of at least 1 metre

(vi)

Maximum and minimum depth from the Mine Surface

(vii)

State;

(a)

the rate of emission of inflammable gas per tonne of coal raised

(b)

The percentage of inflammable gas in the general body of air

(viii)

Is there any history of fire in the seam

(a)

in the same mine or

(b)

in the adjoining mines? Give details.

(ix)

What is the known or expected incubation period of the seam?

3. Condition of overlying and underlying seam

(i)

Give section of the strata separately.

(ii)

Are the seams free from water? If not, give details regarding position of water level.

(iii)

Are the seams extracted /split/ standing on pillars / virgin?

(iv)

If the seams have been extracted/split state if by caving method or by hydraulic stowing or dry stowing.

(v)

Is there any fire in any overlying or underlying seams /sections or at the surface? If so, give detailed history about the same, and the present condition of the fire.

(vi)

State:

(a)

the rate of emission inflammable gas per tonne of coal raised.

(b)

the percentage of inflammable gas in the general body air.

4. Proposed method of Development

(a)

Explain in detail the proposed layout of workings:

A layout plan also should be submitted in duplicate showing the area proposed to be worked, and all other features (including surface features) required to be shown on an underground plan maintained under regulation 59.

(b)

Type of machinery to be used for coal cutting/coal getting and for the transport of coal from the face to the surface.

5. Proposed method of extraction

(a)

By hydraulic stowing of sand, crushed material and / or pneumatic stowing or by caving method.

(b)

BY longwall retreating or longwall advancing method or

(c)

By any other special method, like working by inclined slices, horizontal Slices and sub-level caving etc.

Note - In each case, illustrate the manner of extraction in detail and with suitable sketches.

(d)

Type of machinery to be used for coal cutting /coal getting and for the transport of coal from the face to the surface.

6. Support

(a)

Proposed method of support during

(i)

Development

(ii)

Depillaring /final extraction

Note - In each case, illustrate with sketches.

(b)

Material to be used for support whether timber or steel. If steel supports are to be used, state the type whether rigid, friction or hydraulic type giving the trade-name, if any, of the type of supports.

7. Precautions against coal dust

(a)

Within 60 metres of working faces.

(b)

In the haulage road and airways.

(c)

In other parts of workings.

(d)

Whether stone dust barriers would be provided? If not give reasons,

8. Precautions against danger from water

(1)

If provisions of regulation 126 are applicable, state precautions that are proposed to be taken against danger from surface water.

(2)

If provisions of regulation 127 are applicable, state precautions that are proposed to be taken against danger from underground water.

9. Ventilation

(1)

Surface fan

(a)

(i)

Type

(ii)

Capacity (state the range) Min., Max.

(iii)

Water-gauge

(b)

Whether the same fan will meet the ventilation requirement of the mine during its different stages of development and depillaring or any other fanes) will be installed. In the latter case give details stating the equivalent orifice of the mine at the different stages of its life.

(2)

Underground fans, if any: Auxiliary or Booster.

(i)

Type

(ii)

capacity

(3)

Explain the proposed system of ventilation and also indicate on the layout plan:

(i)

Ventilating District

(ii)

Quantity of air in cubic metres-cubic feet

(4)

Minimum quantity of air available per person employed in the largest shift or per daily tonne output whichever is greater passing along the last ventilating connection for each ventilating district.

(5)

(i)

What is the anticipated rate of emission of gas per tones of coal.

(ii)

Maximum percentage [actual or planned] of gas in the return of any ventilating district.

10. Any other relevant details. - Certified that the information given above is correct to the best of my knowledge & belief.

Signature:

Designation : Owner / Agent / Manager

Date:

Instructions

1. Separate sheets may be used in case the space against any of the columns is insufficient for the information required.
2. This form should be submitted in duplicate accompanied by a layout plan showing the area proposed to be worked

and all other features (including the surface features) required to be shown on an underground plan maintained under regulation 59.

3. Equivalent Orifice should be calculated in square metres.

Related AI tags, queries and research notes

Translation